EMC404S — Behavioural Emergencies (Section G) and Toxicology (Section H)

Comprehensive study guide for the Emergency Care Practitioner (B.EMC) — South African prehospital practice

Prepared October 2026

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How to use this study guide

This study guide covers every learning outcome from **Outcome 39 to Outcome 65** of the EMC404S study guide, written out under the exact outcome wording, in full sentences, with in-text citations and a single consolidated reference list at the end of the document. Each outcome is written from the point of view of how an Emergency Care Practitioner (ECP) in South Africa would actually assess and manage the patient in the prehospital environment, within the South African legal framework (the Constitution, the Mental Health Care Act 17 of 2002, the National Health Act 61 of 2003, the Children's Act 38 of 2005, the Criminal Law (Sexual Offences and Related Matters) Amendment Act 32 of 2007, the Domestic Violence Act 116 of 1998 as amended, and the HPCSA ethical rules and guidelines).

The documents that you uploaded have been used as primary sources throughout and are cited by author and year, namely Evans, Geduld and Stassen (2018); Tilley et al. (2026); the HPCSA Booklet 7 on withholding and withdrawing treatment (HPCSA, 2023); Saucedo Uribe et al. (2025); Desai, Kulkarni and Verma (n.d.); the Royal College of Emergency Medicine guideline on Acute Behavioural Disturbance (RCEM, 2025); Mesthrie (2025); Hüser et al. (2025); Kaswa (2024); and the three EMC404S tutorial memoranda. Additional trusted sources (legislation, DSM-5-TR, peer-reviewed reviews in journals such as *The Lancet* and *The New England Journal of Medicine*, WHO guidance, and standard emergency medicine and toxicology texts) have been added where your documents do not cover an outcome.

Important note on drug doses and scope of practice

Your scope of practice as an ECP in South Africa is defined by the HPCSA Professional Board for Emergency Care (PBEC). The PBEC's Clinical Practice Guidelines (CPGs) were published in 2018, and the PBEC has since released the **Clinical Decision Support Tools for Emergency Care Providers (CDST), South Africa 2024, version 1.3, October 2025** (HPCSA PBEC, 2025). The CDST was commissioned and funded by the PBEC, developed by the Knowledge Translation Unit of the University of Cape Town, and is based on the 2018 CPGs together with the latest synthesised evidence. In the CDST, black text applies to all registered categories (BAA, AEA, ECA, ECT, ANT, ECP); *green text with stars* applies to AEA, ECA, ECT, ANT and ECP (not all green medicines are within AEA/ECA scope); ^orange text with carets^ applies only to ANT and ECP; and ~purple text with curvy lines~ applies only to ECPs (HPCSA PBEC, 2025, p. 3).

In this guide, every prehospital drug and dose for South African practice is taken from the CDST v1.3 (HPCSA PBEC, 2025) wherever the CDST addresses the condition, and is labelled "CDST". Where the CDST does not address a condition (for example lithium, iron or paracetamol overdose, envenomation or sexual assault PEP), doses come from the other documents you supplied (for example RCEM, 2025; Hüser et al., 2025) or from peer-reviewed and South African national guidance, and are labelled as **hospital-level or reference information** so that you know they are not prehospital ECP scope. If your service's medication protocol differs from the CDST, follow your employer's protocol in practice. In the examination, answer according to the CDST and what your lecturers have taught.

Corrections you should know about

1. Outcome 50.1 in the study guide refers to the "Metal Health Care Act (Act 15 of 2004)". The correct citation is the **Mental Health Care Act 17 of 2002**, which came into operation on 15 December 2004 (Mental Health Care Act 17 of 2002; Evans, Geduld and Stassen, 2018). The Act was subsequently amended by the Mental Health Care Amendment Act 12 of 2014; the section numbers and wording quoted in this guide (ss 9, 32, 33, 34 and 40) are those of the current consolidated Act.
2. The Toxicology Tutorial 1 memo describes sorbitol-containing charcoal as "emetic". Sorbitol is an osmotic **cathartic** (laxative), not an emetic; it is avoided because it causes diarrhoea, fluid loss and electrolyte disturbance, particularly in infants, and because cathartics provide no proven benefit (Hüser et al., 2025). Kaswa (2024) also uses the word "emetic", so note that your lecturers may accept it, but the pharmacologically correct term is cathartic.
3. The Behavioural Emergencies Tutorial 1 memo refers to the "Mental Capacity Act" in the context of restraint. That is the **United Kingdom** Mental Capacity Act 2005 that the RCEM (2025) guideline refers to. **It does not apply in South Africa.** In South Africa the equivalent legal authority for treating a patient without consent comes from section 9(1)(c) of the Mental Health Care Act, section 7(1)(e) of the National Health Act, the common-law defence of necessity, and the HPCSA ethical guidelines on consent.

Five-day study plan

You have five days. The plan below divides the material so that every outcome is covered once in detail and the highest-yield material is revisited on the final day. Each day assumes roughly six to eight hours of focused study, broken into 50-minute blocks with 10-minute breaks.

Day	Material	Active-recall task at the end of the day
Day 1	Outcomes 39–44: definition and causes of behavioural emergencies, psychiatric signs and symptoms, interviewing, MSE and MMSE, principles of care, restraint ethics, violent patients	Without notes, write out the MSE headings, the ten domains of verbal de-escalation, and the organic "red flags" that suggest a medical cause of behaviour. Draw a flow diagram of your scene approach to a violent patient.
Day 2	Outcomes 45–51: suicide risk assessment and interview questions, legal duties in self-harm, abscondment, children and adolescents, older persons, the Mental Health Care Act, consent on behalf of incompetent patients, rights	Write out sections 9, 32, 33, 34 and 40 of the Mental Health Care Act in your own words. Write ten suicide-risk interview questions from memory. Answer the Behavioural Emergencies Tutorial 1 and 2 questions

Day	Material	Active-recall task at the end of the day
	violations	(Appendix A) without looking.
Day 3	Outcome 52 (DSM-5-TR disorders) and Outcomes 53–57 (sexual assault and domestic violence)	Make a one-page table of each DSM category: key features, emergency risks, prehospital management. Write out the elements of rape and sexual assault (sections 3 and 5 of the Sexual Offences Act), the 72-hour PEP rule, evidence preservation steps, and the protective measures of the Domestic Violence Act.
Day 4	Outcomes 58, 59, 61, 62 (routes, epidemiology, emesis, general management, toxidromes, activated charcoal, unknown/mixed overdose) and Outcomes 60.1–60.16	Draw the toxidrome table from memory. List the contraindications to activated charcoal. Write the antidote for each poison in 60.1–60.16.
Day 5	Outcomes 60.17–60.32 and Outcomes 63–65 (plants, mushrooms, traditional medicine, envenomation, scombroid). Final revision of Appendix A (tutorial questions) and the "High-yield points" boxes throughout	Answer every Toxicology Tutorial 1 question (Appendix A) under timed conditions. Re-read every "High-yield points" box in the guide.

Study technique. After reading each outcome, close the guide and write down or say aloud everything you remember (the "blurting" or free-recall method); this active recall produces much better retention than re-reading. Revisit material at increasing intervals (spaced repetition) by re-testing Day 1 material briefly on Day 3 and Day 5.

Abbreviations used

Abbreviation	Meaning
ABD	Acute behavioural disturbance
ALS	Advanced life support
BLS / ILS	Basic / intermediate life support

Abbreviation	Meaning
CDST	Clinical Decision Support Tool (HPCSA PBEC, 2025)
CPG	Clinical Practice Guidelines (HPCSA PBEC)
DSH	Deliberate self-harm
DSM-5-TR	*Diagnostic and Statistical Manual of Mental Disorders*, 5th edition, text revision (APA, 2022)
DVA	Domestic Violence Act 116 of 1998 (as amended by Act 14 of 2021)
ECP	Emergency Care Practitioner
EMS	Emergency Medical Services
HPCSA	Health Professions Council of South Africa
IM / IV / IO	Intramuscular / intravenous / intraosseous
MDAC / SDAC	Multiple-dose / single-dose activated charcoal
MHCA	Mental Health Care Act 17 of 2002
MHCU	Mental health care user
MMSE	Mini-Mental State Examination
MSE	Mental State Examination
NHA	National Health Act 61 of 2003
PEP	Post-exposure prophylaxis
PBEC	Professional Board for Emergency Care (HPCSA)
PRF	Patient report form
SAPS	South African Police Service
SOA	Criminal Law (Sexual Offences and Related Matters) Amendment Act 32 of 2007
TCC	Thuthuzela Care Centre

SECTION G — BEHAVIOURAL EMERGENCIES

Outcome 39. Discuss the identification and causes of behavioural emergencies

39.1 Explain what constitutes a behavioural emergency

Definition

A **behaviour** is any observable action or response of a person to their environment. A **behavioural emergency** is a situation in which a patient's behaviour has become so abnormal, disordered or dangerous that it is unacceptable or intolerable to the patient, the family or the community, and in which there is an **immediate risk of harm** to the patient or to others, or in which the behaviour indicates an underlying condition that requires urgent intervention (NAEMT, 2021; Tintinalli et al., 2020). The key elements of the definition are therefore (1) a change in behaviour that deviates markedly from what is expected or from the person's own baseline, (2) a threat to the safety of the patient, bystanders or emergency personnel, and (3) the need for immediate intervention, whether that intervention is medical, psychiatric, social or legal.

It is important to understand that a behavioural emergency is **a presentation, not a diagnosis**. The RCEM guideline makes exactly this point about acute behavioural disturbance (ABD), stating that ABD "should not be considered a diagnosis or syndrome, but rather a clinical picture with a variety of presenting features and potential causes" (RCEM, 2025, p. 2). Similarly, Desai, Kulkarni and Verma (n.d.) emphasise that agitation in an older person "is rarely a diagnosis in itself and is more often the behavioral surface of delirium, dementia, pain, psychosis, medication toxicity, withdrawal, infection, or environmental distress". The same principle applies to every age group: the behaviour is the *surface* and the ECP's job is to look for what lies beneath it.

What makes behaviour "abnormal"?

There is no single objective line between normal and abnormal behaviour, and judgements about behaviour are influenced by culture, religion, age and context. Behaviour is generally regarded as abnormal or disordered when it is:

1. **Statistically unusual** for the person's culture and context (for example, talking to people who are not present, or undressing in public).
2. **A marked change from the person's baseline**, as reported by family or caregivers (for example, a previously calm grandmother who has become confused and aggressive over two days, which strongly suggests delirium).
3. **Maladaptive**, in that it interferes with the person's ability to function, care for themselves, or maintain relationships.
4. **Distressing** to the person or to those around them.
5. **Dangerous** to the person or to others.

In South Africa, cultural context is especially important. Behaviour such as communication with ancestors, or the experience of *ukuthwasa* (the calling to become a traditional healer), may be culturally normative and should not automatically be labelled as psychosis (Tilley et al., 2026, citing Kirmayer, 2022). Tilley et al. (2026) note that South Africa's multicultural context "can aggravate the need for urgent western mental health care, and complicate health seeking behaviour further". The ECP should ask family members whether the behaviour is considered normal within their cultural or religious framework.

Types of behavioural emergency encountered prehospitally

In practice, the ECP will encounter the following broad presentations:

- **Acute agitation and aggression**, including violent behaviour directed at others, property, or emergency personnel.
- **Acute behavioural disturbance (ABD)** — a severe agitated state with abnormal physiology, which may include agitation, constant physical activity, bizarre behaviour (including paranoia and hypervigilance), fear and panic, unusual strength, sustained non-compliance with police or ambulance staff, insensitivity to pain, hyperthermia and sweating, tachypnoea and tachycardia (RCEM, 2025, adapted from Gonin et al., 2018). ABD is a medical emergency because of the risk of sudden cardiac arrest from metabolic acidosis, hyperthermia and catecholamine excess.
- **Suicidal ideation, suicide attempts and deliberate self-harm.**
- **Acute psychosis** (hallucinations, delusions, disorganised behaviour).
- **Severe anxiety and panic states.**
- **Withdrawn, catatonic or mute states.**
- **Confusion and delirium**, especially in older persons.
- **Intoxication and withdrawal states.**
- **Behavioural disturbance in children and adolescents**, and in persons with intellectual disability or autism.

The scale of the problem in South African EMS

Behavioural emergencies form a large and growing part of the South African prehospital caseload. In a census of three years (2017–2019) of Western Cape Government EMS incident records from the rural Garden Route District, Tilley et al. (2026) identified 2,976 mental-health-related incidents. Of these, 52% were overdose or deliberate self-poisoning, 7.6% suicidal ideation, 3.6% substance abuse, 3.4% death by suicide, 3% depression, 2.7% attempted suicide, 2% anxiety, 1.3% cutting self-harm, 0.7% bipolar disorder, 0.6% schizophrenia and 0.1% post-traumatic stress disorder. Tilley et al. (2026) argue that, because deinstitutionalisation under the Mental Health Care Act 17 of 2002 was not accompanied by adequate community mental health services, "EMS is de facto relied upon as a primary health care point for all poor mental health sequelae and related emergencies". This is an important point to make in an examination answer about the role of EMS: the ECP is often the first, and sometimes the only, health professional that a person in a mental health crisis will see (Evans, Geduld and Stassen, 2018).

High-yield points — 39.1

- A behavioural emergency = abnormal behaviour + risk of harm + need for immediate intervention.
- It is a **presentation, not a diagnosis**. Always look for the cause, especially an organic (medical) cause.
- Behaviour must be judged against the person's baseline and cultural context.
- ABD (agitation, strength, pain insensitivity, hyperthermia, tachycardia) is a life-threatening medical emergency.
- In the Western Cape, 52% of EMS mental-health incidents were overdose/self-poisoning (Tilley et al., 2026).

39.2 Compare potential causes for behavioural emergencies, including situational causes, organic causes and psychiatric illnesses

The causes of behavioural emergencies are classically divided into three groups: **organic (biological/medical) causes**, **psychiatric (functional) causes**, and **situational (psychosocial/environmental) causes** (NAEMT, 2021; Tintinalli et al., 2020). In reality these groups overlap: a person with schizophrenia may become agitated because of a urinary tract infection, an older person with dementia may become aggressive because of pain and an unfamiliar environment, and a person who has just been assaulted may be intoxicated and have a head injury. The ECP must therefore consider all three groups in every patient and must **always exclude organic causes first**, because they are frequently life-threatening and are treatable.

Organic (medical) causes

An organic cause is a physical disease, injury or substance that alters brain function and thereby produces abnormal behaviour. The RCEM (2025) lists the following factors that may lead to an ABD presentation, noting that substance intoxication and mental health conditions are the most common:

- substance intoxication
- sepsis
- mental health conditions
- head injury
- substance withdrawal
- anticholinergic syndrome
- serotonin syndrome
- neuroleptic malignant syndrome
- hypoxia
- thyroid storm
- hypoglycaemia
- heat stroke
- electrolyte disturbance
- seizures (including post-ictal states).

A useful way to remember the organic causes in the field is to think systematically:

Category	Examples
Hypoxia and hypercapnia	Respiratory failure, pneumonia, asthma, COPD exacerbation, carbon monoxide poisoning, pulmonary embolism, shock
Metabolic and endocrine	Hypoglycaemia (the single most important reversible cause to exclude — check the glucose in every patient), hyperglycaemic states (DKA, HHS), hyponatraemia and other electrolyte disturbances, uraemia, hepatic encephalopathy, thyrotoxicosis/thyroid storm, hypothyroidism, Wernicke's encephalopathy (thiamine deficiency)
Neurological	Head injury (concussion, intracranial haemorrhage), stroke, seizures and post-ictal states, CNS infections (meningitis, encephalitis, cerebral malaria, HIV-associated infections such as cryptococcal meningitis and toxoplasmosis), brain tumours, dementia
Infection and sepsis	Urinary tract infection and pneumonia in the elderly, sepsis of any cause, HIV and its opportunistic infections, neurosyphilis
Toxicological	Intoxication with stimulants (methamphetamine "tik", cocaine), alcohol, cannabis, hallucinogens, inhalants; withdrawal from alcohol (delirium tremens) or benzodiazepines; anticholinergic toxicity (for example *Datura stramonium*); serotonin syndrome; neuroleptic malignant syndrome; medication side effects
Thermal	Heat stroke, hypothermia
Cardiovascular	Hypotension and hypoperfusion, arrhythmias, hypertensive encephalopathy

Clinical clues that suggest an organic cause (sometimes called "red flags") include: (1) acute onset over hours to days; (2) first presentation of "psychiatric" symptoms after the age of about 40 years with no previous psychiatric history; (3) a fluctuating level of consciousness and disorientation to time and place; (4) visual, tactile or olfactory hallucinations (in contrast, primary psychiatric illness more typically causes auditory hallucinations); (5) abnormal vital signs (fever, tachycardia, hypotension, hypoxia, hypertension with bradycardia); (6) focal neurological signs, abnormal pupils, or a recent head injury; (7) evidence of drug or alcohol use; and (8) poor attention and impaired memory (Tintinalli et al., 2020; Inouye, Westendorp and Saczynski, 2014). Desai, Kulkarni and Verma (n.d.) summarise the single most important

principle as follows: “behavior change should be treated as medically caused until a more specific explanation proves otherwise”.

Psychiatric (functional) causes

Psychiatric causes are mental disorders in which no structural or metabolic brain abnormality explains the behaviour on routine examination (the older term “functional” reflects this). The principal psychiatric causes of behavioural emergencies are:

- **psychotic disorders** (schizophrenia and its spectrum, brief psychotic disorder), which may produce agitation driven by paranoid delusions or command hallucinations;
- **bipolar disorder**, particularly mania with grandiosity, irritability, disinhibition and agitation;
- **depressive disorders**, with suicidal ideation, self-harm, psychomotor retardation, or agitated depression;
- **anxiety disorders**, including panic attacks;
- **trauma- and stressor-related disorders** such as acute stress disorder and PTSD;
- **personality disorders**, particularly borderline and antisocial personality disorders;
- **substance use disorders** (which overlap with the organic group);
- **neurodevelopmental disorders** such as intellectual disability and autism spectrum disorder, where behavioural distress is often triggered by a change in routine, sensory overload or pain that the person cannot communicate.

Psychomotor agitation occurs in approximately 11–13% of patients with mood disorders or schizophrenia, 20–30% of those with anxiety disorders, 25% of those with alcohol use disorder, 24–45% of those with dementia and 35% of those with substance use disorder (Saucedo Uribe et al., 2025). Clinical features that favour a primary psychiatric cause include a known psychiatric history, a gradual onset, a clear sensorium with orientation preserved, auditory rather than visual hallucinations, normal vital signs, and an age at first presentation of between about 15 and 35 years for schizophrenia.

Situational (psychosocial and environmental) causes

Situational causes are external stressors that overwhelm a person’s coping mechanisms and produce an acute behavioural reaction in a person who may have no underlying mental illness. Examples include:

- the sudden death of a loved one, or receiving devastating news;
- being the victim of violent crime, rape, domestic violence or a motor-vehicle collision;
- witnessing a traumatic event (an acute stress reaction);
- loss of a job, financial crisis or eviction;
- relationship breakdown;
- natural disasters and mass-casualty incidents;
- exposure to a frightening or overstimulating environment, such as a crowded emergency department or the back of an ambulance — which is a particularly important trigger in older persons with dementia and in persons with autism (Desai, Kulkarni and Verma, n.d.).

South Africa's social context produces an exceptionally heavy burden of situational stress. Tilley et al. (2026) note that 63% of South Africans live in poverty, 31.9% are unemployed, a rape is estimated to occur every 12 minutes, there are 86 murders and 595 assaults reported daily, and 18.9% of the population abuse alcohol and/or drugs. They describe these as "social antecedents for suicidality and suicide", and introduce the concept of suicide as a **syndemic** — a situation in which poverty, violence, trauma, substance use, chronic illness, stigma and mental illness interact with and amplify each other (Tilley et al., 2026, citing Caulkins, 2022).

Comparison of the three groups

Feature	Organic	Psychiatric (functional)	Situational
Onset	Usually acute (hours–days)	Usually gradual, may relapse acutely	Acute, linked to an identifiable event
Age	Any; new-onset behaviour >40 years strongly suggests organic	Schizophrenia typically 15–35 years	Any
Level of consciousness	Often altered or fluctuating	Usually clear	Clear (may be distressed)
Orientation	Often disoriented	Usually oriented	Oriented
Attention	Impaired	Usually intact (may be distracted)	Usually intact
Hallucinations	Visual, tactile, olfactory more common	Auditory most common	Usually absent
Vital signs	Often abnormal	Usually normal	May show sympathetic arousal
Physical signs	Often present (focal signs, fever, smell of alcohol, track marks, trauma)	Usually absent	Injuries if assaulted
Psychiatric history	Often absent	Often present	Often absent
Response to reassurance	Poor	Variable	Usually good
Priority for the ECP	Identify and treat the cause; this may be life-threatening	Safety, de-escalation, transport for psychiatric assessment	Psychological first aid, support, referral

High-yield points — 39.2

- Three causes: organic, psychiatric, situational. **Always exclude organic causes first.**
- Check the **blood glucose**, oxygen saturation, temperature and pupils in every behavioural emergency.

- Red flags for organic cause: acute onset, age >40 at first presentation, fluctuating consciousness, disorientation, visual hallucinations, abnormal vital signs, focal neurological signs.
- Substance intoxication and mental illness are the most common factors in ABD (RCEM, 2025).
- In South Africa, poverty, violence, unemployment and substance abuse are powerful situational drivers (Tilley et al., 2026).

39.3 Explain psychiatric symptoms and signs associated with disorders of consciousness, motor activity, speech, thinking, mood and affect, memory, orientation, perception and intelligence

This outcome tests your vocabulary of descriptive psychopathology — the terms used to describe what you observe and what the patient tells you. Using correct terminology on the PRF and at handover communicates precisely and protects you legally. The definitions below follow the standard descriptions in *Kaplan & Sadock's Synopsis of Psychiatry* (Boland, Verduin and Ruiz, 2021) and the DSM-5-TR (American Psychiatric Association, 2022).

Disorders of consciousness

Consciousness is a state of awareness of the self and the environment. Disorders of consciousness strongly suggest an **organic** cause.

- **Clouding of consciousness**: incomplete clear-mindedness with disturbances in perception and attitudes.
- **Delirium**: an acute, fluctuating disturbance of attention and awareness with disorganised thinking, often with restlessness, disorientation and hallucinations (see 52.9.1).
- **Drowsiness, stupor and coma**: progressively reduced responsiveness. Stupor is a state in which the patient responds only to vigorous stimulation; coma is complete unresponsiveness. Note that **psychiatric stupor** (for example catatonic stupor or depressive stupor) also exists, in which the patient is unresponsive yet the eyes may be open and track, the patient may resist eye opening, and there are no neurological signs.
- **Hypervigilance**: excessive attention to and focus on all internal and external stimuli, seen in paranoid states, stimulant intoxication and PTSD.
- **Distractibility**: inability to concentrate; attention is drawn to unimportant stimuli (mania, delirium).
- **Dissociation**: a disruption in the normal integration of consciousness, memory, identity and perception (for example depersonalisation, derealisation, dissociative fugue), seen after severe trauma.

Disorders of motor activity (behaviour)

- **Psychomotor agitation**: excessive motor and cognitive activity, usually non-productive and in response to inner tension (for example pacing, wringing of hands, pulling at clothing, inability to sit still) (Saucedo Uribe et al., 2025).

- **Psychomotor retardation**: slowing of movement, speech and thought; seen in severe depression.
- **Catatonia**: a syndrome of motor abnormalities including stupor, mutism, negativism (resistance to instructions), posturing (holding a posture against gravity), catalepsy, waxy flexibility (limbs remain where placed by the examiner), echolalia and echopraxia (mimicking speech or movements), stereotypy, mannerisms and grimacing (see 52.3).
- **Stereotypies**: repetitive, purposeless movements (for example rocking), seen in autism and schizophrenia.
- **Mannerisms**: odd but goal-directed movements.
- **Compulsions**: repetitive behaviours performed in response to an obsession (for example handwashing in OCD).
- **Tics**: sudden, rapid, recurrent non-rhythmic movements or vocalisations (Tourette's disorder).
- **Akathisia**: subjective feeling of restlessness with an inability to sit still, commonly a side effect of antipsychotics — important because it can be mistaken for worsening agitation (RCEM, 2025).
- **Dystonia**: sustained muscle contraction producing twisting, abnormal postures (for example oculogyric crisis, torticollis), an acute side effect of haloperidol and other dopamine antagonists.
- **Tremor, rigidity and bradykinesia**: suggest Parkinson's disease, drug-induced parkinsonism or neuroleptic malignant syndrome.
- **Aggression and violence**: forceful, goal-directed action that may be verbal or physical.

Disorders of speech

Speech is described in terms of its **rate, volume, quantity, tone and fluency**.

- **Pressure of speech**: rapid, increased amount of speech that is difficult to interrupt (mania, stimulant intoxication).
- **Poverty of speech**: restricted amount of speech, with monosyllabic answers (depression, negative symptoms of schizophrenia).
- **Poverty of content**: adequate amount of speech but conveying little information because it is vague or repetitive.
- **Mutism**: complete absence of speech (catatonia, severe depression, conversion disorder, selective mutism in children).
- **Dysarthria**: difficulty articulating words (a neurological or intoxication sign, not a thought disorder).
- **Aphasia (dysphasia)**: a disturbance in language comprehension or expression (expressive/Broca's, receptive/Wernicke's) due to brain damage, for example stroke — an organic sign that is easily mistaken for confusion or psychosis.
- **Echolalia**: repetition of the examiner's words.
- **Neologisms**: invented new words (schizophrenia).
- **Clang associations**: words chosen for their sound rather than their meaning, for example rhyming (mania).
- **Word salad**: an incoherent mixture of words and phrases.

Disorders of thinking (thought)

Thinking is assessed in terms of **form (process)** — how thoughts are linked — and **content** — what the person is thinking about.

Disorders of thought form (formal thought disorder):

- **Flight of ideas**: rapid, continuous jumping from one idea to another with understandable connections (mania).
- **Loosening of associations (derailment)**: ideas shift from one subject to an unrelated one without the speaker being aware (schizophrenia).
- **Tangentiality**: the person drifts away from the point and never returns to it.
- **Circumstantiality**: the person gives excessive, unnecessary detail but eventually reaches the point.
- **Thought blocking**: an abrupt interruption in the train of thought before an idea is finished (schizophrenia).
- **Perseveration**: persistent repetition of the same response beyond the point of relevance (organic brain disease).
- **Incoherence**: thought that is generally not understandable.

Disorders of thought content:

- **Delusion**: a fixed, false belief, out of keeping with the person's cultural and educational background, that is held with conviction despite evidence to the contrary. Types include **persecutory** (paranoid) delusions (being followed, poisoned, bewitched), **grandiose** delusions (special powers or identity), **delusions of reference** (the television or radio is talking about the person), **nihilistic** delusions (the person is dead or their organs are rotting — seen in psychotic depression), **somatic** delusions, **delusional jealousy**, **erotomania**, and **delusions of control**, including **thought insertion, thought withdrawal and thought broadcasting** (classic features of schizophrenia).
- **Overvalued idea**: an unreasonable, sustained false belief held with less than delusional intensity (for example the overvalued idea of being fat in anorexia nervosa).
- **Obsession**: a recurrent, intrusive, unwanted thought, image or urge that the person recognises as the product of their own mind and tries to resist (OCD).
- **Phobia**: persistent, irrational, exaggerated fear of a specific stimulus or situation.
- **Suicidal and homicidal ideation**: thoughts of killing oneself or others — **always ask about these directly** (see Outcome 45).
- **Preoccupation**: thoughts focused on a particular idea, often with strong emotion (for example hopelessness, guilt).

Disorders of mood and affect

Mood is the pervasive and sustained emotion described by the patient (subjective, for example "I feel hopeless"). **Affect** is the outward expression of emotion as observed by the examiner (objective). A useful analogy is that mood is the "climate" and affect is the "weather".

- **Euthymic**: normal mood.

- **Depressed / dysphoric**: low, sad, unpleasant mood.
- **Elevated / euphoric / expansive**: abnormally high mood (mania, stimulant intoxication).
- **Irritable**: easily annoyed and provoked to anger (mania, intoxication, withdrawal, personality disorders).
- **Anxious**: apprehension and tension.
- **Anhedonia**: loss of interest in and enjoyment of usual activities (core symptom of depression).

Affect is described according to its **range** (full, restricted, **blunted** — severely reduced intensity, **flat** — virtually absent), its **appropriateness** (whether it matches the content of speech; for example **inappropriate affect** — laughing while describing a tragedy, seen in schizophrenia), its **reactivity**, and its **stability** (**labile affect** — rapid, abrupt changes in emotional expression unrelated to external stimuli, seen in mania, delirium, intoxication, frontal lobe damage and borderline personality disorder).

Disorders of memory

Memory is the ability to store and retrieve information. It is described as **immediate** (registration — recall within seconds, for example repeating three words), **recent/short-term** (minutes to days, for example what the patient ate for breakfast), and **remote/long-term** (years, for example date of birth, childhood events).

- **Amnesia**: partial or total inability to recall past experiences. **Anterograde amnesia** is the loss of memory for events after an incident (for example after a head injury or after benzodiazepines/GHB — relevant to drug-facilitated sexual assault). **Retrograde amnesia** is the loss of memory for events before the incident.
- **Confabulation**: unconscious filling of memory gaps with imagined or untrue experiences that the patient believes — classically seen in **Korsakoff syndrome** (chronic thiamine deficiency due to alcohol use disorder).
- **Dissociative amnesia**: inability to recall important personal information, usually of a traumatic nature, that is inconsistent with ordinary forgetting.
- Progressive loss of **recent** memory with relative preservation of remote memory is typical of **Alzheimer's disease**.

Disorders of orientation

Orientation is the awareness of **time, place and person** (and some include situation). Disorientation is classically lost in the order **time first, then place, then person**. Disorientation is a feature of **organic** brain disease (delirium, dementia, intoxication, head injury, hypoxia, hypoglycaemia). A patient with a primary psychotic illness is usually fully oriented, even when severely deluded. Note that the Glasgow Coma Scale verbal "confused" response is not a substitute for a structured assessment of orientation or capacity; Stander, Hodkinson and Dippenaar (2021) found that South African prehospital providers frequently used the GCS to assess mental capacity, which is an inappropriate use of a scale designed for acute cerebral injury.

Disorders of perception

Perception is the process of transferring physical stimulation into psychological information.

- **Hallucination**: a false sensory perception occurring in the absence of any external stimulus. Hallucinations may be **auditory** (most common in schizophrenia; for example voices commenting on the person's actions or **command hallucinations** instructing the person to harm themselves or others, which represent a major risk factor for violence and suicide), **visual** (more typical of organic causes such as delirium, alcohol withdrawal, hallucinogens and Lewy body dementia), **tactile** (for example the feeling of insects crawling under the skin — formication — in cocaine/methamphetamine intoxication and alcohol withdrawal), **olfactory** and **gustatory** (temporal lobe epilepsy and other organic causes).
- **Illusion**: a misperception or misinterpretation of a **real** external stimulus (for example mistaking a coat on a hook for an intruder in a darkened room), common in delirium.
- **Hypnagogic and hypnopompic hallucinations**: hallucinations occurring while falling asleep or waking up; these are normal phenomena.
- **Depersonalisation and derealisation**: the sense that oneself or the environment is unreal or changed.

Disorders of intelligence

Intelligence is the ability to understand, recall, mobilise and integrate previously learned knowledge constructively in meeting new situations.

- **Intellectual disability (intellectual developmental disorder)**: deficits in intellectual functioning (reasoning, problem solving, planning, abstract thinking, judgement, academic and experiential learning) **and** in adaptive functioning, with onset during the developmental period (APA, 2022) (see 52.1.1).
- **Dementia**: an acquired, global decline of cognitive function from a previously higher level, without impairment of consciousness (see 52.9.2).
- **Pseudodementia**: cognitive impairment that mimics dementia but is due to depression, and which improves when the depression is treated.

Intelligence is relevant prehospitally because it affects the patient's ability to communicate symptoms (including pain), to understand information, and therefore their **capacity to consent or refuse**. The ECP must adapt communication to the person's level of understanding, use simple language, and involve a trusted caregiver.

Insight and judgement

Although not listed in the outcome, you should also be able to describe **insight** (the patient's awareness and understanding of their own illness — poor insight is typical of psychosis and mania and is a major reason why patients refuse care) and **judgement** (the ability to assess a situation correctly and act appropriately). Both are essential components of capacity (Evans, Geduld and Stassen, 2018, note that patients who have attempted suicide "may have limited insight into their illness and a restricted ability to cooperate with treatment").

High-yield points — 39.3

- Disorientation and fluctuating consciousness = organic until proven otherwise.
- Visual/tactile hallucinations suggest organic or toxic causes; auditory (especially command) hallucinations suggest psychosis and increase the risk of violence and suicide.
- Delusion = fixed false belief out of keeping with culture; illusion = misinterpretation of a real stimulus; hallucination = perception without a stimulus.
- Mood (subjective) versus affect (objective).
- Confabulation = Korsakoff syndrome.
- Akathisia and dystonia are drug side effects that can mimic agitation.

39.4 Apply effective techniques for interviewing a patient during a behavioural emergency

Scene safety comes before the interview

Before any interview begins, the ECP must ensure that the scene is safe. Evans, Geduld and Stassen (2018) found that more than half of South African prehospital providers in their survey identified threats to personal safety during the care of suicidal patients; three had been injured (bite wounds, facial scratching and blunt assault), two had witnessed patients stabbing themselves, and one reported a patient jumping from a moving ambulance. Another unpublished South African study cited by the same authors found that 56% of 158 prehospital personnel had been assaulted on duty, consistent with international reports of violence against prehospital providers of 61.0–87.5%. The following safety principles therefore apply to every behavioural interview:

1. Obtain as much information as possible from dispatch (Is there a weapon? Is the patient known? Has SAPS been dispatched?) and **stage away from the scene until SAPS secure it** if there is any suggestion of a weapon or violence.
2. On arrival, scan for weapons and potential weapons (knives, bottles, ligatures, medication), and remove them from the area where it is safe to do so.
3. Position yourself so that **both you and the patient have an exit**, never stand between the patient and the door in a way that makes them feel trapped, and never allow the patient to stand between you and your only exit (RCEM, 2025).
4. Stay **out of arm's reach** — approximately two arm-lengths away (Richmond et al., 2012).
5. Never turn your back on the patient and never interview alone if the patient is potentially violent; have a colleague present who is not directly engaged but who observes.
6. Be aware of your own body language and keep your hands visible.

General principles of the interview

The aim of the interview is to (1) establish rapport and calm the patient, (2) gather enough information to identify the likely cause (organic, psychiatric or situational) and assess risk, (3) assess capacity, and (4) plan safe management and transport. The following techniques are

recommended in emergency psychiatric practice (Richmond et al., 2012; Tintinalli et al., 2020; NAEMT, 2021):

- **Introduce yourself** clearly by name and role, explain why you are there, and tell the patient that your aim is to keep them safe (RCEM, 2025).
- **Have only one person speak** to the patient; multiple voices increase confusion and agitation (RCEM, 2025).
- **Choose a quiet, private, low-stimulus environment** where possible — turn off the television or radio, ask unnecessary bystanders to leave, and switch off sirens and flashing lights near the patient.
- **Speak calmly, slowly and clearly**, in short sentences, and give the patient time to respond; repetition may be needed (RCEM, 2025).
- **Use the patient's preferred language** or a trusted interpreter. In South Africa, language barriers between provider and patient are common and are a frequent cause of misunderstanding; the HPCSA (2023, para 8.1.7) states that where a linguistic or cultural barrier exists, "an interpreter who is culturally sensitised and fluent in the language used by the patient should be present". Avoid using children as interpreters, and be cautious when using a family member who may be the abuser in cases of domestic violence.
- **Use open-ended questions first** ("Tell me what has been happening today"), followed by closed questions to clarify specific facts ("Have you taken any tablets?").
- **Listen actively**: show that you are listening by nodding, paraphrasing ("So you are saying that…"), summarising, and using clarifying statements (RCEM, 2025).
- **Acknowledge and validate feelings** ("It sounds like you are very frightened") without necessarily agreeing with delusional content.
- **Do not argue with delusions or hallucinations**, and do not pretend to share them. A suitable response is: "I can't hear the voices, but I can see they are really upsetting you."
- **Be honest.** Do not lie to or trick the patient. Stander, Hodkinson and Dippenaar (2021) found that South African prehospital providers frequently used deception to get mental health care users to agree to transport; this damages trust, may escalate the situation when the deception is discovered, and undermines future engagement with health services.
- **Be non-judgemental.** Avoid sarcasm, threats, accusations and moralising. Evans, Geduld and Stassen (2018) documented negative attitudes among South African providers, with some describing suicidal patients as "a waste of time", "attention seeking" or "pretending". Such attitudes are unprofessional, are associated with poorer care, and can directly increase the risk of a patient refusing help.
- **Maintain non-threatening body language**: stand at an angle rather than face-on, keep your hands open and visible, avoid prolonged direct eye contact (which may be perceived as threatening) but do not avoid eye contact entirely, and avoid sudden movements.
- **Explain everything before you do it**, especially before touching the patient or taking vital signs.
- **Offer choices** where possible (for example "Would you like to walk to the ambulance or sit on the stretcher?"), as this gives the patient a sense of control (RCEM, 2025).

- **Set clear limits** in a matter-of-fact, non-threatening way (for example "I can't let you hurt yourself or anyone else") (RCEM, 2025).
- **Involve family or friends** if their presence calms the patient; ask them to leave if their presence escalates the situation. Evans, Geduld and Stassen (2018) reported that some providers found family helpful in convincing patients to accept transport, but others found that family "get in the way" or were "more difficult than the patient".

Content of the interview — what to ask

A structured interview in a behavioural emergency should cover:

1. **The presenting problem**: what happened, when it started, what triggered it, and whether there is a history of similar episodes.
2. **Collateral history** from family, friends, bystanders and SAPS (this is often the most valuable information, particularly regarding baseline behaviour, onset, substances taken, and recent stressors) (RCEM, 2025; Desai, Kulkarni and Verma, n.d.).
3. **Medical history**: diabetes, epilepsy, head injury, HIV status and ARV use, thyroid disease, recent illness or infection, chronic medication and recent changes in medication.
4. **Psychiatric history**: previous diagnoses, admissions, psychiatric medication, compliance with medication, and previous self-harm or violence.
5. **Substance use**: alcohol, cannabis ("dagga"), methamphetamine ("tik"), methaqualone ("mandrax"), heroin or "nyaope/whoonga", cocaine, prescription medications; time of last use (to anticipate withdrawal).
6. **Risk assessment**: **ask directly about suicidal and homicidal thoughts, intent and plans** (see Outcome 45 for specific questions); ask about access to weapons, firearms, medication and other means.
7. **Social history**: living arrangements, support system, recent losses, domestic violence and safety at home.
8. **The patient's own understanding and wishes** (insight), which forms part of the capacity assessment.

Capacity assessment during the interview

Capacity (the ability to make a specific decision at a specific time) is central to every behavioural emergency, because it determines whether a refusal of treatment or transport can be accepted. A patient has decision-making capacity for a particular decision if they can: (1) **understand** the relevant information (the nature of their condition, the proposed treatment, alternatives, and the consequences of refusal); (2) **retain** that information long enough to make a decision; (3) **use and weigh** the information to arrive at a choice; and (4) **communicate** that choice. The HPCSA (2023, para 8.1.2) states that the decision of a competent adult patient to refuse a particular medical intervention "must be respected, even where this would result in serious harm to them or in their own death unless the patient's state of mind is such that he/she is incapable of taking or participating in the decision". The ECP should document the capacity assessment explicitly on the PRF, including the questions asked and the patient's responses.

Special interviewing situations

- **The paranoid patient**: maintain greater physical distance, avoid touching, be transparent about everything you do, and do not whisper to colleagues.
- **The manic patient**: be firm and brief, avoid being drawn into arguments, and reduce stimulation.
- **The depressed or withdrawn patient**: allow silences, be patient, use open questions, and always ask about suicide.
- **The intoxicated patient**: keep the interview short and focused on safety; repeat the assessment as intoxication wears off; do not assume that intoxication is the only problem (head injury, hypoglycaemia and co-ingestion are common).
- **The patient with intellectual disability or autism**: speak to someone known to the patient, use simple concrete language, reduce sensory stimulation, and consult any available care plan (RCEM, 2025).
- **The older patient with suspected dementia or delirium**: approach from the front, introduce yourself repeatedly if necessary, restore hearing aids and glasses, and obtain collateral history about baseline cognition (Desai, Kulkarni and Verma, n.d.).

High-yield points — 39.4

- Safety first: exit route, two arm-lengths, no weapons, SAPS if violent, never alone.
- One voice, calm, short sentences, open questions, active listening, validate feelings.
- Never argue with or collude with delusions; never lie to the patient.
- Always obtain collateral history and always ask directly about suicidal and homicidal intent.
- Assess and document capacity: understand, retain, weigh, communicate.

Outcome 40. Discuss the utility of MMSE in an emergency care setting

What the MMSE is

The **Mini-Mental State Examination (MMSE)** is a brief, structured, 30-point screening test of cognitive function that was developed by Folstein, Folstein and McHugh (1975). It takes approximately 5–10 minutes to administer and assesses the following cognitive domains:

Domain	Task	Points
Orientation to time	Year, season, date, day, month	5
Orientation to place	Country, province, town, building/place, floor/room	5
Registration (immediate memory)	Repeat three unrelated words	3
Attention and calculation	Serial sevens (100 – 7 = 93, etc., five times) or spell "WORLD" backwards	5
Recall (short-term memory)	Recall the three words	3
Language — naming	Name two objects (for example a pen and a watch)	2
Language — repetition	Repeat "No ifs, ands or buts"	1
Language — three-stage command	"Take this paper in your right hand, fold it in half and put it on the floor"	3
Reading	Read and obey "Close your eyes"	1
Writing	Write a sentence	1
Visuospatial	Copy intersecting pentagons	1
Total		**30**

A score of **24–30** is generally regarded as normal, **18–23** suggests mild cognitive impairment, and **0–17** suggests severe impairment, although the commonly used single cut-off is **≤23 or ≤24** out of 30 (Folstein, Folstein and McHugh, 1975). Scores must be interpreted in the light of the patient's **education, literacy, language and culture**, because the MMSE is strongly affected by these factors.

Utility (usefulness) of the MMSE in the emergency care setting

The outcome asks you to *discuss the utility*, which means you must give both the **advantages** and the **limitations**, and reach a balanced conclusion.

Advantages and appropriate uses

1. **Objective, standardised and reproducible.** The MMSE provides a numerical score that can be documented and compared over time, allowing the receiving facility to detect improvement or deterioration (for example as delirium resolves or as intoxication wears off).
2. **Screens multiple cognitive domains quickly**, including orientation, memory, attention, language and visuospatial ability.
3. **Helps differentiate organic from functional presentations.** Impaired orientation, attention and recall suggest an organic cause (delirium, dementia, intoxication), whereas a patient with a primary psychotic illness usually scores relatively well despite severe abnormalities of thought content.
4. **Contributes to capacity assessment.** A very low score supports a conclusion that the patient may lack capacity for complex decisions, although the MMSE is **not** itself a test of capacity.
5. **Widely known** and used in hospitals, so a prehospital score is meaningful at handover.
6. **Supports documentation** of the patient's mental state when a refusal of treatment or transport is being considered, or when involuntary care under the MHCA is being considered.

The Behavioural Emergencies Tutorial 1 memo summarises this well: cognitive screens such as the MMSE and the Montreal Cognitive Assessment (MoCA) "aid in differentiating delirium, dementia, or intoxication and guide treatment planning" and influence decisions about "psychiatric hold, urgent imaging or neurology consult, pharmacological intervention, or liaison with mental health services".

Limitations in the emergency (and especially the prehospital) setting

1. **Time.** Five to ten minutes is long in an emergency, and the MMSE is not suitable for a patient who is unstable, violent or severely agitated. Scene safety and resuscitation always take priority.
2. **Requires cooperation and adequate consciousness.** It cannot be performed on an agitated, aggressive, intoxicated, sedated or drowsy patient, which describes many behavioural emergency patients.
3. **Strongly influenced by education, literacy, language and culture.** Items such as serial sevens, reading, writing and spelling "WORLD" backwards disadvantage patients with limited formal education, patients who are not English-speaking, and patients from different cultural backgrounds. This is a major limitation in South Africa, which has eleven official languages (twelve, since South African Sign Language was recognised in 2023) and large disparities in educational attainment. A low score in a patient with little schooling may be falsely interpreted as cognitive impairment.
4. **Affected by sensory impairment** (poor hearing or vision), aphasia and physical disability (for example inability to write).
5. **Insensitive to mild impairment and to executive (frontal lobe) dysfunction.** The MoCA (Nasreddine et al., 2005) is more sensitive for mild cognitive impairment.
6. **It does not diagnose delirium.** The MMSE measures cognitive performance at one point in time; delirium is defined by **acute onset, a fluctuating course and**

inattention, which a single MMSE cannot capture. A dedicated delirium tool such as the **Confusion Assessment Method (CAM)** (Inouye et al., 1990), or its brief version, the **4AT**, is more appropriate.

7. **It does not assess thought content, perception, mood, suicidality, insight or judgement**, which are often the most important issues in a behavioural emergency.
8. **It does not assess capacity.** Capacity is decision-specific; a patient with a low MMSE may still have capacity for a simple decision, and a patient with a normal MMSE may lack capacity because of delusions or severe depression.
9. **Copyright** — the MMSE is protected by copyright, which limits its free use in some services.
10. **Practical environment** — noise, lack of privacy, poor lighting and the presence of bystanders on scene all reduce validity.

Conclusion on utility

The MMSE has **limited utility as a routine prehospital tool** in acute behavioural emergencies, because the patients who most need assessment are often too agitated, intoxicated or unstable to complete it, and because the test is biased by education, language and culture. However, it is **useful in the stable, cooperative patient** (for example an older person with suspected dementia or delirium, or a patient who is refusing transport and whose capacity is in question) as an objective, documentable baseline measure of cognition, provided its results are interpreted cautiously and supplemented by the full Mental State Examination, a delirium screen such as the CAM or 4AT, collateral history, and a decision-specific capacity assessment. In practice, many ECPs use an abbreviated screen in the field (orientation to time, place and person; registration and recall of three words; attention, for example months of the year backwards) and document the findings in MSE format.

40.1 Apply key elements of the prehospital examination during a behavioural emergency, including the Mental State Examination (MSE)

Overall structure of the prehospital examination

The prehospital examination of a patient with a behavioural emergency follows the standard systematic approach, with the addition of the MSE:

1. **Scene size-up and safety** (see 39.4 and 43.4), including number of patients, mechanism (for example an empty pill container, a ligature, a firearm), need for SAPS and additional resources.
2. **Primary survey (ABCDE)** — behavioural presentations do not exempt the patient from a full primary survey. Look specifically for airway compromise (intoxication, hanging, burns), breathing problems (hypoxia, Kussmaul breathing in DKA or salicylate poisoning), circulation (shock, bleeding from self-inflicted wounds), disability (GCS, pupils, **blood glucose**, seizure activity, lateralising signs), and exposure (temperature, rashes, track marks, injuries, signs of self-harm, signs of assault, signs of restraint injury).

3. **Vital signs**: pulse, blood pressure, respiratory rate, SpO_2, temperature, blood glucose, GCS and, where appropriate, end-tidal CO_2 and ECG (particularly with overdose, stimulants, or before and after sedation).
4. **Focused history** (SAMPLE/AMPLE plus the behavioural history in 39.4) with collateral information.
5. **Secondary survey** looking for occult trauma (particularly head injury), signs of infection, toxidromes, signs of chronic disease, and signs of abuse or neglect.
6. **Mental State Examination.**
7. **Risk assessment** (suicide, violence, abscondment, vulnerability).
8. **Capacity assessment.**

The Mental State Examination (MSE)

The MSE is a structured description of the patient's psychological functioning **at the time of examination**. It is the psychiatric equivalent of the physical examination: it records what the clinician observes and what the patient reports, using the standard terminology from 39.3 (Boland, Verduin and Ruiz, 2021). The MSE can be remembered with the mnemonic **"ASEPTIC"** or by following the headings below.

MSE heading	What to observe and record	Prehospital examples
1. Appearance	Apparent age, gender, build, dress (appropriate to weather and situation), grooming and hygiene, signs of self-neglect, injuries, scars from self-harm, track marks, smell of alcohol, distinguishing features	"Unkempt man appearing older than stated age, wearing a winter coat on a hot day, strong smell of alcohol, fresh superficial lacerations to left forearm"
2. Behaviour	Level of activity (agitated, restless, retarded), eye contact, rapport, cooperation, posture, abnormal movements (tremor, tics, dystonia, catatonic features), response to internal stimuli (looking around as if hearing voices), aggression	"Pacing, unable to sit, poor eye contact, appears to be responding to unseen stimuli, guarded"
3. Speech	Rate, volume, quantity, tone, fluency, articulation, latency	"Pressured, loud, difficult to interrupt" or "slow, soft, monosyllabic, long latency of response"
4. Mood (subjective, in the patient's words)	Ask "How are you feeling in yourself?"	"'I feel like there is no point anymore'"
5. Affect (objective)	Range, reactivity, appropriateness, stability	"Flat, restricted, congruent with low mood" or "labile,

MSE heading	What to observe and record	Prehospital examples
		tearful then laughing"
6. Thought form	Flow and organisation: flight of ideas, loosening of association, tangentiality, blocking	"Flight of ideas, clang associations"
7. Thought content	Delusions, obsessions, preoccupations, overvalued ideas, **suicidal and homicidal ideation (with intent and plan)**	"Persecutory delusion that neighbours are poisoning his food; denies suicidal ideation; expresses thoughts of harming neighbour"
8. Perception	Hallucinations (type and content, especially command hallucinations), illusions, depersonalisation	"Reports voices telling him to jump in front of a train"
9. Cognition	Level of consciousness, orientation (time, place, person), attention and concentration, memory (immediate, recent, remote); MMSE/abbreviated test if appropriate	"Alert, oriented to person only, unable to name months backwards; MMSE not possible due to agitation"
10. Insight	Does the patient recognise that they are unwell and need help?	"No insight — believes nothing is wrong"
11. Judgement	Ability to make reasonable decisions about their own safety and care	"Poor judgement — plans to walk home along the highway"

Prehospital application: putting the examination together

In the prehospital environment, the MSE is often completed **during** the interview and observation rather than as a separate formal test, and is documented using the headings above. The key prehospital questions that the examination must answer are:

1. **Is there an organic cause that requires immediate treatment?** (hypoglycaemia, hypoxia, head injury, sepsis, overdose, withdrawal). If yes, treat it.
2. **What is the immediate risk** to the patient (suicide, self-neglect, physical deterioration) and to others (violence)?
3. **Does the patient have capacity** to accept or refuse care and transport?
4. **What is the appropriate destination?** Patients with possible organic causes or physical injuries require an emergency centre with medical facilities, not a psychiatric facility; the MHCA requires assessment at a health establishment administered under the auspices of the State for involuntary assessments (MHCA, s 40(1)(a)).

5. **What needs to be communicated at handover and documented on the PRF?** The RCEM (2025) recommends that the prehospital history should ideally include observed behaviours, attempts at verbal and environmental de-escalation, assessments of mental capacity, restraint applied (with duration and indication), security or police involvement (including use of force or Tasers), and the sedative strategy with any adverse events.

High-yield points — Outcome 40 and 40.1

- MMSE = 30 points; ≤23–24 suggests impairment; adjust for education/language.
- Utility: objective, quick, documentable, helps separate organic from functional, supports capacity assessment — **but** needs cooperation, is culturally and educationally biased, does not diagnose delirium, does not assess suicidality or capacity.
- Use the CAM/4AT for delirium; the MoCA is more sensitive for mild impairment.
- MSE headings: Appearance, Behaviour, Speech, Mood, Affect, Thought form, Thought content, Perception, Cognition, Insight, Judgement.
- Always complete a full ABCDE and check glucose before attributing behaviour to a psychiatric cause.

Outcome 41. Analyse the principles which should be considered in the prehospital care of a patient with a behavioural emergency

The principles below form the framework that you should apply to every behavioural emergency. They are drawn from the CDST "Aggression or agitation" tool (HPCSA PBEC, 2025, pp. 46–48), the RCEM (2025) guideline, South African legislation and the South African prehospital literature (Evans, Geduld and Stassen, 2018; Stander, Hodkinson and Dippenaar, 2021; Tilley et al., 2026).

Principle 1: Safety of the crew, the patient and bystanders comes first

The CDST instructs the provider to "ensure the safety of all EMS personnel and the patient", to "be vigilant for unpredictable behaviour", and to "contact SAPS for help if the potential risk to the patient, family or staff is significant" (HPCSA PBEC, 2025, p. 46). A crew member who is injured cannot help the patient and becomes a second casualty. Practical measures include staging until SAPS has secured the scene, scanning for weapons, maintaining an exit, staying out of arm's reach, never turning your back, never allowing yourself to be cornered, and removing any "weapon/mechanism for self-harm" (HPCSA PBEC, 2025, p. 46).

Principle 2: Assume an organic cause until proven otherwise

The CDST states explicitly: "Do not assume aggression/agitation is due to a psychiatric condition", and directs the provider to "look for and manage injury" (head injury) and to "ask about drug misuse and possible overdose" (HPCSA PBEC, 2025, p. 47). Once the patient is contained, a rapid primary survey must be done, including oxygen saturation (give oxygen if SpO_2 <92%), blood pressure (manage shock if BP <90/60), and **blood glucose (treat if <3.5 mmol/L)** (HPCSA PBEC, 2025, p. 47). This principle is reinforced by Desai, Kulkarni and Verma (n.d.), who warn that "if agitation is treated as a purely psychiatric event when it is actually hypoxia, urinary retention, severe pain, sepsis, intracranial bleeding, or medication toxicity, temporary quiet may be purchased at the cost of diagnostic delay".

Principle 3: Use the least restrictive intervention, in a stepwise manner

Care should escalate in a stepwise way, only moving to a more restrictive step when the less restrictive step has failed or is unsafe:

1. Verbal and environmental de-escalation.
2. Offer of **voluntary oral or buccal/intranasal sedation** (the CDST asks: "Is the patient willing to accept voluntary sedation for the transfer?").
3. Brief physical restraint, only to facilitate sedation.
4. Parenteral rapid sedation (IM or IV).
5. Mechanical restraint, only on the instruction of SAPS.
6. Advanced airway management if sedation causes airway compromise.

The RCEM (2025) states that any interventions "must be using the least restrictive intervention possible and should be applied for the shortest duration possible", and that offering oral

sedation “is an important step in establishing that you are using the least restrictive practice in a patient who lacks capacity”.

Principle 4: Respect the patient’s dignity, rights and autonomy

Section 10 of the Constitution of the Republic of South Africa (1996) guarantees everyone inherent dignity, section 12 guarantees freedom and security of the person (including bodily and psychological integrity), and section 27(3) states that no one may be refused emergency medical treatment. Section 8 of the MHCA requires that the person, human dignity and privacy of every mental health care user be respected, and that care be provided in a manner that facilitates community care. The ECP must treat the patient with kindness, respect and empathy (HPCSA PBEC, 2025, p. 46), avoid stigmatising language, protect privacy (for example by not discussing the patient in front of bystanders), and keep the patient informed — “explain to the patient what is about to happen” (HPCSA PBEC, 2025, p. 46).

Principle 5: Assess capacity and act within the law

Every intervention without consent must have a lawful basis. In South Africa, the main lawful bases are the patient’s own informed consent; the emergency provision of the MHCA, which allows care without consent where, due to mental illness, any delay may result in death or irreversible harm to the health of the user, the user inflicting serious harm to themselves or others, or serious damage to property (MHCA, s 9(1)(c)); the emergency provision of the NHA, which allows treatment without consent where any delay might result in death or irreversible damage to health and the patient has not expressly, impliedly or by conduct refused (NHA, s 7(1)(e)); and the common-law defence of necessity. The ECP must assess and document capacity (see 39.4 and Outcome 51).

Principle 6: The obligation to transport

The CDST states that the provider is **obligated to transport** the patient to the nearest or most appropriate emergency centre (with a family member), and to request SAPS instruction for mechanical restraint, if any of the following are present (HPCSA PBEC, 2025, p. 48):

- the patient is a danger to him/herself or may be a suicide risk;
- the patient is a danger to family or community;
- the patient is causing serious damage or destruction to property;
- SAPS are requesting hospitalisation;
- the patient has been sedated and needs monitoring.

These criteria mirror section 9(1)(c) of the MHCA. This principle is critical because Evans, Geduld and Stassen (2018) found that South African providers frequently, and **erroneously**, believed that “patients cannot be transported to hospital without consent under any circumstances”, and that “the importance of the patient signing for their refusal of care was commonly reported and many expressed the belief that this documentation absolved them from legal responsibility”. Some providers reported returning to find patients dead after being allowed to sign a refusal. A refusal form signed by a patient who lacks capacity or who meets the criteria above has no legal value and does not protect the provider.

Principle 7: Monitor continuously, especially after sedation or restraint

The CDST requires BP, HR, RR, SpO_2 and GCS to be monitored and recorded **every 10 minutes** after sedation, with immediate reassessment of airway, breathing and circulation if BP <92 systolic, HR >100 or <50, RR >20 or <9, SpO_2 <92% or a drop in GCS (HPCSA PBEC, 2025, p. 47). If the patient is physically or mechanically restrained, the provider must check and document that the pulses are present in each limb (HPCSA PBEC, 2025, p. 47). The RCEM (2025) recommends continuous monitoring including end-tidal CO_2 (capnography) during and after rapid tranquillisation, because respiratory depression and airway obstruction may be detected earlier by capnography than by pulse oximetry, particularly when the patient is receiving oxygen.

Principle 8: Choose the appropriate destination

Patients with a possible medical or traumatic cause, those who have been sedated, and those who are injured must go to an **emergency centre** with medical capability, not directly to a psychiatric hospital. The MHCA requires that persons apprehended by SAPS be taken to "an appropriate health establishment administered under the auspices of the State" for assessment (MHCA, s 40(1)(a)). Interfacility transfers to specialised psychiatric services are a second common EMS role (Evans, Geduld and Stassen, 2018).

Principle 9: Accurate, contemporaneous documentation

The CDST requires documentation of the history (mental illness, medical illness, epilepsy, medication, substance misuse or possible poisoning, fever or neck stiffness, trauma — especially head trauma), the examination (responsiveness, seizures, injuries, vital signs), and management (medications, routes, doses and times, response, and procedures performed) (HPCSA PBEC, 2025, p. 47). In addition, the RCEM (2025) recommends recording the features that supported the decision, all de-escalation attempts, capacity assessments, restraint (duration and indication), police involvement and use of force, and the sedative strategy with any adverse events. HPCSA (2023, para 12) requires that records be made "at the time of the events described or soon thereafter", be "legible, clear, accurate and unambiguous", avoid confusing abbreviations, and comply with the Protection of Personal Information Act 4 of 2013.

Principle 10: Teamwork, communication and staff welfare

Behavioural emergencies require coordination between EMS, SAPS, family, the receiving facility and mental health services. South African providers report difficulties with SAPS members described as reluctant and lacking understanding of the MHCA (Stander, Hodkinson and Dippenaar, 2021), and a lack of support, counselling and debriefing after adverse events such as a patient's suicide (Evans, Geduld and Stassen, 2018). Tilley et al. (2026) highlight the risk of direct and vicarious traumatisation of emergency care providers who attend suicide scenes, and recommend trauma-informed care. Debriefing the patient and the staff after any restrictive intervention is the final domain of de-escalation in the RCEM (2025) model.

41.1 Critically appraise the use and ethics of physical and chemical restraints in the context of patients with psychosis in the emergency care setting

Definitions

- **Physical restraint** is the use of manual force by staff to restrict a person's movement (for example holding the limbs).
- **Mechanical restraint** is the use of a device (for example soft limb restraints, straps, a spine board) to restrict movement. The CDST permits mechanical restraint **only on the instruction of SAPS** (HPCSA PBEC, 2025, p. 46).
- **Chemical restraint (rapid tranquillisation)** is the use of medication with the primary intent of calming a patient who is exhibiting acutely disturbed or severely agitated behaviour (RCEM, 2025). The RCEM prefers the term "rapid tranquillisation", noting that the intent is "to counteract excessive psychomotor stimulation at the earliest opportunity". Uribe and colleagues emphasise that many clinicians identify **"tranquillisation" (calming)** rather than complete sedation as the ideal target, considering complete sedation "as a last resource" (Saucedo Uribe et al., 2025).

Why patients with psychosis become agitated

Patients with acute psychosis may become agitated or aggressive because of persecutory delusions (for example the belief that the ambulance crew are kidnappers or that medication is poison), command hallucinations, fear, overstimulation, akathisia from antipsychotic medication, intoxication, or because they experience being restrained as an assault, which increases their fear (RCEM, 2025). Evans, Geduld and Stassen (2018) note that patients who have had previous negative experiences of police restraint often react poorly to it.

The case for restraint (benefits)

1. **Prevention of harm.** Restraint and sedation may be necessary to prevent the patient from seriously injuring themselves (for example running into traffic, jumping from a height or a moving vehicle) or others.
2. **Enabling assessment and life-saving treatment.** Sedation allows a primary survey, glucose measurement, IV access, monitoring and treatment of underlying conditions such as hypoglycaemia, head injury or overdose (HPCSA PBEC, 2025; RCEM, 2025).
3. **Preventing physiological collapse.** In severe ABD, "rapid tranquilisation… is important to prevent further sympathetic over-stimulation and excessive muscular activity from causing a metabolic storm and subsequent cardiovascular collapse", and its use "for more severe ABD presentations is associated with reduced mortality" (RCEM, 2025).
4. **Shortening the duration of physical restraint.** Prolonged physical restraint is itself dangerous; the RCEM (2025) states that "prolonged restraint should prompt consideration of rapid tranquilisation", and the CDST says restraint should be limited "to a brief period, only to facilitate sedation" (HPCSA PBEC, 2025, p. 46).
5. **Safe transport.** An agitated patient in a moving ambulance is a danger to themselves and the crew; the CDST frames sedation in terms of "voluntary sedation for the transfer" (HPCSA PBEC, 2025, p. 46).

The case against restraint (risks and harms)

Physical and mechanical restraint:

- **Death.** Deaths during or shortly after restraint have been reported, particularly in agitated patients with stimulant intoxication or ABD. The RCEM (2025) notes that "continued exertion under restraint can contribute to poor outcomes (likely due to increasing catecholamine levels, worsening hyperthermia, and metabolic acidosis)", and that "there is no high-quality data to suggest that there are risk-free methods to restrain an undifferentiated, potentially co-morbid patient with ABD". Although the theory that prone restraint causes positional asphyxia "has not been supported by recent research", it remains prudent "to ensure that there is no obstruction to ventilation and to minimise the risk of asphyxiation" — **never kneel on the chest, neck or back, and never restrain prone with weight on the torso**.
- Injuries to the patient (fractures, nerve injuries, skin damage, compartment syndrome, rhabdomyolysis, aspiration) and to staff.
- **Psychological trauma**, re-traumatisation (especially in people with histories of abuse), humiliation, loss of trust in health services, and future avoidance of care.
- **Masking of deterioration** — a restrained patient may be falsely assumed to be "calm" when in fact they are exhausted, hypoxic or in pre-arrest.

Chemical restraint:

- **Respiratory depression, airway obstruction and apnoea** — "most sedative agents for ABD have been associated with apnoea, airway obstruction, or a requirement for subsequent intubation", and "respiratory adverse events appear to be more common in patients sedated with benzodiazepines than antipsychotics" (RCEM, 2025). Both midazolam and lorazepam cause respiratory depression (Saucedo Uribe et al., 2025).
- **Hypotension**, particularly in volume-depleted patients and older persons.
- **Antipsychotic adverse effects**: acute dystonia (haloperidol caused significantly more dystonia than placebo and aripiprazole), akathisia, QT prolongation and torsade de pointes, neuroleptic malignant syndrome, and lowering of the seizure threshold (Saucedo Uribe et al., 2025; RCEM, 2025). Haloperidol and droperidol are contraindicated in Parkinson's disease and Lewy body dementia (RCEM, 2025).
- **Ketamine adverse effects**: emergence phenomena, hypersalivation, laryngospasm, vomiting, worsening tachycardia and hypertension (synergistic with stimulants), and high rates of intubation in prehospital series, which the RCEM attributes partly to the need to secure safe transport (RCEM, 2025).
- **Paradoxical agitation** with benzodiazepines, particularly in children, older persons and those with intellectual disability.
- **Loss of diagnostic information** — sedation obscures the neurological examination, the MSE and the evolving picture of an overdose or head injury.
- **Interaction with other substances** — alcohol and other CNS depressants potentiate benzodiazepines.

Evidence for the choice of drug in psychosis-related agitation

The umbrella review by Saucedo Uribe et al. (2025), which included 20 systematic reviews and meta-analyses covering 51,486 participants, reported that:

- loxapine 10 mg (inhaled) was more effective than 5 mg at reducing agitation within 120 minutes;
- aripiprazole was effective and caused less sedation than olanzapine, although haloperidol needed fewer additional doses;
- intramuscular ziprasidone had a faster onset and was better tolerated than haloperidol;
- lorazepam was effective, with fewer side effects than antipsychotics and greater efficacy when combined with them;
- midazolam provided rapid sedation but carried a risk of serious side effects, especially in older adults;
- droperidol was as effective as olanzapine but sedated faster;
- haloperidol plus promethazine reduced aggression effectively with fewer adverse effects. In one trial, haloperidol alone was stopped early because it was "considered too toxic" (ten cases of acute dystonia versus none in the combination group), and the authors concluded that "haloperidol used alone without something to compensate for its frequent and serious adverse effects seems difficult to justify".

The RCEM (2025) recommends **IM ketamine (4 mg/kg) or IM droperidol (5–10 mg)** as first-line agents for severe ABD in UK emergency departments, with midazolam (5–10 mg IM), lorazepam (4 mg IM) or haloperidol (5 mg IM ± lorazepam 2 mg IM) as alternatives. Midazolam may be preferred in patients at risk from sympathomimetic effects (for example ischaemic heart disease or possible cocaine-induced psychosis).

What the South African CDST directs (prehospital practice)

The CDST "Aggression or agitation" tool (HPCSA PBEC, 2025, pp. 46–47) directs the following:

If the patient remains aggressive or severely agitated despite verbal de-escalation, the patient "should be sedated". If the patient is willing to accept voluntary sedation (adult):

- no IV access: *midazolam 7.5 mg orally, **or** midazolam 10 mg buccal/intranasal, **or** diazepam 10 mg PO, **or** lorazepam 2 mg PO*;
- IV access: *midazolam 2 mg IV (may repeat in 15 minutes)*.

If the patient refuses sedation and remains dangerous: restrain physically and prepare to sedate; explain to the patient what is about to happen; remove any weapon or mechanism for self-harm; restrain with **five people (one for each limb and one for the head) and a sixth person to administer medication**; limit restraint to a brief period, only to facilitate sedation; and apply mechanical restraint only on the instruction of SAPS. Then **give rapid sedation**:

- *midazolam 0.2 mg/kg IM (maximum 15 mg), undiluted; if older than 60 years, give 0.1 mg/kg*;

- if midazolam is not available: *lorazepam 4 mg IM* **or** ^ketamine 4 mg/kg IM (for more rapid onset)^ (ANT/ECP only).

Review after 15 minutes (IV) or 30 minutes (IM/PO). If the patient is still aggressive or severely agitated, give a further rapid sedation dose: *midazolam 0.1 mg/kg IM (maximum 5 mg); if older than 60 years, 0.05 mg/kg*; if midazolam is not available, lorazepam 2 mg IM.

Children (<13 years): use the dose charts. The CDST drug appendix gives midazolam for aggression in children as 0.2 mg/kg buccal/IN (maximum 10 mg per dose), 0.15 mg/kg IM (maximum total dose 10 mg), 0.05 mg/kg IV (maximum total 0.5 mg/kg), or 0.5 mg/kg PO (maximum total 20 mg); lorazepam 0.1 mg/kg (IM, or PO maximum 4 mg); diazepam 0.2 mg/kg PO (maximum 10 mg per dose); and ketamine 4–5 mg/kg IM (ANT/ECP) (HPCSA PBEC, 2025, pp. 69–72).

Note that the CDST does **not** include haloperidol or droperidol for aggression; the prehospital agents in South Africa are benzodiazepines (midazolam, lorazepam, diazepam) and, for ANT/ECP, ketamine. This reflects the prehospital priorities of a rapid onset, availability, the IM route, and the fact that an undifferentiated prehospital patient may have a medical or toxicological cause for which antipsychotics may be harmful (for example anticholinergic toxicity, alcohol withdrawal, or seizures).

Ethical analysis

The use of restraint involves a conflict between the four principles of biomedical ethics:

- **Autonomy**: restraint overrides the patient's freedom and choices. It can be ethically justified only where the patient **lacks capacity** because of their mental state (for example severe psychosis), or where the risk to others is such that the law permits intervention.
- **Beneficence**: restraint may be the only way to deliver life-saving assessment and treatment and to prevent self-harm.
- **Non-maleficence**: restraint carries significant physical and psychological risks; the harm of restraint must be proportionate to, and less than, the harm it prevents.
- **Justice**: restraint must never be used because of discrimination, prejudice, convenience, punishment, or staff frustration. Persons with mental illness are vulnerable to stigma and to being restrained more readily than others; the MHCA prohibits unfair discrimination (s 10) and requires protection from exploitation, abuse and degrading treatment (s 11); in addition, the MHCA General Regulations (GN R1467 of 2004) regulate the use of mechanical restraint and seclusion of mental health care users in health establishments, so that these may be used only when necessary to prevent harm and must be recorded.

Ethical conditions for restraint, which you should state in an examination answer, are therefore that restraint must be:

1. **Necessary** — there is an imminent risk of serious harm.
2. **A last resort** — less restrictive measures have been tried and have failed, or are clearly unsafe.

3. **Proportionate** — the least restrictive effective method, for the shortest possible time.
4. **Lawful** — within the MHCA, NHA, common law and the provider's scope of practice (CDST).
5. **Performed safely** — by enough trained people, with no pressure on the neck, chest or abdomen, with continuous monitoring.
6. **Never punitive** and always respectful of dignity.
7. **Documented** fully, and followed by debriefing.
8. **Reviewed continuously** — removed as soon as it is no longer necessary.

Critical appraisal — conclusion

Physical and chemical restraint are high-risk interventions with a weak and largely extrapolated evidence base, particularly in the prehospital environment and in older adults (Saucedo Uribe et al., 2025; Desai, Kulkarni and Verma, n.d.). Yet in severe agitation the risks of **not** intervening (injury, ABD-related cardiac arrest, missed hypoglycaemia or head injury, suicide) may be greater than the risks of intervening. Restraint is therefore ethically defensible only when it is necessary, proportionate, lawful, applied for the shortest possible time, and accompanied by continuous monitoring and the capacity to manage complications such as airway compromise. South African evidence shows that prehospital providers lack training in the MHCA and in the management of mental health care users, rely on deception, and receive inadequate support from SAPS and their services (Evans, Geduld and Stassen, 2018; Stander, Hodkinson and Dippenaar, 2021); improved training, clear national protocols such as the CDST, and dedicated response teams are needed to make restraint both safer and more ethical.

41.2 Discuss the social, ethical and legal implications of management in the emergency care setting

Social implications

1. **Stigma.** Mental illness and suicide carry strong social stigma in South Africa. Stigma discourages help-seeking, leads to refusal of treatment, and may lead families to conceal illness (Evans, Geduld and Stassen, 2018). Arrival of an ambulance and SAPS, and visible restraint in front of neighbours, can increase stigma and shame.
2. **Provider attitudes.** Negative attitudes among providers ("waste of time", "attention seeking") are documented in South Africa (Evans, Geduld and Stassen, 2018) and are associated with reduced commitment to transporting patients to definitive care. Tilley et al. (2026) call for further research into "EC provider stigmatisation towards death by suicide and suicidality".
3. **Social determinants.** Poverty, unemployment, violence, substance abuse and the legacy of apartheid contribute to mental illness and suicide, and limit access to care (Tilley et al., 2026). Many patients cannot afford transport to follow-up appointments or to collect medication.
4. **The mental health treatment gap and deinstitutionalisation.** Deinstitutionalisation under the MHCA was not matched by adequate community services, so EMS has become a de facto primary point of access for mental health care (Tilley et al., 2026). Long waiting times and bed shortages at facilities discourage providers from transporting patients (Evans, Geduld and Stassen, 2018).

5. **Cultural and religious beliefs.** Beliefs about causation (for example witchcraft or ancestral displeasure) influence help-seeking, and families may first consult traditional healers or religious leaders. Respectful exploration of these beliefs improves engagement.
6. **Impact on families and communities.** Families may be exhausted, frightened, or themselves victims of violence by the patient. Conversely, they are often the patient's main support system.
7. **Impact on providers.** Behavioural emergencies are physically dangerous and emotionally taxing; providers describe fear, frustration and feeling drained (Stander, Hodkinson and Dippenaar, 2021; Evans, Geduld and Stassen, 2018), and are at risk of PTSD, depression and burnout (Tilley et al., 2026).

Ethical implications

1. **Autonomy versus beneficence.** The central dilemma: respecting a patient's refusal versus acting to prevent harm. The HPCSA (2023, para 8.1.2) requires that a competent adult's refusal be respected "even where this would result in serious harm to them or in their own death unless the patient's state of mind is such that he/she is incapable of taking or participating in the decision". In behavioural emergencies the patient's state of mind is frequently the very issue in question, so a careful, documented capacity assessment is essential.
2. **Non-maleficence.** Restraint, sedation and involuntary transport all carry the potential for harm and must be proportionate.
3. **Justice and non-discrimination.** Persons with mental illness must receive the same standard of care as other patients. The HPCSA (2023, para 2.5) states that practitioners "must not allow their views about, for example, a patient's age, disability, race, colour, culture, beliefs, sexuality, gender, lifestyle, social or economic status or other irrational grounds to prejudice the choices of treatment offered or the general standard of care provided".
4. **Confidentiality versus duty to warn and report.** Information about mental health is particularly sensitive (MHCA, s 13; POPIA). Disclosure is permitted where required by law (for example the duty to report sexual offences against vulnerable persons under s 54 of the SOA, the duty to report child abuse, or the duty to report domestic violence against children, older persons and persons with disabilities under the DVA) or where necessary to protect others from serious harm.
5. **Truth-telling.** Deceiving patients to secure transport (Stander, Hodkinson and Dippenaar, 2021) violates the principle of respect for persons and damages trust.
6. **Dual loyalty.** Providers may feel caught between the patient's interests and the demands of SAPS, family, or their employer (for example pressure to clear the scene quickly).
7. **The HPCSA ethical tension in suicidal patients** (see 45.1): the HPCSA Booklet 7 is written for end-of-life decisions in terminal illness. Its emphasis on patient autonomy, advance directives and best interests must not be misapplied to a suicidal patient, whose wish to die is usually a symptom of a treatable mental disorder and whose capacity is often impaired.

Legal implications

1. **Constitutional rights**: dignity (s 10), freedom and security of the person (s 12), access to health care and the right not to be refused emergency medical treatment (s 27), and children's rights (s 28) (Constitution of the Republic of South Africa, 1996).
2. **Mental Health Care Act 17 of 2002** (see Outcome 50): emergency care without consent (s 9(1)(c)), the duty to report to the Review Board and the 24-hour limit (s 9(2)), voluntary, assisted and involuntary care (ss 25–38), and SAPS intervention (s 40).
3. **National Health Act 61 of 2003**: informed consent (s 7), the right to participate in decisions (s 8), the duty to inform the patient (s 6), confidentiality (s 14), and the right to refuse treatment.
4. **Criminal and civil liability**: treating a competent patient without consent may constitute assault, and restraining or transporting a person without lawful authority may constitute unlawful detention (deprivation of liberty) or kidnapping — which is why some providers wrongly believe that they can never transport without consent (Evans, Geduld and Stassen, 2018). Conversely, **leaving a patient who lacks capacity and is at risk** may constitute negligence, and a resulting death could lead to civil claims, criminal charges (culpable homicide) and HPCSA disciplinary action. The legal standard is that of the reasonable practitioner in the same circumstances.
5. **Scope of practice**: drugs and procedures must be within the provider's registered scope (CDST; HPCSA PBEC, 2025).
6. **Documentation as legal evidence**: the PRF is a legal document and may be used in court, in HPCSA inquiries and in Review Board processes.
7. **Mandatory reporting duties** (SOA s 54; Children's Act s 110; DVA ss 2A and 2B; Older Persons Act 13 of 2006).

High-yield points — Outcome 41

- Ten principles: safety, organic cause first, least restrictive, dignity, capacity and law, obligation to transport, monitoring, destination, documentation, teamwork and welfare.
- CDST adult rapid sedation: **midazolam 0.2 mg/kg IM (max 15 mg; 0.1 mg/kg if >60 years)**; if unavailable lorazepam 4 mg IM or ^ketamine 4 mg/kg IM^; repeat midazolam 0.1 mg/kg IM (max 5 mg) after 30 minutes if needed.
- Voluntary sedation options: midazolam 7.5 mg PO, 10 mg buccal/IN, diazepam 10 mg PO, lorazepam 2 mg PO, midazolam 2 mg IV.
- Restraint: five people plus one, brief, only to facilitate sedation; mechanical restraint only on SAPS instruction; check distal pulses.
- Monitor every 10 minutes after sedation.
- Restraint is ethical only if necessary, a last resort, proportionate, lawful, safe, non-punitive, documented and reviewed.

Outcome 42. Differentiate key symptoms and management techniques for selected behavioural and psychiatric disorders

This outcome requires you to distinguish the common behavioural presentations from one another and to match each to the appropriate prehospital management. The detailed DSM-5-TR descriptions are in Outcome 52; the table below is a comparative, field-oriented summary for South African prehospital practice.

Presentation	Key differentiating symptoms and signs	Key prehospital management
Panic attack / acute anxiety	Abrupt surge of intense fear peaking within minutes; palpitations, sweating, trembling, dyspnoea, chest pain, choking sensation, dizziness, paraesthesia (often perioral and carpopedal from hyperventilation), fear of dying or "going crazy"; oriented; normal SpO_2	**Exclude organic causes first** (ACS, PE, asthma, hypoglycaemia, thyrotoxicosis, arrhythmia, stimulant use) with vital signs, SpO_2, glucose and a 12-lead ECG. Calm reassurance, remove from the stressor, coach slow breathing. **Do not use paper-bag rebreathing** (risk of hypoxia, especially if the cause is actually PE or ACS). Benzodiazepines are rarely needed. Transport if any doubt about the cause.
Depression with suicidal ideation	Low mood, anhedonia, hopelessness, psychomotor retardation, poor eye contact, sleep and appetite change, guilt, suicidal thoughts	Direct suicide risk assessment (Outcome 45); remove means; never leave the patient alone; transport (MHCA s 9(1)(c) if at risk and refusing); search for and treat any overdose or self-injury.
Mania	Elevated, expansive or irritable mood; pressured speech; flight of ideas; grandiosity; decreased need for sleep; distractibility; disinhibition, reckless spending or sexual behaviour; poor insight	Low-stimulus environment, firm brief communication, avoid arguing, set limits; exclude stimulant intoxication and thyrotoxicosis; sedation per CDST if dangerous; transport.
Acute psychosis (schizophrenia spectrum)	Hallucinations (usually auditory), delusions (persecutory, reference, control), disorganised speech	Safety, de-escalation, do not argue with delusions, ask specifically about **command hallucinations** and violent

Presentation	Key differentiating symptoms and signs	Key prehospital management
	and behaviour, often clear consciousness and preserved orientation; poor insight	or suicidal intent; exclude organic causes (glucose, head injury, drugs, infection); CDST sedation if dangerous; transport under MHCA provisions if lacking capacity and at risk.
Delirium (see 52.9.1)	**Acute onset, fluctuating course, inattention**, disorientation, altered level of consciousness (hyperactive, hypoactive or mixed); visual hallucinations, illusions; often older patients with infection, hypoxia, medication changes, or withdrawal	This is a **medical emergency**. ABCDE, oxygen if SpO_2 <92%, glucose, temperature, look for infection, trauma, toxins; reorientation, family presence, glasses and hearing aids; avoid benzodiazepines unless withdrawal or seizures; use the lowest effective sedative dose (0.1 mg/kg midazolam IM if >60 years in the CDST) only if dangerous; transport to an emergency centre.
Dementia with behavioural disturbance	Chronic, progressive memory loss with intact consciousness; agitation often triggered by pain, unfamiliar surroundings, constipation, urinary retention, infection	Treat triggers (pain, retention, fear), calm approach, involve a familiar caregiver; consider superimposed delirium; minimal sedation.
Acute stress reaction / situational crisis	Recent traumatic event, distress, crying, numbness, anger, dissociation, oriented	Psychological first aid: safety, calm, connect with support, practical help, information; screen for injuries and for suicidal ideation; refer.
Alcohol or drug intoxication	Smell of alcohol, slurred speech, ataxia, nystagmus, toxidrome features (see 62.3), history of use	ABCDE, airway protection, glucose, thiamine 100 mg IV/IM if malnourished or chronic alcohol use (CDST); look for head injury and co-ingestion; treat the toxidrome; do not assume intoxication explains everything.

Presentation	Key differentiating symptoms and signs	Key prehospital management
Alcohol withdrawal / delirium tremens	6–24 h after last drink: tremor, sweating, anxiety, tachycardia, hypertension; 12–48 h seizures; 48–96 h delirium tremens with visual and tactile hallucinations, disorientation, hyperthermia	Benzodiazepines (the agents of choice for withdrawal), thiamine, glucose, fluids, cooling, seizure management per the CDST seizure tool.
Conversion (functional neurological) disorder	Neurological symptoms (paralysis, blindness, seizures, unresponsiveness) inconsistent with known neurological disease, often after stress	Treat as genuine until organic disease (stroke, seizure, hypoglycaemia) has been excluded; do not use painful stimuli to "prove" the patient is pretending; calm reassurance; transport.
Catatonia (see 52.3)	Stupor, mutism, negativism, posturing, waxy flexibility, staring, echolalia	Medical emergency (risk of dehydration, aspiration, VTE, malignant catatonia); ABCDE, glucose, temperature; transport; benzodiazepines (lorazepam) are the specific treatment in hospital; avoid antipsychotics until NMS is excluded.
Personality disorder crisis (for example borderline)	Intense, unstable emotions, impulsivity, self-harm, fear of abandonment, anger	Calm, consistent, non-judgemental; set clear limits; assess and treat self-harm; take every threat of suicide seriously.
Acute behavioural disturbance (ABD)	Extreme agitation, unusual strength, pain insensitivity, hyperthermia, sweating, tachycardia, tachypnoea, sustained non-compliance; usually stimulants or psychosis	**Life-threatening.** Early rapid sedation (CDST), minimise duration of physical struggle and restraint, never compress the chest or neck, cool, high-flow oxygen, IV fluids, continuous monitoring, rapid transport; be prepared for cardiac arrest (RCEM, 2025).

High-yield points — Outcome 42

- Delirium versus dementia versus psychosis: **delirium = acute, fluctuating, inattention, altered consciousness**; dementia = chronic, progressive, clear consciousness; psychosis = clear consciousness and usually preserved orientation.

- Panic attack is a diagnosis of exclusion in the field — always do vital signs, a glucose and an ECG.
- Benzodiazepines are the agents of choice for alcohol/sedative withdrawal and stimulant toxicity.

Outcome 43. Discuss the management of violent patients in the emergency care setting

43.1 Describe the assessment of a potentially violent patient

Predicting violence

The single best predictor of future violence is a **history of past violence**. Other risk factors include: male sex; young age; substance intoxication or withdrawal; acute psychosis (particularly persecutory delusions and command hallucinations); mania; certain personality disorders (antisocial, borderline); organic brain disease (head injury, delirium, dementia, intellectual disability); access to weapons; a history of being a victim of abuse; and situational factors such as being restrained, being in pain, long waits, crowding, or feeling disrespected (Tintinalli et al., 2020; NAEMT, 2021).

Warning signs of impending violence (escalation)

The ECP must recognise the behavioural cues that violence may be imminent:

- **Posture and movement**: standing rather than sitting, pacing, clenched fists or jaw, tense muscles, rapid movements, invading personal space, kicking or throwing objects, gripping furniture, adopting a fighting stance.
- **Speech**: loud, rapid, threatening, abusive, profane or sarcastic speech, or sudden silence.
- **Facial expression and eyes**: staring, scanning the room, glaring, flushed face, or avoiding eye contact.
- **Intoxication cues**: smell of alcohol, slurred speech, dilated pupils, agitation.
- **Verbal threats**: any threat should be taken seriously.
- **Holding or reaching for a potential weapon.**

Structured assessment

1. **Dispatch information and scene size-up**: nature of the call, history of violence at the address, presence of weapons, need for SAPS.
2. **Collateral history before approaching**: the CDST advises that the provider "obtain a brief history from the family/friends, if possible, before you approach the patient" (HPCSA PBEC, 2025, p. 46).
3. **Observation from a safe distance** of behaviour, posture, speech and evidence of intoxication or injury.
4. **Agitation severity scoring**. The **Sedation Assessment Tool (SAT)** (Calver, Stokes and Isbister, 2011), included in the RCEM (2025) guideline, scores both responsiveness and speech:

Score	Responsiveness	Speech
+3	Combative, violent, out of control	Continual loud outbursts
+2	Very anxious and agitated	Loud outbursts
+1	Anxious, restless	Normal/talkative

Score	Responsiveness	Speech
0	Awake and calm, cooperative	Speaks normally
–1	Asleep but rouses if name is called	Slurring or prominent slowing
–2	Responds to physical stimulation	Few recognisable words
–3	No response to stimulation	None

The RCEM (2025) suggests that the most potent parenteral strategies be reserved for severe agitation with an SAT score of +2 or +3.

5. **Medical assessment** once safe: primary survey, vital signs including temperature and glucose, pupils, signs of head injury, toxidromes, signs of infection (fever, neck stiffness) (HPCSA PBEC, 2025, p. 47). The Tutorial 1 memo summarises the important differential diagnoses in a violent patient with altered mental status as intoxication (alcohol, cannabis, amphetamines, sedatives), withdrawal (especially alcohol withdrawal), hypoglycaemia, hypoxia, infection (sepsis, meningitis), head injury and acute psychiatric crisis.
6. **Capacity and risk assessment** (to self and others).

43.2 Evaluate measures that may be used to try to safely defuse a potentially violent situation

Verbal de-escalation

Verbal de-escalation is the first-line, least restrictive intervention, and the RCEM (2025) emphasises that it "should always be attempted", even if it "may feel futile", because it is "an important step in ensuring that the use of restraint and rapid tranquilisation are justified" and "is a continuous process". The RCEM (2025) model, adapted from the American Association for Emergency Psychiatry Project BETA consensus (Richmond et al., 2012), has ten domains:

1. **Respect personal space** — identify exits; stay out of arm's reach.
2. **Do not be provocative** — non-confrontational body language; keep hands visible; do not challenge, insult or argue.
3. **Establish verbal contact** — only one staff member should talk to the patient; introduce yourself; explain why you are there; reassure the patient that your aim is to keep them safe.
4. **Be concise** — short sentences; give the patient time to respond; repeat as needed.
5. **Identify wants and feelings** — identify expectations; empathise.
6. **Listen closely** — use clarifying statements.
7. **Agree, or agree to disagree** — consider "fogging" (agree with the truth, agree in principle, or agree with the odds).
8. **Set clear limits** — state limits as matter-of-fact information, not as a threat.
9. **Offer choices and optimism** — offer acts of kindness (for example a blanket, a drink if sedation is not planned), and offer oral sedative medication.

10. **Debrief the patient and staff** — explain why any intervention was necessary, restore the therapeutic relationship, and identify potential improvements.

The CDST (HPCSA PBEC, 2025, p. 46) gives a practical South African summary:

Try to	Avoid
Approach the patient from a safe distance	Turning your back or allowing yourself to be cornered
Talk in a quiet, calm area if possible	Making direct eye contact
Listen to the patient's concerns	Touching the patient unnecessarily
Be calm and reassuring	Arguing or confronting any delusions
Show kindness, respect and empathy	Attempting to reason with the patient
Set clear boundaries regarding the interaction	Engaging in prolonged conversation

Environmental de-escalation

Features of an ideal environment for de-escalation (RCEM, 2025) include a quiet, low-stimulus space that is not too warm; exits located so that staff cannot be trapped; doors that open outwards; the absence of furniture or objects that could be used as weapons or to barricade an exit; no ligature points; constant observability; and an easy way for staff to summon help. In the prehospital setting this translates into: turning off sirens, radios and televisions; moving bystanders away; removing potential weapons; keeping the ambulance door accessible; and positioning so that both the provider and the patient have an exit. If sedation or anaesthesia is likely, food or drink should not be offered (RCEM, 2025).

Other measures

- **Show of concern, not show of force**: a large number of staff and SAPS members visible together may either deter violence or escalate fear; their presence should be calm and in the background until needed.
- **Involving trusted family or friends** if they calm the patient (RCEM, 2025).
- **Treating reversible discomforts** — pain, hunger, thirst, cold, a full bladder, fear (Desai, Kulkarni and Verma, n.d.).
- **Offering voluntary oral or intranasal sedation** (CDST).
- **Tactical withdrawal**: if the situation becomes unsafe, withdraw to safety, request SAPS assistance, and maintain observation from a safe distance where possible.

Evaluation

The strength of verbal de-escalation is that it is non-invasive, preserves dignity and the therapeutic relationship, preserves diagnostic information, and can avoid restraint and sedation entirely. Its limitations are that it requires time and training, is less effective in patients who are severely intoxicated, delirious or in ABD (who may be unable to process communication), and may delay necessary treatment in a physiologically unstable patient. In severe ABD, the RCEM (2025) supports early sedation. The evidence base for de-escalation

consists mainly of expert consensus rather than randomised trials, but it is universally recommended (Richmond et al., 2012; RCEM, 2025).

43.3 Formulate protocols for the use of patient restraint (physical/chemical), taking into account legal and ethical considerations

The protocol below is formulated for South African prehospital practice, based on the CDST (HPCSA PBEC, 2025), the RCEM (2025) guideline and the legal framework.

Protocol: Restraint and rapid sedation of the violent or severely agitated patient (adult)

Step 1 — Indications (all must be present):

- the patient is a danger to themselves or others, or is causing serious damage to property, or urgently needs assessment or treatment that cannot otherwise be delivered;
- verbal and environmental de-escalation have failed or are unsafe;
- voluntary oral, buccal or intranasal sedation has been offered and refused, or is inappropriate;
- the patient lacks capacity to refuse **or** the MHCA s 9(1)(c) / NHA s 7(1)(e) emergency criteria are met.

Step 2 — Preparation (safety brief):

- request SAPS assistance (the CDST says that "if restraint is considered, SAPS to be contacted");
- assign roles: **five people for restraint (one per limb and one for the head), and a sixth to administer medication** (HPCSA PBEC, 2025, p. 46); one team leader directs and talks to the patient; ideally one "safety person" monitors the patient's airway and breathing and is not involved in the restraint (RCEM, 2025);
- prepare drugs (correct concentration for IM use), airway equipment (suction, BVM, airway adjuncts, oxygen), monitoring (SpO_2, ECG, BP, $ETCO_2$) and the glucometer;
- explain to the patient what is about to happen (HPCSA PBEC, 2025, p. 46).

Step 3 — Physical restraint technique:

- approach together on a pre-agreed signal;
- control the large joints (shoulders, elbows, hips, knees) rather than the hands and feet;
- place the patient **supine** (or in the lateral position if vomiting is likely), **not prone**; if prone positioning cannot be avoided momentarily, apply **no weight to the back, chest or neck**, and turn the patient supine or lateral as soon as possible;
- never apply pressure to the neck, chest or abdomen; never use choke holds or "head-locks" (Evans, Geduld and Stassen, 2018, reported providers using a head-lock, which is dangerous and inappropriate);
- limit physical restraint to a brief period, only to facilitate sedation (HPCSA PBEC, 2025, p. 46);
- apply mechanical restraint only on SAPS instruction (HPCSA PBEC, 2025, p. 46), using padded devices, never handcuffs applied by health care workers, and never "hog-tying".

Step 4 — Rapid sedation (CDST):

- adult: *midazolam 0.2 mg/kg IM, undiluted (maximum 15 mg); **0.1 mg/kg if older than 60 years***;
- if midazolam is unavailable: *lorazepam 4 mg IM* **or** ^ketamine 4 mg/kg IM^ (ANT/ECP) for a more rapid onset;
- children <13 years: use the CDST dose charts (for example midazolam 0.15 mg/kg IM, maximum total 10 mg);
- IM is preferred in the actively restrained patient, because attempting IV cannulation during restraint risks needle-stick injury (RCEM, 2025).

Step 5 — Immediately after sedation:

- rapid primary survey: airway (manage if the patient cannot maintain the airway, SpO_2 <92% or breathing very slow or gasping), oxygen if SpO_2 <92%, BP (manage shock if <90/60), **glucose (treat if <3.5 mmol/L)** (HPCSA PBEC, 2025, p. 47);
- place the patient in the recovery (lateral) position if the airway is at risk;
- obtain IV access once contained;
- check and document distal pulses in all restrained limbs;
- look for and manage injury (especially head injury) and possible poisoning.

Step 6 — Reassessment and repeat dosing:

- review after 15 minutes (IV) or 30 minutes (IM/PO); if still severely agitated, give *midazolam 0.1 mg/kg IM (maximum 5 mg; 0.05 mg/kg if older than 60 years)* or lorazepam 2 mg IM (HPCSA PBEC, 2025, p. 47);
- avoid "stacking" doses before the previous dose has had time to act (Desai, Kulkarni and Verma, n.d.);
- if sedation fails after the full doses, request ALS/ECP or medical advice and consider whether the patient needs advanced airway management.

Step 7 — Monitoring:

- BP, HR, RR, SpO_2 and GCS every 10 minutes; reassess immediately if BP <92 systolic, HR >100 or <50, RR >20 or <9, SpO_2 <92% or GCS falls (HPCSA PBEC, 2025, p. 47);
- continuous ECG and, where available, $ETCO_2$ (RCEM, 2025);
- monitor temperature (hyperthermia in ABD).

Step 8 — Transport: to the nearest or most appropriate emergency centre, with a family member if possible, with SAPS escort where needed (HPCSA PBEC, 2025, p. 48).

Step 9 — Documentation (HPCSA PBEC, 2025, p. 47; RCEM, 2025): indication for restraint and sedation; de-escalation attempts; capacity assessment and the legal basis; number of people, method, start and end times of restraint; SAPS involvement and any use of force or Taser; drugs, doses, routes and times; response and adverse events; pulse checks; vital signs; handover details.

Step 10 — Debrief the patient (once calm), the crew and SAPS; report any injury or adverse event through the service's incident reporting system.

Legal and ethical safeguards built into the protocol

- **Lawful authority**: patient consent, MHCA s 9(1)(c) (followed by the receiving facility's report to the Review Board and the 24-hour limit in s 9(2)), NHA s 7(1)(e), or common-law necessity; SAPS authority under MHCA s 40.
- **Scope of practice**: only CDST/HPCSA-approved drugs within the provider's registration category.
- **Proportionality and least restriction** at every step.
- **Dignity and privacy** (MHCA s 8; Constitution s 10).
- **No punishment or convenience restraint** (MHCA s 11).
- **Contemporaneous documentation** (HPCSA, 2023, para 12).

43.4 Describe the safety measures that should be taken when patient violence is anticipated

1. **Before arrival**: obtain as much information as possible from dispatch; request SAPS early; **stage** at a safe distance until SAPS confirms that the scene is safe if there is any report of a weapon, violence or a hostile crowd; plan an approach and an escape route; agree on a code word with your partner to signal danger and the need to withdraw.
2. **On arrival**: park the ambulance facing the exit route, out of the line of sight of the house where possible, and do not block yourself in; scan for hazards (weapons, dogs, crowds, intoxicated bystanders); approach cautiously, standing to the side of doors rather than in front.
3. **Personal protective measures**: wear appropriate PPE (gloves; eye protection, because patients may spit; consider a mask); remove items that could be grabbed or used as a weapon (stethoscope around the neck, ties, lanyards, scissors, pens in breast pockets); tie back long hair.
4. **Positioning**: two arm-lengths away; at an angle to the patient; with a clear exit for both you and the patient; never turn your back; never allow yourself to be cornered (HPCSA PBEC, 2025, p. 46).
5. **Teamwork**: never enter alone; one person talks, the partner observes and maintains radio contact; establish clear roles.
6. **Remove weapons and means of self-harm** (HPCSA PBEC, 2025, p. 46); SAPS should search the patient for weapons before the patient enters the ambulance.
7. **In the ambulance**: the patient should be secured on the stretcher (sedated and/or restrained if required); the crew member in the back should sit nearest the door, with clear access to the exit; remove sharp objects and loose equipment; lock the side and rear doors from the inside where the design allows it, to prevent the patient jumping from a moving vehicle (Evans, Geduld and Stassen, 2018, reported a patient jumping from a moving ambulance); have a SAPS member travel in the back for high-risk patients; keep the driver informed and maintain radio contact with control.
8. **Withdraw if unsafe**: personal safety takes priority; withdraw, call for SAPS and maintain observation from a safe place.
9. **After an incident**: seek medical attention for injuries (including post-exposure prophylaxis for blood-borne virus exposure from bites or spitting), report the incident,

document fully, lay a criminal charge if appropriate, and access debriefing and counselling (Evans, Geduld and Stassen, 2018, found such support lacking).

High-yield points — Outcome 43

- Best predictor of violence: past violence. Watch for escalation cues.
- RCEM/Project BETA ten domains of de-escalation; CDST "Try to / Avoid" list.
- Protocol: indications → preparation → restraint (5 + 1, brief, supine) → sedation (CDST midazolam 0.2 mg/kg IM) → primary survey and glucose → monitor every 10 minutes → transport → document → debrief.
- Mechanical restraint only on SAPS instruction; check distal pulses.
- Stage until SAPS secures the scene; never enter alone; secure the ambulance doors.

Outcome 44. Critically appraise the role and practice of emergency care providers in the context of violent patients

The role of the ECP

The ECP's role with a violent patient has several dimensions:

1. **Clinical**: to identify and treat the underlying cause (medical, toxicological, traumatic or psychiatric), to protect the patient's airway and physiology during agitation and after sedation, and to transport the patient to definitive care. The CDST is explicit that the ECP should "not assume aggression/agitation is due to a psychiatric condition" (HPCSA PBEC, 2025, p. 47).
2. **Safety**: to protect the patient, the public and the crew.
3. **Legal**: to act within the MHCA, the NHA and the provider's scope of practice; to understand when transport without consent is permitted and obligatory; and to document appropriately.
4. **Advocacy**: to protect the dignity and rights of a vulnerable person, including against inappropriate use of force by others.
5. **Leadership and clinical supervision**: as the highest-qualified prehospital practitioner, the ECP is frequently called by BLS and ILS crews for sedation and decision support. Stander, Hodkinson and Dippenaar (2021) found that BLS providers viewed themselves as "an extra set of hands", ILS providers felt they could try to defuse situations but could not sedate, and ALS providers felt that their main expected role was sedation and that they were sometimes called "prematurely before sedation was clinically indicated". Evans, Geduld and Stassen (2018) found that providers frequently sought to shift decision-making responsibility to a more senior role player.
6. **Liaison**: with SAPS, family, the receiving facility and mental health services.

Critical appraisal of current practice in South Africa

Strengths

- The ECP is uniquely placed as the first health professional on scene, often the only one, and can interrupt suicidality and limit access to methods of harm (Tilley et al., 2026; Evans, Geduld and Stassen, 2018).
- ECPs have advanced skills, including IM, IV and IN sedation, advanced airway management, and the clinical reasoning to recognise organic causes.
- The CDST (HPCSA PBEC, 2025) now provides a clear, national, stepwise algorithm for aggression and agitation, specifying de-escalation, voluntary sedation, restraint, rapid sedation, monitoring, documentation and the obligation to transport.

Weaknesses and challenges

1. **Inadequate training.** Eighty per cent of surveyed Cape Town providers had no training in managing psychiatric patients, and only 7% had specific training in the MHCA (Evans, Geduld and Stassen, 2018). There was no association between qualification level and training, which means that higher qualifications did not guarantee competence in this area.

2. **Legal confusion.** Many providers wrongly believed that patients could never be transported without consent, regarded involuntary transport as "a form of kidnap", and relied on refusal forms as legal protection (Evans, Geduld and Stassen, 2018). The MHCA assigns involuntary transport to SAPS (s 40) and does not clearly address the role of EMS; the HPCSA and MHCA have provided no explicit guidance on how involuntary care should be provided prehospitally (Stander, Hodkinson and Dippenaar, 2021). The CDST's "obligated to transport" criteria (HPCSA PBEC, 2025, p. 48) go some way to address this.
3. **Negative attitudes and stigma**, with providers describing these patients as a "waste of time" (Evans, Geduld and Stassen, 2018), which risks substandard care.
4. **Use of deception** to obtain cooperation (Stander, Hodkinson and Dippenaar, 2021).
5. **Unsafe restraint practices**, including head-locks and Taser use reported by providers (Evans, Geduld and Stassen, 2018).
6. **Inappropriate tools**, for example the use of the GCS to assess mental capacity (Stander, Hodkinson and Dippenaar, 2021).
7. **Risks of chemical restraint in the field**: limited monitoring in a moving vehicle, limited staff, and no immediate back-up for airway complications; the RCEM (2025) notes that high rates of intubation after ketamine are "predominantly seen in pre-hospital rather than ED settings".
8. **System factors**: long waiting times and bed shortages at facilities, performance pressure (providers feeling "penalised" for time spent), poor dispatch information, difficult relationships with SAPS, and limited community mental health services (Evans, Geduld and Stassen, 2018; Stander, Hodkinson and Dippenaar, 2021; Tilley et al., 2026).
9. **Staff safety and welfare**: 56% of surveyed South African EMS personnel had been assaulted on duty (Evans, Geduld and Stassen, 2018), and support after traumatic incidents is lacking.

Recommendations

- Mandatory undergraduate and continuing education in the MHCA, capacity assessment, verbal de-escalation, suicide risk assessment and safe restraint (Evans, Geduld and Stassen, 2018).
- National prehospital guidelines on involuntary transport and refusal by mental health care users (now partly addressed by the CDST).
- Dedicated mental health response teams, joint EMS–SAPS training and protocols (Evans, Geduld and Stassen, 2018).
- Trauma-informed care and a syndemic understanding of suicidality (Tilley et al., 2026).
- Structured debriefing and psychological support for staff.
- Audit of restraint and sedation events.

Conclusion

The ECP plays a pivotal clinical, legal and ethical role in managing violent patients. Current South African practice is undermined by gaps in training and legal knowledge, negative attitudes, unsafe practices and system pressures. Safe and ethical practice requires the ECP to prioritise safety, look for organic causes, use the least restrictive effective intervention, follow

the CDST sedation pathway, act within the law, monitor closely, transport when obligated, document thoroughly, and advocate for the dignity of a highly vulnerable patient group.

High-yield points — Outcome 44

- Roles: clinical, safety, legal, advocacy, leadership, liaison.
- South African evidence: 80% no psychiatric training, 7% MHCA training (Evans, Geduld and Stassen, 2018); deception, use of GCS for capacity, poor SAPS support (Stander, Hodkinson and Dippenaar, 2021).
- Recommendations: training, national guidelines, dedicated teams, trauma-informed care, debriefing.

Outcome 45. Reflect on the factors the paramedic must consider when assessing a suicide risk and formulate appropriate interview questions for determining suicidal intent

Definitions

- **Suicide**: death caused by self-directed injurious behaviour with an intent to die as a result of the behaviour.
- **Suicide attempt**: a non-fatal, self-directed, potentially injurious behaviour with an intent to die. It may or may not result in injury.
- **Suicidal ideation**: thinking about, considering or planning suicide. Tilley et al. (2026) note that callers to EMS may describe this as a person "having thoughts of death".
- **Deliberate self-harm (DSH)**: intentional self-poisoning or self-injury, irrespective of motive or degree of suicidal intent; **non-suicidal self-injury (NSSI)** is self-injury without intent to die (for example cutting to relieve emotional distress). Both are strong predictors of later suicide.
- **Parasuicide**: an older term for a suicide attempt with no intention of death; Tilley et al. (2026) used it to label repeated attempts by the same person.

The burden of suicide — why this matters

Globally, suicide caused about 727,000 deaths in 2021, with an estimated 20 attempts for every suicide; the global age-standardised suicide rate in 2021 was 8.9 per 100,000, compared with 11.5 per 100,000 in the WHO Africa region and **23.5 per 100,000 in South Africa** — almost 14,000 deaths per year, placing South Africa third highest in Africa (WHO, 2025, cited in Tilley et al., 2026). Suicide is the third leading cause of death among people aged 15–29 years (Tilley et al., 2026). In the Garden Route District study, there were on average **2.8 deaths by suicide and 2.3 attempted suicides per month** attended by EMS, and about 6.3 patients a month with suicidal ideation (Tilley et al., 2026).

Evans, Geduld and Stassen (2018) highlight three facts that are directly relevant to prehospital decision-making:

1. "There is a deep-seated belief that people who threaten suicide are not likely to follow through with their threat… the data assert that this is incorrect."
2. About 10% of those who threaten or attempt suicide eventually kill themselves.
3. **Eighty per cent of those who die by suicide have previously verbally stated their intention to do so.**

Every suicidal statement must therefore be taken seriously.

Factors to consider in assessing suicide risk

Suicide risk assessment is the process of identifying the factors that increase or decrease a person's risk, assessing the immediacy and seriousness of the current suicidal thinking or behaviour, and using this information to plan management. No single factor or tool reliably

predicts suicide; risk assessment is a clinical judgement that integrates the following domains (WHO, 2014; Tintinalli et al., 2020).

1. Demographic and historical risk factors (static factors)

- **Male sex**: in the Garden Route data, the odds of males dying by suicide were **5.049 times** those of females, and the odds of males attempting suicide were 2.053 times those of females (Tilley et al., 2026). Women more often attempt suicide and more commonly use overdose or poisoning; men more commonly use lethal methods such as hanging and firearms.
- **Age**: in the Garden Route study, the risk of death by suicide increased with age to a **peak at 41 years**, with median ages of 36 years for death by suicide and 30 years for attempted suicide (Tilley et al., 2026). Globally, rates are also high in older men and in young people aged 15–29 years.
- **Previous suicide attempt** — the single strongest predictor of future suicide.
- **Previous self-harm**, including NSSI (Tilley et al., 2026).
- **Family history of suicide** or exposure to the suicide of a friend.
- **History of childhood abuse or trauma**; Tilley et al. (2026) highlight the role of past trauma and advocate trauma-informed care.

2. Clinical risk factors

- **Mental disorders**: depression (especially with hopelessness), bipolar disorder, schizophrenia (especially young men early in the illness and those with command hallucinations), PTSD, personality disorders (especially borderline), eating disorders, and substance use disorders. Tilley et al. (2026) note that PTSD, bipolar disorder, depression, substance abuse and suicidal ideation are associated with deaths by suicide.
- **Alcohol and drug use** — intoxication causes disinhibition and impulsivity.
- **Chronic or terminal physical illness, chronic pain, disability**, and HIV (particularly around the time of diagnosis).
- **Hopelessness** — a strong independent predictor.
- **Recent discharge from a psychiatric facility**, or discharge against medical advice (Evans, Geduld and Stassen, 2018, citing Kuo et al., 2010).

3. Psychosocial and situational risk factors

- Recent loss: death, divorce or relationship breakdown, job loss, financial crisis, imprisonment.
- Social isolation, living alone, lack of support.
- Unemployment and poverty (Tilley et al., 2026).
- Domestic violence, sexual violence, bullying (especially in adolescents).
- Legal problems, shame, humiliation.
- Access to lethal means (firearms, pesticides, large quantities of medication, ligature points, heights).
- Stigma and barriers to accessing health care.

4. Features of the current episode (dynamic factors) — the most important in the field

- **Intent**: Does the person want to die? Do they still want to die now?

- **Planning**: Was the act planned in advance or impulsive? Was there preparation (for example buying tablets, writing a will or note, giving away possessions, saying goodbye)?
- **Lethality of the method** and the person's **belief** about its lethality (a person who believed that ten paracetamol tablets would kill them had high intent even if the dose was not lethal).
- **Precautions against discovery**: was the act timed or located so that rescue was unlikely?
- **Suicide note**.
- **Reaction to survival**: regret at being alive is a very serious sign; relief and a request for help are more reassuring.
- **Ongoing suicidal ideation** and **access to means**.
- **Command hallucinations** instructing self-harm.
- **Agitation, severe anxiety, insomnia** — acute risk factors.

5. Protective factors

Protective factors that reduce risk include strong family and social support, responsibility for children, religious or cultural beliefs that discourage suicide, engagement with health services, problem-solving skills, future plans, and restricted access to lethal means. The absence of protective factors increases risk.

Clinical reflection

Evans, Geduld and Stassen (2018) found that patients with "less dramatic" suicidal histories were more likely to be discharged on scene, that few providers used formal suicide evaluation tools, and that patients with minimal traditional risk factors were managed more conservatively — yet such patients "are still inherently at risk", and respondents described patients left on scene who later died. The lesson is that an apparently "low-risk" presentation does not mean "no risk", and that **risk factors guide judgement but cannot be used to exclude risk**.

Risk assessment tools

Several tools exist, but none is sufficiently accurate to be used on its own to decide whether a patient can safely be left at home; they should support, not replace, clinical judgement.

- **SAD PERSONS** (Patterson et al., 1983): Sex (male), Age (<19 or >45), Depression, Previous attempt, Ethanol/drug use, Rational thinking loss (psychosis), Social support lacking, Organised plan, No spouse, Sickness (chronic illness). Its predictive accuracy is poor, but it is a useful memory aid for risk factors.
- **Columbia Suicide Severity Rating Scale (C-SSRS) screener** (Posner et al., 2011): a series of structured questions about wish to be dead, suicidal thoughts, methods, intent, plan and recent behaviour, widely used by first responders.

Interview questions for determining suicidal intent

Principles

- **Ask directly.** Asking about suicide **does not** put the idea into someone's head and does not increase risk; most people experience relief when asked.
- Use a **private, calm** setting and a non-judgemental, empathic manner.
- Move from **general to specific** ("gradual approach") — from feelings, to thoughts of death, to suicidal thoughts, to intent, to plans, to preparation and past behaviour.
- Use clear words such as "kill yourself" or "end your life"; avoid euphemisms such as "do something silly".
- Interview the patient **and** obtain collateral history.

Formulated questions

Opening and rapport

1. "I'm [name], an emergency care practitioner. I'm here to help you and to keep you safe. Can you tell me what has been happening today?"
2. "It sounds like things have been very difficult for you. How have you been feeling in yourself lately?"

Mood and hopelessness

3. "Have you been feeling sad, down or hopeless most of the time?"
4. "Do you feel that things will ever get better?"
5. "Have you lost interest in things you used to enjoy?"

Passive and active suicidal ideation

6. "Have you felt that life is not worth living, or wished you could go to sleep and not wake up?"
7. "Have you had thoughts of killing yourself or ending your life?"
8. "How often do these thoughts come? How long do they last? Can you control them?"

Intent

9. "Did you want to die when you [took the tablets / tied the rope]?"
10. "Do you still want to die now?"
11. "How do you feel about the fact that you are still alive?"

Plan, method and preparation

12. "Have you thought about how you would do it?"
13. "Do you have access to [tablets, poison, a firearm, a rope]? Where are they now?"
14. "Have you made any preparations, such as writing a note, giving away belongings or saying goodbye to people?"
15. "When were you planning to do it?"

The current act (if an attempt has occurred)

16. “What exactly did you take or do, how much, and at what time?”
17. “Was this planned, or did it happen on the spur of the moment?”
18. “Did you think it would kill you?”
19. “Did you do anything to avoid being found or stopped?”
20. “Did you leave a note or message for anyone?”
21. “Who found you, and how did that happen?”

History

22. “Have you ever tried to harm or kill yourself before? What happened?”
23. “Have you been diagnosed with any mental health condition, or are you taking any medication for your nerves or mood?”
24. “Do you drink alcohol or use any drugs? Have you used any today?”

Psychotic symptoms

25. “Are you hearing voices that other people can’t hear? Are they telling you to do anything, such as harm yourself or someone else?”

Harm to others

26. “Have you had thoughts of harming anyone else?”

Protective factors and support

27. “What has stopped you from acting on these thoughts?”
28. “Who do you have to support you? Is there someone you trust whom we can contact?”
29. “What do you have to look forward to? Are there people who depend on you, such as children?”

Collaboration

30. “Would you be willing to come with us to the hospital so that a doctor can help you?”

High-yield points — Outcome 45

- South Africa’s suicide rate is 23.5/100,000 (WHO 2021 data); males are 5 times more likely to die by suicide; hanging is the most common method in men; poisoning/overdose in women; peak risk age 41 in the Garden Route data (Tilley et al., 2026).
- 80% of people who die by suicide previously stated their intent; 10% of those who threaten or attempt eventually die by suicide (Evans, Geduld and Stassen, 2018).
- Strongest predictor: previous attempt. Hopelessness, intent, planning, lethality, precautions against discovery, notes, regret at survival.
- Ask directly; asking does not increase risk.
- Tools (SAD PERSONS, C-SSRS) support but never replace clinical judgement.

45.1 Critically discuss the legal and ethical obligations of ECPs in the management of patients and scenarios where deliberate self-harm / suicide has occurred

Legal obligations

1. The duty of care and the duty to treat. Once an ECP has responded and established a practitioner–patient relationship, a duty of care exists. Section 27(3) of the Constitution provides that no one may be refused emergency medical treatment. The ECP must provide care that meets the standard of a reasonable practitioner, and leaving an at-risk patient who lacks capacity may constitute negligence.

2. Consent and capacity. A competent adult may refuse treatment (NHA, s 7; HPCSA, 2023, para 8.1.2). However, the HPCSA qualifies this: a refusal must be respected "unless the patient's state of mind is such that he/she is incapable of taking or participating in the decision" (HPCSA, 2023, para 8.1.2). Patients who have attempted suicide "may have limited insight into their illness and a restricted ability to cooperate with treatment", which may require measures "that, in line with local legislation, restrict their personal freedom and allow transportation to hospital against their will" (Evans, Geduld and Stassen, 2018).

3. The Mental Health Care Act. Section 9(1)(c) permits a health care provider to provide care without consent where, **due to mental illness**, any delay may result in (i) death or irreversible harm to the health of the user, (ii) the user inflicting serious harm to themselves or others, or (iii) the user causing serious damage to or loss of property. The provider or establishment that does so must report this in writing to the Review Board and may not continue for longer than 24 hours unless an application under Chapter V is made within that period (MHCA, s 9(2)). Section 40 empowers and obliges SAPS to apprehend a person who, due to mental illness, is likely to inflict serious harm to self or others, and to take that person to a State health establishment for assessment. The formal involuntary application (s 33) requires a written application by a relative, partner, associate or guardian (or a health care provider if they are unwilling, incapable or unavailable) and examination by two mental health care practitioners, which "is not achievable at the scene" (Evans, Geduld and Stassen, 2018). The prehospital role is therefore emergency assessment and **transport for evaluation**.

4. The CDST obligation to transport. The provider is "obligated to transport" if the patient is a danger to themselves or may be a suicide risk, a danger to family/community, causing serious damage to property, if SAPS request hospitalisation, or if the patient has been sedated and needs monitoring (HPCSA PBEC, 2025, p. 48).

5. Refusal of hospital treatment (RHT) forms. An RHT form is valid only if signed by a **competent** patient who has been fully informed of the risks of refusal. A form signed by a suicidal patient who lacks capacity, or whose refusal is driven by a mental illness, does not absolve the provider from legal responsibility (Evans, Geduld and Stassen, 2018). One participant in that study commented that "more thought could be put towards actually helping the patient in need instead of only worrying about paperwork".

6. Documentation. The PRF must record the capacity assessment, the risk assessment, the legal basis for any action without consent, the people consulted, and all interventions (HPCSA, 2023, para 12).

7. Confidentiality and disclosure. Information about a suicide attempt is confidential (MHCA, s 13; NHA, s 14; POPIA). It may be shared with the receiving facility, and with family or SAPS to the extent necessary to protect the patient's life. If the patient is a child, information may be shared with parents or guardians in the child's best interests.

8. Death by suicide. A death by suicide is an **unnatural death** in terms of the Inquests Act 58 of 1959; it must be reported to SAPS, and the scene must be treated as a potential crime scene. The ECP should not move the body or disturb the scene unnecessarily once death has been confirmed, should preserve any note or ligature as found (cutting a ligature away from the knot if it must be removed for resuscitation), should document the position of the body and findings, and should hand over to SAPS. The body will undergo a medicolegal post-mortem in terms of the Inquests Act.

9. Children and adolescents. A child aged 12 or older who is of sufficient maturity may consent to medical treatment (Children's Act 38 of 2005, s 129(2)); younger children require parental consent. In an emergency, treatment necessary to preserve life may proceed (Children's Act, s 129(6); NHA, s 7(1)(e)). An involuntary MHCA application for a user under 18 must be made by a parent or guardian (MHCA, s 33(1)(a)(i)). Child abuse that precipitated the self-harm must be reported (Children's Act, s 110; SOA, s 54).

Ethical obligations

1. **Beneficence and the duty to preserve life** — the ECP must act to prevent avoidable death. Because suicidal ideation is usually a symptom of a treatable mental disorder and is frequently transient, intervention is ethically justified.
2. **Respect for autonomy** — the patient must be involved in decisions as far as possible, informed, and treated as a person with dignity. Autonomy presupposes capacity; an impaired decision is not an autonomous one.
3. **Non-maleficence** — coercive measures (restraint, sedation, involuntary transport) must be proportionate and must not cause unnecessary harm or humiliation.
4. **Justice** — patients who self-harm must receive the same standard of care as other patients. Negative attitudes ("waste of time", "attention seeking", "chose to die") documented by Evans, Geduld and Stassen (2018) are ethically unacceptable and may breach the HPCSA requirement not to allow views about lifestyle or other irrational grounds to prejudice care (HPCSA, 2023, para 2.5).
5. **Fidelity and truth-telling** — do not deceive the patient.
6. **Care for the carer** — exposure to suicide is traumatic for providers, and services have an ethical obligation to provide debriefing and counselling (Evans, Geduld and Stassen, 2018; Tilley et al., 2026).

The ethical tension in the HPCSA Guidelines for Withholding and Withdrawing Treatment (Booklet 7)

The Behavioural Emergencies Tutorial 1 asks you to critically assess the ethical tensions in the HPCSA guidelines when they are applied to suicidal patients. The key points are:

- **The guideline's context is terminal illness, not suicide.** It is intended to provide "an ethical framework of good practice for health practitioners in circumstances where they are faced with making a decision on whether to withhold or withdraw life-prolonging treatment" for patients "for whom there is little hope of recovery" (HPCSA, 2023, paras 2.1–2.2). It explicitly prohibits "killing, active euthanasia, and assisted suicide" (HPCSA, 2023, para 2.2), and its annexure (the WMA Declaration on Terminal Illness) states that "physicians are ethically prohibited from actively assisting patients in suicide".
- **Autonomy versus incapacity.** The guideline requires respect for a competent adult's refusal "even where this would result in serious harm to them or in their own death" (para 8.1.2), but with the explicit exception where the patient's "state of mind is such that he/she is incapable of taking or participating in the decision". In a suicidal patient, capacity is often impaired by depression, psychosis, intoxication or the acute crisis. Misapplying the autonomy principle (for example accepting a refusal because "the patient has chosen to die", as some providers in the Evans study believed) could result in avoidable death.
- **Emergencies.** Paragraph 7.1 states that, in an acute life-threatening emergency "where any delay might prejudice the outcome and where it is impossible to obtain all relevant information or hold any consultations required or where there is uncertainty about the diagnosis or the likelihood of recovery, health practitioners should start treatment which may be of some benefit to the patient until a clearer assessment can be made". This supports initiating treatment for a patient who has self-harmed, even when the patient's wishes are unclear. Paragraph 7.2 requires that the patient be informed as soon as they are mentally capable.
- **Advance directives.** The guideline gives weight to directives and living wills (paras 3.3, 9.1.4). A "directive" or note written by a person in a suicidal crisis is **not** an advance directive in this sense, because it was not made by a competent person in contemplation of terminal illness, and a directive "cannot compel the healthcare team to provide treatments that lie outside acceptable standards of care" (para 9.1.4).
- **Best interests and consultation.** The guideline emphasises consultation with colleagues, family and proxies, second opinions, and documentation (paras 3.5, 3.7, 5.2, 12). These are achievable prehospitally through on-line medical advice, consultation with family, and thorough PRF documentation — but time pressure, limited information and uncertainty make them difficult, which is a genuine tension.
- **Resource considerations.** The guideline permits limitation of treatment based on scarcity (para 10), but this can never justify withholding emergency care from a suicidal patient; at most it guides triage in mass-casualty situations.
- **Prolonging suffering versus relieving it.** In the rare case of a patient with a genuine terminal illness and a valid advance directive who has also self-harmed, the tension between preventing death and respecting a previously expressed wish is real; such cases

require senior medical consultation, and the default in an emergency is to treat until a clearer assessment can be made (para 7.1).

Critical discussion — conclusion

The law and ethics converge on a clear position: the ECP must take every suicide attempt seriously, assess and document capacity and risk, provide emergency treatment, and transport patients who are at risk or who lack capacity, using the least restrictive means and with SAPS assistance when needed. The South African evidence shows a gap between this position and practice, with many providers believing they cannot transport without consent, relying on refusal forms, holding negative attitudes, and receiving little training (Evans, Geduld and Stassen, 2018). Closing this gap requires training, clear protocols (the CDST obligation-to-transport criteria), and supportive systems.

High-yield points — 45.1

- MHCA s 9(1)(c): care without consent due to mental illness if delay may cause death, serious harm to self/others, or serious property damage; report to Review Board; 24-hour limit.
- MHCA s 40: SAPS must apprehend and take to a State facility.
- CDST: obligated to transport if a suicide risk.
- RHT form signed by an incompetent patient has no legal value.
- HPCSA Booklet 7 is for terminal illness; it prohibits assisted suicide; para 7.1 supports emergency treatment under uncertainty.
- Death by suicide = unnatural death → SAPS, Inquests Act, preserve scene.

Outcome 46. Assess prehospital management techniques for a patient who has attempted suicide

Step-by-step prehospital management

1. Scene safety and size-up

- Look for hazards: weapons (knives, firearms), toxic gases (car exhaust — carbon monoxide; burning charcoal; fires in poorly ventilated rooms — the CDST advises checking for these and for cyanide or carbon monoxide poisoning; HPCSA PBEC, 2025, p. 54), chemicals and pesticides (organophosphates require PPE because of secondary contamination), electricity, heights, railway lines, water.
- Request SAPS (always for a firearm, an ongoing threat, or a death), and fire/rescue for heights, entrapment, or hazardous materials.
- Look for clues: empty pill containers, suicide notes, ligatures, alcohol, drug paraphernalia — take containers and tablets with you to hospital (HPCSA PBEC, 2025, p. 55).

2. Primary survey and resuscitation, according to the method

- **Hanging/strangulation** (the most common method in the Garden Route data, accounting for 82 of 102 deaths and 51 of 83 attempts; Tilley et al., 2026): cut the ligature away from the knot to preserve it as evidence, support the body while cutting it down, assume possible cervical spine injury (particularly with a drop), manage the airway (laryngeal and tracheal injury, airway oedema), give high-flow oxygen, look for petechiae and neurological deficit, and anticipate seizures, pulmonary oedema and hypoxic brain injury.
- **Overdose and self-poisoning** (52% of EMS mental-health incidents and the most common method in women; Tilley et al., 2026): follow the CDST Poisoning tool (Outcome 62): ABCDE, oxygen if SpO_2 <92%, RR >30 or <9, or GCS ≤14; airway management; glucose; IV access and fluids; identify the toxidrome; give specific antidotes within scope (for example naloxone, atropine, sodium bicarbonate); and give activated charcoal **only** if the patient is fully conscious and has taken an oral overdose of a harmful substance within the past hour, and the substance is suitable (HPCSA PBEC, 2025, pp. 54–57).
- **Cutting and stabbing**: haemorrhage control (direct pressure, tourniquets for life-threatening limb haemorrhage), shock management, and assessment for deeper injury (tendons, nerves, vessels, penetrating chest or abdominal injury).
- **Jumping from height or in front of a vehicle/train**: major trauma management with spinal precautions.
- **Firearm**: major trauma management; SAPS scene.
- **Drowning, burns, electrocution**: manage according to the relevant CDST tool.

3. Psychological care during treatment

- Talk to the patient throughout, explaining what you are doing; be calm, compassionate and non-judgemental.
- Avoid comments that blame or minimise (“Why would you do something so stupid?”).

- Do not leave the patient alone at any time, including in the bathroom.
- Remove all potential means of further self-harm from the patient and the environment (sharp objects, medication, ligatures, belts, shoelaces where appropriate).

4. Risk assessment and capacity assessment

Using the questions in Outcome 45, assess intent, plan, lethality and ongoing risk, and assess capacity to refuse care. Document both.

5. Disposition and transport

- **All patients who have attempted suicide should be transported** for medical and psychiatric assessment. Even an apparently minor overdose may be life-threatening (for example paracetamol, which causes delayed liver failure after an asymptomatic period, or TCAs, which cause rapid deterioration), and the act itself indicates significant risk.
- If the patient refuses: attempt persuasion and involve family; if the patient lacks capacity or meets the MHCA s 9(1)(c) criteria, or meets the CDST obligation-to-transport criteria (danger to self or suicide risk), the patient must be transported, using the least restrictive means, with SAPS assistance and sedation per the CDST if necessary (HPCSA PBEC, 2025, pp. 46–48).
- Destination: an **emergency centre** first, because medical clearance is required before psychiatric admission.
- During transport: continuous observation, monitoring every 10 minutes (more often if unstable or sedated), the attending provider seated near the door, doors secured to prevent the patient from jumping out, and SAPS escort if high risk.
- Bring family if possible, as well as any notes, tablets and containers.

6. Handover

Hand over the method, time, quantity, the circumstances, the risk assessment, capacity, the legal basis for transport, interventions and response, and collateral information.

7. After the call

Debrief with your partner; seek support if the incident was distressing.

Assessment (critical evaluation) of these techniques

- **Strengths**: a structured approach ensures that physical life threats are treated first, that psychiatric risk is assessed, and that transport decisions are lawful.
- **Gaps in South African practice**: few providers use formal suicide evaluation tools; transport decisions are influenced by how “dramatic” the presentation appears; decision-making is frequently shifted to family or seniors; less than half of respondents would transport a refusing patient; and patients have died after being left on scene (Evans, Geduld and Stassen, 2018).
- **Evidence for follow-up**: patients who receive active suicide prevention contact and follow-up after a suicide attempt have a reduced risk of repeat attempts over the following year (Evans, Geduld and Stassen, 2018, citing Inagaki et al., 2014). EMS is

therefore a crucial link in the chain to mental health care, and “to be left alone is like saying nobody cares” (participant quoted in Evans, Geduld and Stassen, 2018).

- **Means restriction**: limiting access to methods is one of the most effective suicide prevention interventions (Tilley et al., 2026, citing Florentine and Crane, 2010). ECPs can contribute by removing means on scene and advising families (for example locking away medication and pesticides and removing firearms).

High-yield points — Outcome 46

- Treat the physical injury first (method-specific), then assess risk and capacity.
- Never leave the patient alone; remove means.
- Transport every suicide attempt to an emergency centre; obligated to transport if at risk (CDST).
- Preserve ligatures, notes and containers; cut ligatures away from the knot.

Outcome 47. Discuss the management of patient abscondment in the emergency care setting

Definition

Abscondment is the unauthorised departure of a patient from care — for example running away from the scene, leaving the ambulance, or leaving the emergency centre — when the patient is at risk and should not leave. In the MHCA, the term is used for an **assisted or involuntary mental health care user** who leaves a health establishment without authorisation. Section 40(4) of the MHCA states that if such a user "has absconded or is deemed to have absconded", the head of the health establishment may request SAPS assistance to "locate, apprehend and return the user to the health establishment concerned"; SAPS "must comply with the request" (s 40(5)); SAPS must be informed of "the estimated level of dangerousness" of the user (s 40(6)); and SAPS may use "such constraining measures as may be necessary and proportionate in the circumstances" (s 40(8)). The prescribed form for notifying SAPS of an abscondment is Form MHCA 25 (Mental Health Care Regulations, 2004).

47.1 Reflect on the factors the paramedic must consider when assessing a patient's risk of abscondment in a behavioural emergency

Patient factors

- **Psychiatric state**: psychosis (persecutory delusions that the crew intend harm), mania, agitation, delirium, intoxication or withdrawal.
- **Lack of insight** into illness and the need for care.
- **Fear** — of hospitalisation, of being "locked up", of SAPS, of stigma; previous negative experiences of psychiatric admission or police restraint (Evans, Geduld and Stassen, 2018).
- **Expressed intent to leave**, refusal of care, or repeated questions about when they can go.
- **Previous abscondment** or previous involuntary admission.
- **Active suicidal intent** — the patient may abscond to complete suicide.
- **Substance craving or withdrawal**.
- **Young age** (adolescents), and dementia with wandering behaviour in older persons.
- **Responsibilities or concerns** (children at home, pets, work, fear of losing a job).

Environmental and situational factors

- An unsecured scene with open exits, crowds, or proximity to roads, railways or water.
- Delays on scene or long waiting times at the facility (Evans, Geduld and Stassen, 2018, noted long waits as a frustration for providers).
- Inadequate staff numbers or the absence of SAPS.
- An unlocked ambulance, or the patient seated near the door.
- Lack of family support, or family who encourage refusal.

Provider factors

- Confrontational or dismissive communication.
- Deception that is discovered (Stander, Hodkinson and Dippenaar, 2021).
- Failure to explain what will happen.
- Leaving the patient unsupervised.

Consequences of abscondment

The consequences are serious: suicide or self-harm (Evans, Geduld and Stassen, 2018, describe a patient jumping from a moving ambulance), injury in traffic, deterioration of an untreated medical condition (for example an overdose), harm to others, and legal and disciplinary consequences for the provider. Patients who leave against advice have poorer psychiatric and psychosocial outcomes, overuse emergency services, are readmitted sooner and have a higher risk of suicide (Evans, Geduld and Stassen, 2018, citing Brook et al., 2006, and Kuo et al., 2010).

47.2 Formulate prehospital management techniques for a patient in a behavioural emergency who attempts abscondment

Prevention

1. **Assess risk early** using the factors above, and plan accordingly.
2. **Build rapport**: calm, honest, respectful communication; explain what will happen and why; listen to concerns; offer choices; involve trusted family (RCEM, 2025; HPCSA PBEC, 2025).
3. **Address practical concerns** (for example arranging care for children or pets, informing an employer).
4. **Minimise time on scene** once the decision to transport has been made.
5. **Position** yourself between the patient and the exit without making the patient feel trapped, keep the patient within sight at all times, and never leave a high-risk patient alone.
6. **Secure the environment**: close doors; in the ambulance, secure the side and rear doors from the inside where possible; the attending provider sits closest to the door; the patient is secured on the stretcher with the seatbelts.
7. **Request SAPS** early for high-risk patients, and have SAPS escort the ambulance or travel in the patient compartment.
8. **Offer voluntary sedation** for transfer where indicated (CDST).

When the patient attempts to leave

1. **Do not physically chase the patient into danger** (for example onto a highway), and do not endanger yourself or your crew.
2. **Attempt verbal de-escalation** immediately; ask the patient to stop and talk.
3. If the patient **has capacity** and does not meet MHCA s 9(1)(c) or CDST obligation-to-transport criteria, the patient may lawfully leave; explain the risks, offer alternatives, document the refusal on an RHT form, advise the patient to call back at any time, and inform family where appropriate.

4. If the patient **lacks capacity or is a danger to self or others**: notify SAPS immediately (giving a description of the patient, clothing, direction of travel, the level of risk and dangerousness); SAPS has the authority and the duty under MHCA s 40(1) to apprehend a person who is likely to inflict serious harm to themselves or others due to mental illness; notify your control centre; if safe and practicable, keep the patient in sight from a safe distance; once the patient is apprehended, proceed with assessment, sedation per the CDST if necessary, and transport.
5. **Restraint and sedation** may be used in accordance with the CDST protocol (Outcome 43.3) if the patient is contained and remains dangerous.
6. **If the patient absconds from a health establishment**, it is the responsibility of the head of that establishment to request SAPS assistance (MHCA s 40(4); Form MHCA 25). EMS may be requested to assist with the transfer back.

Documentation

Document the risk assessment, capacity assessment, the attempts to persuade, the time and circumstances of abscondment, the direction of travel, who was notified (SAPS, control, family, facility), and the outcome.

High-yield points — Outcome 47

- MHCA s 40(4): head of establishment may request SAPS to locate, apprehend and return an absconded assisted or involuntary user; SAPS must comply; Form MHCA 25.
- Risk factors: psychosis, lack of insight, fear, intoxication, suicidal intent, previous abscondment, unsecured environment, long delays.
- Prevent: rapport, honesty, supervision, secure doors, SAPS, voluntary sedation.
- Do not chase into danger; notify SAPS with description and risk level; document.

Outcome 48. Assess variations in approach to behavioural emergencies in children and adolescents

Why children and adolescents are different

Children and adolescents differ from adults developmentally, physiologically, legally and socially. Their ability to describe distress depends on their developmental stage; distress often presents as behaviour (tantrums, aggression, withdrawal, regression, school refusal, somatic complaints) rather than words; they depend on caregivers who may themselves be the source of harm; their drug doses are weight-based; and their legal status (consent, confidentiality, mandatory reporting) is different. The CDST defines a child as **younger than 13 years** for clinical purposes, following the South African hospital definition, but advises clinical judgement about whether to use weight-based calculations in older children and adolescents (HPCSA PBEC, 2025, p. 3). Legally, a child is a person under 18 years (Children's Act 38 of 2005, s 1; Constitution, s 28(3); HPCSA, 2023, para 1).

48.1 Reflect on the factors the paramedic must consider when assessing a patient's risk of behavioural emergencies in children

Medical and organic factors

Organic causes must be excluded first, as in adults. In children these include hypoglycaemia (including in neonates and children with diabetes), hypoxia, febrile delirium and CNS infections (meningitis, encephalitis, cerebral malaria), seizures and post-ictal states, head injury (including non-accidental injury), poisoning (accidental poisoning in young children, deliberate overdose in adolescents), substance intoxication in adolescents, and pain the child cannot verbalise. The RCEM (2025) notes that **pre-adolescent** children "are unlikely to have a similar differential diagnosis" to adults, while **adolescents** "are likely to have a similar differential diagnosis to adults and pose similar management challenges".

Developmental and psychiatric factors

- **Neurodevelopmental disorders** — autism spectrum disorder, ADHD, intellectual disability, communication disorders (see 52.1) — in which distress is often triggered by sensory overload, a change in routine, pain or frustration at not being understood.
- **Mood disorders and suicidality**: suicide is the third leading cause of death in people aged 15–29 (Tilley et al., 2026); risk factors include depression, bullying (including cyber-bullying), academic pressure, family conflict, sexual identity-related stigma, abuse, substance use, and exposure to a peer's suicide (contagion).
- **Self-harm**: lifetime and 12-month self-harm prevalence in adolescents is high worldwide (Tilley et al., 2026, citing Lim et al., 2019).
- **Eating disorders**, typically in adolescent girls (see 52.7).
- **Early-onset psychosis and bipolar disorder**, which commonly emerge in late adolescence.
- **Trauma-related disorders** (acute stress disorder, PTSD).
- **Substance use**: alcohol, cannabis, inhalants, methamphetamine and "nyaope/whoonga" among South African adolescents.

Social and environmental factors: violence against children in South Africa

Mesthrie (2025) describes violence against children in South Africa as "a widespread and pressing problem in need of urgent attention". Key facts from her analysis are:

- The estimated child homicide rate in South Africa (5.5 per 100,000) is nearly twice the global average.
- 45% of South African households with children reported some form of violence against children in 2021–2022 (Statistics South Africa, 2024, cited by Mesthrie, 2025).
- One in five South African children are likely to be victims of sexual abuse, and two in five male school pupils are likely to be sexually abused (Cloete, 2022, cited by Mesthrie, 2025).
- At least 250,000 homeless children and young adults live on the streets.
- Children's exposure to violence "is often not a once-off event, but rather part of a condition of life" (Falconer et al., 2021, cited by Mesthrie, 2025).
- Exposure to violence changes brain structure and function, producing "cognitive deficits, behavioural and emotional problems, social and educational difficulties and high-risk behaviour such as engaging in unsafe sexual practices and substance abuse", and children exposed to violence have a higher risk of later perpetrating violence (Mesthrie, 2025).
- There is a clear gender dynamic: sexual violence is more pervasive in girls, while adolescent boys are more likely to be victims and perpetrators of physical violence and homicide (Mesthrie, 2025).

The paramedic must therefore consider **child abuse and neglect** as a cause of, or contributor to, every paediatric behavioural emergency. Indicators include an explanation of injury inconsistent with the injury or the child's developmental stage, delay in seeking care, injuries of different ages, patterned injuries (burns, slap or bite marks, loop marks), injuries in non-mobile infants, sexualised behaviour or genital injuries, fearful or "frozen watchfulness" behaviour, and a caregiver who is hostile, intoxicated or evasive.

Children with disabilities

The Behavioural Emergencies Tutorial 2 memo, drawing on Mesthrie and the disability rights framework (the UN Convention on the Rights of Persons with Disabilities and the African Disability Protocol), notes that children with disabilities face disproportionate exposure to abuse, neglect and exploitation because of communication barriers, stigma and dependence on caregivers. The National Child Care and Protection Policy itself acknowledges a past "lack of adequate support for children with disabilities" (NCPP, 2021, cited by Mesthrie, 2025). The paramedic must use accessible communication, involve a trusted person, and maintain a high index of suspicion for abuse.

48.2 Formulate prehospital management techniques for a patient in a behavioural emergency in children

General approach

1. **Safety** of the child, the family and the crew; SAPS if there is a risk of violence or abuse.

2. **Primary survey (ABCDE)** with age-appropriate vital signs (CDST Quick Reference Chart), **glucose** (treat if <3.5 mmol/L: oral glucose smeared on the gums, or 3 mL/kg 10% dextrose IV, or glucagon 0.5 mg IM if no IV access; HPCSA PBEC, 2025, p. 47), temperature, pupils, and a search for injuries and signs of poisoning.
3. **Developmentally appropriate communication**:
   - get down to the child's eye level; speak calmly and simply; use the child's name;
   - allow the parent or caregiver to stay and hold the child **unless the caregiver is the source of distress or a suspected abuser**;
   - use distraction (toys, a favourite object, a phone video) for younger children;
   - for adolescents, offer privacy and the opportunity to speak without parents present, explain confidentiality and its limits honestly (for example "What you tell me stays with the medical team unless I'm worried that you or someone else is in danger"), and treat them with respect.
4. **Verbal and environmental de-escalation** — low stimulation, fewer people, a quiet space. The RCEM (2025) observes that in pre-adolescent children "brief restraint interventions are more likely to be successful if verbal/environmental de-escalation fails", meaning that a parent holding a young child briefly is often enough, without medication.
5. **Pharmacological sedation (CDST, children <13 years)** — only if the child remains dangerous after de-escalation, using the CDST dose charts (HPCSA PBEC, 2025, pp. 69–72):
   - *midazolam 0.2 mg/kg buccal or intranasal (maximum 10 mg per dose)*;
   - *midazolam 0.15 mg/kg IM (maximum total dose 10 mg)*;
   - *midazolam 0.05 mg/kg IV (maximum total 0.5 mg/kg)*;
   - *midazolam 0.5 mg/kg PO (maximum total 20 mg)*;
   - alternatives: *lorazepam 0.1 mg/kg (PO maximum 4 mg; or IM)*, diazepam 0.2 mg/kg PO (maximum 10 mg per dose), ^ketamine 4–5 mg/kg IM^ (ANT/ECP);
   - monitor for paradoxical agitation (more common with benzodiazepines in children) and respiratory depression; monitor age-appropriate vital signs every 10 minutes (HPCSA PBEC, 2025, p. 47).
   - For adolescents (13 years and older), the adult CDST doses apply, using clinical judgement for weight.
6. **Specific situations**:
   - **suicide attempt or self-harm**: as in Outcome 46, always transport; ask directly about suicidal intent; consider abuse;
   - **accidental ingestion in a young child**: CDST Poisoning tool (Outcome 62);
   - **child with autism**: reduce noise and lights, allow the caregiver to lead, avoid unnecessary touch, use a known comfort object, allow extra time, and consult any care plan (RCEM, 2025);
   - **suspected abuse**: treat injuries, avoid repeated questioning (which can contaminate later forensic interviews — ask open, non-leading questions only as needed for care), document the child's own words verbatim, preserve evidence, and report (see 48.3).

7. **Transport** with a parent or guardian where appropriate and safe; if abuse is suspected, inform SAPS and the receiving facility, and ensure the child is not released back into an unsafe environment.

48.3 Critically appraise emergency care practice in relation to legal and ethical implications related to behavioural emergencies in children

The constitutional and international framework

- **Section 28 of the Constitution** guarantees every child the right, among others, to family or parental care, to basic nutrition, shelter, basic health care services and social services, and to be protected from maltreatment, neglect, abuse or degradation; and provides that **a child's best interests are of paramount importance in every matter concerning the child** (s 28(2)).
- **Article 19 of the UN Convention on the Rights of the Child** (ratified by South Africa in June 1995) obliges States to "take all appropriate legislative, administrative, social and educational measures to protect the child from all forms of physical or mental violence, injury or abuse, neglect or negligent treatment, maltreatment or exploitation, including sexual abuse", including procedures for "identification, reporting, referral, investigation, treatment and follow-up" (Mesthrie, 2025).
- The CRC's four general principles are non-discrimination (Article 2), the best interests of the child (Article 3), the right to life, survival and development (Article 6), and the right to be heard (Article 12). General Comment 13 adopts a **child rights-based approach**, which treats children as active rights-holders rather than passive objects of welfare, and emphasises child participation (Mesthrie, 2025).

Consent

- **Children's Act 38 of 2005, s 129(2)**: a child may consent to his or her own medical treatment (or that of his or her child) if the child is **over the age of 12 years** and of **sufficient maturity and mental capacity** to understand the benefits, risks, social and other implications of the treatment.
- **s 129(3)**: a child may consent to a surgical operation if over 12, of sufficient maturity, **and** duly assisted by a parent or guardian.
- **s 129(4)–(5)**: for children younger than 12, or older children lacking sufficient maturity, a parent, guardian or care-giver may consent.
- **s 129(6)**: the superintendent of a hospital or the person in charge (in their absence) may consent to the medical treatment of or surgical operation on a child if the treatment is necessary to preserve the life of the child or to save the child from serious or lasting physical injury or disability, and the need is so urgent that it cannot be deferred for the purpose of obtaining consent.
- In the prehospital emergency, treatment necessary to preserve life may be given without consent under the emergency provisions (NHA, s 7(1)(e)) and the common-law doctrine of necessity.
- **Refusal by a child**: the HPCSA (2023, para 15.2) states that practitioners "should respect the decisions of children who have the legal capacity to make decisions about refusing health care – except in cases where the practitioners believe that it is not in the child's best interests", in which case they should approach the parents or guardians, the

person in charge of the health establishment, the Minister, the High Court or the Children's Court. A suicidal adolescent's refusal is therefore not determinative.

- **Children who lack legal capacity** but have sufficient maturity to understand the procedures "should be consulted during the decision-making process" (HPCSA, 2023, para 15.4), which reflects the CRC right to be heard.
- **Mental Health Care Act**: an application for involuntary care of a user under 18 must be made by the parent or guardian (MHCA, s 33(1)(a)(i)).
- **Parental refusal** of necessary treatment that endangers a child may be overridden through s 129(6)–(10) of the Children's Act (including by the Minister or a court), and may itself indicate that the child is in need of care and protection (s 150).

Mandatory reporting

- **Children's Act, s 110(1)**: a defined list of professionals (including medical practitioners, nurses, psychologists, social workers, teachers, traditional health practitioners and others) who, on reasonable grounds, conclude that a child has been abused in a manner causing physical injury, sexually abused or deliberately neglected, **must** report that conclusion in the prescribed form to a designated child protection organisation, the provincial department of social development, or a police official. **Section 110(2)** allows **any person** who believes on reasonable grounds that a child is in need of care and protection to report. Once a report is made to the police, the police must ensure the child's safety and notify the provincial department of social development or a designated child protection organisation within 24 hours (s 110(4); Mesthrie, 2025). As an ECP, you should regard yourself as bound to report.
- **Sexual Offences Act, s 54(1)** (as amended by Act 13 of 2021): **any person** who has knowledge, reasonable belief or suspicion that a sexual offence has been committed against a person who is vulnerable (which includes a child) must report it **immediately to a police official**; failure to do so is an offence punishable by a fine or imprisonment not exceeding five years, or both (s 54(2)).
- **Domestic Violence Act, s 2A and s 2B** (inserted by Act 14 of 2021, in force from 14 April 2023): functionaries (medical practitioners, health care personnel, social workers, officials in public health establishments, educators and caregivers) who believe or suspect on reasonable grounds that a child may be suffering domestic violence must complete the prescribed report and risk assessment (Form 2) and submit it to a social worker or SAPS; any adult with knowledge, reasonable belief or suspicion of domestic violence against a child must report it to a social worker or SAPS (s 2B). **Exposing a child to domestic violence** (causing a child to see or hear it, or experience its effects) is itself defined as domestic violence (DVA, s 1 as amended).
- **Children in need of care and protection** (Children's Act, s 150) include children who have been abandoned, who live on the street or beg, who are addicted to substances without support, who are exploited, who live in circumstances that may seriously harm their physical, mental or social well-being, or who are physically or mentally neglected. A police official or social worker may remove a child to temporary safe care without a court order in an emergency (s 152).

Confidentiality

Adolescents are entitled to confidentiality in the same way as adults, but confidentiality is limited by the child's best interests, the duty to report abuse and sexual offences, and serious risk of harm. The paramedic should be honest with the adolescent about these limits.

Critical appraisal of current practice

1. **Over-reliance on parental decision-making.** Evans, Geduld and Stassen (2018) found that in vignettes involving minors who had attempted suicide, prehospital providers "would allow family to make the ultimate decision about transport". This is problematic: the child's best interests are paramount (Constitution, s 28(2)); a parent's refusal of necessary care may be overridden; and the family may be part of the problem (for example abuse or conflict). The provider must act in the child's best interests, involve SAPS where needed, and transport an at-risk child.
2. **Failure to recognise and report abuse.** Abuse is often missed when the presenting problem is behavioural. Under-reporting is common, driven by uncertainty, fear of confrontation, lack of training, and doubt about whether EMS providers are legally obliged to report. As shown above, the SOA s 54 duty and the DVA s 2B duty apply to everyone, so the ECP is legally obliged to report.
3. **Inadequate child participation.** Mesthrie (2025) argues that South African law and policy pay insufficient attention to child participation, and that "fulfilling the right to protection from all forms of violence necessitates the fulfilment of the right to be heard". In emergency care, this means explaining what is happening in age-appropriate language, asking the child's views, and taking them into account (HPCSA, 2023, para 15.4).
4. **Gender blindness.** Mesthrie (2025) criticises the Children's Act and the NCPP for not taking an explicit gendered stance, despite the gendered pattern of violence. The ECP should be aware of the high risk of sexual violence against girls and of physical violence against adolescent boys.
5. **Limits of protectionism.** The Tutorial 2 memo (drawing on Chikoko's work on street children in Harare) notes that purely protectionist approaches — removing children from the streets without addressing poverty, abuse and exclusion — risk perpetuating harm. Prehospital care should link children to holistic services (social workers, child protection organisations, health and education services).
6. **Use of sedation and restraint in children** carries particular risks (paradoxical reactions, airway compromise, psychological trauma) and should be a last resort, with caregiver involvement and weight-based CDST dosing.
7. **Lack of paediatric mental health resources and training** across the system.

High-yield points — Outcome 48

- Child (legal) <18 years; CDST child <13 years.
- Children's Act s 129: consent from age 12 if mature; s 129(6) emergency consent by superintendent; HPCSA para 15.2: a child's refusal may be overridden in their best interests.
- Mandatory reporting: Children's Act s 110 (professionals; any person may report), SOA s 54 (any person — immediately to SAPS; offence if not), DVA ss 2A/2B.

- Best interests of the child are paramount (Constitution s 28(2)).
- Mesthrie (2025): child homicide rate twice the global average; 45% of households report violence against children; need for child participation and a gendered approach.
- CDST paediatric midazolam: 0.2 mg/kg buccal/IN, 0.15 mg/kg IM, 0.05 mg/kg IV, 0.5 mg/kg PO.

Outcome 49. Assess variations in the approach to behavioural emergencies in geriatrics

Why older adults are different

Desai, Kulkarni and Verma (n.d.) state that "older adults are not simply younger agitated patients with more years attached to them. Frailty, sensory impairment, multimorbidity, polypharmacy, sarcopenia, lower physiologic reserve, and preexisting cognitive impairment all alter how agitation emerges and how treatment behaves once it is given." Age-related changes in pharmacokinetics (reduced renal and hepatic clearance, increased body fat, reduced albumin) and pharmacodynamics (increased brain sensitivity to sedatives) "narrow the margin between useful calming and preventable harm".

49.1 Reflect on the factors the paramedic must consider when assessing a patient's risk of behavioural emergencies in geriatric patients

1. Delirium — the most important cause to identify

Delirium is common in older persons, is frequently missed, and carries a high mortality. It is characterised by acute onset, a fluctuating course, inattention, disorganised thinking and an altered level of consciousness, and may be **hyperactive** (agitated, restless), **hypoactive** (quiet, withdrawn, drowsy — easily mistaken for depression or "tiredness") or **mixed** (Inouye, Westendorp and Saczynski, 2014). Desai, Kulkarni and Verma (n.d.) emphasise that "medically ill older adults with fluctuating consciousness and new inattention should be presumed delirious until shown otherwise". New visual hallucinations or abrupt changes in arousal should heighten concern for delirium (except in Lewy body dementia, where visual hallucinations are part of the disease).

2. Underlying medical causes

A cause-oriented assessment should begin with physiology and safety. Desai, Kulkarni and Verma (n.d.), citing Shenvi et al. and Kennedy et al., advise immediate evaluation for **hypoxia, hypoglycaemia, shock, acute coronary syndromes, stroke, intracranial haemorrhage (including subdural haematoma after minor falls), sepsis and severe pain**, and stress that **urinary retention, constipation, medication effects and environmental triggers** are also common and easily missed. Common infections presenting as behavioural change are urinary tract infection and pneumonia, often without fever.

3. Medications and polypharmacy

Medication effects are "often decisive": anticholinergics, recent opioid escalation, abrupt sedative withdrawal, dopamine-blocking agents causing akathisia, corticosteroids and polypharmacy interactions "can each be sufficient to produce or amplify agitation" (Desai, Kulkarni and Verma, n.d.). The 2023 American Geriatrics Society Beers Criteria list benzodiazepines, antipsychotics and anticholinergic drugs among those that can induce or worsen delirium, cognitive impairment, falls and fractures in older adults (American Geriatrics

Society Beers Criteria Update Expert Panel, 2023). Alcohol use and withdrawal are frequently overlooked in older people.

4. Dementia

Around 97% of people with dementia experience at least one behavioural or psychological symptom during their illness, and agitation affects about 30–50% of people with Alzheimer's disease and related dementias (Kales et al. and Carrarini et al., cited by Desai, Kulkarni and Verma, n.d.). Agitation in dementia is often triggered by pain, unfamiliar surroundings, overstimulation, hunger, thirst, the need to use the toilet, sleep disruption, missing hearing aids or glasses, and the way staff approach the person. The **dementia subtype** matters: patients with **Lewy body dementia and Parkinson's disease are extremely sensitive to dopamine-blocking antipsychotics** (haloperidol and droperidol are contraindicated — RCEM, 2025).

5. Depression and suicide in older persons

Older men have high suicide rates in many countries; risk factors include bereavement, social isolation, chronic pain and illness, disability, loss of independence, and alcohol use. Older people who attempt suicide use more lethal methods and have higher intent. Tilley et al. (2026) cite evidence linking mental disorders and cognitive impairment with suicide risk in older adults (Kułak-Bejda, Bejda and Waszkiewicz, 2021).

6. Sensory impairment and communication

Hearing and visual impairment cause misinterpretation, fear and apparent confusion; aphasia after stroke may be mistaken for confusion or psychosis.

7. Elder abuse and neglect

Elder abuse includes physical, psychological, sexual and financial abuse and neglect, and is frequently committed by family members or caregivers (Lachs and Pillemer, 2015). Behavioural change, fearfulness, unexplained injuries, poor hygiene, malnutrition, pressure sores, missing medication or the inappropriate use of sedation by caregivers may be signs. The amended DVA expressly includes **elder abuse** within the definition of domestic violence.

8. Environmental factors

The emergency setting itself — "bright, noisy, crowded, sleep-disrupting, and cognitively disorienting" — may trigger agitation (Desai, Kulkarni and Verma, n.d.); the ambulance and the emergency centre are such settings.

49.2 Formulate prehospital management techniques for a patient in a behavioural emergency in geriatric patients

A cause-oriented, layered assessment

The Tutorial 2 memo, based on Desai, Kulkarni and Verma (n.d.), describes a layered evaluation:

1. **Reversible acute threats**: hypoxia, hypoglycaemia, shock, stroke, sepsis, severe pain.
2. **Baseline cognitive status and dementia type** (from collateral history).
3. **Precipitating stressors**: environmental overstimulation, toileting needs, sensory impairment.
4. **Iatrogenic contributors**: polypharmacy, recent dose changes, immobilisation.

Collateral history is crucial to establish baseline cognition, onset, fluctuation and previous similar episodes. Prehospitally, this means: primary survey; full vital signs including temperature and glucose; SpO_2 and oxygen if <92% (CDST); 12-lead ECG if possible; neurological examination including FAST for stroke; a search for injuries from falls; checking the abdomen for a distended bladder; a medication list (bring all medication to hospital); and a delirium screen (for example the CAM or 4AT) where practicable.

Non-pharmacological management first

Desai, Kulkarni and Verma (n.d.) argue that non-pharmacological care "is often discussed as if it were a humane but secondary add-on to the real intervention", but that "the literature does not support that hierarchy" — these measures "frequently determine whether medication becomes necessary at all". Techniques include:

- **one voice**, slow and simple speech, short concrete statements, and simple choices rather than repeated commands;
- **approach from the front**, introduce yourself (repeatedly if necessary), and avoid unnecessary touching of frightened or perceptually disorganised patients;
- **do not confront delusional content**;
- **restore sensory aids** (hearing aids, glasses, dentures);
- **reduce noise and light**, dim lights where possible, turn off unnecessary alarms and sirens;
- **involve a familiar caregiver** at the bedside or in the ambulance;
- **treat comfort needs**: pain (analgesia per the CDST "Patient in pain" tool), thirst, hunger, toileting;
- **minimise restraints** and immobilising devices;
- offer simple purposeful activity (a blanket to fold, family photographs, familiar music) (Desai, Kulkarni and Verma, n.d.).

An agitated patient with an ischaemic limb or a displaced fracture "may become cooperative after analgesia without receiving any antipsychotic at all", and a hypoxaemic patient "may settle when respiratory support is improved rather than when cognition is suppressed" (Desai, Kulkarni and Verma, n.d.).

Pharmacological management — only when danger exceeds what non-drug care can control

The key geriatric principles are summarised by Desai, Kulkarni and Verma (n.d.): **start low, avoid stacking (delay repeat dosing), match the drug to the syndrome, monitor continuously, prefer calm over deep sedation, and plan discontinuation**. The goal is "calm enough for care, not suppression for its own sake".

CDST doses for older persons (>60 years) — note the halved doses (HPCSA PBEC, 2025, pp. 46–47, 72):

- rapid sedation: *midazolam **0.1 mg/kg IM** (instead of 0.2 mg/kg)*, maximum 15 mg;
- repeat dose after 30 minutes if needed: *midazolam **0.05 mg/kg IM*** (instead of 0.1 mg/kg), maximum 5 mg;
- voluntary sedation options as for adults, but use the lowest effective dose.

The RCEM (2025) similarly recommends halving doses in people over 65, a "start low, go slow" strategy, and notes that the differential diagnosis in this group "is less likely to feature stimulant drugs, and more likely to feature hyperactive delirium".

Evidence on specific drug classes in older adults:

- **Benzodiazepines** have a limited role: they are appropriate for **alcohol or benzodiazepine withdrawal** and **seizures**, but are discouraged as routine first-line agents for undifferentiated delirium or dementia-related agitation because of respiratory depression, worsening delirium, falls and prolonged cognitive impairment; **midazolam** in particular raises "substantial safety concern" in older adults (Casey et al., cited by Desai, Kulkarni and Verma, n.d.; Saucedo Uribe et al., 2025). This is a genuine tension with the prehospital CDST, in which midazolam is the main sedative available; the resolution is to use non-drug measures first, use the reduced CDST dose, avoid stacking, and monitor closely.
- **Antipsychotics** (hospital-level agents): the Beers Criteria advise avoiding antipsychotics for behavioural problems of dementia or delirium unless non-pharmacological options have failed or are not possible and the patient threatens substantial harm. Haloperidol carries extrapyramidal and QT risks and is associated with increased mortality in older adults; low-dose olanzapine may have the most favourable short-term profile; quetiapine is preferred in Parkinson's disease and Lewy body dementia; haloperidol and droperidol are contraindicated in Parkinson's disease and Lewy body dementia (Desai, Kulkarni and Verma, n.d.; RCEM, 2025).
- **Ketamine**: although effective in younger profoundly agitated patients, ketamine "should not be imported into routine geriatric agitation protocols", because older adults "face greater cardiovascular variability, aspiration concern, and post-sedation diagnostic complexity", and age-specific evidence is lacking (Desai, Kulkarni and Verma, n.d.). It should be reserved for exceptional cases.

Monitoring after sedation must include airway, vital signs, cardiac rhythm (QT), extrapyramidal symptoms, sedation level, falls risk and cognition (Desai, Kulkarni and Verma, n.d.), and the CDST 10-minute observation cycle.

Cumulative exposure: "a single emergency dose is rarely truly single by the time the whole chain of care is examined. An ambulance crew may have given midazolam. A nursing facility may have administered quetiapine before transfer" (Desai, Kulkarni and Verma, n.d.). The ECP must ask what sedatives have already been given and must hand over clearly what was given, when, and why.

Transport and handover

Transport to an emergency centre (delirium is a medical emergency), with a familiar caregiver where possible, and hand over the baseline cognition, onset, medication list, any sedation given, and concerns about abuse.

49.3 Critically appraise emergency care practice in relation to legal and ethical implications of incapacity related to behavioural emergencies in geriatric patients

Capacity in older adults

Capacity is **decision-specific and time-specific**, and must not be assumed to be absent because of age or a diagnosis of dementia. A person with mild dementia may have capacity to refuse transport, while a person with delirium may lack it today but regain it tomorrow. The HPCSA (2023, para 9.1.1.1) requires that where patients have difficulty retaining information, communicating, or are only intermittently competent, practitioners "should provide any assistance a patient might need to enable him or her to reach and communicate a decision" — for example restoring hearing aids, using simple language, choosing a time of better orientation, and involving a familiar person.

Who can consent when an older person lacks capacity?

South Africa does not have a single "Mental Capacity Act". The legal routes are:

1. **NHA, s 7(1)(a)–(b)**: if the user is unable to give consent, consent may be given by a person mandated **in writing** by the user to grant consent on their behalf, or authorised by law or court order; if no such person exists, consent may be given by the **spouse or partner**, or in their absence a **parent, grandparent, adult child, brother or sister**, in that order of priority.
2. **NHA, s 7(1)(e)**: in an emergency, treatment may be provided without consent where any delay might result in death or irreversible damage to health and the user has not expressly, impliedly or by conduct refused the service.
3. **Curatorship**: a court may appoint a curator to the person and/or a curator bonis (for property) for a person who is incapable of managing their own affairs.
4. **Advance directives / living wills**: not specifically legislated in South Africa, but recognised by the HPCSA as an expression of autonomy that "should be taken into consideration" (HPCSA, 2023, paras 3.3 and 9.1.4). A directive "cannot compel the healthcare team to provide treatments that lie outside acceptable standards of care". The ECP should respect a valid advance refusal of specific treatment (HPCSA, 2023, para 9.2.1), but in an emergency with uncertainty should start treatment that may be of benefit until a clearer assessment can be made (HPCSA, 2023, para 7.1).
5. **MHCA assisted care (ss 26–27)**: for a mental health care user who is incapable of making informed decisions due to mental illness or severe or profound intellectual disability but who **does not refuse** care, an application for assisted care can be made — relevant to some older persons with dementia requiring admission. If the user refuses, the involuntary pathway (ss 32–34) applies.

6. **Older Persons Act 13 of 2006**: protects older persons from abuse and neglect, requires persons who deal with older persons in a professional capacity to report suspected abuse (s 26), and provides for the protection of older persons in need of care and protection.

Ethical issues

1. **Autonomy versus protection (beneficence)**: older persons have the right to make choices that others consider unwise, including refusing hospital admission. Overriding a competent refusal because of age is ageist and unlawful. Conversely, accepting the "refusal" of a delirious patient who cannot understand the risks is negligent.
2. **Ageism and justice**: the HPCSA (2023, para 2.5) prohibits allowing views about age or disability to prejudice care. Older patients with dementia may be undertreated (for example pain not assessed) or over-sedated.
3. **Chemical restraint as a substitute for care**: sedating an older person for convenience, or because the environment is unsuitable, violates non-maleficence and dignity; Desai, Kulkarni and Verma (n.d.) emphasise that medication "can make the room safer without making the patient better in any deep sense", and that drugs started in a crisis "have a persistent tendency to survive the situation that justified them".
4. **Best interests and family involvement**: where the patient lacks capacity and there is no advance directive, close family must be consulted and a decision made in the patient's best interests (HPCSA, 2023, para 3.4). The HPCSA also requires that cultural practices related to death and dying be respected (para 4.4).
5. **Withholding and withdrawing treatment at the end of life**: some older patients with advanced dementia or terminal illness may have advance directives or "do not resuscitate" decisions. The HPCSA Booklet 7 guides such decisions (consultation, documentation, palliative care); the CDST includes a "Palliative end-of-life care" tool (HPCSA PBEC, 2025).
6. **Elder abuse and the duty to report**: the Older Persons Act (s 26) and the amended DVA (s 2A requires functionaries to report suspected domestic violence against older persons on Form 2; s 2B requires any adult with knowledge or suspicion to report) create reporting obligations.

Critical appraisal of current practice

- **Strengths**: the CDST explicitly halves midazolam doses for patients older than 60 years, reflecting age-specific risk; the CDST and RCEM both emphasise looking for medical causes.
- **Weaknesses**: prehospital tools for delirium screening are rarely used; providers frequently rely on the GCS, which is inappropriate for capacity (Stander, Hodkinson and Dippenaar, 2021); the available prehospital sedative (midazolam) is the agent with the greatest safety concern in older adults (Desai, Kulkarni and Verma, n.d.); there is a lack of geriatric-specific evidence (major trials and guidelines such as the 2024 ACEP policy excluded patients over 65; Desai, Kulkarni and Verma, n.d.); and handover of cumulative sedation is often incomplete.
- **Improvements**: routine collateral history, delirium screening, non-pharmacological first-line care, reduced doses with no stacking, explicit capacity assessment and

documentation, involvement of family and advance directives, and vigilance for elder abuse.

High-yield points — Outcome 49

- Presume delirium in an older person with acute confusion and inattention; hypoactive delirium is easily missed.
- Look for hypoxia, hypoglycaemia, sepsis (UTI, pneumonia), stroke, subdural haematoma, pain, urinary retention, constipation, medications.
- Non-drug measures first: one voice, front approach, sensory aids, family, comfort.
- CDST: midazolam **0.1 mg/kg IM if >60 years**, repeat **0.05 mg/kg**.
- Avoid haloperidol/droperidol in Parkinson's/Lewy body dementia; benzodiazepines mainly for withdrawal and seizures; ketamine not routine in older adults.
- Consent for incapacitated adults: NHA s 7(1)(b) hierarchy — spouse/partner, parent, grandparent, adult child, sibling; emergency s 7(1)(e).
- Report elder abuse (Older Persons Act s 26; DVA ss 2A/2B).

Outcome 50. Analyse relevant South African legislation and legal principles affecting the management of mentally ill persons

The legislative framework

The management of mentally ill persons in South Africa is governed by the following:

1. **The Constitution of the Republic of South Africa, 1996** — the supreme law. Relevant rights include equality and non-discrimination (s 9, which lists disability as a prohibited ground), human dignity (s 10), life (s 11), freedom and security of the person, including the right not to be deprived of freedom arbitrarily or without just cause, the right not to be treated in a cruel, inhuman or degrading way, and the right to bodily and psychological integrity (s 12), privacy (s 14), access to health care services and the right not to be refused emergency medical treatment (s 27), and children's rights (s 28). Any limitation of these rights must be reasonable and justifiable in terms of the limitation clause (s 36), which is why involuntary care must be lawful, necessary and proportionate.
2. **The Mental Health Care Act 17 of 2002 (MHCA)** and its General Regulations (GN R1467 of 2004).
3. **The National Health Act 61 of 2003 (NHA)** — consent, information, participation, confidentiality and the rights of users.
4. **The Children's Act 38 of 2005** — consent by and for children; protection of children.
5. **The Criminal Procedure Act 51 of 1977** (ss 77–79) — assessment of accused persons who may lack the mental capacity to stand trial or who may not be criminally responsible, leading to State patient status under Chapter VI of the MHCA.
6. **The Prevention of and Treatment for Substance Abuse Act 70 of 2008** — voluntary and involuntary admission for substance dependence.
7. **The Older Persons Act 13 of 2006**, the **Domestic Violence Act 116 of 1998** (as amended) and the **Sexual Offences Act 32 of 2007** (as amended) — protection and reporting duties.
8. **The Protection of Personal Information Act 4 of 2013 (POPIA)** — health information is special personal information.
9. **The Inquests Act 58 of 1959** — unnatural deaths, including suicide.
10. **The Health Professions Act 56 of 1974** and the **HPCSA ethical rules and guidelines** (including Booklet 7, HPCSA, 2023), and the **PBEC scope of practice and CDST** (HPCSA PBEC, 2025).
11. **International instruments**: the UN Convention on the Rights of Persons with Disabilities (CRPD) and the UN Principles for the Protection of Persons with Mental Illness (1991), which strongly influenced the MHCA.

Key legal principles

- **Least restrictive care**: care should be provided in the least restrictive manner and environment.
- **Consent as the norm; involuntary care as the exception**, permitted only on statutory criteria.

- **Capacity is presumed** in adults unless shown otherwise; it is decision-specific.
- **Best interests**, where a person lacks capacity.
- **Review and oversight**: Mental Health Review Boards review involuntary and assisted care decisions, and the High Court exercises judicial review.
- **Non-discrimination and dignity.**
- **Confidentiality**, subject to statutory exceptions.
- **Necessity**: the common-law defence that justifies otherwise unlawful conduct (for example restraining or treating without consent) where it is necessary to avert an imminent danger to life or health, and the means used are reasonable and proportionate.

The structure of the Mental Health Care Act

The MHCA (in force from 15 December 2004) replaced the Mental Health Act 18 of 1973 and introduced a rights-based, community-oriented system in which mental health care is integrated into general health services. Its main chapters are:

- **Chapter I** — Introduction (definitions, objects).
- **Chapter II** — Fundamental provisions (including the integration of mental health care into general health services).
- **Chapter III** — Rights and duties relating to mental health care users: respect, human dignity and privacy (s 8); consent to care, treatment and rehabilitation services and admission (s 9); unfair discrimination (s 10); exploitation and abuse (s 11); determinations concerning mental health status (s 12); disclosure of information (s 13); limitation on intimate adult relationships (s 14); right to representation (s 15); discharge reports (s 16); knowledge of rights (s 17).
- **Chapter IV** — Mental Health Review Boards.
- **Chapter V** — Voluntary, assisted and involuntary mental health care (ss 25–40).
- **Chapter VI** — State patients.
- **Chapter VII** — Mentally ill prisoners.
- **Chapter VIII** — Care and administration of property of mentally ill persons.
- **Chapters IX–X** — Regulations and general provisions.

Definitions that matter for EMS

- **Mental health care user (MHCU)**: a person receiving care, treatment and rehabilitation services, or using a health service at a health establishment aimed at enhancing their mental health status (the definition also extends to certain persons acting on the user's behalf).
- **Mental health care practitioner**: a psychiatrist or registered medical practitioner, or a nurse, occupational therapist, psychologist or social worker who has been trained to provide prescribed mental health care, treatment and rehabilitation services. **Emergency care practitioners are not included in this definition**, which is one reason why ECPs cannot perform the s 33 examinations or initiate a formal involuntary admission.

- **Health care provider**: a person providing health care services, which includes EMS practitioners. Health care providers may act under s 9(1)(c) in an emergency and may make a s 33 application if relatives are unwilling, incapable or unavailable.

The three categories of user in Chapter V

Category	Who	Key sections
Voluntary	A user who **consents** to care	s 25 — entitled to appropriate care; may be referred to an appropriate establishment
Assisted	A user who is **incapable of making informed decisions** due to mental illness or severe or profound intellectual disability, and who **does not refuse** care	ss 26–31 — written application by a spouse, next of kin, partner, associate, parent or guardian (or a health care provider); examination by two mental health care practitioners; review by the Review Board
Involuntary	A user who is **incapable** of making an informed decision **and unwilling (refuses)**, and who is likely to inflict serious harm to self or others, or for whom care is necessary to protect their financial interests or reputation	ss 32–38 — s 33 application; two practitioners; s 34 72-hour assessment; Review Board; judicial review by the High Court (s 36)

Sections you must know verbatim or nearly verbatim

Section 9 — consent and emergency care without consent. Section 9(1) provides that a health care provider or a health establishment may provide care, treatment and rehabilitation services to, or admit, a mental health care user **only if** (a) the user has consented; (b) it is authorised by a court order or a Review Board; **or (c) due to mental illness, any delay in providing care, treatment and rehabilitation services or admission may result in the (i) death or irreversible harm to the health of the user; (ii) user inflicting serious harm to himself or herself or others; or (iii) user causing serious damage to or loss of property belonging to him or her or others.** Section 9(2) provides that any person or health establishment that provides care or admits a user under s 9(1)(c) (a) must report this fact in writing in the prescribed manner to the relevant Review Board, and (b) may not continue to provide care for longer than **24 hours** unless an application in terms of Chapter V is made within the 24-hour period.

Section 32 — criteria for involuntary care. A user must be provided with care without consent at a health establishment on an outpatient or inpatient basis if (a) a written application is made to the head of the health establishment and granted; (b) at the time of the

application there is a reasonable belief that the user has a mental illness of such a nature that (i) the user is likely to inflict serious harm to himself or herself or others, or (ii) care is necessary for the protection of the financial interests or reputation of the user; **and** (c) at the time of the application the user is incapable of making an informed decision on the need for the care and is unwilling to receive it.

Section 33 — application for involuntary care. The application may only be made by the user's spouse, next of kin, partner, associate, parent or guardian; if the user is under 18, it must be made by the parent or guardian; if those persons are unwilling, incapable or unavailable, it may be made by a **health care provider**. Applicants must have seen the user within **seven days** before the application (s 33(1)(b)). A health care provider applicant must state why they are making the application and what steps were taken to locate relatives (s 33(2)(b)). On receipt, the head of the establishment must cause the user to be examined by **two mental health care practitioners**, neither of whom may be the applicant, and at least one of whom must be qualified to conduct physical examinations (s 33(4)). They submit written findings on whether the s 32 criteria apply (s 33(5)).

Section 34 — 72-hour assessment. If the head of the establishment grants the application, the user is admitted for a **72-hour assessment** of physical and mental health status by a medical practitioner and another mental health care practitioner. Within 24 hours after the 72 hours, the findings are made available to the applicant. If involuntary care is not warranted, the user must be discharged immediately unless they consent to care; if further involuntary outpatient care is warranted, the user is discharged subject to conditions and the Review Board is informed; if further involuntary inpatient care is warranted, a written request must be submitted to the Review Board within seven days.

Section 40 — intervention by SAPS. Section 40(1): if a member of SAPS has reason to believe, **from personal observation or from information obtained from a mental health care practitioner**, that a person due to mental illness or severe or profound intellectual disability is likely to inflict serious harm to himself or herself or others, the member **must apprehend** the person and cause that person to be (a) taken to an appropriate health establishment administered under the auspices of the State for assessment of the person's mental health status, and (b) handed over into the custody of the head of the establishment or a designated person. Section 40(2): if the mental health care practitioner who assesses the person considers that the person is likely to inflict serious harm, the person must be admitted for a period **not exceeding 24 hours** for an application to be made in terms of s 33; if unlikely to cause harm, the person must be released immediately. Section 40(3): if no application is made within 24 hours, the person must be discharged immediately. Section 40(4)–(7): on request of the head of a health establishment, SAPS must locate, apprehend and return an absconded assisted or involuntary user, or assist with prescribed transfers, and must be informed of the estimated level of dangerousness. Section 40(8): SAPS members may use "such constraining measures as may be necessary and proportionate in the circumstances".

50.1 Discuss the implications of the Mental Health Care Act (Act 17 of 2002) on the ECP and EMS systems in South Africa

(The study guide lists "Act 15 of 2004"; the correct citation is Act 17 of 2002, in force from December 2004.)

1. The ECP may provide emergency care without consent — but within limits

Section 9(1)(c) applies to "a health care provider", which includes EMS practitioners. An ECP may therefore assess, treat (including sedation within scope), and transport a mental health care user without consent where, due to mental illness, any delay may result in death or irreversible harm, serious harm to self or others, or serious damage to property (Stander, Hodkinson and Dippenaar, 2021, confirm that s 9 allows prehospital providers to provide involuntary care during a behavioural emergency). The CDST operationalises this through its "obligated to transport" criteria (HPCSA PBEC, 2025, p. 48). However:

- the power exists only "due to mental illness" — for a patient whose risk arises from intoxication or a medical condition without mental illness, the ECP relies instead on the NHA s 7(1)(e) emergency provision and the common-law doctrine of necessity;
- the power is limited to what is necessary to deal with the emergency, i.e. to get the person safely to a health establishment for assessment;
- s 9(2) requires a written report to the Review Board and limits care without consent to 24 hours unless a Chapter V application is made. In practice the receiving health establishment completes the formal processes, but the ECP's PRF must clearly document the s 9(1)(c) grounds so that the establishment can rely on them.

2. The ECP cannot initiate a formal involuntary admission on scene

The s 33 application must be made in writing to the head of a health establishment, normally by a relative, and requires examination by two mental health care practitioners. Evans, Geduld and Stassen (2018) correctly note that "for prehospital providers, this is not achievable at the scene and therefore requires the rapid assessment and transport of these patients to health facilities for evaluation and possible emergency admission on an involuntary basis".

3. SAPS holds the primary statutory role for apprehension and transport

Section 40 places the duty to apprehend and convey persons likely to cause serious harm on SAPS. The MHCA "describes such involuntary transport to be a function of the South African Police Services and does not address the role of EMS", yet "in reality… EMS are often the first to arrive at the scene of a suicide attempt and have to manage these patients without police assistance" (Evans, Geduld and Stassen, 2018). Stander, Hodkinson and Dippenaar (2021) note that this places clinical decisions about mental state in the hands of SAPS members who lack medical training, and that the MHCA "fails to consider the responsibilities of prehospital emergency care providers". Implications for EMS systems include:

- the need for **joint EMS–SAPS protocols**, so that SAPS apprehends and EMS provides medical assessment, sedation and monitored transport;
- the need for SAPS to escort, and where necessary travel in, the ambulance;
- the CDST requirement that SAPS be contacted if restraint is considered, and that mechanical restraint be applied only on SAPS instruction (HPCSA PBEC, 2025, p. 46);
- delays and conflict when SAPS are unavailable, reluctant or poorly informed about the MHCA (Stander, Hodkinson and Dippenaar, 2021).

4. Destination

Section 40 requires conveyance to "an appropriate health establishment administered under the auspices of the State", and the 72-hour assessment takes place at designated establishments. EMS systems therefore need clear referral pathways to designated public facilities, and must ensure **medical clearance at an emergency centre** before transfer to psychiatric units. EMS is also heavily involved in **interfacility transfers** of assisted and involuntary users between establishments, which must comply with the prescribed transfer provisions (ss 27(10), 33(9), 34(4)(b), 34(6), 39 and 40(4)).

5. Rights of the user during EMS care

The ECP must respect the user's dignity and privacy (s 8), must not unfairly discriminate (s 10), must protect the user from exploitation, abuse and degrading treatment (s 11), and must keep information confidential except where disclosure is permitted (s 13). Restraint must be necessary and proportionate.

6. Deinstitutionalisation and EMS demand

The MHCA's deinstitutionalisation agenda was not matched by adequate community services, and EMS has become a de facto entry point for mental health care (Tilley et al., 2026). EMS systems must plan for this demand, including training, dedicated response capacity and pre-arranged agreements with facilities and community services (Evans, Geduld and Stassen, 2018, citing WHO, 2009).

7. Training and policy gaps

Only 7% of surveyed providers had training in the MHCA (Evans, Geduld and Stassen, 2018), and neither the MHCA nor the HPCSA provided explicit guidance on prehospital involuntary care (Stander, Hodkinson and Dippenaar, 2021). The CDST (HPCSA PBEC, 2025) now gives prehospital direction on sedation, restraint and obligation to transport, but national EMS policies on refusal by mental health care users and on involuntary transport are still recommended, together with curriculum changes (Evans, Geduld and Stassen, 2018).

8. Legal risk to the ECP

Acting outside the Act (for example detaining a competent person without grounds) may expose the ECP to claims of unlawful detention or assault; failing to act when the s 9(1)(c) criteria are met may expose the ECP to negligence claims and disciplinary action. Thorough documentation of the risk assessment, capacity assessment and legal grounds is the ECP's best protection.

High-yield points — Outcome 50

- MHCA 17 of 2002 (not 15 of 2004); in force December 2004.
- s 9(1)(c): emergency care without consent due to mental illness; s 9(2): report to Review Board, 24 hours.
- s 32: involuntary criteria (serious harm OR financial interests/reputation) AND incapable AND unwilling.

- s 33: application by relative (or health care provider); seen within 7 days; two mental health care practitioners.
- s 34: 72-hour assessment.
- s 40: SAPS must apprehend and take to a State facility; 24-hour admission pending s 33; abscondment.
- ECPs are health care providers but **not** mental health care practitioners.

Outcome 51. Apply the law insofar as it applies to consent to medical treatment on behalf of mentally incompetent patients

The elements of valid consent

For consent to be valid in South African law, it must be (1) given by a person with **legal capacity** (age) and **mental capacity** (competence) to consent; (2) **informed** — the person must know and understand the nature and extent of the risk and the consequences (the doctrine of informed consent confirmed in *Castell v De Greef* 1994 (4) SA 408 (C), and codified in ss 6 and 7 of the NHA); (3) **voluntary**, free from coercion or undue influence; (4) given **before** the intervention; and (5) **specific** to the intervention. Consent may be express (written or verbal) or implied by conduct.

The NHA requires that the user be informed of their health status, the range of diagnostic procedures and treatment options, the benefits, risks, costs and consequences of each option, and the user's **right to refuse** health services and the implications, risks and obligations of such refusal (NHA, s 6(1)), in a language the user understands and in a manner that takes the user's level of literacy into account (s 6(2)).

Assessing competence (mental capacity)

A patient is competent to consent or refuse if they can understand the information, retain it, weigh it in reaching a decision, and communicate the decision (see 39.4). Incompetence may be permanent (severe intellectual disability, advanced dementia), temporary (intoxication, delirium, head injury, hypoglycaemia, post-ictal state), or fluctuating (psychosis, mania, delirium). The HPCSA (2023, para 9.1.1.2) requires practitioners who doubt a patient's capacity to consult other relevant health practitioners.

Who may consent on behalf of an incompetent adult?

1. Under the National Health Act (general health care)

Section 7(1) of the NHA provides that a health service may not be provided to a user without the user's informed consent, unless:

- **(a)** the user is unable to give informed consent and such consent is given by a person **mandated by the user in writing** to grant consent on his or her behalf, or **authorised to give such consent in terms of any law or court order**;
- **(b)** the user is unable to give informed consent and no person is mandated or authorised to give such consent, and the consent is given by the **spouse or partner** of the user or, in the absence of such spouse or partner, a **parent, grandparent, an adult child or a brother or a sister** of the user, **in the specific order as listed**;
- **(c)** the provision of a health service without informed consent is authorised in terms of any law or a court order;
- **(d)** failure to treat the user, or group of people which includes the user, will result in a serious risk to public health; or

- **(e)** any delay in the provision of the health service to the user might result in his or her death or irreversible damage to his or her health and the user has not expressly, impliedly or by conduct refused that service.

The HPCSA (2023, para 3.1) encourages patients to appoint, in writing, a person to make decisions when they are no longer able to, and states that this "patient-selected proxy" must be regarded as the substitute for the patient.

2. Under the Mental Health Care Act (mental health care)

For mental health care, the MHCA rather than the NHA is the specific statute:

- **Assisted care** (ss 26–27): for a user incapable of making informed decisions who does not refuse care, an application by a spouse, next of kin, partner, associate, parent or guardian (or a health care provider if they are unwilling, incapable or unavailable) authorises care.
- **Involuntary care** (ss 32–34): for a user who is incapable and refuses, as above.
- **Emergency care** (s 9(1)(c)): no substitute consent is required in an emergency due to mental illness.

3. Curators

A High Court may appoint a **curator ad personam** (to the person, for decisions about the person's welfare and health care) or a **curator bonis** (for property and finances) for a person who is unable to manage their affairs. Chapter VIII of the MHCA also provides a process for the appointment of an **administrator** to manage the property of a mentally ill person or a person with severe or profound intellectual disability.

4. Children

See 48.3: Children's Act s 129 (consent from age 12 with sufficient maturity; parents/guardians for younger or less mature children; superintendent consent in an emergency; Minister or court in specified circumstances).

5. Emergency and necessity

In an emergency, the ECP may provide treatment necessary to prevent death or irreversible harm without consent (NHA, s 7(1)(e)), provided the patient has not validly refused. The HPCSA (2023, para 7.1) supports starting treatment that may be of benefit "until a clearer assessment can be made", and requires that the patient be informed afterwards when capable (para 7.2). The common-law defence of necessity provides additional justification.

6. Advance directives

An advance refusal of a specific treatment, made by the patient while competent, should be respected (HPCSA, 2023, para 9.2.1), although its legal status is not specifically legislated in South Africa. A directive cannot compel treatment outside acceptable standards (para 9.1.4).

Applying the law: worked prehospital scenarios

Scenario 1. A 70-year-old woman with known dementia has a fractured hip and is confused and refusing to go to hospital. Her husband is present. *Application*: she lacks capacity to make this decision (cannot understand and weigh the information). Under NHA s 7(1)(b), her spouse may consent on her behalf. If no relative were present and the fracture required urgent treatment, s 7(1)(e) and necessity would apply. Use non-drug calming measures and analgesia, involve the husband, document capacity and the source of consent.

Scenario 2. A 28-year-old man with known schizophrenia, off medication, is hearing voices telling him to kill himself and refuses transport. *Application*: due to mental illness, delay may result in serious harm to himself, so s 9(1)(c) of the MHCA applies; the CDST obliges transport. Request SAPS (s 40); de-escalate; offer voluntary sedation; if he remains dangerous, use restraint and CDST rapid sedation; transport to a State emergency centre; document the s 9(1)(c) grounds so that the facility can report to the Review Board and start a s 33 application within 24 hours.

Scenario 3. A 35-year-old woman took 20 paracetamol tablets four hours ago. She is calm, oriented, understands the risk of liver failure and death, and refuses transport. *Application*: assess capacity rigorously. The overdose itself, the context (suicidal intent), and possible underlying depression strongly suggest that her decision may be driven by mental illness, so her refusal is unlikely to be a capacitous, autonomous decision, and s 9(1)(c) is likely to apply (risk of death due to mental illness). The CDST obliges transport of a patient who "may be a suicide risk". Seek senior or medical advice, involve family, request SAPS if needed. Leaving her on scene with a signed refusal would be indefensible if she later died (Evans, Geduld and Stassen, 2018).

Scenario 4. An unconscious 45-year-old man is found hypoglycaemic, with no family present. *Application*: NHA s 7(1)(e) — treat immediately (CDST: dextrose 50% 20 mL IV or glucagon 1 mg IM).

Scenario 5. A competent 50-year-old man with a chronic mental illness who is currently stable refuses transport for a minor laceration. *Application*: a diagnosis of mental illness does not remove capacity. If he understands and weighs the information, his refusal must be respected (HPCSA, 2023, para 8.1.2). Document the refusal and advise.

51.1 Discuss how EMS systems and the actions of EMS practitioners have the potential to violate the rights of mentally ill persons

EMS systems and practitioners can violate the rights of mentally ill persons in the following ways:

Violations through excessive or unlawful intervention

1. **Unlawful detention and deprivation of liberty** (Constitution, s 12): restraining or transporting a competent person against their will without lawful grounds.
2. **Assault and violation of bodily integrity** (Constitution, s 12(2)): treating or sedating a competent patient without consent; excessive force; unsafe restraint (head-locks,

prone restraint with weight on the back, Tasers), as reported by South African providers (Evans, Geduld and Stassen, 2018).

3. **Cruel, inhuman or degrading treatment** (Constitution, s 12(1)(e); MHCA, s 11): punitive restraint, humiliating treatment, restraint in public view, leaving a patient soiled.
4. **Chemical restraint for convenience** rather than necessity.

Violations through neglect and discrimination

5. **Refusal or neglect of emergency treatment** (Constitution, s 27(3)): leaving at-risk patients on scene, refusing to transport, or delaying responses to "psych calls" because of stigma; some providers described suicidal patients as "a waste of time" (Evans, Geduld and Stassen, 2018).
6. **Unfair discrimination** (Constitution, s 9; MHCA, s 10): providing a lower standard of care to mental health care users, for example failing to assess for physical illness ("diagnostic overshadowing") and attributing all symptoms to psychiatric illness, contrary to the CDST instruction not to assume a psychiatric cause.
7. **Failure to treat pain** in patients who cannot express it (dementia, intellectual disability).
8. **Accepting invalid refusals** from patients who lack capacity, leading to preventable death — a violation of the right to life (s 11) and to access to health care (s 27).

Violations of dignity, autonomy and information

9. **Disrespect and stigmatising language** (Constitution, s 10; MHCA, s 8).
10. **Deception** — lying to obtain cooperation (Stander, Hodkinson and Dippenaar, 2021).
11. **Failure to inform the patient** of what is happening and why (NHA, s 6; HPCSA, 2023, para 7.2).
12. **Failure to involve the patient** in decisions to the extent possible.
13. **Breach of privacy and confidentiality** (Constitution, s 14; MHCA, s 13; POPIA): discussing the patient in front of bystanders, sharing information or images on social media, leaving PRFs unsecured.
14. **Failure to use an interpreter**, so that the patient cannot understand or communicate (HPCSA, 2023, para 8.1.7).

System-level violations

15. **Inadequate training** of practitioners in the MHCA and mental health care (Evans, Geduld and Stassen, 2018).
16. **Lack of protocols** for involuntary transport and refusal (Stander, Hodkinson and Dippenaar, 2021).
17. **Prolonged response times**, long waits at facilities and lack of suitable beds.
18. **Performance management systems** that penalise time spent with mental health care users (Evans, Geduld and Stassen, 2018).
19. **Inadequate staffing**, so that safe de-escalation and restraint are impossible.
20. **Failure to report abuse** of vulnerable mentally ill persons (SOA, s 54; DVA, ss 2A–2B; Older Persons Act).

Preventing rights violations

Training; adherence to the CDST and MHCA; capacity assessment and documentation; least restrictive practice; respectful communication; confidentiality; audit and complaints mechanisms; joint EMS–SAPS protocols; and advocacy for the patient.

High-yield points — Outcome 51

- Valid consent: capacity, informed, voluntary, prior, specific.
- NHA s 7(1): (a) written mandate/law/court; (b) spouse/partner → parent → grandparent → adult child → sibling; (c) law/court; (d) public health; (e) emergency.
- MHCA: assisted (ss 26–27), involuntary (ss 32–34), emergency (s 9(1)(c)).
- Rights at risk: liberty, bodily integrity, dignity, equality, privacy, access to emergency care, life.
- Diagnostic overshadowing is a form of discrimination.

Outcome 52. Discuss disorders listed in the DSM-5 relevant to emergency care practice

The *Diagnostic and Statistical Manual of Mental Disorders*, fifth edition, text revision (DSM-5-TR), published by the American Psychiatric Association (APA) in 2022, is the current classification. For each disorder below you will find the core diagnostic features (APA, 2022; Boland, Verduin and Ruiz, 2021), the emergency presentations and risks, and the South African prehospital management. You are not expected to diagnose psychiatric disorders in the field, but you must recognise them, understand their risks, and manage the emergency safely.

52.1 Neurodevelopmental disorders

Neurodevelopmental disorders have their onset in the developmental period, typically before school entry, and produce deficits in personal, social, academic or occupational functioning (APA, 2022).

52.1.1 Intellectual disabilities (intellectual developmental disorder)

Core features. Three criteria must be met: (A) deficits in intellectual functions — reasoning, problem solving, planning, abstract thinking, judgement, academic learning and learning from experience — confirmed by clinical assessment and standardised testing; (B) deficits in **adaptive functioning** that result in failure to meet developmental and sociocultural standards for personal independence and social responsibility, in the conceptual, social and practical domains; and (C) onset during the developmental period (APA, 2022). Severity (mild, moderate, severe, profound) is based on **adaptive functioning**, not IQ alone. Causes include genetic conditions (for example Down syndrome, fragile X syndrome), **fetal alcohol spectrum disorder** (South Africa, particularly the Western Cape, has some of the highest reported rates of fetal alcohol spectrum disorder in the world), perinatal hypoxia, infections (for example congenital infections, meningitis), malnutrition and traumatic brain injury.

Emergency relevance.

- Difficulty communicating pain or symptoms, so that **medical illness presents as behavioural change** (constipation, dental pain, urinary tract infection, otitis, fractures, seizures).
- Higher rates of epilepsy, which may cause post-ictal behaviour.
- Vulnerability to abuse, exploitation and sexual assault (SOA s 54 duty to report applies to persons with mental disability).
- Distress from a change in routine or environment.
- The term “severe or profound intellectual disability” appears in the MHCA (for example s 40), which means that the involuntary and assisted pathways can apply.
- Capacity to consent varies and must be assessed for each decision.

Prehospital management. Speak to the caregiver and consult any care plan; use simple language, visual cues and a calm, unhurried approach; allow the caregiver to accompany the patient; look carefully for a medical cause and pain; minimise restraint; use CDST sedation only if necessary, remembering that paradoxical reactions to benzodiazepines are more common;

the RCEM (2025) notes that "the differential diagnosis in a patient with learning difficulties is less likely to feature stimulant drugs", and recommends that "efforts should be made to have someone known to the patient communicate with them".

52.1.2 Communication disorders

Core features. Communication disorders include **language disorder** (persistent difficulties in acquiring and using language — reduced vocabulary, limited sentence structure, impaired discourse), **speech sound disorder** (difficulty producing speech sounds), **childhood-onset fluency disorder (stuttering)**, and **social (pragmatic) communication disorder** (difficulty with the social use of verbal and non-verbal communication, without the restricted and repetitive behaviours of autism) (APA, 2022).

Emergency relevance. Patients may be unable to explain symptoms, may become frustrated and agitated when not understood, and may be mistaken for being confused, intoxicated or uncooperative. Stuttering worsens under stress. Acquired communication problems (aphasia after stroke, dysarthria from intoxication or neurological disease) must be distinguished from developmental disorders — new-onset speech disturbance is a stroke until proven otherwise.

Prehospital management. Allow extra time; ask closed (yes/no) questions; use gestures, pictures, writing or communication devices; do not finish the patient's sentences; involve caregivers; do not interpret communication difficulty as lack of capacity without proper assessment; and use the FAST assessment for any new speech difficulty.

52.1.3 Autism spectrum disorder (ASD)

Core features. (A) Persistent deficits in **social communication and social interaction** across multiple contexts — deficits in social-emotional reciprocity, in non-verbal communication (eye contact, gestures, facial expressions), and in developing, maintaining and understanding relationships; and (B) **restricted, repetitive patterns of behaviour, interests or activities** — stereotyped or repetitive movements or speech (for example echolalia, hand-flapping, rocking), insistence on sameness and inflexible adherence to routines, highly restricted fixated interests, and **hyper- or hypo-reactivity to sensory input** (for example distress at noise, lights or touch, apparent indifference to pain); with onset in early development and clinically significant impairment (APA, 2022). Severity is graded in three levels (requiring support, substantial support, very substantial support), with or without accompanying intellectual or language impairment (Lord et al., 2018).

Emergency relevance.

- **"Meltdowns"** (involuntary responses to overwhelming sensory or emotional overload) may look like aggression or tantrums; they are not deliberate.
- Sirens, flashing lights, uniforms, touch, strangers and the confined ambulance are powerful triggers.
- Altered pain response — a serious injury may not be reported.
- Self-injurious behaviour (head-banging, biting).
- High rates of epilepsy, anxiety, ADHD and gastrointestinal problems.
- Wandering ("elopement") and drowning risk in children.

- Adults with ASD have increased rates of depression and suicidality.

Prehospital management. Reduce sensory stimulation (switch off sirens and lights near the patient, dim lights, speak quietly, limit the number of people); let the caregiver lead and stay with the patient; explain each step in concrete, literal language and allow processing time; avoid light touch and warn before touching; allow comfort objects, headphones or stimming behaviours; avoid restraint where possible (it may escalate fear); examine carefully for injury and pain; use CDST sedation only if necessary.

52.1.4 Motor disorders

Core features. Motor disorders include **developmental coordination disorder** (motor skills substantially below expected for age), **stereotypic movement disorder** (repetitive, seemingly driven, purposeless motor behaviour such as body-rocking, head-banging or self-biting, which may cause injury), and **tic disorders** — **Tourette's disorder** (multiple motor tics and at least one vocal tic present for more than one year, with onset before age 18), persistent (chronic) motor or vocal tic disorder, and provisional tic disorder (APA, 2022).

Emergency relevance. Self-injury from stereotypic movements; tics (including coprolalia — involuntary obscene utterances — in a minority with Tourette's) may be misinterpreted as intoxication, aggression or deliberate provocation; tics worsen with stress. Motor disorders must also be distinguished from **acute dystonia** (antipsychotics, metoclopramide), akathisia, chorea, seizures, and stimulant-induced movements.

Prehospital management. Reassure the patient and bystanders, protect from injury, do not punish or restrain tics, reduce stress, and look for an acute medical cause of any new movement disorder (for example a drug-induced dystonia, which responds to an anticholinergic such as biperiden or benztropine in hospital).

52.2 Psychotic disorders

Psychosis refers to a loss of contact with reality characterised by one or more of: **delusions, hallucinations, disorganised thinking (speech), grossly disorganised or abnormal motor behaviour (including catatonia), and negative symptoms** (diminished emotional expression and avolition) (APA, 2022).

52.2.1 Schizophrenia and its spectrum

Schizophrenia — DSM-5-TR criteria. (A) Two or more of the following, each present for a significant portion of time during a one-month period (or less if successfully treated), **at least one of which must be (1), (2) or (3)**: (1) delusions; (2) hallucinations; (3) disorganised speech; (4) grossly disorganised or catatonic behaviour; (5) negative symptoms. (B) Functioning in work, interpersonal relations or self-care is markedly below the level achieved before onset. (C) Continuous signs of disturbance persist for **at least six months**, including at least one month of active-phase symptoms (which may include prodromal or residual periods). (D) Schizoaffective disorder and depressive or bipolar disorder with psychotic features have been ruled out. (E) The disturbance is not attributable to a substance or another medical condition (APA, 2022).

The schizophrenia spectrum includes:

- **schizotypal personality disorder** (see 52.10);
- **delusional disorder** — one or more delusions for at least one month, without other prominent psychotic symptoms and with relatively preserved functioning;
- **brief psychotic disorder** — psychotic symptoms lasting at least one day but less than one month, with full return to premorbid functioning, often after a marked stressor or postpartum;
- **schizophreniform disorder** — as for schizophrenia, but lasting at least one month and less than six months;
- **schizoaffective disorder** — a major mood episode concurrent with schizophrenia criterion A, plus delusions or hallucinations for two weeks or more without a mood episode;
- **substance/medication-induced psychotic disorder** and **psychotic disorder due to another medical condition** (below);
- **catatonia** (52.3).

Epidemiology and pathophysiology. Lifetime prevalence is about 0.3–0.7%; onset is typically in late adolescence to the mid-30s, slightly earlier in men; the dopamine hypothesis (excess mesolimbic dopamine producing positive symptoms) underpins antipsychotic treatment, and cannabis use, urbanicity, migration, childhood adversity and obstetric complications are risk factors (McCutcheon, Reis Marques and Howes, 2020).

Emergency presentations and risks.

- Agitation and aggression driven by persecutory delusions or command hallucinations.
- **Suicide** — lifetime suicide risk is substantially elevated (about 5%), especially in young men early in the illness, after discharge, and with depression, command hallucinations or substance use.
- Self-neglect, malnutrition, exposure.
- **Medication adverse effects**: acute dystonia, akathisia, parkinsonism, **neuroleptic malignant syndrome** (hyperthermia, rigidity, altered consciousness, autonomic instability, raised CK), QT prolongation, clozapine-associated agranulocytosis (infection) and myocarditis, metabolic syndrome, and anticholinergic toxicity.
- Relapse due to non-adherence, substance use (cannabis, methamphetamine) or stress.
- Victimisation and comorbid physical illness (higher rates of cardiovascular disease, diabetes, HIV, TB).

Prehospital management. Scene safety; calm, honest de-escalation without challenging delusions (HPCSA PBEC, 2025, p. 46); ask about command hallucinations and suicidal and homicidal intent; exclude organic causes (glucose, SpO_2, temperature, head injury, intoxication); check for medication side effects (dystonia, NMS); CDST sedation if dangerous; MHCA s 9(1)(c) and s 40 if the patient lacks capacity and is at risk; transport to an emergency centre.

52.2.2 Substance/medication-induced psychotic disorder

Core features. Prominent delusions and/or hallucinations that develop **during or soon after** substance intoxication or withdrawal or after exposure to a medication, where the substance is capable of producing the symptoms and the disturbance is not better explained by an independent psychotic disorder (APA, 2022). Evidence of an independent disorder would include symptoms preceding substance use, persisting for a substantial period (about a month) after cessation of withdrawal or intoxication, or a history of non-substance-related episodes.

Common causes.

- **Stimulants**: methamphetamine ("tik" — a major problem in the Western Cape), cocaine, methcathinone ("cat").
- **Cannabis**, particularly high-potency products and synthetic cannabinoids.
- **Hallucinogens** (LSD, psilocybin mushrooms), **phencyclidine** and **ketamine**.
- **Alcohol** — alcoholic hallucinosis and delirium tremens (withdrawal).
- **Sedative-hypnotic withdrawal**.
- **Anticholinergic** agents (including *Datura stramonium*).
- **Medications**: corticosteroids, levodopa and dopamine agonists, **efavirenz** (an ARV historically widely used in South Africa, associated with vivid dreams, insomnia and occasional psychosis), isoniazid, anticholinergics, some antimalarials (mefloquine), and others.

Emergency relevance. Stimulant psychosis is frequently accompanied by **sympathomimetic toxicity** (tachycardia, hypertension, hyperthermia, seizures, arrhythmias, stroke, rhabdomyolysis) and may progress to **acute behavioural disturbance** with a risk of sudden death (RCEM, 2025). Tactile hallucinations (formication) and paranoia are typical.

Prehospital management. As for the sympathomimetic toxidrome (Outcome 60.1): de-escalation, early sedation per CDST (benzodiazepines are preferred because they treat both the agitation and the sympathomimetic toxicity), cooling if hyperthermic (CDST: remove clothing, spray water, fan or air-conditioning if temperature ≥38 °C; HPCSA PBEC, 2025, p. 57), aspirin, benzodiazepine and nitrates for stimulant chest pain within scope (HPCSA PBEC, 2025, p. 57), ECG, glucose, and transport.

52.2.3 Psychosis due to another medical condition

Core features. Prominent hallucinations or delusions, with evidence from the history, examination or investigations that the disturbance is the **direct pathophysiological consequence of another medical condition**, which is not better explained by another mental disorder and does not occur exclusively during delirium (APA, 2022).

Causes.

- **Neurological**: epilepsy (especially temporal lobe), brain tumours, stroke, head injury, Parkinson's disease and Lewy body dementia, Huntington's disease, multiple sclerosis.

- **Infections**: HIV (HIV-associated psychosis and opportunistic CNS infections), neurosyphilis, encephalitis (including autoimmune anti-NMDA receptor encephalitis, particularly in young women), cerebral malaria, sepsis.
- **Endocrine and metabolic**: hyperthyroidism and hypothyroidism, Cushing's syndrome, hypo- and hyperglycaemia, hypo- and hypercalcaemia, hyponatraemia, hepatic and uraemic encephalopathy, porphyria, vitamin B12 and thiamine deficiency (Wernicke–Korsakoff), pellagra (niacin deficiency).
- **Autoimmune**: systemic lupus erythematosus.
- **Postpartum** (postpartum psychosis — a psychiatric emergency with a risk of suicide and infanticide).

Clues. Late onset (>40 years), sudden onset, visual/tactile/olfactory hallucinations, altered consciousness, abnormal vital signs, focal neurological signs, and the absence of a psychiatric history.

Prehospital management. Identify and treat what you can (hypoglycaemia, hypoxia, seizures), protect the patient, sedate per CDST only if dangerous, and transport to an emergency centre for medical work-up.

52.3 Catatonia and its associated disorders and presentations

Definition. Catatonia is a marked psychomotor disturbance. DSM-5-TR requires **three or more of twelve features**: (1) **stupor** (no psychomotor activity; not actively relating to the environment); (2) **catalepsy** (passive induction of a posture held against gravity); (3) **waxy flexibility** (slight, even resistance to positioning by the examiner); (4) **mutism**; (5) **negativism** (opposition or no response to instructions or external stimuli); (6) **posturing** (spontaneous, active maintenance of a posture against gravity); (7) **mannerism** (odd caricature of normal actions); (8) **stereotypy** (repetitive, abnormally frequent, non-goal-directed movements); (9) **agitation** not influenced by external stimuli; (10) **grimacing**; (11) **echolalia**; (12) **echopraxia** (APA, 2022).

Associated disorders. In DSM-5-TR, catatonia may be specified as:

- **catatonia associated with another mental disorder** — most commonly **bipolar disorder and major depression** (more often than schizophrenia), schizophrenia, brief psychotic disorder, autism and post-traumatic states;
- **catatonic disorder due to another medical condition** — for example encephalitis (including anti-NMDA receptor encephalitis), non-convulsive status epilepticus, metabolic disorders (hyponatraemia, hepatic encephalopathy), head injury, and HIV-related disease;
- **unspecified catatonia**;
- **substance/medication-related catatonia**, including withdrawal from benzodiazepines.

Malignant (lethal) catatonia is catatonia with autonomic instability and hyperthermia, which can be fatal and closely resembles **neuroleptic malignant syndrome** (NMS is regarded by many as a drug-induced form of malignant catatonia).

Emergency risks. Dehydration, malnutrition, aspiration pneumonia, venous thromboembolism (pulmonary embolism), pressure injuries, contractures, rhabdomyolysis and acute kidney injury, and death in malignant catatonia. Excited catatonia may produce sudden violent agitation.

Recognition. The Bush–Francis Catatonia Rating Scale is a standard instrument (Bush et al., 1996). A useful bedside clue is a patient who is mute and immobile with eyes open, resisting passive movement or holding unusual postures, and who does not respond to commands, but who has no focal neurological deficit.

Management.

- **Prehospital**: ABCDE (protect the airway, risk of aspiration), glucose, temperature (hyperthermia suggests malignant catatonia/NMS), assess hydration, look for medical causes and recent antipsychotic exposure, IV fluids if dehydrated, pressure-area protection, transport.
- **Hospital**: a **lorazepam challenge** (for example 1–2 mg IV, with improvement within minutes to hours supporting the diagnosis) and ongoing benzodiazepines are first-line; **electroconvulsive therapy (ECT)** is highly effective for benzodiazepine-resistant and malignant catatonia; **antipsychotics should be avoided** in the acute phase because they may precipitate NMS (Boland, Verduin and Ruiz, 2021).

52.4 Bipolar disorders

Definitions (DSM-5-TR).

- **Manic episode**: a distinct period of abnormally and persistently elevated, expansive or irritable mood **and** abnormally and persistently increased activity or energy, lasting **at least one week** and present most of the day, nearly every day (or any duration if hospitalisation is necessary), with **three or more** of the following (**four** if the mood is only irritable): (1) inflated self-esteem or grandiosity; (2) decreased need for sleep; (3) more talkative than usual or pressure to keep talking; (4) flight of ideas or racing thoughts; (5) distractibility; (6) increased goal-directed activity or psychomotor agitation; (7) excessive involvement in activities with a high potential for painful consequences (spending sprees, sexual indiscretions, foolish investments). The episode causes marked impairment, necessitates hospitalisation, or has psychotic features, and is not due to a substance or medical condition (APA, 2022). The mnemonic **DIGFAST** (Distractibility, Impulsivity/Indiscretion, Grandiosity, Flight of ideas, Activity increased, Sleep decreased, Talkativeness) is useful.
- **Hypomanic episode**: the same symptoms lasting **at least four consecutive days**, with an observable change in functioning but **not** severe enough to cause marked impairment or require hospitalisation, and **without** psychotic features.
- **Major depressive episode**: see 52.5.1.

Bipolar I disorder requires **at least one manic episode**; hypomanic and major depressive episodes may occur but are not required. **Bipolar II disorder** requires **at least one hypomanic episode and at least one major depressive episode**, and **no manic episode ever**. **Cyclothymic disorder** involves at least two years (one year in children and adolescents)

of numerous periods of hypomanic and depressive symptoms that do not meet full criteria for an episode (APA, 2022). Bipolar disorder may also be substance/medication-induced (for example corticosteroids, stimulants, antidepressant-induced switching) or due to a medical condition (for example hyperthyroidism).

Clinical implications. Bipolar I tends to have more severe manic phases with a higher risk of psychosis and functional impairment; bipolar II is often dominated by depressive episodes. Antidepressants used without a mood stabiliser can precipitate mania or rapid cycling. Treatment includes mood stabilisers (**lithium**, valproate, carbamazepine, lamotrigine) and atypical antipsychotics, plus psychotherapy (Vieta et al., 2018; Tutorial 1 memo).

Emergency presentations and risks.

- Manic agitation, aggression, disinhibition, reckless behaviour (traffic accidents, sexual risk, assault), exhaustion, dehydration.
- **Suicide** — bipolar disorder carries one of the highest suicide risks of any psychiatric disorder, especially in depressive and mixed episodes.
- **Lithium toxicity** (Outcome 60.27) — narrow therapeutic index; precipitated by dehydration, renal impairment, NSAIDs, ACE inhibitors and diuretics.
- **Valproate toxicity** (CNS depression, hyperammonaemia) and **carbamazepine toxicity** (ataxia, nystagmus, coma, seizures, cardiac conduction effects).
- Psychotic features.

Prehospital management. Low-stimulus environment, brief and firm communication, avoid arguing, set limits; exclude stimulant intoxication, thyrotoxicosis and medication toxicity; ask about lithium and check for toxicity signs (tremor, vomiting, diarrhoea, ataxia, confusion); CDST sedation if dangerous; assess suicide risk in depressive or mixed states; transport.

52.5 Mood disorders

52.5.1 Depressive disorders

Major depressive disorder (MDD) — DSM-5-TR criteria. Five or more of the following during the same **two-week** period, representing a change from previous functioning, with **at least one being (1) depressed mood or (2) loss of interest or pleasure (anhedonia)**: (1) depressed mood most of the day, nearly every day; (2) markedly diminished interest or pleasure in all or almost all activities; (3) significant weight loss or gain, or change in appetite; (4) insomnia or hypersomnia; (5) psychomotor agitation or retardation (observable by others); (6) fatigue or loss of energy; (7) feelings of worthlessness or excessive or inappropriate guilt; (8) diminished ability to think or concentrate, or indecisiveness; (9) recurrent thoughts of death, recurrent suicidal ideation, a suicide attempt or a specific plan (APA, 2022). Symptoms cause clinically significant distress or impairment, are not due to a substance or medical condition, and there has never been a manic or hypomanic episode. The mnemonic **SIG E CAPS** (Sleep, Interest, Guilt, Energy, Concentration, Appetite, Psychomotor, Suicidality) plus depressed mood is useful.

Other depressive disorders.

- **Persistent depressive disorder (dysthymia)**: depressed mood for most of the day, for more days than not, for **at least two years** (one year in children).
- **Premenstrual dysphoric disorder**.
- **Disruptive mood dysregulation disorder** (children 6–18 years with severe recurrent temper outbursts and persistently irritable mood).
- **Substance/medication-induced depressive disorder** (for example alcohol, benzodiazepine withdrawal, corticosteroids, interferon, efavirenz) and **depressive disorder due to another medical condition** (hypothyroidism, stroke, Parkinson's disease, cancer, HIV).
- Specifiers include **with psychotic features** (often mood-congruent nihilistic or guilty delusions), **with peripartum onset**, with melancholic features, with anxious distress, and with catatonia.

Emergency relevance. Suicide and self-harm (the major risk); self-neglect, dehydration and malnutrition; psychotic depression; depressive stupor; overdose of antidepressants (**tricyclic antidepressants are highly dangerous in overdose** — Outcome 60.18; SSRIs can cause serotonin syndrome); and the **hypoactive delirium/dementia mimic** in older people (pseudodementia). Depression is common in South Africa and is closely linked to poverty, trauma, HIV and substance use (Tilley et al., 2026).

Prehospital management. Empathic communication; **always ask about suicidal ideation, intent and plan** (Outcome 45); remove means; look for overdose and self-injury; never leave the patient alone; transport if at risk (CDST; MHCA s 9(1)(c)). Treatment in hospital and the community includes psychotherapy (CBT) and antidepressants (SSRIs first-line) (Malhi and Mann, 2018).

52.5.2 Anxiety disorders

(Note: in DSM-5-TR, anxiety disorders form their own chapter rather than being "mood disorders", but they are grouped here as your study guide lists them.)

Core features. Anxiety disorders share **excessive fear** (an emotional response to a real or perceived imminent threat) and **anxiety** (anticipation of a future threat), with related behavioural disturbances such as avoidance (APA, 2022; Craske and Stein, 2016).

Disorder	Key DSM-5-TR features
Generalised anxiety disorder (GAD)	Excessive anxiety and worry about a number of events or activities, more days than not for **at least six months**, which the person finds difficult to control, with **three or more** of: restlessness, being easily fatigued, difficulty concentrating, irritability, muscle tension, sleep disturbance
Panic disorder	**Recurrent unexpected panic attacks**, with at least one month of persistent concern about further attacks or their consequences, or a significant maladaptive change in

Disorder	Key DSM-5-TR features
	behaviour (for example avoidance)
Panic attack (a specifier, not a disorder)	An abrupt surge of intense fear or discomfort that **peaks within minutes**, with **four or more** of: palpitations/tachycardia; sweating; trembling; shortness of breath; feelings of choking; chest pain; nausea/abdominal distress; dizziness/light-headedness; chills or heat sensations; paraesthesias; derealisation or depersonalisation; fear of losing control or "going crazy"; fear of dying
Agoraphobia	Marked fear or anxiety about two or more of: public transport, open spaces, enclosed places, standing in line or being in a crowd, being outside the home alone — because escape might be difficult or help unavailable
Specific phobia	Marked fear or anxiety about a specific object or situation (for example needles, blood, heights, animals), which is out of proportion to the actual danger and is actively avoided, for six months or more. The **blood-injection-injury type** can cause **vasovagal syncope**
Social anxiety disorder	Marked fear of social situations in which the person may be scrutinised by others, with fear of negative evaluation
Separation anxiety disorder and **selective mutism**	Mainly in children
Substance/medication-induced anxiety and **anxiety due to another medical condition**	For example caffeine, stimulants, alcohol or benzodiazepine withdrawal; hyperthyroidism, phaeochromocytoma, hypoglycaemia, arrhythmias, asthma, PE

Obsessive-compulsive disorder (OCD), which the Tutorial 1 memo discusses alongside anxiety disorders, is classified in DSM-5-TR in a separate chapter of obsessive-compulsive and related disorders. OCD consists of **obsessions** (recurrent, intrusive, unwanted thoughts, urges or images causing marked anxiety, which the person tries to suppress or neutralise) and/or **compulsions** (repetitive behaviours or mental acts that the person feels driven to perform in response to an obsession or according to rigid rules), which are time-consuming (more than one hour a day) or cause significant distress or impairment, with insight specified as good/fair, poor, or absent/delusional (APA, 2022).

Emergency relevance.

- **Panic attacks mimic life-threatening conditions** (ACS, PE, asthma, arrhythmias, hypoglycaemia, thyrotoxicosis, stimulant toxicity) — panic is a diagnosis of exclusion.
- Hyperventilation causes respiratory alkalosis with perioral and carpopedal paraesthesia and carpopedal spasm.
- Anxiety disorders are associated with suicidal ideation, particularly with comorbid depression or substance use.
- Benzodiazepine dependence and withdrawal.
- Needle phobia and claustrophobia may complicate procedures; OCD rituals may interfere with care (Tutorial 1 memo).

Prehospital management. Full assessment including vital signs, SpO_2, glucose and a 12-lead ECG; calm reassurance; coached slow breathing; remove from the stressor; avoid paper-bag rebreathing; explain procedures carefully for phobic patients and allow them to lie down for injections (blood-injection-injury phobia); respect OCD rituals where safe; transport if any doubt about the diagnosis or if the patient has suicidal thoughts. First-line long-term treatments for GAD and panic disorder are SSRIs/SNRIs and CBT (with exposure for panic disorder); benzodiazepines are limited to short-term use because of dependence (Craske and Stein, 2016; Tutorial 1 memo).

52.6 Disorders related to trauma and stressors

These disorders require **exposure to a traumatic or stressful event** as a diagnostic criterion (APA, 2022). Exposure to a traumatic event is defined as exposure to actual or threatened death, serious injury, or sexual violence through: (1) directly experiencing it; (2) witnessing it in person as it occurs to others; (3) learning that it occurred to a close family member or friend (violent or accidental); or (4) repeated or extreme exposure to aversive details of traumatic events — which includes **first responders** (for example collecting human remains, or repeated exposure to details of child abuse) (APA, 2022). South Africa's high levels of violence, rape and road trauma make these disorders very common, both in patients and in emergency care providers (Tilley et al., 2026, citing Padmanabhanunni and Pretorius, 2025).

The symptom clusters (common to both disorders below) are:

- **Intrusion**: distressing memories, nightmares, **flashbacks** (dissociative re-experiencing), and psychological or physiological distress at cues.
- **Avoidance** of memories, thoughts, feelings or external reminders.
- **Negative alterations in cognition and mood**: inability to remember aspects of the event, persistent negative beliefs ("I am bad", "the world is completely dangerous"), distorted blame, persistent negative emotions (fear, horror, anger, guilt, shame), diminished interest, detachment, inability to feel positive emotions.
- **Alterations in arousal and reactivity**: irritability and angry outbursts, reckless or self-destructive behaviour, **hypervigilance**, exaggerated startle response, poor concentration, sleep disturbance.
- **Dissociative symptoms**: depersonalisation, derealisation, a sense of being in a daze.

52.6.1 Acute stress disorder

Core features. Following exposure to a traumatic event, the presence of **nine or more** symptoms from any of the five categories (intrusion, negative mood, dissociation, avoidance, arousal), with onset after the trauma and a duration of **three days to one month** after the trauma (APA, 2022). It causes clinically significant distress or impairment and is not due to a substance or medical condition.

Emergency relevance. Victims of assault, rape, collisions, fires and disasters may be in an acute stress state at the scene — dazed, dissociated, numb, agitated, or unable to give a history. Acute stress reactions are normal responses to abnormal events, and most people recover.

Prehospital management — Psychological First Aid (PFA) (WHO principles: look, listen, link): ensure safety and treat injuries; provide calm, comforting presence; **do not force the person to talk** about the event or to give detailed accounts ("psychological debriefing" that forces recounting may be harmful); listen if they wish to talk; meet basic needs (warmth, water, information); help them connect with family and social supports; provide accurate information; screen for injuries, intoxication, suicidal ideation and the need for protection (for example in domestic violence or sexual assault); and refer for follow-up.

52.6.2 Post-traumatic stress disorder (PTSD)

Core features. Exposure to a traumatic event plus symptoms from **all four clusters** — at least **one intrusion**, at least **one avoidance**, at least **two negative alterations in cognition and mood**, and at least **two alterations in arousal and reactivity** — lasting **more than one month**, causing clinically significant distress or impairment, and not attributable to a substance or medical condition (APA, 2022). Specifiers include "with dissociative symptoms" and "with delayed expression" (full criteria not met until at least six months after the event). A preschool subtype exists for children aged six years or younger.

Emergency relevance. Flashbacks can cause sudden agitation, fear and aggression (for example when a patient with PTSD from an assault is restrained or touched unexpectedly); comorbid depression, substance use and suicide; self-harm; and re-traumatisation through restraint, examinations (especially genital examinations after sexual assault), uniforms and sirens. PTSD is also an **occupational hazard for EMS staff**, with providers describing "lasting visions" from suicide scenes (Tilley et al., 2026).

Prehospital management. Trauma-informed care: explain every action before doing it, seek permission before touching, give the patient choice and control, avoid restraint where possible, offer a same-gender provider where possible (especially after sexual violence), use a calm and predictable approach, ground a patient experiencing a flashback ("You are safe, you are in an ambulance, today is…, my name is…"), assess suicide risk, and refer. Evidence-based long-term treatments include trauma-focused psychotherapy (prolonged exposure, cognitive processing therapy, EMDR) and SSRIs (Shalev, Liberzon and Marmar, 2017).

52.7 Eating disorders

52.7.1 Anorexia nervosa

DSM-5-TR criteria. (A) Restriction of energy intake relative to requirements, leading to a **significantly low body weight** in the context of age, sex, developmental trajectory and physical health; (B) **intense fear of gaining weight** or becoming fat, or persistent behaviour that interferes with weight gain, even though at a significantly low weight; (C) disturbance in the way body weight or shape is experienced, undue influence of weight or shape on self-evaluation, or persistent lack of recognition of the seriousness of the low weight (APA, 2022). Subtypes are **restricting type** and **binge-eating/purging type**. Severity in adults is based on BMI (mild ≥17; moderate 16–16.99; severe 15–15.99; extreme <15 kg/m^2).

Emergency relevance — anorexia has the highest mortality of any psychiatric disorder, from medical complications and suicide (Treasure, Duarte and Schmidt, 2020). Life-threatening complications include:

- **bradycardia, hypotension, arrhythmias, QT prolongation** and sudden cardiac death;
- **hypokalaemia, hypophosphataemia, hypomagnesaemia, hypoglycaemia**;
- **hypothermia**;
- syncope and falls; osteoporosis and fractures;
- **refeeding syndrome** — when nutrition is restarted, insulin release drives phosphate, potassium and magnesium into cells, causing severe hypophosphataemia with cardiac failure, arrhythmias, respiratory failure, seizures and death; thiamine deficiency may precipitate Wernicke's encephalopathy;
- suicide.

Prehospital management. ABCDE, ECG monitoring (bradycardia, QT prolongation, arrhythmias), glucose (treat hypoglycaemia), temperature (prevent heat loss), cautious IV fluids in shock (risk of cardiac failure), thiamine 100 mg IV/IM for malnourished patients within CDST scope (HPCSA PBEC, 2025, p. 73), suicide risk assessment, and a non-judgemental approach (do not comment on weight or appearance). Transport to hospital.

52.7.2 Bulimia nervosa

DSM-5-TR criteria. (A) Recurrent episodes of **binge eating** (eating, in a discrete period, an amount of food definitely larger than most people would eat in a similar period, with a **sense of lack of control**); (B) recurrent inappropriate **compensatory behaviours** to prevent weight gain — self-induced vomiting, misuse of laxatives, diuretics or other medications, fasting or excessive exercise; (C) both occurring on average **at least once a week for three months**; (D) self-evaluation unduly influenced by body shape and weight; (E) not occurring exclusively during episodes of anorexia nervosa (APA, 2022). Body weight is usually normal or above normal.

Emergency relevance.

- **Electrolyte disturbances from purging**: **hypokalaemic, hypochloraemic metabolic alkalosis** from vomiting (and metabolic acidosis from laxative abuse), causing weakness, arrhythmias and sudden death.
- Dehydration.
- **Oesophageal complications**: Mallory–Weiss tear (haematemesis), and rarely **Boerhaave syndrome** (oesophageal rupture — severe chest pain, subcutaneous emphysema, shock).
- Gastric dilatation or rupture after binges.
- Ipecac abuse (cardiomyopathy).
- Physical signs: **Russell's sign** (calluses on the knuckles from self-induced vomiting), parotid gland enlargement, dental enamel erosion.
- Comorbid depression, substance use, self-harm and suicide.

Prehospital management. ABCDE, ECG monitoring for arrhythmias, IV fluids for dehydration, management of haematemesis and suspected oesophageal rupture (keep nil by mouth), suicide risk assessment, non-judgemental approach, and transport.

52.8 Substance-related and addictive disorders

DSM-5-TR classifies **substance use disorders** and **substance-induced disorders** (intoxication, withdrawal, and substance/medication-induced mental disorders) for ten classes of substances: alcohol, caffeine, cannabis, hallucinogens (phencyclidine and other hallucinogens), inhalants, opioids, sedatives/hypnotics/anxiolytics, stimulants, tobacco, and other/unknown substances. **Gambling disorder** is the only behavioural addiction included (APA, 2022).

52.8.1 Substance intoxication and withdrawal

- **Substance intoxication**: the development of a reversible, substance-specific syndrome due to recent ingestion of a substance, with clinically significant problematic behavioural or psychological changes (for example belligerence, mood lability, impaired judgement) attributable to the effects of the substance on the CNS, developing during or shortly after use, and not attributable to another medical condition or mental disorder (APA, 2022).
- **Substance withdrawal**: a substance-specific problematic behavioural change, with physiological and cognitive concomitants, due to the cessation of, or reduction in, heavy and prolonged substance use. Withdrawal syndromes are generally the **opposite** of the intoxication effects (for example sedative intoxication causes sedation; sedative withdrawal causes agitation, tremor, seizures).

Prehospital approach to intoxication and withdrawal: ABCDE; airway protection; glucose; thiamine for alcohol users (CDST: thiamine 100 mg IV/IM when there is concern about vitamin deficiency, for example malnourishment or chronic alcohol use; HPCSA PBEC, 2025, p. 73); identify the toxidrome (Outcome 62.3); treat specific toxicity (CDST Poisoning tool); manage seizures (CDST Seizures tool); look for head injury and co-ingestion; do not assume intoxication explains everything.

52.8.2 Substance use disorders

DSM-5-TR criteria. A problematic pattern of use leading to clinically significant impairment or distress, with **at least two of eleven criteria within a 12-month period**, grouped as follows (APA, 2022):

- **Impaired control**: (1) taking larger amounts or for longer than intended; (2) persistent desire or unsuccessful efforts to cut down; (3) a great deal of time spent obtaining, using or recovering; (4) **craving**.
- **Social impairment**: (5) failure to fulfil major role obligations; (6) continued use despite social or interpersonal problems; (7) giving up important activities.
- **Risky use**: (8) recurrent use in physically hazardous situations; (9) continued use despite knowledge of physical or psychological problems caused or worsened by the substance.
- **Pharmacological criteria**: (10) **tolerance**; (11) **withdrawal**.

Severity: **mild** = 2–3 criteria; **moderate** = 4–5; **severe** = 6 or more. (Tolerance and withdrawal alone do not count when the substance is taken as prescribed under medical supervision.)

South African context. Alcohol is the most widely used substance; cannabis ("dagga"), methamphetamine ("tik"), methaqualone ("Mandrax", often smoked with cannabis as a "white pipe"), heroin-based mixtures ("nyaope" or "whoonga"), cocaine and prescription opioids and benzodiazepines are important (data from the South African Community Epidemiology Network on Drug Use, SACENDU, of the South African Medical Research Council). Tilley et al. (2026) cite an estimate that 18.9% of the population abuse alcohol and/or drugs. The Prevention of and Treatment for Substance Abuse Act 70 of 2008 provides for voluntary and involuntary admission to treatment centres.

52.8.3 Alcohol-related disorders

- **Alcohol intoxication**: slurred speech, incoordination, unsteady gait, nystagmus, impaired attention or memory, stupor or coma, with behavioural changes (disinhibition, aggression, labile mood). Severe intoxication causes respiratory depression, aspiration, hypothermia and **hypoglycaemia** (especially in children and malnourished adults).
- **Alcohol withdrawal**: begins about 6–24 hours after the last drink, with tremor, sweating, tachycardia, hypertension, anxiety, nausea, insomnia and agitation; **withdrawal seizures** typically at 12–48 hours; **alcoholic hallucinosis** (usually visual, sometimes auditory or tactile, with a clear sensorium); and **delirium tremens** typically at 48–96 hours — disorientation, fluctuating consciousness, vivid visual and tactile hallucinations, severe autonomic hyperactivity and hyperthermia, with significant mortality if untreated (Schuckit, 2014). Severity can be scored in hospital with the CIWA-Ar (Sullivan et al., 1989).
- **Wernicke's encephalopathy**: acute thiamine deficiency with the classic (but often incomplete) triad of **confusion, ataxia and ophthalmoplegia/nystagmus**; give thiamine **before or with** glucose in at-risk patients where possible (CDST thiamine 100 mg IV/IM). Untreated, it progresses to **Korsakoff syndrome** (anterograde amnesia with confabulation).

- **Other complications**: head injury (subdural haematoma), GI bleeding (varices, Mallory–Weiss), pancreatitis, alcoholic ketoacidosis, hepatic encephalopathy, cardiomyopathy and arrhythmias ("holiday heart"), aspiration pneumonia, trauma, interpersonal violence and suicide.
- **Fetal alcohol spectrum disorder** (prenatal exposure).

Prehospital management: airway and aspiration prevention (lateral position), glucose, thiamine, temperature, look for injury, benzodiazepines for withdrawal seizures and agitation (CDST seizure and aggression tools), IV fluids.

52.8.4 Cannabis-related disorders

- **Intoxication**: euphoria, anxiety, sensation of slowed time, impaired judgement and coordination, **conjunctival injection**, increased appetite, dry mouth, **tachycardia**; at high doses, paranoia, panic, psychosis. Children who ingest edibles may present with drowsiness, ataxia, hypotonia, and occasionally coma or seizures.
- **Withdrawal**: irritability, anger, anxiety, insomnia, decreased appetite, restlessness, depressed mood, and physical symptoms (abdominal pain, tremor, sweating, headache) beginning within about a week of cessation of heavy use.
- **Cannabinoid hyperemesis syndrome**: cyclical severe vomiting and abdominal pain in chronic heavy users, characteristically relieved by **hot showers or baths**.
- **Synthetic cannabinoids** cause much more severe toxicity: agitation, psychosis, seizures, tachycardia, myocardial ischaemia, acute kidney injury and death.
- Cannabis use is a risk factor for psychosis and schizophrenia, especially with adolescent onset and high-potency products.

Prehospital management: reassurance in a calm environment, supportive care, benzodiazepines for severe agitation (CDST), treatment of dehydration from vomiting, and consideration of co-ingestion.

52.8.5 Hallucinogen-related disorders

Hallucinogens include LSD, psilocybin ("magic mushrooms"), mescaline, MDMA ("ecstasy", which has both stimulant and hallucinogenic effects), and the dissociatives phencyclidine (PCP) and ketamine (which DSM-5-TR classifies separately under phencyclidine-related disorders).

- **Intoxication**: perceptual changes (intensified perceptions, illusions, hallucinations, synaesthesia), depersonalisation, derealisation, anxiety or panic ("bad trip"), paranoia, impaired judgement, with sympathomimetic signs (**mydriasis**, tachycardia, sweating, palpitations, blurred vision, tremor, incoordination). PCP and ketamine cause dissociation, **vertical and horizontal nystagmus**, analgesia, hypertension, agitation and violence.
- **MDMA** specifically causes hyperthermia, **hyponatraemia** (from excessive water intake and SIADH — seizures and cerebral oedema), serotonin syndrome, rhabdomyolysis, DIC and multi-organ failure.
- **Hallucinogen persisting perception disorder** ("flashbacks").
- There is **no recognised withdrawal syndrome** for most hallucinogens.

Prehospital management: a calm, quiet environment and "talking down" with reassurance; prevent injury (patients may act on perceptions, for example believing they can fly); benzodiazepines for agitation (CDST); cool if hyperthermic; check glucose; consider hyponatraemia in MDMA users with seizures or altered consciousness; transport.

52.8.6 Opioid-related disorders

- **Intoxication**: initial euphoria followed by apathy and sedation; **the opioid toxidrome — depressed level of consciousness, pinpoint pupils (miosis), and respiratory depression**, with bradycardia, hypotension and hypothermia (HPCSA PBEC, 2025, p. 55). Note that pupils may be dilated with severe hypoxia or co-ingestants, and some opioids (for example tramadol, pethidine) may not cause miosis. Tramadol lowers the seizure threshold and may cause serotonin syndrome.
- **Withdrawal** (6–12 hours after short-acting opioids such as heroin; longer for methadone): dysphoria, craving, nausea and vomiting, muscle aches, **lacrimation, rhinorrhoea, mydriasis, piloerection ("gooseflesh")**, sweating, diarrhoea, yawning, fever and insomnia. Opioid withdrawal is extremely unpleasant but rarely life-threatening in adults (neonatal withdrawal can be severe).
- **Complications**: non-cardiogenic pulmonary oedema, aspiration, rhabdomyolysis and compartment syndrome from prolonged immobility, infections from injecting (abscesses, endocarditis, HIV, hepatitis B and C).
- South Africa has seen growing use of heroin-based "nyaope/whoonga".

Prehospital management (CDST, HPCSA PBEC, 2025, p. 56): airway and BVM ventilation first; **naloxone if respiratory depression**: adult *naloxone 0.4 mg IV/IM*; if a **known opioid abuser**, *naloxone 0.04 mg IV/IM* to limit withdrawal; reassess every two minutes and give increasing doses (0.8 mg, 2 mg, 4 mg every two minutes) to a maximum total of 10 mg; child *naloxone 0.1 mg/kg IV/IM (up to 2 mg)*, repeated every two minutes to a maximum cumulative 10 mg. If the respiratory rate remains low, consider other poisoning. **Naloxone wears off quickly** (its duration of action is shorter than that of most opioids), so monitor the respiratory rate closely and give further doses if needed. Titrate to adequate breathing, not full awakening, to avoid precipitating acute withdrawal, agitation and vomiting.

52.8.7 Sedative-, hypnotic- or anxiolytic-related disorders

This class includes **benzodiazepines** (diazepam, lorazepam, alprazolam, midazolam), **Z-drugs** (zolpidem, zopiclone), **barbiturates** (phenobarbital), **methaqualone** ("Mandrax") and **GHB/GBL**.

- **Intoxication**: slurred speech, incoordination, unsteady gait, nystagmus, impaired cognition, stupor or coma, and (with co-ingestion of alcohol or opioids, or with barbiturates) respiratory depression. The CDST sedative-hypnotic toxidrome is a **depressed level of consciousness, slurred speech and respiratory depression** with relatively normal vital signs (HPCSA PBEC, 2025, p. 55). Benzodiazepines alone rarely cause death unless combined with other depressants.
- **Withdrawal**: autonomic hyperactivity, tremor, insomnia, anxiety, nausea, hallucinations, agitation, **seizures** and delirium — **potentially life-threatening**, like alcohol withdrawal.

- **Anterograde amnesia** with benzodiazepines and GHB (relevant to drug-facilitated sexual assault, 60.14).

Prehospital management: airway and ventilatory support; the CDST does **not** include flumazenil for prehospital use; flumazenil is a hospital-level antidote that is contraindicated in mixed overdoses (particularly with TCAs), in benzodiazepine-dependent patients and in patients with epilepsy, because it can precipitate seizures and withdrawal (Hüser et al., 2025; RCEM, 2025). Manage withdrawal seizures with benzodiazepines (CDST seizure tool).

52.8.8 Stimulant-related disorders

Stimulants include amphetamine-type substances (**methamphetamine "tik"**, MDMA, methcathinone "cat"), **cocaine** (including crack cocaine), and prescription stimulants (methylphenidate).

- **Intoxication**: euphoria, hypervigilance, grandiosity, anxiety, anger, impaired judgement, repetitive behaviour, paranoia, psychosis; with the **sympathomimetic toxidrome** — agitation, hallucinations, seizures, sweating, hyperthermia, **dilated pupils**, tachycardia and hypertension (HPCSA PBEC, 2025, p. 55). Complications include chest pain and myocardial infarction (cocaine-induced coronary vasospasm), arrhythmias, aortic dissection, stroke and intracranial haemorrhage, seizures, hyperthermia, rhabdomyolysis, acute kidney injury, and **acute behavioural disturbance with sudden death** (RCEM, 2025).
- **Withdrawal ("crash")**: dysphoric mood, fatigue, vivid unpleasant dreams, insomnia or hypersomnia, increased appetite, psychomotor retardation or agitation, and **suicidal ideation**.

Prehospital management: see Outcome 60.1.

52.8.9 Situations in which the substance related to the disorder is not known

DSM-5-TR provides the category **"other (or unknown) substance-related disorders"** for situations in which the substance is unknown or not covered by the other classes (APA, 2022). In emergency practice this is common: patients may be unconscious, unwilling or unable to give a history; street drugs are frequently adulterated or misrepresented (for example fentanyl in heroin, synthetic cannabinoids sold as cannabis, mixtures such as "nyaope"); and mixed ingestions occur in about half of poisoning admissions (Hüser et al., 2025). The approach is to **treat the patient, not the poison**:

1. ABCDE stabilisation (airway, breathing, circulation, disability including glucose, exposure including temperature) (Hüser et al., 2025; HPCSA PBEC, 2025).
2. History and collateral information, a scene search (containers, paraphernalia, notes), and bringing all substances to hospital (HPCSA PBEC, 2025, pp. 54–55).
3. **Toxidrome recognition** to identify the likely class (Outcome 62.3).
4. Empirical treatment: naloxone for an opioid picture with respiratory depression; benzodiazepines for agitation and seizures; cooling for hyperthermia; glucose for hypoglycaemia; sodium bicarbonate for a widened QRS (ANT/ECP).

5. Contact the **Poisons Information Helpline of the Western Cape (0861 555 777)** (HPCSA PBEC, 2025, p. 55).
6. Transport every patient poisoned with an unknown substance to the closest emergency centre (HPCSA PBEC, 2025, p. 55).

52.9 Neurocognitive disorders

Neurocognitive disorders (NCDs) are disorders in which the primary clinical deficit is in cognitive function and which are **acquired** rather than developmental (APA, 2022). DSM-5-TR assesses six cognitive domains: complex attention, executive function, learning and memory, language, perceptual-motor function, and social cognition. **Major NCD** (previously "dementia") involves significant decline that interferes with independence in everyday activities; **mild NCD** involves modest decline without loss of independence. The aetiological subtypes include Alzheimer's disease, frontotemporal lobar degeneration, Lewy body disease, vascular disease, traumatic brain injury, substance/medication use, HIV infection, prion disease, Parkinson's disease, Huntington's disease, another medical condition, and multiple aetiologies.

52.9.1 Delirium

DSM-5-TR criteria. (A) A disturbance in **attention** (reduced ability to direct, focus, sustain and shift attention) and **awareness** (reduced orientation to the environment); (B) the disturbance **develops over a short period** (usually hours to a few days), represents a change from baseline, and tends to **fluctuate** in severity during the day; (C) an additional disturbance in cognition (memory deficit, disorientation, language, visuospatial ability or perception); (D) the disturbances are not better explained by a pre-existing NCD and do not occur in the context of a severely reduced level of arousal such as coma; (E) there is evidence that the disturbance is a direct physiological consequence of another medical condition, substance intoxication or withdrawal, exposure to a toxin, or multiple aetiologies (APA, 2022). Subtypes are **hyperactive, hypoactive and mixed**.

Epidemiology and importance. Delirium is common in older inpatients and emergency patients, is often missed (especially the hypoactive form), and is associated with increased mortality, longer hospital stay, functional decline, institutionalisation and later dementia (Inouye, Westendorp and Saczynski, 2014).

Causes — think of the "**I WATCH DEATH**" or "**PINCH ME**" mnemonics: infection (UTI, pneumonia, sepsis, meningitis), withdrawal (alcohol, benzodiazepines), acute metabolic disturbance (glucose, sodium, calcium, uraemia, hepatic failure), trauma (head injury, burns, fractures), CNS pathology (stroke, haemorrhage, seizures), hypoxia, deficiencies (thiamine, B12), endocrine (thyroid), acute vascular (hypertensive encephalopathy, shock), toxins and drugs (anticholinergics, opioids, benzodiazepines, polypharmacy, poisoning), heavy metals; and pain, constipation, urinary retention, sleep deprivation and environmental change (Desai, Kulkarni and Verma, n.d.).

Assessment tools. The **Confusion Assessment Method (CAM)** (Inouye et al., 1990) requires (1) acute onset and fluctuating course **and** (2) inattention, plus **either** (3) disorganised thinking **or** (4) altered level of consciousness.

Prehospital management. Delirium is a **medical emergency**. ABCDE; oxygen if SpO_2 <92%; glucose; temperature; look for infection, trauma, dehydration and toxins; bring medications; non-pharmacological measures (reorientation, family presence, glasses and hearing aids, quiet environment); avoid benzodiazepines except for alcohol/benzodiazepine withdrawal or seizures; reduced CDST doses if sedation is unavoidable; transport to an emergency centre.

52.9.2 Alzheimer's disease (major or mild NCD due to Alzheimer's disease)

Core features. Insidious onset and gradual progression of impairment in one or more cognitive domains (two or more for major NCD), with prominent early impairment of **learning and memory** (particularly recent memory), later involving language (word-finding difficulty), visuospatial function (getting lost), executive function and behaviour; with evidence of a causative genetic mutation or a steadily progressive decline without extended plateaus, and no evidence of mixed aetiology (APA, 2022). Pathology involves extracellular amyloid-β plaques and intracellular tau neurofibrillary tangles; it is the most common cause of dementia; and age is the main risk factor (Scheltens et al., 2021).

Emergency relevance. Superimposed **delirium** (from infection, dehydration, medications); behavioural and psychological symptoms (agitation, aggression, wandering, psychosis, sundowning); falls and injuries; neglect and abuse; inability to report pain; questions of capacity and consent (Outcome 49).

Prehospital management. Collateral history to establish baseline; look for an acute cause of any change; non-pharmacological calming; analgesia; minimal sedation (CDST >60-year doses); capacity assessment and substitute consent (NHA s 7(1)(b)); consider elder abuse.

52.9.3 Traumatic brain injuries (major or mild NCD due to TBI)

Core features. Evidence of a traumatic brain injury (an impact to the head or rapid movement or displacement of the brain within the skull) with one or more of: loss of consciousness, post-traumatic amnesia, disorientation and confusion, or neurological signs (neuroimaging abnormality, new seizures, visual field deficits, anosmia, hemiparesis); with the NCD presenting immediately after the injury or after recovery of consciousness and persisting past the acute post-injury period (APA, 2022).

Behavioural features of TBI. Acutely: confusion, agitation, combativeness, disinhibition, amnesia — the agitated head-injured patient is often mistaken for being drunk or psychotic. Chronically: personality change, impulsivity, irritability and aggression (frontal lobe injury), depression, anxiety, PTSD, substance use and **increased suicide risk**, together with cognitive deficits (Maas et al., 2022). South Africa's very high rates of interpersonal violence and road traffic injuries make TBI common.

Prehospital management. CDST Head injury tool: manage the airway and spine, avoid hypoxia (SpO_2 >92%) and hypotension (both worsen secondary brain injury), check glucose, assess and trend GCS and pupils, look for signs of raised intracranial pressure, and transport to an appropriate facility. **Never attribute agitation to intoxication without excluding head injury** (the CDST instructs the provider to "look for and manage injury"). If sedation is required for agitation, the CDST notes ketamine 0.5–1 mg/kg IV for agitation due to hypoxia or TBI where the patient cannot be adequately pre-oxygenated for airway management (HPCSA

PBEC, 2025, p. 14 — Airway management tool, drug-assisted intubation; ketamine is ANT/ECP scope).

52.9.4 HIV infections (major or mild NCD due to HIV infection)

Core features. Documented HIV infection, with an NCD not better explained by non-HIV conditions such as secondary brain diseases (for example progressive multifocal leukoencephalopathy or cryptococcal meningitis) (APA, 2022). HIV-associated neurocognitive disorders (HAND) range from asymptomatic neurocognitive impairment to mild neurocognitive disorder to HIV-associated dementia (the Frascati criteria; Antinori et al., 2007). Typical features are "subcortical": psychomotor slowing, poor concentration and memory retrieval, apathy, executive dysfunction, and motor problems (gait disturbance, tremor).

South African relevance. South Africa has one of the largest HIV epidemics in the world (roughly 13% of the population living with HIV). Although antiretroviral therapy has reduced severe HIV-associated dementia, milder HAND remains common, and **acute behavioural change in a person living with HIV demands a search for opportunistic infections**: cryptococcal meningitis, toxoplasmosis, tuberculous meningitis and tuberculoma, PML, CMV encephalitis, neurosyphilis, primary CNS lymphoma, as well as HIV-related psychosis, depression, and **ARV neuropsychiatric side effects** (efavirenz: vivid dreams, insomnia, depression, psychosis; dolutegravir: insomnia, headache, occasionally depression).

Prehospital management. Treat as an organic presentation: ABCDE, glucose, temperature, neck stiffness, focal signs, seizures; ask about CD4 count, viral load, ARV adherence and recent changes in ARV regimen; universal precautions; transport to an emergency centre.

52.9.5 Parkinson's disease (major or mild NCD due to Parkinson's disease)

Core features. The NCD occurs in the setting of established Parkinson's disease, with insidious onset and gradual progression, and is not attributable to another condition (APA, 2022). Parkinson's disease is a neurodegenerative disorder caused by loss of dopaminergic neurons in the substantia nigra, with the cardinal motor features of **bradykinesia, rest tremor, rigidity and postural instability**, and non-motor features including depression, anxiety, apathy, sleep disorders, autonomic dysfunction (orthostatic hypotension, constipation, urinary problems), hallucinations and dementia (Bloem, Okun and Klein, 2021). Parkinson's disease dementia and **dementia with Lewy bodies** share fluctuating cognition, visual hallucinations and extreme sensitivity to antipsychotics.

Emergency relevance.

- **Psychosis and visual hallucinations**, often precipitated by dopaminergic medications or infection.
- Delirium.
- **Falls and fractures**; aspiration pneumonia (dysphagia).
- **Parkinsonism-hyperpyrexia syndrome** — an NMS-like syndrome (rigidity, hyperthermia, altered consciousness) caused by **abrupt withdrawal** of levodopa or dopamine agonists (for example when the patient cannot swallow or is kept nil by mouth).

- **Dopamine antagonists (haloperidol, droperidol, metoclopramide) are contraindicated** — they severely worsen parkinsonism and may precipitate NMS (RCEM, 2025).

Prehospital management. Look for infection, dehydration and medication changes; avoid dopamine antagonists; use non-pharmacological calming; CDST benzodiazepine sedation at reduced doses only if necessary; protect the airway (aspiration risk); bring the medication list and report the time of the last levodopa dose (missed doses must be avoided); transport.

52.10 Briefly discuss cluster A, B and C categories of personality disorders

A **personality disorder** is "an enduring pattern of inner experience and behaviour that deviates markedly from the expectations of the individual's culture, is pervasive and inflexible, has an onset in adolescence or early adulthood, is stable over time, and leads to distress or impairment" (APA, 2022). The pattern is manifested in two or more of: cognition, affectivity, interpersonal functioning and impulse control. DSM-5-TR groups ten personality disorders into three clusters.

Cluster A — "odd or eccentric"

- **Paranoid personality disorder**: pervasive distrust and suspiciousness; others' motives are interpreted as malevolent; reluctance to confide; bears grudges; perceives attacks on character; recurrent unjustified suspicion of partner's fidelity.
- **Schizoid personality disorder**: detachment from social relationships and a restricted range of emotional expression; prefers solitary activities; appears indifferent to praise or criticism; emotional coldness.
- **Schizotypal personality disorder**: acute discomfort in close relationships, cognitive or perceptual distortions, and eccentric behaviour — ideas of reference, odd beliefs or magical thinking, unusual perceptual experiences, odd speech, suspiciousness, inappropriate affect. Genetically related to schizophrenia.

Emergency relevance: suspicion and mistrust of the crew; transient psychotic episodes under stress; refusal of care. **Approach**: be transparent, respect personal space, explain everything, avoid excessive familiarity.

Cluster B — "dramatic, emotional or erratic"

- **Antisocial personality disorder**: pervasive disregard for and violation of the rights of others since age 15, with evidence of conduct disorder before age 15, in a person at least 18 years old — failure to conform to laws, deceitfulness, impulsivity, irritability and aggressiveness (repeated fights or assaults), reckless disregard for safety, irresponsibility, and **lack of remorse**.
- **Borderline personality disorder**: pervasive instability of interpersonal relationships, self-image and affect, with marked impulsivity — frantic efforts to avoid abandonment; unstable, intense relationships alternating between idealisation and devaluation; identity disturbance; impulsivity in potentially self-damaging areas (spending, sex, substance use, reckless driving, binge eating); **recurrent suicidal behaviour, gestures, threats or self-mutilating behaviour**; affective instability; chronic feelings of emptiness; inappropriate, intense anger; transient stress-related paranoid ideation or

dissociation. Up to about 10% of people with borderline personality disorder die by suicide.

- **Histrionic personality disorder**: excessive emotionality and attention-seeking — discomfort when not the centre of attention, inappropriately seductive or provocative behaviour, rapidly shifting and shallow emotions, use of physical appearance to draw attention, impressionistic speech, theatricality, suggestibility, considering relationships more intimate than they are.
- **Narcissistic personality disorder**: grandiosity, need for admiration and lack of empathy — sense of entitlement, interpersonal exploitation, envy, arrogance, fragile self-esteem with rage when criticised.

Emergency relevance: Cluster B disorders are the most frequently encountered in emergency care. They are associated with **self-harm and suicide** (especially borderline), **violence** (especially antisocial), substance use, interpersonal crises, domestic violence (as perpetrators or victims), and frequent EMS contact. Staff may experience strong negative emotions (frustration, anger), and patients may be labelled "attention seeking" — this label is dangerous, because the suicide risk is real (Evans, Geduld and Stassen, 2018). **Approach**: calm, consistent, non-judgemental, set clear boundaries; take every threat of self-harm seriously; assess and treat injuries; maintain safety; avoid being drawn into arguments or "splitting" between team members.

Cluster C — "anxious or fearful"

- **Avoidant personality disorder**: social inhibition, feelings of inadequacy and hypersensitivity to negative evaluation; avoids interpersonal contact for fear of criticism or rejection; restraint in intimate relationships; views self as socially inept.
- **Dependent personality disorder**: a pervasive and excessive need to be taken care of, leading to submissive and clinging behaviour and fears of separation — difficulty making everyday decisions without excessive advice; needs others to assume responsibility; difficulty expressing disagreement; urgently seeks another relationship when one ends. Dependent individuals may remain in abusive relationships.
- **Obsessive-compulsive personality disorder (OCPD)**: preoccupation with orderliness, perfectionism and mental and interpersonal control at the expense of flexibility, openness and efficiency — preoccupation with details, rules and lists; perfectionism that interferes with task completion; excessive devotion to work; inflexibility about morality; inability to discard worthless objects; reluctance to delegate; miserliness; rigidity and stubbornness. Unlike OCD, OCPD does not involve true obsessions and compulsions, and the traits are usually experienced as consistent with the self (ego-syntonic).

Emergency relevance: anxiety, comorbid depression and anxiety disorders, difficulty making decisions about care (dependent), and potential vulnerability to domestic violence (dependent). **Approach**: reassurance, clear information, encouragement to participate in decisions, and screening for abuse.

General principles for personality disorders in emergency care

Personality disorders are diagnosed only after thorough assessment over time — **never on the basis of a single emergency encounter**. The ECP should describe behaviour objectively on the PRF rather than labelling the patient, treat medical and psychiatric emergencies on their merits, and be aware of their own emotional reactions.

High-yield points — Outcome 52

- Schizophrenia: two or more of delusions, hallucinations, disorganised speech, disorganised/catatonic behaviour, negative symptoms (at least one of the first three), for ≥6 months including ≥1 month active.
- Mania ≥7 days (or hospitalisation), three or more DIGFAST symptoms (four if only irritable); hypomania ≥4 days; Bipolar I = ≥1 mania; Bipolar II = hypomania + major depression, never mania.
- MDD: five or more of nine symptoms for two weeks, including depressed mood or anhedonia.
- Catatonia: three or more of twelve features; treat with lorazepam/ECT; avoid antipsychotics.
- Acute stress disorder 3 days–1 month; PTSD >1 month; first responders count as exposed.
- Anorexia: highest psychiatric mortality; refeeding syndrome; bulimia: hypokalaemic metabolic alkalosis, Russell's sign.
- Substance use disorder: two or more of eleven criteria in 12 months (mild 2–3, moderate 4–5, severe ≥6).
- CDST naloxone: 0.4 mg (0.04 mg in known abusers), increase 0.8, 2, 4 mg to 10 mg total; child 0.1 mg/kg (max 2 mg per dose).
- Delirium: acute, fluctuating, inattention; CAM.
- Parkinson's/Lewy body: no haloperidol, droperidol or metoclopramide.
- Personality clusters: A odd (paranoid, schizoid, schizotypal); B dramatic (antisocial, borderline, histrionic, narcissistic); C anxious (avoidant, dependent, obsessive-compulsive).

Outcome 53. Comprehensively discuss the approach and management of sexual assault in the emergency care setting

Context

South Africa has one of the highest reported rates of sexual violence in the world. A rape is estimated to occur every 12 minutes (Tilley et al., 2026, citing Action Society, 2024), and one in five South African children are likely to be victims of sexual abuse (Mesthrie, 2025, citing Cloete, 2022). Sexual violence is grossly under-reported because of stigma, fear of the perpetrator, distrust of the police, and economic dependence on the perpetrator. The ECP may be the first person to whom a survivor discloses, and the quality of that first contact influences the survivor's physical recovery, psychological recovery, access to HIV post-exposure prophylaxis (PEP), and whether justice can be pursued.

Use the term **"survivor"** (or "victim" in legal contexts, as the SOA does) and avoid language that implies blame.

53.1 Describe the common traumatic injuries associated with sexual assault

Genital and anal injuries

- **Female genital injuries** most commonly involve the **posterior fourchette**, **fossa navicularis**, **labia minora** and **hymen**, typically at the 3 o'clock to 9 o'clock positions (the lower half when the patient is supine). The injury types are often summarised by the acronym **TEARS**: **T**ears (lacerations), **E**cchymosis (bruising), **A**brasions, **R**edness, and **S**welling.
- Vaginal and cervical lacerations, and occasionally severe lacerations with significant bleeding (particularly in children, postmenopausal women and assaults involving objects), and rarely vaginal vault perforation with intraperitoneal injury.
- **Anal and perianal injuries**: fissures, lacerations, bruising, sphincter injury, rectal injury, in both female and male survivors.
- **Male genital injuries**: bruising, abrasions and lacerations of the penis and scrotum.
- **Children**: hymenal transections, perineal tears, and injuries that may extend to the rectum or peritoneum; but note that many sexually abused children have **normal** examinations.

Very important: the **absence of genital injury does not mean that rape did not occur**. Many survivors — including the majority of adult women in many series — have no visible genital injury, because of the elasticity of genital tissue, the victim's submission under threat (which is not consent — SOA s 1(3)), delays in presentation, or the use of lubricant. The ECP must never imply disbelief because injuries are absent.

Non-genital (extragenital) injuries

Non-genital injuries are more common than genital injuries and include:

- **Bruising of the inner thighs, upper arms (grip marks or fingertip bruises), wrists and ankles** (restraint).

- **Facial injuries**: bruising, black eyes, lacerations, fractures (nose, mandible, orbit, zygoma), dental injuries.
- **Neck injuries from strangulation** — finger-pad bruises, ligature marks, abrasions from the victim's own fingernails, **petechiae** of the face, conjunctivae and mouth, voice change, dysphagia, stridor, loss of consciousness, incontinence. **Non-fatal strangulation is a major red flag**: it can cause delayed airway oedema, carotid or vertebral artery dissection with stroke, and hypoxic brain injury, and it is a strong predictor of future homicide in intimate partner violence.
- **Bite marks** (breasts, neck, face, genitals) — these are forensic evidence (DNA from saliva, dental impressions).
- **Defence injuries** on the forearms and hands.
- **Breast injuries**: bruising, bite marks.
- **Head injuries** from blows or being pushed.
- **Abrasions on the back, buttocks and knees** from being forced to the ground (with embedded soil or grass, which is evidence).
- Stab and gunshot wounds and blunt trauma in violent assaults; injuries from falls or attempts to escape.
- **Signs of drug-facilitated assault**: altered consciousness, amnesia, intoxication (see 60.14).

Associated non-injury harms

Pregnancy, sexually transmitted infections (including HIV, hepatitis B, syphilis, gonorrhoea, chlamydia, trichomoniasis), acute stress disorder, PTSD, depression, substance use, and suicide.

53.2 Formulate general principles for the assessment and management of survivors of rape and sexual assault

Principle 1 — Safety

Ensure the safety of the survivor and the crew; the perpetrator may still be present. Request SAPS. Move the survivor to a private, safe place (for example the back of the ambulance with the doors closed).

Principle 2 — Treat life-threatening injuries first

A full primary survey takes priority over evidence preservation. Control haemorrhage, manage the airway (especially after strangulation), treat shock, manage head injuries, and assess for intoxication or drug-facilitated assault (check glucose and GCS). **Evidence preservation never takes precedence over life-saving care**; but life-saving care can usually be delivered in a way that minimises the loss of evidence.

Principle 3 — Trauma-informed, survivor-centred care

- Introduce yourself, explain your role, and **ask permission** before any touch, examination or intervention; explain why it is needed.
- Offer a **same-gender** provider where possible, and offer to have a trusted support person present.

- Believe the survivor; use a calm, non-judgemental, empathic tone; never ask "why" questions that imply blame ("Why were you walking there?", "Why didn't you fight back?").
- Give the survivor **choice and control** — the right to decline any part of the examination, and the choice about whether to report to SAPS (an adult survivor's choice, subject to the duties to report sexual offences against vulnerable persons — see 53.4 and 54).
- Respect privacy and confidentiality; do not discuss details in front of bystanders.
- Do not leave the survivor alone.

Principle 4 — Limit the prehospital history to what is clinically necessary

The detailed forensic history will be taken by the forensic examiner and SAPS; repeated questioning is traumatising and may introduce inconsistencies that defence attorneys can exploit. The ECP should ask only what is needed for clinical care and safety:

- time and date of the assault (critical for PEP and emergency contraception windows and for evidence collection);
- the nature of injuries and pain;
- whether there was loss of consciousness, strangulation, or suspected drugging;
- the possibility of pregnancy;
- whether the survivor has bathed, changed, urinated, vomited or douched since the assault;
- medical history, medications and allergies;
- immediate safety concerns (is the perpetrator a household member?).

Record **any spontaneous statements in the survivor's own words**, in quotation marks.

Principle 5 — Focused examination only

Do **not** perform a genital or internal examination in the prehospital setting unless there is life-threatening haemorrhage that requires direct pressure. Inspect visible injuries, document them carefully, and cover wounds with sterile dressings.

Principle 6 — Preserve evidence (53.3)

Principle 7 — Time-critical medical care: transport to a designated facility

Transport the survivor promptly to a **designated health establishment**, ideally a **Thuthuzela Care Centre (TCC)** or a hospital with a designated sexual assault service, because:

- **HIV PEP must be started as soon as possible and within 72 hours** (Horak et al., 2023), and the SOA provides that a victim who lays a charge with SAPS **or** reports the incident at a designated health establishment **within 72 hours** may receive PEP at State expense (SOA, s 28(1)–(2));
- **emergency contraception** is time-dependent;
- forensic evidence degrades with time.

Even if the survivor declines to lay a charge with SAPS, they should still be encouraged to go to the TCC or hospital for medical care, PEP, emergency contraception, and the option of having evidence collected and stored should they decide to report later.

Principle 8 — Psychological first aid and suicide risk

Provide psychological first aid (52.6.1), assess for acute stress reactions, and screen for suicidal ideation.

Principle 9 — Special populations

- **Children**: use the parent or caregiver if not suspected to be involved; minimal questioning; mandatory reporting to SAPS (SOA s 54) and child protection (Children's Act s 110); consider the possibility of abuse by a family member and do not return the child to an unsafe environment.
- **Persons with mental disability**: mandatory reporting under SOA s 54; adapted communication.
- **Male survivors**: face particular stigma; provide the same care and evidence preservation.
- **LGBTQI+ survivors**: may fear discrimination; provide respectful, non-judgemental care.
- **Older survivors**: higher risk of injury; consider elder abuse.

Principle 10 — Documentation and handover (53.4)

Principle 11 — Team-based care, referral and follow-up (Outcome 54)

53.3 Discuss and apply specific procedures for the preservation of evidence in cases of sexual assault

Why evidence matters

Physical evidence (semen, saliva, blood, hair, fibres, skin cells under fingernails, soil, condoms, objects) can identify the perpetrator through DNA, corroborate the survivor's account, and support prosecution. Evidence is fragile, is easily lost through washing, urination, eating, drinking or changing clothes, and is easily contaminated. In South Africa, forensic evidence is collected by a trained medical practitioner or forensic nurse using the SAPS **Sexual Assault Evidence Collection Kit (SAECK)**, and the medical findings are recorded on the **J88 form** (the medicolegal report used in court) (National Department of Health, 2003). The ECP does not collect forensic samples, but the ECP's actions in the first hours determine whether evidence survives.

Procedures for the ECP

Advise the survivor (gently, explaining why) not to do the following before the forensic examination, if possible:

- bath, shower, wash or douche;
- change clothes or wash clothing;

- urinate or defecate (if the survivor must urinate, ask them to use a clean, dry, sealable container and keep the urine, which may be tested for drugs in suspected drug-facilitated assault, and **not to wipe** — or to keep the toilet paper in a paper bag);
- eat, drink, smoke, brush teeth or rinse the mouth (especially after oral assault);
- comb hair;
- remove tampons or sanitary pads (or, if removed, keep them in a paper bag).

These are requests, not orders: if the survivor needs to urinate or cannot wait, respect their choice and preserve what can be preserved.

Clothing and items:

- If clothing must be removed for treatment, **do not cut through holes, tears or stains** made by the assault; cut along seams or around damaged areas.
- Place **each item of clothing in a separate paper bag** (never plastic, which retains moisture and allows bacterial and fungal degradation of biological evidence). Do not shake clothing.
- If the survivor is lying on a sheet or blanket, fold it inwards to retain trace evidence and bag it.
- Bag any items such as condoms, tampons or objects found, separately, in paper.
- Seal, label (survivor's name, date, time, item, your name) and sign each bag.

Handling the survivor:

- Wear gloves at all times and change them between handling different items; minimise handling.
- **Do not clean wounds or the genital area** unless necessary for haemorrhage control or life-saving treatment; if cleaning is unavoidable, document it.
- **Do not remove anything from the survivor's body** (for example from under fingernails, from the hair, or from the skin) except to save life; if hands must be protected, paper bags may be placed over them.
- Use a clean sheet on the stretcher, and keep the sheet with the survivor's belongings.
- If the survivor vomits, collect the vomitus if practicable (especially in suspected drug-facilitated assault).

Scene preservation:

- The location of the assault is a crime scene: minimise entry, disturb nothing unnecessarily, note anything you had to move, and hand over to SAPS.
- Do not allow bystanders to clean up.

Chain of custody:

- Evidence must be accounted for continuously from collection to court. Hand over bagged items directly to SAPS (preferably) or to the forensic examiner, and record on the PRF what was handed over, to whom (name, rank, SAPS number), and at what time, obtaining a signature where possible. Never leave evidence unattended.

Time limits: forensic evidence is best collected within 72 hours, although some evidence may still be recoverable up to about five to seven days after the assault, so a delay should never be a reason to discourage the survivor from attending the forensic examination.

53.4 Identify and apply key criteria for the completion of patient report forms in relation to sexual assault

The PRF is a legal document that may be subpoenaed and used in criminal proceedings; you may be called to testify. The HPCSA requires that records be accurate, made at the time or soon after, legible, clear and unambiguous, and without confusing abbreviations (HPCSA, 2023, para 12). For sexual assault, the PRF should meet the following criteria:

1. **Accurate times**: time of call, arrival, patient contact, departure and handover; and the **reported time of the assault** (critical for PEP and evidence).
2. **Objective, factual language**: describe what you saw and did. Avoid opinions and conclusions such as "the patient was raped" (rape is a legal finding for the court) — write "patient states that she was sexually assaulted", or "alleged sexual assault".
3. **The survivor's own words in quotation marks** for key statements (for example: Patient states, "He held a knife to my throat and pushed me to the ground").
4. **The identity of the alleged perpetrator only as reported** (for example "patient reports that the assailant was her uncle"), without speculation.
5. **Detailed description of injuries**: type (bruise, abrasion, laceration, bite mark, petechiae), size (in centimetres), shape, colour, location (using anatomical landmarks and body diagrams), and any pattern. Do not estimate the age of bruises.
6. **Clinical findings**: vital signs, GCS, mental state (for example "tearful, withdrawn, shaking"), signs of intoxication or drugging, and strangulation signs.
7. **Pre-arrival actions by the survivor** (bathed, changed clothes, urinated, etc.).
8. **Treatment given** and the survivor's consent to (or refusal of) each intervention.
9. **Evidence handling**: items collected, bagged and handed over, to whom and when (chain of custody).
10. **Who was present**, including SAPS (names, ranks, station, case number if available).
11. **Notifications and reporting**: SAPS notified (and the mandatory reports for children and vulnerable persons under SOA s 54 and Children's Act s 110); information given to the survivor about PEP and the 72-hour window.
12. **Destination and handover**: name of the receiving TCC or facility and the person receiving the patient.
13. **Confidentiality**: the PRF must be protected under POPIA and the NHA; do not leave it unattended or share it with anyone not entitled to it.
14. **Completion**: legible, signed, with your name, HPCSA registration number and date; corrections made by a single line through the error with initials and date — never obliterate or alter entries after the fact.

53.5 Describe applicable public sector protocols in relation to post-exposure prophylaxis for survivors of sexual assault and rape

(PEP is not prehospital ECP scope in the CDST. The ECP's role is to inform the survivor about PEP, emphasise the 72-hour window, and transport promptly to a designated facility. The details below are hospital/clinic-level information for your knowledge.)

Legal entitlement

In terms of the SOA (s 28), a victim who has been exposed to the risk of HIV infection as a result of a sexual offence may receive **PEP for HIV at State expense** at a designated public health establishment, receive **free medical advice** about PEP before it is administered, and apply to a magistrate for an order that the alleged offender be **tested for HIV** at State expense (ss 28 and 30–31). To receive PEP under the Act, the victim must lay a charge with SAPS **or** report the incident in the prescribed manner at a designated health establishment **within 72 hours** after the alleged offence (s 28(2)). The police official, medical practitioner or nurse to whom the offence is reported must inform the victim of the importance of obtaining PEP within 72 hours, the need for medical advice about other STIs, and the available services (s 28(3)).

HIV PEP — current South African recommendations

According to the Southern African HIV Clinicians Society 2023 PEP guideline (Horak et al., 2023), which updates the South African National 2020 PEP guidelines:

- **Start as soon as possible, and within 72 hours** of exposure; PEP is not recommended after 72 hours.
- **Baseline HIV test** (rapid test plus fourth-generation ELISA; qualitative HIV PCR in children under two years). PEP is not given if the survivor is already HIV-positive (they are linked to antiretroviral therapy instead).
- **Regimen for persons ≥10 years and ≥30 kg**: **TDF 300 mg + 3TC 300 mg + DTG 50 mg (TLD)** as a once-daily fixed-dose combination tablet **for 28 days** (with an additional dose of dolutegravir 50 mg if on rifampicin or carbamazepine). If dolutegravir is contraindicated: TDF + FTC + ATV/r, or TDF + FTC + LPV/r. Adjusted regimens are used for impaired renal function.
- **Regimen for persons <10 years or <30 kg**: **AZT + 3TC + DTG** (paediatric weight-based dosing), with AZT + 3TC + a protease inhibitor if dolutegravir is not available, and ABC + 3TC + DTG as an alternative.
- **Follow-up HIV testing** at **six weeks** and **three months**.
- Adherence counselling and management of side effects.

Other components of post-rape care (public sector)

- **Emergency contraception** must be offered to women and girls of reproductive age (Horak et al., 2023): commonly **levonorgestrel 1.5 mg orally as a single dose**, as soon as possible (it is most effective within 72 hours and may be used up to 120 hours), or a **copper intrauterine device** within five days; a pregnancy test is done at baseline.
- **STI prophylaxis**: should be considered for adults and adolescents, and all children with suspected penetrative injuries should receive STI prophylaxis, according to the details in

the National 2020 PEP guidelines (Horak et al., 2023). South African treatment guidelines have commonly used a single-dose combination covering gonorrhoea, chlamydia and trichomoniasis (for example ceftriaxone IM, azithromycin PO and metronidazole PO); confirm the current regimen in the national guideline used at your facility.

- **Hepatitis B**: vaccination and/or hepatitis B immunoglobulin according to the survivor's vaccination and immune status (Horak et al., 2023).
- **Tetanus** vaccination where there are wounds (Horak et al., 2023).
- **Syphilis** serology.
- **Forensic examination** (SAECK and J88) with consent.
- **Psychological support and counselling**, with referral for ongoing care.
- **Follow-up visits** for HIV testing, adherence, pregnancy testing, and psychological support.

High-yield points — Outcome 53

- Absence of genital injury does not exclude rape; TEARS; strangulation is a red flag.
- Life before evidence, but preserve evidence: no washing, eating, drinking, urinating (collect if necessary), changing; paper bags, separate items; do not cut through holes; chain of custody.
- PRF: objective, survivor's own words, times, injury descriptions, evidence handover, notifications.
- PEP: within 72 hours, TLD for 28 days (≥10 years and ≥30 kg); follow-up HIV tests at 6 weeks and 3 months; free at State expense if reported to SAPS or a designated facility within 72 hours (SOA s 28).

Outcome 54. Discuss the intersections between emergency care, health systems and law enforcement in the context of sexual assault

54.1 Explain the key elements of the main criminal offences in terms of the Criminal Law (Sexual Offences and Related Matters) Amendment Act 2007, including rape and sexual assault

The Criminal Law (Sexual Offences and Related Matters) Amendment Act 32 of 2007 (the SOA) repealed the common-law crimes of rape and indecent assault and replaced them with gender-neutral statutory offences. It has since been amended, including by Act 8 of 2017 and Act 13 of 2021.

Rape (section 3)

"Any person ('A') who **unlawfully** and **intentionally** commits an **act of sexual penetration** with a complainant ('B'), **without the consent of B**, is guilty of the offence of rape" (SOA, s 3). The elements are therefore:

1. **An act of sexual penetration** — defined in s 1 as "any act which causes penetration **to any extent whatsoever** by (a) the genital organs of one person into or beyond the genital organs, anus, or mouth of another person; (b) any other part of the body of one person or, any object, including any part of the body of an animal, into or beyond the genital organs or anus of another person; or (c) the genital organs of an animal, into or beyond the mouth of another person". Important implications: penetration "to any extent whatsoever" is sufficient (full penetration and ejaculation are not required); oral and anal penetration and penetration with fingers or objects constitute rape; and the offence is **gender-neutral** — any person (male or female) can be a perpetrator or a victim, and marital rape is included.
2. **Unlawfulness** — the absence of a ground of justification.
3. **Intention** — the perpetrator acted intentionally (including knowing, or foreseeing the possibility, that B did not consent).
4. **Absence of consent**.

Consent (section 1(2) and (3))

"**Consent**" means "**voluntary or uncoerced agreement**" (s 1(2)). Section 1(3) lists circumstances (not exhaustive) in which B does **not** voluntarily or without coercion agree, including where B submits or is subjected to the sexual act as a result of:

- (a)(i) the **use of force or intimidation** by A against B, a third person or another person, or against their property; or (ii) a **threat of harm** against them or their property;
- (b) an **abuse of power or authority** by A to the extent that B is inhibited from indicating unwillingness or resistance;
-

(c) **false pretences or fraudulent means**, including where B is led to believe that A is a different person, or that the act is something other than a sexual act;

- (d) where B is **incapable in law of appreciating the nature of the sexual act**, including where B is at the time (i) **asleep**; (ii) **unconscious**; (iii) **in an altered state of consciousness, including under the influence of any medicine, drug, alcohol or other substance**, to the extent that B's consciousness or judgement is adversely affected; (iv) **a child below the age of 12 years**; or (v) **a person with a mental disability**.

This is directly relevant to the ECP: a person who is heavily intoxicated or drugged (60.14), unconscious, asleep, or mentally disabled cannot consent. Submission is not consent.

Compelled rape (section 4)

A person who unlawfully and intentionally compels a third person (C), without C's consent, to commit an act of sexual penetration with B, without B's consent, is guilty of compelled rape.

Sexual assault (section 5)

"A person ('A') who unlawfully and intentionally **sexually violates** a complainant ('B'), without the consent of B, is guilty of the offence of sexual assault" (s 5(1)). A person who unlawfully and intentionally **inspires the belief** in B that B will be sexually violated is also guilty of sexual assault (s 5(2)). **Sexual violation** includes any act which causes direct or indirect contact between the genital organs, anus or female breasts of one person and any part of the body of another person, or an animal, or any object; contact between the mouth of one person and the genital organs, anus or breasts of another; and similar acts **without penetration** (s 1). Sexual assault is therefore sexual violation without penetration (for example groping, forced kissing of a sexual nature, or touching of the genitals).

Related offences include **compelled sexual assault** (s 6), **compelled self-sexual assault** (s 7), and offences of exposure, display of child pornography, sexual grooming, and **sexual intimidation** (introduced by the 2021 amendment).

Offences against children

- Children **below 12 years** are incapable of consenting (s 1(3)(d)(iv); s 57).
- **Consensual sexual penetration** with a child aged **12 to 15 years** (statutory rape, s 15) and **consensual sexual violation** of such a child (statutory sexual assault, s 16) are offences; following the Constitutional Court's judgment in *Teddy Bear Clinic for Abused Children v Minister of Justice and Constitutional Development* (2013), the Act was amended in 2015 so that consensual sexual activity between children aged 12–15, or between a child of 12–15 and a person of 16–17 who is not more than two years older, is not criminalised. The age of consent to sex is **16 years**.
- Other offences against children include sexual exploitation, grooming, exposure to pornography and child pornography.

Duty to report (section 54)

Section 54(1) (as amended in 2021) provides that "a person who has knowledge, reasonable belief or suspicion that a sexual offence has been committed against a **person who is vulnerable** … must report such knowledge, reasonable belief or suspicion **immediately to a police official**". "Persons who are vulnerable" include children and persons with mental disability (among others defined in s 40). Failure to report is an offence punishable by a fine or imprisonment of up to **five years**, or both (s 54(2)). This applies to **every person**, including the ECP. (For adult survivors without vulnerability, the decision to report to SAPS rests with the survivor, although the ECP should encourage and facilitate reporting.)

Other key provisions

- Services for victims, PEP and compulsory HIV testing of alleged offenders (ss 27–39; see 53.5).
- The **National Register for Sex Offenders** (Chapter 6), which prevents convicted sex offenders from working with children and persons with mental disability.

54.2 Reflect on the importance of a team-based approach to the management of sexual assault cases

The needs of a survivor are medical, psychological, social, legal and safety-related, and no single professional can meet them. A **multidisciplinary, coordinated team** is essential, and this is the principle behind the **Thuthuzela Care Centres** (TCCs) — one-stop facilities, located at public hospitals, established by the **National Prosecuting Authority's Sexual Offences and Community Affairs (SOCA) Unit** with the Departments of Health, Social Development, Justice and SAPS. "Thuthuzela" is an isiXhosa word meaning "comfort". At a TCC, the survivor can receive medical care, forensic examination, PEP, counselling and assistance with reporting to SAPS in one place, and is supported through to court by case managers and prosecutors, with the aims of reducing secondary victimisation, improving conviction rates and reducing the time to finalise cases.

The team typically includes:

Team member	Role
EMS / ECP	Safety, life-saving care, trauma-informed first contact, evidence preservation, information about PEP and the 72-hour window, transport to a designated facility, documentation, mandatory reporting
SAPS (including Family Violence, Child Protection and Sexual Offences (FCS) Units)	Scene control, statement taking, case number, evidence chain of custody, investigation, arrest, protection
Forensic medical practitioner / forensic nurse	Medical examination, SAECK collection, J88 completion, testimony in court
Nurses and doctors	Treatment of injuries, PEP, emergency contraception, STI prophylaxis, follow-up

Team member	Role
Social workers	Psychosocial support, safety planning, child protection, placement of children
Counsellors / psychologists / psychiatrists	Trauma counselling, management of acute stress disorder, PTSD and depression, suicide prevention
NPA prosecutors and case managers	Court preparation, victim support through the criminal justice process
NGOs	Advocacy, shelter, long-term counselling and support

Why the team approach matters:

1. **Reduces secondary victimisation** (repeated questioning, being sent from place to place, insensitive treatment).
2. **Preserves evidence and the chain of custody** through coordinated handovers.
3. **Ensures time-critical care** (PEP, emergency contraception) is delivered.
4. **Improves prosecution outcomes**.
5. **Addresses the full range of needs**, including safety, housing and mental health.
6. **Supports staff** through shared responsibility.

The ECP's contribution is critical because the first responder's attitude and actions influence whether the survivor engages with the rest of the system. Poor coordination (for example SAPS and EMS disagreeing on scene, or the survivor being taken to a facility without forensic services) can lead to lost evidence and missed PEP.

54.3 Identify key agencies for referral of victims of sexual assault

- **Thuthuzela Care Centres (TCCs)** — one-stop centres for rape survivors at public hospitals (NPA SOCA Unit).
- **SAPS** — any police station; **SAPS Family Violence, Child Protection and Sexual Offences (FCS) Units**; emergency number **10111**.
- **Designated public health establishments** (hospitals and community health centres designated for sexual assault care and PEP).
- **Department of Social Development** and the **GBV Command Centre** — 24-hour helpline **0800 428 428** (and *120*7867# "please call me" USSD).
- **Childline South Africa** — **116** (toll-free, for children).
- **National Prosecuting Authority** (court preparation programmes and case managers).
- **Rape Crisis Cape Town Trust** and similar NGOs — counselling and court support.
- **Lifeline South Africa** — counselling.
- **South African Depression and Anxiety Group (SADAG)** — mental health and suicide crisis support.
- **Designated child protection organisations** (for example Child Welfare South Africa) and the provincial departments of social development (for children, under the Children's Act).

- **Shelters** for survivors of gender-based violence.

(Telephone numbers change; verify current numbers on the official websites of the agencies before relying on them in practice.)

High-yield points — Outcome 54

- Rape (s 3): unlawful, intentional sexual penetration (to any extent) without consent; gender-neutral; includes oral, anal, digital and object penetration.
- Sexual assault (s 5): unlawful, intentional sexual violation without consent (no penetration), or inspiring the belief that one will be violated.
- Consent = voluntary or uncoerced agreement; no consent if force, threats, abuse of power, fraud, asleep, unconscious, intoxicated, <12 years, mentally disabled.
- s 54: everyone must immediately report sexual offences against vulnerable persons (children, mentally disabled) to SAPS; up to 5 years' imprisonment for failure.
- TCCs: NPA SOCA one-stop centres; multidisciplinary approach reduces secondary victimisation.

Outcome 55. Discuss the intersections between the ethical, legal and professional obligations of emergency care providers in the management of domestic violence

Overview of the intersecting obligations

Domain	Obligations of the ECP
Ethical	Beneficence (identify and respond to abuse; prevent further harm), non-maleficence (avoid actions that increase the danger to the survivor, for example confronting the abuser or disclosing information to the abuser), respect for autonomy (an adult survivor's right to make their own decisions about leaving, reporting and accepting help), justice (equal, non-judgemental care for all survivors regardless of gender, sexual orientation or circumstances), confidentiality, and truthfulness
Legal	Treat injuries (Constitution s 27(3)); obtain consent (NHA s 7); maintain confidentiality (NHA s 14; POPIA) subject to statutory duties; **mandatory reporting** of domestic violence against children, older persons and persons with disabilities (DVA ss 2A–2B), of child abuse (Children's Act s 110), of sexual offences against vulnerable persons (SOA s 54), and of abuse of older persons (Older Persons Act s 26); provide accurate documentation that may be used in court (protection order and criminal proceedings)
Professional	Practise within the HPCSA ethical rules and scope of practice; maintain competence in recognising and responding to domestic violence; document fully and objectively; refer appropriately; collaborate with SAPS, social workers and NGOs; protect personal safety; seek support for the emotional impact

These obligations intersect and can conflict. For example, respect for an adult survivor's autonomy (the survivor may not want SAPS involved) may conflict with the ECP's wish to protect them; confidentiality may conflict with the statutory duty to report abuse of a child exposed to the violence; and the ethical duty to help may conflict with the ECP's own safety when the perpetrator is present. The general resolution is: **treat injuries, ensure immediate safety, respect the autonomy of a competent adult survivor, inform them of their**

rights and the available help, report where the law requires it (children, older persons, persons with disabilities, and sexual offences against vulnerable persons), and document thoroughly.

55.1 Analyse the definition of domestic violence in terms of the Domestic Violence Act

The Act

The **Domestic Violence Act 116 of 1998** (DVA), as substantially amended by the **Domestic Violence Amendment Act 14 of 2021** (in force from **14 April 2023**), aims to afford victims of domestic violence the maximum protection that the law can provide.

Definition of domestic violence (s 1, as amended)

"Domestic violence" means:

- (a) **physical abuse**;
- (b) **sexual abuse**;
- (c) **emotional, verbal or psychological abuse**;
- (d) **economic abuse**;
- (e) **intimidation**;
- (f) **harassment**;
- (fA) **sexual harassment**;
- (fB) **related person abuse**;
- (g) **spiritual abuse**;
- (h) **damage to property**;
- (hA) **elder abuse**;
- (hB) **coercive behaviour**;
- (hC) **controlling behaviour**;
- (hD) **to expose a child to domestic violence**;
- (i) **entry into the complainant's residence** (where the parties do not share it) or **workplace or place of study** (where they do not share it) **without consent**; or

(j) **any other behaviour of an intimidating, threatening, abusive, degrading, offensive or humiliating nature** towards a complainant, where such conduct harms, or inspires the reasonable belief that harm may be caused to, the complainant.

Analysis of key terms

- **Physical abuse**: any act or threatened act of physical violence (hitting, kicking, strangling, burning, use of weapons).
- **Sexual abuse**: any conduct that abuses, humiliates, degrades or otherwise violates the sexual integrity of the complainant, including marital rape.
- **Emotional, verbal or psychological abuse**: "degrading, manipulating, threatening, offensive, intimidating or humiliating conduct towards a complainant that causes mental or psychological harm", including insults, ridicule and name-calling; threats to cause emotional pain; obsessive possessiveness or jealousy that seriously invades privacy, liberty, integrity or security; wilful destruction of property near the complainant; harming or threatening to harm a household pet; disclosing or threatening to disclose the complainant's sexual orientation or private information; threatening the death or injury of another person; and **threats to commit suicide or self-harm** (DVA, s 1).
- **Economic abuse**: the deprivation of economic or financial resources to which the complainant is entitled or which they require out of necessity (including education expenses, household necessities, and rent or bond payments); the disposal of household effects or property without permission; the use of the complainant's financial resources without permission; or coercing the complainant to relinquish control over assets or income or to sign documents enabling another person to manage their finances (DVA, s 1).
- **Elder abuse**: abuse of an older person as contemplated in s 30(2) of the Older Persons Act 13 of 2006, occurring within a domestic relationship.
- **Exposing a child to domestic violence**: intentionally causing a child to see or hear domestic violence, or to experience its effects.
- **Harassment** now explicitly includes **technology-facilitated abuse**: repeated electronic contact, unauthorised access to communications, **monitoring or tracking movements by technology**, and disclosure of private or intimate electronic communications.
- **Coercive and controlling behaviour** recognise that domestic violence is a **pattern of power and control**, not only isolated incidents of physical violence.

The domestic relationship

The Act applies only where there is a **domestic relationship** between the complainant and the respondent. The domestic relationships covered include persons who are or were married to each other (including customary and religious marriages); persons who live or lived together in a relationship in the nature of marriage; parents of a child, or persons who have or had parental responsibility for a child; family members related by blood, marriage or adoption; persons in an engagement, dating or customary relationship, including an actual or perceived romantic, intimate or sexual relationship of any duration; and persons who share or recently

shared the same residence (DVA, s 1). The definition is therefore broad, covering same-sex relationships, dating relationships, and abuse of older parents by adult children.

Critical analysis of the definition

Strengths: the definition is among the most comprehensive in the world; it is **gender-neutral**; it recognises non-physical forms of abuse (emotional, economic, spiritual, coercive and controlling behaviour) that often precede and accompany physical violence; it protects children exposed to violence and older persons; and it addresses technology-facilitated abuse.

Challenges: broad definitions require skilled assessment and may be difficult to prove; implementation depends on SAPS, court and health system capacity; survivors may not recognise non-physical abuse as domestic violence; and economic dependence on the abuser, cultural norms and fear limit use of the Act.

55.2 Assess the prevalence of domestic violence in South Africa and analyse the social context in which it takes place

Prevalence

- **Globally**, about one in four (26%) ever-partnered women aged 15 years and older have experienced physical and/or sexual violence by an intimate partner at least once in their lifetime (WHO, 2021).
- In **South Africa**, the **South African Demographic and Health Survey 2016** found that about **21% of ever-partnered women** aged 18 and older reported having ever experienced physical violence by a partner, and about **8%** had experienced it in the 12 months before the survey; about **6%** reported lifetime sexual violence by a partner, and emotional violence was also commonly reported (National Department of Health et al., 2019). These figures are likely to underestimate the true burden because of under-reporting.
- **Intimate partner femicide**: South Africa's female homicide rate is several times the global average, and Abrahams et al. (2013) found that **more than half of women murdered in South Africa in 2009 (about 57%) were killed by an intimate partner** — one woman every eight hours.
- **Violence against children** in the home is widespread: 45% of households with children reported some form of violence against children in 2021–2022 (Statistics South Africa, 2024, cited by Mesthrie, 2025).
- Tilley et al. (2026) cite an estimate that a rape occurs every 12 minutes in South Africa, and associate intimate partner violence with health behaviours and chronic illness among South African women (Gass et al., 2010).
- **Men** also experience domestic violence, and same-sex intimate partner violence occurs, but both are under-reported because of stigma.

Social context

Domestic violence in South Africa is rooted in multiple, interacting social factors:

1. **Patriarchy and gender norms.** Mesthrie (2025) notes that "the dominating patriarchal nature of society in South Africa has legitimized violence against women

and children as a way of holding power and maintaining control". Norms that accept male dominance, male entitlement to sex within relationships, and the use of violence to "discipline" partners and children sustain domestic violence.

2. **The legacy of apartheid and colonialism.** Apartheid "entrenched poverty and inequality in marginalized communities", cultivating "social conditions in which violence could thrive" (Panday et al., cited by Mesthrie, 2025), including the migrant labour system, which disrupted family structures, and the normalisation of state and interpersonal violence.
3. **Poverty, unemployment and inequality.** With 63% of South Africans living in poverty and 31.9% unemployed (Tilley et al., 2026), economic stress fuels conflict, and economic dependence prevents survivors from leaving.
4. **Alcohol and substance abuse** (18.9% of the population, according to Tilley et al., 2026, citing Myers et al., 2022), which is strongly associated with perpetration.
5. **The intergenerational cycle of violence.** "Consistently witnessing violence in the home has been linked with aggressive behaviour later in life and especially rape and domestic violence perpetration" (Mathews and Bonvenuti, cited by Mesthrie, 2025), and growing up in a violent environment leads children to normalise violence as a way of dealing with conflict.
6. **Community violence and the normalisation of violence**, including high rates of murder and assault (Tilley et al., 2026).
7. **Access to firearms.**
8. **Weak institutional responses**: under-resourced SAPS, courts and social services; secondary victimisation; and poor enforcement of protection orders.
9. **Cultural and religious beliefs** that discourage separation or reporting, practices such as ilobolo when misinterpreted as ownership, and stigma attached to divorce.
10. **Overcrowded housing**, which limits privacy and escape.
11. **HIV**, which interacts with intimate partner violence in both directions (violence increases the risk of HIV, and disclosure of HIV status may trigger violence).

These factors interact as a **syndemic**, amplifying each other (Tilley et al., 2026, citing Caulkins, 2022).

55.3 Assess the physical, psychological and economic consequences of domestic violence on survivors and their families

Physical consequences

- **Injuries**: bruises, lacerations, fractures (face, ribs, forearms), burns, stab and gunshot wounds, head injuries and traumatic brain injury, **strangulation injuries** (with risk of hypoxic brain injury, stroke and death), dental injuries, eye injuries, and injuries in various stages of healing.
- **Death**: femicide, homicide of children, and suicide.
- **Sexual and reproductive health**: unintended pregnancy, unsafe abortion, STIs including **HIV**, pelvic inflammatory disease, chronic pelvic pain, sexual dysfunction.

- **Violence in pregnancy**: miscarriage, placental abruption, preterm labour, low birth weight, fetal injury, and maternal death; violence often begins or escalates during pregnancy.
- **Chronic health problems**: chronic pain syndromes, headaches, gastrointestinal disorders (irritable bowel syndrome), cardiovascular disease, and poorer control of chronic illnesses (Gass et al., 2010, cited by Tilley et al., 2026).
- **Health-risk behaviours**: smoking, alcohol and drug use, unsafe sex.

Psychological consequences

- **Depression**, anxiety disorders, **PTSD**, complex trauma, sleep and eating disorders, substance use disorders, low self-esteem and shame.
- **Suicidal ideation, self-harm and suicide** — domestic violence is a major risk factor.
- Psychological entrapment ("traumatic bonding"), learned helplessness, and fear.

Effects on children and families

- Children exposed to domestic violence show emotional and behavioural problems, PTSD, anxiety and depression, poor school performance, and developmental problems; they are at increased risk of being directly abused; and they have an increased risk of later victimisation or perpetration (Mesthrie, 2025).
- Exposure to violence can cause changes in brain structure and function, resulting in cognitive deficits and behavioural and emotional difficulties (Mesthrie, 2025).
- Family breakdown, homelessness, and placement of children in alternative care.

Economic consequences

- **For survivors**: loss of income through absenteeism, job loss and reduced productivity; economic abuse (deprivation of money, control of income, destruction of property); costs of medical care, relocation and legal proceedings; and poverty and homelessness after leaving.
- **For families**: loss of the family's income, education costs for children, and intergenerational poverty.
- **For society**: costs to health services (including EMS), police, courts, social services and shelters, and lost economic productivity. Mesthrie (2025) notes that "the economic and social costs of violence are also significant, and violence can even have intergenerational consequences".

55.4 Explain the key protective measures of the Domestic Violence Act, with a view to counselling survivors of domestic violence

The ECP should be able to explain the following protections in plain language to a survivor.

1. Duties of SAPS and peace officers

- **SAPS must assist the complainant** at the scene or as soon as reasonably possible: help them to find suitable shelter and obtain medical treatment, provide the prescribed list of shelters and public health establishments, hand over and explain the prescribed **notice of rights** (in the official language of the complainant's choice), and explain the remedies available, including the right to lay a criminal charge (DVA, s 2).

- **Arrest without a warrant**: a peace officer **must**, without a warrant, arrest any respondent at the scene of an incident of domestic violence whom they reasonably believe has committed an act of domestic violence that constitutes an offence containing an element of violence against the complainant (DVA, s 3(2)).
- SAPS may enter a private residence without a warrant to obtain statements where an offence with an element of physical violence is reported (s 3A).
- SAPS must seize arms and dangerous weapons in the possession or control of the respondent in specified circumstances (s 9).

2. Protection orders

- **Who may apply**: the complainant, or any other person with a material interest in the complainant's well-being (for example a counsellor, health service provider, social worker or teacher) on the complainant's behalf, with the complainant's written consent — except where the complainant is a minor, mentally incapacitated, unconscious, or unable to give consent. **A child, or anyone on behalf of a child, may apply without the consent or assistance of a parent or guardian** (s 4(4)).
- **Where and when**: at a magistrate's court (including electronically, in terms of the amendments), at any time, including **outside ordinary court hours** if the complainant may suffer harm if the application is not dealt with immediately (s 4(5)).
- **Interim protection order**: the court **must** issue an interim protection order, even without notice to the respondent, if there is prima facie evidence that the respondent is committing or has committed domestic violence and the complainant is suffering or may suffer harm, and an order is immediately necessary to protect the complainant (s 5(2)). It takes effect once served on the respondent (s 5(6)) and calls on the respondent to show cause on a return date (not less than ten days after service) why it should not be made final.
- **Final protection order**: issued after the hearing on the return date.
- **Contents of a protection order** (s 7): the court may prohibit the respondent from committing any act of domestic violence; enlisting the help of another person to commit such acts; entering the shared residence (or a specified part of it), the complainant's residence, workplace or place of study; preventing the complainant from entering or remaining in the shared residence; contacting the complainant (including electronically); and committing any other act specified. The court may order the seizure of weapons, direct SAPS to accompany the complainant to collect personal property, order **emergency monetary relief** (for losses such as medical expenses, loss of earnings, relocation costs and maintenance), refuse or restrict the respondent's contact with children where it is in their best interests, and order that the complainant's physical address not be disclosed.
- **Domestic violence safety monitoring notice** (s 4A): where the parties share a residence and the respondent poses a threat, the court may order SAPS to contact and visit the complainant at regular intervals, and to enter the residence (using reasonable force if necessary) to see the complainant in private.

3. Warrant of arrest

When a protection order (interim or final) is issued, the court **must** also authorise a **warrant of arrest** for the respondent, which is **suspended** subject to compliance with the order (s 8). If the respondent breaches the order, the complainant may hand the warrant, together with an affidavit stating that the respondent has breached the order, to any member of SAPS, who must arrest the respondent if there are reasonable grounds to suspect imminent harm, or otherwise issue a written notice to appear in court.

4. Offences and penalties

Contravening a protection order is a criminal offence punishable by a fine or imprisonment (the Act prescribes imprisonment of up to five years for breach of a protection order) (DVA, s 17). Failure by an adult to report domestic violence against a child, a person with a disability or an older person (s 2B) is also an offence.

5. Reporting and functionary duties (from 14 April 2023)

- **Functionaries** (medical practitioners, health care personnel, social workers, officials in public health establishments, educators and caregivers) who, in the course of their duties, believe or suspect on reasonable grounds that a **child, a person with a disability or an older person** may be a complainant must, without delay, complete the prescribed report (Form 2 — report and risk assessment) setting out the reasons, submit it to a **social worker or SAPS**, conduct a risk assessment, and may refer the complainant for further services (s 2A(2)). A functionary who reports in good faith is not liable to civil, criminal or disciplinary action, and is entitled to confidentiality of identity (s 2A(3)).
- **Any adult** who has knowledge, reasonable belief or suspicion that domestic violence has been committed against a child, a person with a disability or an older person must report it as soon as possible to a social worker or SAPS (s 2B).

Note on EMS: "health care personnel" are functionaries only if they belong to a category designated by directive under the Act. Regardless of whether ECPs are designated, every ECP is bound by the s 2B duty as an adult, by SOA s 54, and should regard the s 2A procedure as best practice.

6. Other protections

- The court may order that the complainant's address be kept confidential, and proceedings are held in camera.
- The complainant may lay a **criminal charge** (for example assault, assault with intent to do grievous bodily harm, rape, attempted murder) **in addition to** applying for a protection order.
- The integrated electronic repository for protection orders allows SAPS and courts to verify orders nationally.

Counselling a survivor: what the ECP should say

1. "What is happening to you is not your fault, and it is against the law."

2. “You have the right to be safe. You can apply for a protection order at the magistrate’s court at any time, even after hours, and the police must help you.”
3. “A protection order can stop the person from hurting, threatening or contacting you, and can order them to stay away from your home and work.”
4. “When a protection order is granted, a warrant of arrest is issued at the same time. If the person breaks the order, you can give the warrant to the police and they can arrest them.”
5. “You can also lay a criminal charge, such as assault.”
6. “The police must give you a list of shelters and help you get medical treatment.”
7. “If there are children in the home, they are also affected, and I have a duty to make sure that they are safe.”
8. “Here are numbers you can call for help: the GBV Command Centre (0800 428 428), SAPS (10111), and…”
9. Develop a simple **safety plan**: a safe place to go, an emergency bag (identity documents, money, medication, children’s documents, keys), a code word with a trusted person, and an emergency number.

The decision to apply for an order, lay a charge or leave belongs to the **competent adult survivor**; leaving is often the most dangerous time, so the survivor’s own assessment of safety must be respected.

High-yield points — Outcome 55

- DVA 116 of 1998, amended by Act 14 of 2021 (in force 14 April 2023).
- Definition: physical, sexual, emotional/verbal/psychological, economic abuse, intimidation, harassment, sexual harassment, related-person abuse, spiritual abuse, damage to property, elder abuse, coercive and controlling behaviour, exposing a child to DV, unauthorised entry, and any other intimidating/abusive conduct.
- SA prevalence: ~21% lifetime physical IPV among ever-partnered women (SADHS 2016); >50% of female homicides by intimate partners (Abrahams et al., 2013).
- Protections: SAPS duties (s 2), mandatory arrest at scene for violent DV offence (s 3), interim and final protection orders (ss 5–7), suspended warrant of arrest (s 8), seizure of weapons (s 9), breach = offence (s 17), functionary report (s 2A, Form 2), adult duty to report for children/disabled/older persons (s 2B).

Outcome 56. Critically appraise current emergency care practice with reference to domestic violence

Current practice and its shortcomings

1. **Under-recognition.** Survivors rarely volunteer that injuries are due to domestic violence, and EMS providers focus on physical injuries without asking about the cause. Calls may be dispatched as "assault", "fall" or "medical" rather than domestic violence.
2. **Lack of training and protocols.** There is no CDST tool specific to domestic violence, and South African EMS curricula have historically given limited attention to gender-based violence. Naidoo (2017, cited by Tilley et al., 2026) argued for strengthening the role and scope of prehospital emergency care in gender-based violence through theory, policy and clinical praxis.
3. **Attitudes and secondary victimisation.** Blaming attitudes ("Why doesn't she just leave?"), dismissiveness, and failure to believe survivors deter disclosure.
4. **Poor documentation**, limiting the usefulness of the PRF in protection order or criminal proceedings.
5. **Failure to address children** in the household.
6. **Confusion about reporting obligations** and confidentiality, particularly after the 2021 amendments.
7. **Safety risks to crews** when perpetrators are present.
8. **Fragmented referral pathways** between EMS, SAPS, social services and NGOs.

The opportunity

EMS providers are often the only health professionals who see the **home environment** — the scene, the dynamics between the people present, the condition of children, signs of substance abuse, weapons, and damage to property — which gives them a unique opportunity to identify domestic violence that would be invisible in a clinic or emergency department.

56.1 Assess the need for routine screening for domestic violence by prehospital care providers

Arguments for routine screening

- **Prevalence is high** in South Africa, and many survivors are not identified.
- **Unique access** to the home environment.
- **Early identification** may allow intervention before serious injury or femicide.
- The **US Preventive Services Task Force** recommends that clinicians screen women of reproductive age for intimate partner violence and refer those who screen positive to ongoing support services (Curry et al., 2018).
- Asking about violence signals that it is an acceptable topic and that help is available, even if the person does not disclose at that time.
- **Mandatory reporting duties** for children, older persons and persons with disabilities require active identification.

Arguments against universal routine screening

- The **WHO (2013)** clinical guidelines on responding to intimate partner and sexual violence **do not recommend universal screening** of all women attending health care. Instead, they recommend **clinical (selective) inquiry** — asking about exposure to violence when assessing conditions that may be caused or complicated by intimate partner violence — and recommend routine inquiry only in antenatal care where safe conditions exist. This is because there is insufficient evidence that universal screening improves outcomes, and because screening must be accompanied by the ability to respond appropriately.
- **Minimum requirements for safe inquiry** may be absent in the prehospital setting: privacy (the perpetrator may be present), a trained provider, a protocol for response, and referral pathways (WHO, 2013).
- **Risks**: asking in the presence of the perpetrator may escalate danger; time pressure; provider discomfort; and the risk of harm if disclosure is not handled well.
- **Prehospital-specific evidence** for screening is limited.

Assessment and recommendation

On balance, **universal routine screening of every EMS patient is not currently supported by evidence or feasible**, but **selective, safe clinical inquiry** is strongly recommended whenever there are indicators of domestic violence (see 56.3), together with **routine screening of the environment** for risks to children, older persons and persons with disabilities, in whom the law imposes reporting duties. This approach is consistent with WHO (2013) recommendations and should be supported by training, a protocol, and established referral pathways.

56.2 Formulate general principles for the assessment and management of survivors of domestic violence

(The study guide repeats the wording of 53.2 here; this section applies the general principles specifically to survivors of domestic violence. The principles in 53.2 also apply where domestic violence includes sexual assault.)

1. **Safety first**: request SAPS if the perpetrator is present or there is ongoing danger; do not confront the perpetrator; separate the survivor from the perpetrator for assessment and questioning ("I need to examine her privately in the ambulance").
2. **Treat injuries** according to the primary and secondary survey; pay particular attention to **strangulation** (airway, voice change, petechiae, neurological signs — transport even if the survivor appears well), head injuries, abdominal injuries in pregnancy, and fractures.
3. **Ask privately and sensitively** (56.3), using an interpreter who is not a family member.
4. **Believe and validate**: "Thank you for telling me. No one deserves to be hurt. It is not your fault."
5. **Respect autonomy**: a competent adult survivor decides whether to report to SAPS, apply for a protection order, go to a shelter or return home. Never force an adult

survivor to leave or to report (except where the law requires reporting for children, older persons and persons with disabilities).

6. **Assess immediate risk**: Is it safe to go home? Has the violence increased in frequency or severity? Has there been strangulation, threats to kill, use or access to weapons, threats to the children, or threats of suicide by the perpetrator? Is the survivor pregnant? These are recognised predictors of severe or lethal violence.
7. **Assess children and other dependents** in the household; report concerns (Children's Act s 110; DVA ss 2A–2B; SOA s 54 where a sexual offence is suspected).
8. **Inform the survivor of their rights and the protection order process** (55.4), and provide referral numbers discreetly (the perpetrator may check the survivor's phone or belongings).
9. **Provide psychological first aid** and assess suicide risk.
10. **Preserve evidence** and document meticulously (objective descriptions, body diagrams, the survivor's own words, photographs only according to service policy and with consent).
11. **Transport** to an emergency centre for treatment and forensic documentation (the medical practitioner completes the J88), and refer to social services.
12. **Confidentiality**: do not disclose information to the perpetrator or other family members.
13. **Self-care and debriefing** for the crew.

56.3 Apply prehospital domestic violence screening procedures

Indicators that should prompt inquiry

- Injuries inconsistent with the explanation, delay in seeking care, repeated injuries, injuries at different stages of healing, or injuries in typical locations (head, face, neck, breasts, abdomen, defensive forearm injuries).
- Strangulation marks, bite marks, burns.
- Injuries during pregnancy.
- A partner who answers for the patient, will not leave the patient alone, appears controlling or hostile, or minimises the injuries.
- A patient who appears fearful, evasive, anxious or depressed, or who looks to the partner before answering.
- Frequent EMS calls to the same address; substance abuse in the home.
- Children who are fearful, withdrawn, injured or neglected.
- Damage to property, weapons, or signs of a struggle at the scene.
- Suicide attempts, depression, anxiety or substance use in the patient.

Procedure

1. **Ensure privacy**: ask questions only when the partner and other family members (including children over about two years old) are not present. Move the patient to the ambulance.
2. **Introduce the topic in a normalising way**: "Because violence at home is very common, I ask many patients about it…"

3. **Ask direct, simple questions**, for example from validated tools such as **HITS** (Sherin et al., 1998) — "How often does your partner physically **H**urt you, **I**nsult you, **T**hreaten you with harm, **S**cream or curse at you?" — or **HARK** (Sohal, Eldridge and Feder, 2007) — "Within the last year, have you been **H**umiliated or emotionally abused by your partner? Have you been **A**fraid of your partner? Have you been **R**aped or forced into any kind of sexual activity by your partner? Have you been **K**icked, hit, slapped or otherwise physically hurt by your partner?"
4. Other useful questions:
   - "Did someone do this to you?"
   - "Do you feel safe at home?"
   - "Has anyone ever tried to strangle or choke you?"
   - "Are you afraid of anyone at home?"
   - "Has anyone threatened to kill you or your children?"
   - "Are there weapons in the house?"
   - "Are the children safe?"
5. **If the patient discloses**: listen, believe, validate; assess immediate safety; offer SAPS involvement, transport and referral; explain rights; document.
6. **If the patient does not disclose** but indicators are present: respect the response, document the indicators objectively, provide information discreetly ("If you ever need help, these are numbers you can call"), and inform the receiving facility.
7. **Report** where required by law (children, older persons, persons with disabilities; sexual offences against vulnerable persons).

High-yield points — Outcome 56

- WHO (2013): no universal screening; selective clinical inquiry when there are indicators; minimum conditions — privacy, training, protocol, referral.
- USPSTF (2018): screen women of reproductive age.
- Recommended prehospital approach: selective, private inquiry + environmental screening for children/older persons/disabled.
- Screening tools: HITS, HARK.
- Strangulation = high lethality risk; always transport.

Outcome 57. Discuss the management of domestic violence incidents with reference to the role of emergency care providers and law enforcement

Roles in a domestic violence incident

Emergency care providers

- Respond safely: stage until SAPS secures the scene if there is any indication of ongoing violence or weapons; approach with caution (domestic violence scenes are among the most dangerous for responders, because emotions are high, weapons are often present, and both parties may turn on the responders).
- Treat injuries and transport; recognise strangulation, head injury and injuries in pregnancy.
- Identify domestic violence through observation and inquiry, including risks to children, older persons and persons with disabilities.
- Provide trauma-informed care, psychological first aid and information about rights.
- Preserve evidence and document objectively.
- Report where required by law (DVA ss 2A–2B; Children's Act s 110; SOA s 54; Older Persons Act s 26) and inform SAPS of concerns.
- Refer to social services and NGOs; hand over to the emergency centre.
- **Do not**: act as investigators, take sides, mediate the dispute, confront the perpetrator, or promise outcomes.

Law enforcement (SAPS)

- Secure the scene and ensure the safety of everyone, including EMS crews.
- Assist the complainant: shelter, medical treatment, the notice of rights, explanation of remedies (DVA s 2).
- **Arrest the respondent without a warrant** where there are reasonable grounds to believe that an act of domestic violence constituting an offence with an element of violence has been committed (DVA s 3(2)).
- Seize weapons (s 9); enter premises for evidence (s 3A); execute warrants of arrest for breach of protection orders (s 8); conduct safety monitoring visits (s 4A).
- Take statements, open a case docket and investigate criminal charges (assault, rape, attempted murder).
- Remove children in need of care and protection to temporary safe care (Children's Act s 152).
- Apprehend persons who are a danger due to mental illness (MHCA s 40).

The intersection

Effective management depends on EMS and SAPS working together: SAPS provides scene safety and legal action, EMS provides medical care and documentation, and both refer to social services. Problems arise when SAPS is delayed or does not respond, when EMS is expected to manage a violent scene alone, or when SAPS members lack training in domestic violence or the

MHCA (Stander, Hodkinson and Dippenaar, 2021). Joint protocols, joint training and clear communication on scene improve outcomes.

Scene management sequence

1. Dispatch information → request SAPS → stage if unsafe.
2. SAPS secures the scene; separate the parties.
3. EMS assesses and treats all injured persons (the alleged perpetrator may also be injured and is also entitled to care — but from a separate crew where possible).
4. Private inquiry and risk assessment; check on children.
5. SAPS takes statements, arrests where indicated, and provides the notice of rights.
6. EMS transports the injured survivor to the emergency centre; evidence handed to SAPS with documentation.
7. Reports submitted (Form 2 to a social worker or SAPS where applicable).
8. Referral to social services, shelters and NGOs.
9. Debrief.

57.1 Identify key agencies for referral of victims of domestic violence

- **SAPS** — emergency number **10111**; any police station; **Family Violence, Child Protection and Sexual Offences (FCS) Units**.
- **Magistrates' courts** — for protection orders (including online and after-hours applications).
- **Gender-Based Violence Command Centre** (Department of Social Development) — **0800 428 428**; "please call me" USSD ***120*7867#**.
- **Department of Social Development** — social workers (who are designated recipients of DVA s 2A and 2B reports).
- **Thuthuzela Care Centres** — where domestic violence includes sexual violence.
- **Childline South Africa** — **116**.
- **Shelters for abused women and children** (for example those coordinated by the National Shelter Movement of South Africa).
- **Lifeline South Africa** and **POWA (People Opposing Women Abuse)** — counselling and legal support.
- **SADAG** — mental health and suicide crisis support.
- **Legal Aid South Africa** — free legal assistance.
- **Designated child protection organisations** and provincial departments of social development (Children's Act).
- **Department of Social Development and older persons' organisations** — for elder abuse (Older Persons Act).

(Verify current helpline numbers on the official agency websites before relying on them in practice.)

High-yield points — Outcome 57

- EMS: safety, treatment, identification, evidence, documentation, reporting, referral — not investigation or mediation.

- SAPS: scene safety, assistance and notice of rights (s 2), mandatory arrest for violent DV offences (s 3), weapons (s 9), warrants (s 8), safety monitoring (s 4A).
- Referral: SAPS 10111, GBV Command Centre 0800 428 428, Childline 116, magistrates' courts, social workers, shelters, TCCs, Legal Aid.

SECTION H — TOXICOLOGY

Note on order. This section covers the general principles first (Outcomes 58, 59, 61 and 62), because you need them to understand the specific poisonings in Outcome 60. Outcomes 60 and 63–65 follow.

Key definitions

- **Poison (toxin/toxicant)**: any substance that causes harm to a living organism through its chemical action when taken in sufficient quantity. Kaswa (2024) defines **poisoning** as "the deliberate or unintentional injury or impairment of a living organism by a substance through its chemical action". Paracelsus' principle, "the dose makes the poison", reminds us that almost any substance can be toxic at a high enough dose.
- **Toxic exposure**: contact with a substance (unintentional). **Overdose**: intentional or accidental exposure to an excessive amount; an overdose "does not typically result in poisoning unless it triggers symptoms" (Kaswa, 2024).
- **Toxidrome**: a constellation of signs and symptoms characteristic of a class of toxins (62.3).
- **Antidote**: a substance that counteracts the effect of a poison.
- **Venom / envenomation**: a toxin injected by an animal through a bite or sting.

Outcome 58. Comprehensively discuss the routes by which poisons may enter the body

58.1 Comprehensively discuss the routes by which poisons may enter the body under the following headings

The route of entry determines the speed of onset, the organs affected first, the decontamination required, and the precautions needed to protect the crew.

58.1.1 Ingestion

Ingestion (swallowing) is the most common route of poisoning, accounting for the majority of both accidental poisonings in children and deliberate self-poisoning in adults (Tintinalli et al., 2020).

- **Examples**: medication overdoses (paracetamol, tricyclic antidepressants, ARVs, anti-epileptics), household products (paraffin, bleach, detergents, drain cleaners), pesticides (organophosphates, carbamates such as aldicarb "Two-step", rat poisons), plants and mushrooms, traditional medicines, contaminated food (scombroid, botulism), alcohol and methanol, and drug packets (body packing and stuffing).
- **Absorption**: mainly in the small intestine (large surface area); weak acids are partly absorbed in the stomach. Onset is usually 30 minutes to two hours, but may be **delayed** by sustained-release or enteric-coated formulations, anticholinergic drugs (which slow gastric emptying), pharmacobezoars (tablet masses, for example iron, salicylates), or food. After absorption, many drugs undergo **first-pass metabolism** in the liver, which can create toxic metabolites (for example the paracetamol metabolite NAPQI, or the toxic metabolites of methanol and ethylene glycol).
- **Local effects**: caustics and corrosives cause burns of the lips, mouth, oropharynx, oesophagus and stomach; hydrocarbons cause aspiration pneumonitis if vomited or aspirated.
- **Prehospital implications**: establish what, how much, when, and the formulation; bring containers; consider activated charcoal only within the strict CDST criteria (62.2); do not induce vomiting (Outcome 61); assess the airway for burns and swelling after caustic ingestion.

58.1.2 Injection

Injection introduces a toxin directly through the skin into subcutaneous tissue, muscle or the bloodstream.

- **Examples**: intravenous drug use (heroin, "nyaope", methamphetamine, cocaine); iatrogenic drug errors and overdoses (for example insulin, opioids, potassium, local anaesthetics); deliberate injection of insulin or other drugs; and **envenomation** — snakebites, scorpion stings, spider bites, bee and wasp stings, and marine stings (Outcome 64).

- **Absorption**: intravenous injection produces the **most rapid onset** (seconds), with no opportunity for decontamination; intramuscular and subcutaneous routes are slower and depend on perfusion.
- **Associated risks**: local tissue damage and necrosis (cytotoxic venoms), infection (abscesses, cellulitis, endocarditis), and blood-borne viruses (HIV, hepatitis B and C) — **needle-stick risk for the crew**.
- **Prehospital implications**: look for track marks and bite or sting marks (the CDST advises checking for "bite/sting/injection marks"; HPCSA PBEC, 2025, p. 55); remove constrictive clothing and jewellery from a bitten limb (HPCSA PBEC, 2025, p. 54); immobilise the limb; dispose of sharps safely; treat the systemic effects.

58.1.3 Absorption (dermal and mucous membrane)

Absorption refers to the entry of a toxin through intact or damaged **skin**, **eyes** or **mucous membranes**.

- **Examples**: **organophosphate and carbamate pesticides** (highly lipid-soluble, readily absorbed through intact skin — a major occupational hazard for farm workers and a risk to EMS crews); hydrofluoric acid (penetrates deeply, causing hypocalcaemia); phenol; solvents; nicotine (green tobacco sickness); fentanyl patches; corrosive acids and alkalis (local burns); and chemical warfare agents (nerve agents, vesicants).
- **Factors increasing absorption**: lipid solubility, high concentration, a large surface area, prolonged contact, damaged skin (burns, abrasions), hydrated or thin skin (children and infants, who also have a larger surface area relative to body weight), and occlusion (contaminated clothing).
- **Eye exposure**: alkalis cause liquefactive necrosis and penetrate deeply, causing severe corneal damage and blindness; acids cause coagulative necrosis.
- **Prehospital implications** (CDST): "If poison on patient's skin: protect yourself: wear gloves, apron, eye protection and mask. **Remove clothes and wash skin with soap and water.** If poison in eye/s: **immediately flush under running water or normal saline for at least 20 min**" (HPCSA PBEC, 2025, p. 54). Removal of clothing alone removes a large proportion of contamination. Bag contaminated clothing. For chemical burns, the CDST Burns tool advises cooling with running water for at least one hour (HPCSA PBEC, 2025, p. 34). Contaminated patients must be decontaminated **before** entering the ambulance to prevent secondary contamination of the crew (Kaswa, 2024, notes that health care workers handling patients exposed to organophosphates or nerve gases "must use appropriate protective equipment").

58.1.4 Inhalation

Inhalation brings toxins into the respiratory tract, where the large alveolar surface area (about 70 m^2) and rich blood supply allow **very rapid absorption** — almost as fast as intravenous injection.

- **Examples**:
  - **asphyxiant gases**: simple asphyxiants that displace oxygen (carbon dioxide, methane, nitrogen); and **chemical asphyxiants** — **carbon monoxide** (from

fires, charcoal braziers ("imbawulas"), paraffin heaters, generators and vehicle exhausts in enclosed spaces) and **hydrogen cyanide** (from burning plastics, wool and silk in house fires, and from industrial sources) (60.9, 60.11);
  - **irritant gases**: highly water-soluble gases (ammonia, chlorine at high concentration) that injure the upper airway, and less soluble gases (phosgene, nitrogen dioxide) that cause delayed pulmonary oedema; **chlorine** (swimming pool chemicals, mixing bleach with acids) (60.10);
  - **volatile substances and solvents** ("glue sniffing", petrol, paint thinners, butane), which cause CNS depression and "sudden sniffing death" from arrhythmias;
  - **smoke** (thermal and chemical injury);
  - **drugs** (smoked methamphetamine, crack cocaine, cannabis, heroin; inhaled loxapine therapeutically);
  - **hydrocarbon aspiration** (paraffin, petrol) — a form of inhalation injury following ingestion (60.16, 60.30);
  - **riot-control agents and chemical warfare agents** (60.32).
- **Prehospital implications**: **scene safety is paramount** — never enter an enclosed space with a suspected toxic atmosphere without appropriate breathing apparatus (fire and rescue services); remove the patient to fresh air; give high-flow oxygen (the CDST Burns tool gives 100% oxygen by non-rebreather mask for suspected carbon monoxide or cyanide poisoning or a fire in an enclosed space; HPCSA PBEC, 2025, p. 34); look for signs of inhalation injury (facial burns, singed nasal hairs, soot, hoarseness, stridor); note that pulse oximetry is **falsely normal** in carbon monoxide poisoning; and anticipate delayed pulmonary oedema with some gases. The CDST Poisoning tool advises checking the scene for "fires in poorly ventilated room (cyanide/carbon monoxide poisoning)" (HPCSA PBEC, 2025, p. 54).

Comparison of routes

Route	Speed of onset	Common examples	Key prehospital action
Ingestion	Minutes to hours (may be delayed)	Medications, paraffin, pesticides, plants	History, containers, consider charcoal (CDST criteria), do not induce vomiting
Injection	Seconds (IV) to minutes (IM/SC)	IV drugs, insulin, envenomation	Look for marks, immobilise bitten limb, treat systemic effects, sharps safety
Absorption	Minutes to hours	Organophosphates, HF acid, solvents, corrosives	PPE, remove clothing, wash skin with soap and water, flush eyes ≥20 min
Inhalation	Seconds to minutes	CO, cyanide, chlorine, solvents, smoke	Scene safety, fresh air, high-flow O_2,

Route	Speed of onset	Common examples	Key prehospital action
			airway assessment

High-yield points — Outcome 58

- Four routes: ingestion (most common), injection (fastest IV), absorption (skin/eyes), inhalation (very rapid).
- CDST: PPE; remove clothes and wash skin with soap and water; flush eyes ≥20 minutes.
- Organophosphates are absorbed through intact skin — protect the crew.
- Pulse oximetry is falsely normal in CO poisoning.

Outcome 59. Discuss the epidemiology of poisoning

Global and South African overview

According to the Global Burden of Disease study, in 2019 unintentional poisoning accounted for 0.14% of all global deaths and self-harm for 1.34%, and the WHO regards poisoning as a significant public health problem in low- and middle-income countries (Kaswa, 2024). Poisonings account for 3–10% of ICU admissions in high-income settings, with a mortality of 2.6–4.5%; about half involve a single substance and half mixed ingestions; pharmaceuticals make up about 67% of toxicants, with psychotropics (~49%) and analgesics (~23%) predominating (Hüser et al., 2025). Kaswa (2024) notes that South African poisoning patterns "mirror the international trends", but that the true number of poisonings is difficult to ascertain, and that chronic environmental exposures are not reported. Health care providers should report notifiable poisonings, which in South Africa include **lead, mercury, and agricultural or stock remedy (pesticide) poisoning**, as notifiable medical conditions (Kaswa, 2024, citing the National Institute for Communicable Diseases).

Two South African studies provide important data. An analysis of 4,771 human exposure consultations to the **Tygerberg Poison Information Centre** found that accidental exposures were more common than intentional poisonings (65.2% versus 34.8%); that 55.8% of cases involved adults, of whom 57.6% were female; that **61.4% of adult cases were intentional**, and 64.3% of these were female; and that **98.8% of exposures in children were accidental**, with a male predominance (59.7%) (Veale, Wium and Müller, 2013). Across all ages, **non-drug chemicals accounted for 52.7%** of exposures, **medicines 35.2%** and **biological toxins (bites, stings, plants) 12.6%**. Among non-drug chemicals, **pesticides (34.8%)**, irritant/corrosive substances (27.7%) and volatile hydrocarbons (8.3%) were most common, and the most commonly ingested were **cholinesterase inhibitors (organophosphates and carbamates, 8.8%)**, anticoagulant rodenticides (7.1%) and pyrethroids (5.0%); **aldicarb** and amitraz were identified as toxicovigilance targets. Among medicines, analgesics were the most common class (26.1%), and **paracetamol (15.8%)**, benzodiazepines (9.2%) and antihistamines (5.2%) were the most common individual agents (Veale, Wium and Müller, 2013). An analysis of 2,872 childhood poisoning incidents at the **Red Cross War Memorial Children's Hospital** in Cape Town found that **paraffin (kerosene) was the commonest agent (24%)**, and concluded that "paraffin and drugs remain the principal agents responsible for paediatric exposures and poisonings, with increasing incidents due to household cleaning products and pesticides" (Balme et al., 2012). In the prehospital setting, **overdose or deliberate self-poisoning made up 52%** of mental-health-related EMS incidents in the Garden Route District, and women more often used poisoning or overdose as a method of suicide (Tilley et al., 2026).

Age patterns:

- **Children** (particularly toddlers aged one to five years) — mainly **accidental (unintentional)** ingestions of household products, paraffin and medications found in the home, usually small amounts, with a low mortality but important exceptions (paraffin aspiration, iron, TCAs, opioids, oral hypoglycaemics, organophosphates and

traditional medicines). Tilley et al. (2026) excluded accidental poisonings of children under eight years from their mental-health dataset because they are not deliberate.

- **Adolescents and adults** — mainly **intentional (deliberate self-poisoning)**, often with multiple agents and alcohol, and higher severity.
- **Older adults** — medication errors, polypharmacy, chronic toxicity (lithium, digoxin), and deliberate overdose with high lethality.

59.1 Discuss epidemiology of poisoning in adults and children under the following headings

59.1.1 Environmental toxins

Environmental toxins are poisons present in the air, water, soil, food or dwelling environment.

- **Carbon monoxide** — a leading cause of poisoning deaths, from indoor coal and wood braziers ("imbawulas"), paraffin heaters and stoves, generators, and fires in informal settlements (where shack fires are common). Exposures peak in winter, and often affect whole families, including children.
- **Smoke inhalation** in house and shack fires (carbon monoxide and cyanide).
- **Lead** — from old lead-based paint, informal battery recycling, mining, contaminated soil and some traditional remedies and cosmetics; children are most vulnerable (neurodevelopmental effects).
- **Mercury** — from artisanal and small-scale gold mining and some skin-lightening creams.
- **Arsenic and other heavy metals** in mining areas and contaminated groundwater.
- **Air pollution** from coal-fired power stations and domestic fuel burning (the Mpumalanga Highveld has some of the highest air pollution levels in the world).
- **Naturally occurring toxins**: poisonous plants and mushrooms (Outcome 63), envenomation by snakes, scorpions and spiders (Outcome 64), and marine toxins (64.4, Outcome 65).
- **Contaminated water and food** (for example cyanobacterial toxins in water, and food-borne toxins such as scombroid).

59.1.2 Industrial and agricultural toxins

- **Pesticides** are a major cause of serious poisoning and death in South Africa, both occupational (farm workers, through skin absorption and inhalation) and intentional (self-poisoning in rural areas). The most important are **organophosphates** and **carbamates**. Illegally decanted agricultural pesticides sold by street vendors as rat poison in townships, especially **aldicarb ("Two-step")** — a highly toxic carbamate — are an important cause of poisoning in children and adults (Rother, 2010; Davies, Hlela and Rother, 2023, cited by Kaswa, 2024). Other important agricultural chemicals include paraquat, rodenticides (anticoagulants, strychnine) and fumigants (aluminium phosphide).
- **Industrial chemicals**: chlorine and ammonia (water treatment, refrigeration), hydrogen sulphide (sewers, mining), **cyanide** (gold extraction in mining,

electroplating), **hydrofluoric acid** (glass etching, rust removal, petroleum industry), solvents, acids and alkalis, and heavy metals.

- **Hazardous materials incidents**: transport accidents (tanker spills on national roads), industrial explosions and leaks.
- **Occupational exposures** affect mainly working-age adults; children are exposed through take-home contamination, storage of chemicals in food or drink containers, and living near farms.

59.1.3 Accidental poisoning

- **Children** account for most accidental poisonings, peaking between one and five years of age, when children are mobile, curious and explore by putting objects in their mouths.
- **Paraffin (kerosene)** ingestion is one of the most common childhood poisonings in South Africa, particularly in low-income households where paraffin is used for cooking, heating and lighting and is often stored in cold-drink bottles (Balme et al., 2012). The main danger is aspiration pneumonitis.
- Other common accidental exposures: medications (adult medications left within reach, including iron tablets, ARVs, TCAs, opioids, antihypertensives and oral hypoglycaemics), household cleaning products (bleach, detergents, dishwasher products), caustic drain cleaners, and pesticides.
- **Adults**: medication errors (especially in older adults with polypharmacy), therapeutic misadventures, occupational exposures, and food poisoning; carbon monoxide.
- **Risk factors**: poor storage, lack of child-resistant packaging, decanting of chemicals into unlabelled containers, overcrowding, poverty and lack of supervision.

59.1.4 Medication overdose

- **Medicines account for about a third (35.2%) of poisoning exposures** reported to the Tygerberg Poison Information Centre, and most adult exposures are intentional (Veale, Wium and Müller, 2013); internationally, pharmaceuticals make up about two-thirds of toxicants in ICU poisoning admissions (Hüser et al., 2025).
- Commonly implicated medications in South Africa include **paracetamol** (the single most common medicine exposure, 15.8%, according to Veale, Wium and Müller, 2013, and widely available over the counter), **benzodiazepines** (9.2%), **antihistamines** (5.2%), **NSAIDs**, **antiretrovirals** (particularly efavirenz), **antidepressants** (including TCAs such as amitriptyline, which is commonly prescribed for pain and depression and is highly toxic in overdose), **anti-epileptics** (carbamazepine, valproate, phenobarbital), **antipsychotics**, **antihypertensives** (beta-blockers, calcium channel blockers), **oral hypoglycaemics** and **isoniazid** (TB treatment — causes refractory seizures in overdose).
- Overdose is more common in **females** for attempts, whereas lethality is often higher in males and older persons (Tilley et al., 2026).
- **Mixed ingestions** are common, frequently with alcohol (Hüser et al., 2025).
- Medication overdose in older adults is often accidental (confusion, polypharmacy) or chronic.

59.1.5 Household chemical overdose

Household products include bleach (sodium hypochlorite), detergents and dishwasher products, drain and oven cleaners (strong alkalis), toilet cleaners (acids), paraffin, methylated spirits, petrol, antifreeze (ethylene glycol), button batteries, cosmetics, mothballs and pesticides stored at home.

- **Children**: most household exposures are accidental and minor, but caustic ingestions (drain cleaners), paraffin aspiration, button-battery ingestion (which can cause oesophageal perforation within hours), and ethylene glycol and methanol are dangerous.
- **Adults**: household chemicals (bleach, paraffin, pesticides, caustics) are used in deliberate self-poisoning, particularly in low-income settings where medications are less accessible.
- **Dangerous mixtures**: mixing bleach with acids releases chlorine gas, and with ammonia releases chloramine gas.

59.1.6 Substance abuse

- **Alcohol** is the most commonly abused substance in South Africa and is involved in a large proportion of injuries, violence, road deaths and co-ingestions.
- **Illicit drugs**: cannabis, methamphetamine ("tik", especially in the Western Cape), methaqualone ("Mandrax"), heroin and heroin-based mixtures ("nyaope/whoonga"), cocaine, MDMA and hallucinogens; prescription opioid (codeine, tramadol) and benzodiazepine misuse; and inhalant use (glue, petrol, thinners) among street children and adolescents (Mesthrie, 2025, notes the large population of homeless children).
- Tilley et al. (2026) cite an estimate that 18.9% of South Africans abuse alcohol and/or drugs (Myers et al., 2022).
- Substance abuse causes acute poisoning, behavioural emergencies (Outcome 52.8), trauma, and withdrawal emergencies; adulterants (for example fentanyl, levamisole in cocaine, ARVs in "nyaope") add unpredictable toxicity.
- **Children** are exposed through accidental ingestion of drugs found at home (for example cannabis edibles, methadone) and, in adolescence, through experimentation.

High-yield points — Outcome 59

- Children: accidental, age 1–5, paraffin and pharmaceuticals; adults: intentional, pharmaceuticals, mixed ingestions with alcohol.
- Pesticides (organophosphates, aldicarb "Two-step") are a major cause of serious poisoning and death in South Africa.
- CO from braziers and paraffin heaters in winter; shack fires.
- Notifiable poisonings: lead, mercury, agricultural/stock remedy poisoning.
- Tygerberg PIC data: non-drug chemicals 52.7% (pesticides most common), medicines 35.2% (paracetamol most common), biological toxins 12.6%; 61.4% of adult exposures intentional; 98.8% of child exposures accidental (Veale, Wium and Müller, 2013).
- Red Cross Children's Hospital: paraffin the commonest agent (24%) (Balme et al., 2012).

Outcome 61. Discuss the role of emesis in the management of ingested poisons

Background

Induced emesis (vomiting), traditionally with **syrup of ipecac** (from *Cephaelis ipecacuanha*, containing emetine and cephaeline, which irritate the gastric mucosa and stimulate the chemoreceptor trigger zone), was once widely recommended for home and prehospital management of ingested poisons. Other methods (salt water, mustard water, fingers in the throat) are dangerous and ineffective; salt water can cause fatal hypernatraemia.

Current evidence and recommendations

The joint position paper of the American Academy of Clinical Toxicology (AACT) and the European Association of Poisons Centres and Clinical Toxicologists (EAPCCT) concluded that **ipecac syrup should not be administered routinely** in the management of poisoned patients, because:

1. there is **no evidence from clinical studies that ipecac improves the outcome** of poisoned patients;
2. the amount of toxin removed is **variable and diminishes rapidly with time** (most experimental studies show limited removal even when given within an hour);
3. vomiting may continue for an hour or more, **delaying the administration of activated charcoal, oral antidotes** (for example oral N-acetylcysteine) and whole bowel irrigation;
4. it causes **complications**: aspiration pneumonia, Mallory–Weiss tears, oesophageal and gastric rupture, diarrhoea, lethargy (which can confuse the clinical picture), and, with chronic abuse (in bulimia), cardiomyopathy (Höjer et al., 2013).

The CDST is explicit: "**Avoid milk/inducing vomiting**" (HPCSA PBEC, 2025, p. 54). Ipecac is not stocked by South African EMS services. Kaswa (2024) and Hüser et al. (2025) do not include emesis among decontamination strategies; gastric decontamination is limited to activated charcoal and whole bowel irrigation, with gastric lavage reserved for exceptional cases.

61.1 Recognise the instances and situations in which emesis can be effectively used as a therapy

Contraindications to induced emesis

Emesis must **never** be induced in a patient with:

- a **reduced level of consciousness** or the likelihood of a rapidly declining level of consciousness, or an unprotected airway (risk of aspiration);
- ingestion of a substance likely to cause **seizures** or **rapid CNS depression** (for example TCAs, isoniazid, camphor, strychnine, opioids, benzodiazepines, calcium channel blockers, beta-blockers);

- ingestion of **corrosives** (acids or alkalis) — re-exposure of the oesophagus causes further burns and risk of perforation;
- ingestion of **hydrocarbons** (paraffin, petrol, turpentine) — vomiting greatly increases the risk of aspiration pneumonitis;
- ingestion of **sharp objects** or button batteries;
- **haemorrhagic diathesis** or recent GI surgery;
- **infants younger than six months**, frail older persons, and patients with significant medical illness;
- prior vomiting.

Situations in which emesis might theoretically be considered

The AACT/EAPCCT position paper states that ipecac might be considered **only in rare circumstances**, when **all** of the following apply (Höjer et al., 2013):

1. there are **no contraindications** to its use;
2. there is a **substantial risk of serious toxicity** to the patient;
3. there is **no alternative therapy available or effective** to decrease gastrointestinal absorption (for example activated charcoal is unavailable, or the toxin is not bound by charcoal, such as iron or lithium);
4. there will be a **delay of more than one hour before the patient arrives at an emergency medical facility** (for example in a very remote rural area); and
5. ipecac can be administered **within 30–90 minutes** of the ingestion.

In South African practice, these conditions are almost never all met, ipecac is not available to EMS, and the CDST directs that vomiting not be induced. **The correct examination answer is therefore that induced emesis no longer has a role in routine emergency care**, and that the rare theoretical exception (an alert patient, in a remote area more than an hour from care, with a potentially life-threatening ingestion of a non-corrosive, non-hydrocarbon substance that is not adsorbed by charcoal and does not cause rapid CNS depression or seizures, within 30–90 minutes, where no alternative exists) is of historical rather than practical relevance. Spontaneous vomiting should be managed by protecting the airway (lateral position, suction) and preventing aspiration.

High-yield points — Outcome 61

- Ipecac/induced emesis: no proven benefit, delays charcoal and antidotes, causes aspiration — not recommended (Höjer et al., 2013).
- CDST: avoid milk/inducing vomiting.
- Contraindications: reduced consciousness, seizure risk, corrosives, hydrocarbons, sharp objects, infants <6 months.
- Theoretical use only if all five conditions are met — virtually never in practice.

Outcome 62. Comprehensively discuss the general management principles of poisoning in the emergency care setting

The general approach (CDST and literature)

Kaswa (2024) summarises the approach to poisoning with the mnemonic **ABCDEFG**: **A**irway, **B**reathing, **C**irculation, **D**econtamination (with PPE to protect health care workers), **E**nhance elimination, **F**ocus therapy (antidotes and specific therapy), and **G**et help from the poison control centre. Hüser et al. (2025) emphasise that "the cornerstone of treatment is rapid stabilization of vital functions — often before the causative pharmaceutical is identified", using the ABCDE approach, and that "while most patients recover with supportive care alone, a significant proportion require targeted interventions and advanced organ support". Kaswa (2024) stresses: "treat the patient first, not the laboratory results".

The CDST Poisoning tool (HPCSA PBEC, 2025, pp. 54–57) is the South African prehospital standard and is summarised below.

CDST Poisoning tool — step by step (adult and child)

"Please ensure the scene and own safety are prioritised."

1. Primary survey and stabilisation. If convulsing, use the Seizures tool. Attach monitors and check SpO_2, HR, BP, RR, GCS, pupils and capillary refill.

- Give oxygen if SpO_2 <92%, RR >30, RR <9 or GCS ≤14.
- Suction airway secretions; if the patient is unable to maintain the airway, use the Airway tool.
- If breathing is inadequate, assist ventilation with a BVM connected to oxygen (one breath every three to five seconds).

2. Prevent ongoing exposure.

- Skin: protect yourself (gloves, apron, eye protection, mask); remove clothes and wash skin with soap and water.
- Eyes: flush under running water or normal saline for at least 20 minutes.
- **Activated charcoal**: "Only if fully conscious and oral overdose of harmful substance in past 1 hour: give activated charcoal **1 g/kg** mixed with water. Avoid if ingested household products, paraffin, petrol, corrosive poisons (acids/bases), iron, lithium or alcohol. Avoid milk/inducing vomiting." (Adult dose in the drug appendix: 50–100 g; child dose by weight from the formulary.)
- Bites and stings: expose the site and cover with a sterile dressing; remove constrictive clothing and jewellery.

3. IV access and circulation.

- Adult: if SBP <90, give 250–500 mL crystalloid IV over 5–10 minutes and reassess; repeat if SBP is still <90; stop IV fluids if the patient becomes breathless or develops new crackles.

- Child: if two or more signs of shock (altered mental status, fast respiratory rate, fast pulse, weak peripheral pulses, delayed capillary refill >2 seconds, or low BP as a late sign), give crystalloid 10 mL/kg IV over 10–30 minutes.

4. Glucose. If <3.5 mmol/L: smear 25 g oral glucose inside the cheek/gum, or give dextrose 50% 20 mL IV (child: dextrose 10% 3 mL/kg IV); if no IV access, glucagon 1 mg IM (child 0.5 mg IM); recheck after 10 minutes and repeat if still <3.5 mmol/L.

5. History from family and friends: time of exposure, amount and type of substance; ask about **all** drugs and toxins that may have been accessible — prescribed, over-the-counter, herbal, chemicals, pesticides, **traditional medications**, recreational substances; ask to see tablets, packets and containers. **Check the scene** for pill bottles, a suicide note, and fires in poorly ventilated rooms (cyanide/carbon monoxide). Check for bite, sting and injection marks.

6. Identify the poison. If the substance is known, give specific management according to its class (Specific poisoning pages). If unknown, note unusual smells, look for sweating, drooling and secretions, check pupils and temperature, and identify the toxidrome (62.3).

7. Get advice. "Wherever possible, contact **Poisons Information Helpline of the Western Cape (0861 555 777)** for specific management recommendations."

8. Determine toxicity and disposition. Calculate the dose ingested in mg/kg (total mg ÷ weight) and check the package insert or seek advice to determine whether the dose is toxic.

- If a toxic dose, if the patient is symptomatic with no response to treatment, or if the poisoning is due to an unknown substance: **transport to the closest emergency centre**, taking remaining tablets, packets and containers if safe to do so.
- If a non-toxic dose and the patient is well with no signs or symptoms: advise the patient to remain at home and monitor for changes. *(Note: this does not apply to deliberate self-poisoning, where the suicide risk itself requires transport — see Outcome 46 and the CDST Aggression tool obligation to transport.)*

9. Monitoring during transport. Monitor BP, HR, RR, SpO_2 and GCS every 10 minutes; recheck glucose if it was initially low; reassess airway, breathing, circulation and level of consciousness if BP <90/60, HR >100 or <50, RR >20 or <9, SpO_2 <92% or GCS falls; reassess for further fluids if shock was present.

Specific poisoning management in the CDST (summary)

Poison class	CDST adult management	CDST child management
Opioid	If respiratory depression: *naloxone 0.4 mg IV/IM* (0.04 mg in a known opioid abuser); reassess every 2 min; increasing doses 0.8 mg, 2 mg, 4 mg every 2 min to a maximum total of 10 mg; if RR remains low, consider other poisoning; naloxone wears off quickly — monitor	*Naloxone 0.1 mg/kg IV/IM (up to 2 mg)*, repeat every 2 min to a maximum cumulative 10 mg

Poison class	CDST adult management	CDST child management
	and repeat	
Cholinergic (organophosphate)	^Atropine 2 mg IV^; look for response after 5 min (clear lung sounds, dilated pupils, increased HR, dry mouth); if no response, ^double the previous dose every 3 min until improvement^; when responding, ^give 20% of the total bolus dose per hour by infusion^; intubate if unable to maintain the airway or toxicity persists; **avoid succinylcholine** for RSI	^Atropine 0.05 mg/kg IV^, doubling every 3 min until response, then 20% of the total per hour
Tricyclic antidepressant	ECG; if QRS widened, ventricular arrhythmia, seizures or hypotension not responsive to fluids: ^sodium bicarbonate 8.4% 100 mL IV over 3 min^; reassess after 10 min; repeat if toxicity persists, then seek advice; if intubated, ^hyperventilate with 100% oxygen at 20–24 breaths/min^, aiming for $ETCO_2$ 20–25 mmHg	^Sodium bicarbonate 8.4% 2 mL/kg IV over 3 min^; repeat once if needed
Beta-blocker / calcium channel blocker	Bradycardia → RCSA bradycardia algorithm; hypotension → 250–500 mL crystalloid; if still hypotensive → ^calcium chloride 10% 10 mL IV over 10 min^ or ^calcium gluconate 10% 30 mL IV over 10 min^ under ECG monitoring; if still hypotensive → ^adrenaline infusion^ (1 mg in 50 mL or 4 mg in 200 mL normal saline; start 5 mL/h = 1.6 mcg/min; target 2–10 mcg/min; titrate to SBP 90/MAP 65)	Fluids 10 mL/kg; ^calcium chloride 10% 0.2 mL/kg or calcium gluconate 10% 0.6 mL/kg at 1 mL/min^; if beta-blocker more likely: *glucagon 100 mcg/kg IV over 1–2 min (max 1 mg/min)*, repeat after 10 min; ^adrenaline infusion^
Sedative-hypnotic, anticholinergic,	If temperature ≥38 °C: remove clothing, water spray,	Cooling; Aggression tool; Airway tool; widened QRS

Poison class	CDST adult management	CDST child management
sympathomimetic	fan or air-conditioning; if aggressive or agitated → Aggression tool; airway → Airway tool; stimulant chest pain: aspirin 300 mg PO, ^diazepam 5 mg or midazolam 1 mg IV slowly^, ^isosorbide dinitrate 5 mg SL or GTN 400 mcg SL^, exclude cardiac cause; widened QRS → ^sodium bicarbonate 8.4%^	(anticholinergic) → ^sodium bicarbonate 8.4%^

62.1 Know how to obtain advice regarding poisons

Sources of advice in South Africa:

1. **Poisons Information Helpline of the Western Cape — 0861 555 777** (24 hours, national), operated by the Tygerberg Poison Information Centre (Stellenbosch University) and the Red Cross War Memorial Children's Hospital Poisons Information Centre (University of Cape Town). This is the number specified in the CDST (HPCSA PBEC, 2025, p. 55).
2. **Afritox** — the South African poisons information database for health professionals, maintained by the poison centres.
3. **Senior medical advice / on-line medical control** — the CDST repeatedly directs the provider to "consult senior medical advisor/Poisons Information Helpline for further guidance" (HPCSA PBEC, 2025, p. 56).
4. **The receiving emergency centre** (pre-alert).
5. **Product information**: package inserts, product labels, and **Safety Data Sheets (SDS)** for industrial and agricultural chemicals; the CDST advises checking the package insert to determine whether a dose is toxic (HPCSA PBEC, 2025, p. 55).
6. **Hazardous materials resources**: the UN number and hazard placards on vehicles, the Emergency Response Guidebook, and fire and rescue HAZMAT teams.
7. **For snakebite**: the poison centres and recognised South African snakebite experts and resources (for example the African Snakebite Institute).

Information to have ready when calling: the patient's age, weight and sex; the substance(s) (name, strength, formulation — take the container); the amount; the time and route of exposure; whether intentional or accidental; symptoms and vital signs; treatment already given; and relevant medical history.

Hüser et al. (2025) note that "poison information centers provide 24/7 access to expert guidance on toxicant identification, risk assessment, and individualized treatment strategies. Early consultation — particularly in unclear or complex cases — can enhance clinical decision-making, avoid unnecessary interventions and improve patient safety."

62.2 Understand and critically appraise the use of activated charcoal

What activated charcoal is and how it works

Activated charcoal is a fine, black, odourless powder produced by heating organic material (for example wood, coconut shells) and then "activating" it with steam or chemicals to create an extremely porous structure with a very large surface area (about 950–2,000 m^2 per gram). Toxins are **adsorbed** (bound to its surface) by weak intermolecular forces, preventing their absorption from the gut (Tintinalli et al., 2020).

Activated charcoal works in two ways:

1. **Single-dose activated charcoal (SDAC)** — prevents absorption of a toxin still in the gut.
2. **Multiple-dose activated charcoal (MDAC)** — enhances elimination of drugs already absorbed by interrupting **enterohepatic recirculation** and by **"gastrointestinal dialysis"** (drug diffuses from the blood back into the gut lumen and is bound by charcoal) (Kaswa, 2024; Hüser et al., 2025).

Dosing

- **CDST**: 1 g/kg orally mixed with water (adult 50–100 g; child by weight band, for example 3.5–7 kg 5 g, 7–11 kg 10 g, 11–17 kg 15 g, 17.5–35 kg 25 g, 35–55 kg 50 g, ≥55 kg 100 g) (HPCSA PBEC, 2025, pp. 54, 68, 74).
- **Hüser et al. (2025)**: SDAC commonly 50 g in adults, aiming for a charcoal-to-drug ratio of about 10:1 (higher ratios up to 40:1 may be warranted); MDAC — an initial dose followed by 12.5 g every one, two or four hours in adults.
- **Kaswa (2024)**: 0.5–1 g/kg in children and 25–100 g in adults.
- Avoid **sorbitol-containing** preparations, particularly in children (a cathartic that causes diarrhoea and dehydration; Kaswa, 2024, describes it as "emetic").

Indications (CDST criteria — all must be met)

1. The patient is **fully conscious** (and able to protect the airway).
2. The overdose was **oral**.
3. The substance is **harmful** (a potentially toxic dose).
4. It was taken **within the past one hour**.
5. The substance is **adsorbed by charcoal** (that is, not one of the exclusions).

Contraindications and substances not adsorbed

- **Reduced level of consciousness or unprotected airway** (unless the patient is intubated, in which case charcoal can be given via a nasogastric tube in hospital) — the main risk is **aspiration**, which can cause severe pneumonitis.
- **Substances poorly adsorbed**: the CDST lists **household products, paraffin, petrol (hydrocarbons), corrosives (acids and alkalis), iron, lithium and alcohol** (HPCSA PBEC, 2025, p. 54). Other non-adsorbed substances include other metals and ions (potassium, lead, mercury, arsenic), cyanide (poorly), methanol and ethylene glycol, and mineral acids. A useful mnemonic for poorly adsorbed substances is "**PHAILS**":

Pesticides (some) and **P**otassium, **H**ydrocarbons and **H**eavy metals, **A**cids, alkalis and **A**lcohols, **I**ron, **L**ithium, **S**olvents.

- **Corrosives**: charcoal obscures endoscopic visualisation and does not bind the agent.
- **Hydrocarbons**: increased aspiration risk with vomiting.
- **Bowel obstruction, perforation or ileus**, and patients at risk of GI haemorrhage or perforation.
- **Patients who will need endoscopy.**
- A **non-toxic ingestion** (no benefit, only risk).
- **Uncooperative patients** — forcing charcoal (for example via nasogastric tube in an unprotected airway) is dangerous.

Indications for MDAC (hospital)

MDAC should be considered for massive ingestions or for drugs with enterohepatic or entero-enteric recirculation, such as **carbamazepine, phenobarbital, dapsone, quinine and theophylline** (Hüser et al., 2025), and also for sustained-release preparations and some others (for example amatoxin mushrooms, 63.1).

Complications

Aspiration pneumonitis (potentially fatal), vomiting (common, especially when given quickly or with sorbitol), constipation and bowel obstruction (especially MDAC), corneal abrasion if spilled in the eyes, black stools, and interference with oral antidotes (for example oral N-acetylcysteine).

Critical appraisal

Arguments for activated charcoal:

- Volunteer studies show that charcoal reduces drug absorption, by about 50% or more if given within the first hour, with the effect declining with time.
- It is cheap, widely available, and relatively safe in an alert patient.
- It may reduce the need for other interventions and the severity of poisoning with highly toxic substances.
- The time window may be **extended** for massive overdoses, sustained-release formulations and anticholinergic drugs (which delay gastric emptying) (Hüser et al., 2025).

Arguments against (limitations):

- **No clinical studies have demonstrated a clear reduction in morbidity or mortality** compared with no intervention (Hüser et al., 2025).
- Most patients present after one hour, when benefit is small.
- Risk of **aspiration**, particularly in patients whose consciousness may deteriorate after charcoal is given (for example TCAs, seizure-inducing drugs).
- Poor palatability and vomiting reduce compliance.

- It is **ineffective for many common South African poisons** (paraffin, corrosives, alcohols, iron, lithium, many household products).
- It may **delay transport** if time is spent persuading a patient to drink it.
- In the prehospital setting, the one-hour limit and the requirement of full consciousness significantly restrict its use.

Conclusion: activated charcoal is **not a routine treatment for all poisoned patients**. It should be used **selectively**: in an alert, cooperative patient who has ingested a potentially toxic amount of a charcoal-adsorbed substance within about an hour (or later for sustained-release or massive ingestions on expert advice), with no contraindications. In the prehospital setting, the CDST criteria should be followed strictly, and charcoal must never delay airway protection or transport. Advice from the Poisons Information Helpline can help decide whether it is warranted.

62.3 Discuss the use of toxidromes in the context of emergency care diagnostics in patients presenting with poisoning

What a toxidrome is

A **toxidrome** (toxic syndrome) is a group of signs and symptoms that consistently occur together with exposure to a particular class of toxins. Recognising a toxidrome allows the clinician to identify the **likely class** of toxin (even when the specific agent is unknown), anticipate complications, and start empirical treatment and specific antidotes (Hüser et al., 2025). The key examination findings that differentiate toxidromes are **level of consciousness, pupil size, heart rate, blood pressure, respiratory rate, temperature, skin (sweaty versus dry), mucous membranes (salivation versus dry mouth), bowel sounds and urinary function, and muscle tone and reflexes** (Kaswa, 2024; Tutorial 1 memo).

The CDST toxidromes (HPCSA PBEC, 2025, p. 55)

Toxidrome	Features	Examples
Sympathomimetic / psychostimulant / hallucinogenic	Agitation, hallucinations, seizures; **sweating**; hyperthermia; **dilated pupils**; tachycardia; hypertension	LSD, cocaine, amphetamines (methamphetamine “tik”), MDMA
Anticholinergic	Agitation, hallucinations, seizures, unresponsiveness; **flushed, dry, warm skin**; hyperthermia; **dilated pupils**; tachycardia; **urinary retention**; **decreased bowel sounds**; blurred vision	Antihistamines, atropine, TCAs, *Datura stramonium*, some antipsychotics
Opioid	Depressed level of consciousness; hypothermia; **pinpoint pupils**;	Morphine, codeine, methadone, heroin, fentanyl, tramadol

Toxidrome	Features	Examples
	bradycardia; hypotension; **respiratory depression**	
Cholinergic	Depressed level of consciousness, seizures; **sweating**; **pinpoint pupils, watery eyes**; **bradycardia**; **drooling**; urinary incontinence; diarrhoea and vomiting; **airway secretions/crackles**	Organophosphates, carbamates (aldicarb "Two-step")
Sedative-hypnotic	Depressed level of consciousness; **slurred speech**; respiratory depression (vital signs otherwise often near normal)	Alcohol, benzodiazepines, barbiturates

Mnemonics:

- **Anticholinergic**: "**Hot as a hare** (hyperthermia), **dry as a bone** (dry skin and mucosa), **red as a beet** (flushing), **blind as a bat** (mydriasis, blurred vision), **mad as a hatter** (delirium), **full as a flask** (urinary retention), and the heart runs alone" (tachycardia).
- **Cholinergic (muscarinic)**: **DUMBELS** — **D**iarrhoea/**D**iaphoresis, **U**rination, **M**iosis, **B**radycardia/**B**ronchorrhoea/**B**ronchospasm, **E**mesis, **L**acrimation, **S**alivation; the "killer Bs" (bradycardia, bronchorrhoea, bronchospasm) cause death. **Nicotinic** effects (muscle fasciculations, weakness, paralysis, tachycardia and hypertension) can be remembered by the days of the week: **M**ydriasis (occasionally), **T**achycardia, **W**eakness, **T**witching/fasciculations, **F**asciculations, **S**ugar (hyperglycaemia).

Other important toxidromes and syndromes (beyond the CDST table)

- **Serotonin toxicity (serotonin syndrome)**: the triad of **altered mental status** (agitation, confusion), **autonomic instability** (tachycardia, hypertension, hyperthermia, sweating, diarrhoea, mydriasis) and **neuromuscular hyperactivity** (**clonus** — inducible, ocular or spontaneous — hyperreflexia and tremor, more marked in the lower limbs), with rapid onset (within hours) after serotonergic drugs (SSRIs, SNRIs, MAOIs, tramadol, MDMA, linezolid, fentanyl, combinations). The **Hunter criteria** (spontaneous clonus; inducible clonus with agitation or diaphoresis; ocular clonus with agitation or diaphoresis; tremor with hyperreflexia; or hypertonia with temperature >38 °C and ocular or inducible clonus) are the most accurate diagnostic tool (Dunkley et al., 2003; Boyer and Shannon, 2005).
- **Neuroleptic malignant syndrome (NMS)**: develops over days after starting or increasing dopamine antagonists (or abrupt withdrawal of dopaminergic drugs), with **"lead-pipe" rigidity**, **hyperthermia**, altered mental status, autonomic instability and raised creatine kinase; **bradyreflexia** and the absence of clonus distinguish it from serotonin toxicity.

- **Withdrawal syndromes** (alcohol, sedatives, opioids) (52.8).
- **Sodium channel blockade**: widened QRS (>100 ms), terminal R wave in aVR, ventricular arrhythmias, seizures, hypotension (TCAs, diphenhydramine, local anaesthetics, flecainide) (Hüser et al., 2025).
- **Metabolic acidosis with a raised anion gap** (salicylates, methanol, ethylene glycol, iron, cyanide, metformin): the mnemonic **MUDPILES** or **CAT MUDPILES** (carbon monoxide/cyanide, alcoholic ketoacidosis, toluene, methanol, uraemia, diabetic ketoacidosis, paraldehyde/phenformin, iron/isoniazid, lactic acidosis, ethylene glycol, salicylates) (Tintinalli et al., 2020); clinically suggested by Kussmaul breathing (Kaswa, 2024).
- **Asphyxiant syndrome** (CO, cyanide): headache, confusion, seizures, coma, cardiovascular collapse, with a normal SpO_2 reading.

How toxidromes differ — key discriminators

Finding	Sympathomimetic	Anticholinergic	Opioid	Cholinergic	Sedative-hypnotic	Serotonin
Mental status	Agitated	Agitated/ delirious	Depressed	Depressed/ seizures	Depressed	Agitated
Pupils	Dilated	Dilated	Pinpoint	Pinpoint	Normal/ small	Dilated
Skin	**Sweaty**	**Dry, hot, flushed**	Normal/ cool	**Sweaty**	Normal	Sweaty
HR	↑	↑	↓	↓ (may ↑ — nicotinic)	Normal/↓	↑
Temperature	↑	↑	↓	Normal	Normal/↓	↑
Bowel sounds	Normal/↑	↓	↓	↑ **(diarrhoea)**	↓	↑
Secretions	—	Dry mouth	—	**Drooling, bronchorrhoea**	—	—
Neuromuscular	Tremor	Myoclonus possible	—	Fasciculations, weakness	—	**Clonus, hyperreflexia**

The single most useful bedside discriminator between the sympathomimetic and anticholinergic toxidromes is the **skin**: sweaty in sympathomimetic toxicity, dry in anticholinergic toxicity (Tutorial 1 memo; Tintinalli et al., 2020).

Value and limitations of toxidromes in emergency diagnostics

Value:

- Allows rapid, bedside identification of the likely class of toxin without laboratory tests, which is essential prehospitally and in resource-limited settings.
- Guides immediate, potentially life-saving treatment (naloxone for opioids, atropine for organophosphates, sodium bicarbonate for sodium channel blockade, cooling and benzodiazepines for sympathomimetic toxicity).
- Helps anticipate complications (seizures, arrhythmias, respiratory failure).
- Helps prioritise investigations and communicate with the poison centre.

Limitations:

- **Mixed ingestions** (about 50% of cases; Hüser et al., 2025) produce mixed or **opposing** features that obscure the toxidrome ("toxidromes may be more difficult to appraise with multiple toxicants especially if they have antagonistic effects").
- **Incomplete toxidromes** are common (for example tramadol may not cause miosis; hypoxia may dilate pupils in opioid overdose).
- Features **evolve over time** (early versus late presentation).
- Some important poisons do **not produce a recognisable toxidrome early** (paracetamol, iron, toxic alcohols, colchicine, amatoxin mushrooms) — patients may appear well initially despite lethal ingestions.
- **Comorbid conditions** (hypoglycaemia, head injury, sepsis, hypothermia) can mimic or mask toxidromes; Hüser et al. (2025) warn clinicians to "avoid attributing altered mental status solely to intoxication".
- Toxidromes identify a **class**, not a specific agent.

Toxidromes should therefore be used together with the history, the scene, the ECG, glucose and other point-of-care tests, and expert advice.

62.4 Comprehensively discuss the approach to the mixed drug overdose

Why mixed overdoses are challenging

About half of poisoned patients admitted to ICU have ingested more than one substance (Hüser et al., 2025), often with **alcohol**. Mixed overdoses are challenging because:

- toxidromes overlap or cancel out (for example an opioid with a stimulant — "speedball" — or an anticholinergic with a sedative);
- **synergistic toxicity** occurs (for example benzodiazepines plus opioids or alcohol — profound respiratory depression; multiple serotonergic drugs — serotonin syndrome; multiple QT-prolonging drugs — torsade de pointes; multiple cardiodepressants);
- the clinical course is unpredictable, with **delayed and sequential toxicity** (for example a patient who appears drowsy from a benzodiazepine may later develop liver failure from paracetamol or cardiotoxicity from a sustained-release calcium channel blocker);
- treatment of one agent may worsen another (for example **flumazenil** to reverse a benzodiazepine can precipitate seizures in a co-ingestion of TCAs; naloxone may unmask stimulant-induced agitation);
- the history is often unreliable (Hüser et al., 2025).

Approach

1. **Assume the worst**: treat as a potentially life-threatening poisoning; assume that dangerous agents (paracetamol, TCAs, cardiotoxic drugs, sustained-release preparations) may be involved until excluded.
2. **ABCDE stabilisation** (CDST): airway (anticipate deterioration; lateral position; BVM; advanced airway if needed), breathing (oxygen; ventilation), circulation (IV access, fluids, ECG monitoring), disability (GCS, pupils, **glucose**, seizures), exposure (temperature, injuries, patches, track marks).
3. **12-lead ECG and continuous cardiac monitoring** — look for QRS widening (>100 ms), QTc prolongation (≥500 ms), bradycardia, AV block and arrhythmias (Hüser et al., 2025).
4. **History and collateral information** — every substance available in the home (prescribed, over-the-counter, herbal, traditional, recreational); time of ingestion; bring all containers (HPCSA PBEC, 2025).
5. **Identify the dominant toxidrome** and treat life threats in order of priority: airway and breathing first; then cardiovascular toxicity; then seizures; then hyperthermia.
6. **Antidotes with caution**: titrate **naloxone** to adequate breathing (not full awakening) to avoid unmasking agitation and withdrawal (CDST); **avoid flumazenil** in mixed overdoses (not in the CDST; contraindicated with TCA co-ingestion and in benzodiazepine-dependent patients — Hüser et al., 2025; RCEM, 2025); give **sodium bicarbonate** for a widened QRS (ANT/ECP).
7. **Decontamination**: activated charcoal only if the CDST criteria are met (conscious, within one hour, adsorbable substances); never if consciousness is impaired.
8. **Contact the Poisons Information Helpline** (0861 555 777).
9. **Transport all mixed overdoses** to an emergency centre (by definition, toxicity is uncertain); monitor every 10 minutes.
10. **Hospital**: paracetamol and salicylate levels in **all** deliberate overdoses (because these may be clinically silent early), blood gas, electrolytes, renal and liver function, CK, and further management (NAC, enhanced elimination, ICU) (Hüser et al., 2025; Kaswa, 2024).
11. **Psychosocial assessment** before discharge for deliberate self-poisoning (Kaswa, 2024).

62.5 Discuss the approach to a poisoning situation where the causative agent is not known

The principle is "**treat the patient, not the poison**". The approach integrates the general management above with a systematic search for clues:

1. **Scene safety** — consider toxic gases (CO, cyanide, hydrogen sulphide), chemical contamination (organophosphates), or multiple patients with similar symptoms (suggesting an environmental, food-borne, or deliberate chemical release).
2. **ABCDE stabilisation** and empirical treatment of reversible causes: oxygen; ventilation; **glucose** (hypoglycaemia mimics many poisonings); naloxone if there is an opioid picture with respiratory depression; thiamine in malnourished or alcohol-dependent

patients; benzodiazepines for seizures and agitation; cooling for hyperthermia; warming for hypothermia (HPCSA PBEC, 2025).

3. **Clues from the history and scene**: containers, pill bottles, syringes, notes, plants, chemicals, the patient's occupation (farm worker, miner, cleaner), recent prescriptions, medications belonging to other household members, traditional medicines, and the time course (HPCSA PBEC, 2025, pp. 54–55).
4. **Clues from the examination**: the toxidrome; **unusual odours** (bitter almonds — cyanide; garlic — organophosphates or arsenic; fruity/acetone — ketoacidosis or isopropanol; alcohol; petrol/paraffin — hydrocarbons; rotten eggs — hydrogen sulphide; mothballs — naphthalene); skin (cherry-red is a late and unreliable sign of CO; cyanosis unresponsive to oxygen suggests methaemoglobinaemia; bullae suggest barbiturates or CO; needle marks); pupils; temperature; breathing pattern (Kussmaul breathing suggests metabolic acidosis — salicylates, methanol, ethylene glycol); mouth (burns from corrosives, tablet residue, charcoal-like residue from previous treatment); and the ECG.
5. **Point-of-care investigations**: glucose, ECG (QRS, QTc), SpO_2 (falsely normal in CO; low and unresponsive in methaemoglobinaemia — Hüser et al., 2025), $ETCO_2$, and temperature.
6. **Contact the Poisons Information Helpline** for help with identification (for example identifying tablets by markings).
7. **Transport all patients poisoned with an unknown substance** to the closest emergency centre (HPCSA PBEC, 2025, p. 55), with all containers and substances found.
8. **In hospital**: blood gas, anion and osmolal gaps, electrolytes, renal and liver function, paracetamol and salicylate levels, CK; retaining a blood sample for later analysis; urine toxicology screens have limited value (false positives and negatives; inability to detect many drugs such as Z-drugs, GHB and fentanyl analogues) (Hüser et al., 2025).

62.6 Describe and discuss the importance of history taking in drug overdose

Why history matters

History is the foundation of the diagnosis in poisoning: Kaswa (2024) states that "history-taking aims to identify the toxic substance that caused the overdose or poisoning", and that "identifying toxic substances in a patient is crucial to obtaining a thorough medical history". The history determines the likely toxin, the expected toxicity and time course, the need for decontamination (time since ingestion), the choice of antidote, the need for enhanced elimination, the risk of delayed toxicity, the disposition, and the psychosocial risk.

Limitations

Hüser et al. (2025) warn that "in patients with poisoning, history is often unreliable". Reasons include altered consciousness, intoxication, intentional concealment or exaggeration (in deliberate self-harm), fear of legal consequences (illicit drugs), embarrassment, unknown contents of street drugs, uncertainty about amounts and timing, and language barriers. **Collateral history** from EMS, family, friends and bystanders is therefore essential (Hüser et al., 2025; Kaswa, 2024).

What to ask — the "WHAT, HOW MUCH, WHEN, HOW, WHY, WHO, WHAT ELSE" framework

1. **What** was taken? Name, strength, formulation (immediate- versus sustained-release, enteric-coated), and whether it could have been a combination; ask to see packets and containers (HPCSA PBEC, 2025). Ask about **all** substances available: prescribed medications (the patient's and other household members'), over-the-counter medicines, herbal and **traditional medicines**, chemicals, pesticides, recreational drugs, and alcohol.
2. **How much?** Count remaining tablets and estimate the maximum possible amount (assume the worst); calculate mg/kg (CDST).
3. **When?** The time of ingestion determines whether charcoal is appropriate (within one hour), whether a drug level is interpretable (for example the paracetamol nomogram requires a level at least four hours after ingestion), and when delayed toxicity may develop. Was it a single ingestion or staggered over hours?
4. **How (route)?** Ingested, injected, inhaled, or applied to the skin.
5. **Why (intent)?** Accidental or deliberate? Suicidal intent determines the psychiatric risk and the need for transport (Outcomes 45–46). In children, consider abuse or neglect.
6. **Who?** Who found the patient, who else was present, and is anyone else affected (for example CO, food poisoning, chemical release)?
7. **What else?**
   - Symptoms since ingestion (vomiting, seizures, collapse);
   - treatment already given (by bystanders, other health care providers — for example milk, salt water, induced vomiting);
   - past medical history (renal and liver disease, epilepsy, cardiac disease, diabetes, HIV, pregnancy);
   - psychiatric history and previous overdoses;
   - allergies;
   - social circumstances;
   - weight (for dose calculations).

The importance of the ECP's history

The ECP is often the **only** person to see the scene, to speak with family and bystanders before they disperse, and to collect containers and notes. Information not gathered on scene is often lost. A thorough prehospital history and the containers handed over at the emergency centre (HPCSA PBEC, 2025, p. 55) can be decisive for management — for example identifying a sustained-release calcium channel blocker, a TCA, or paracetamol in a patient who currently looks well.

High-yield points — Outcome 62

- Kaswa's ABCDEFG; treat the patient first; supportive care is the cornerstone.
- CDST: O_2 if SpO_2 <92%, RR >30 or <9, GCS ≤14; skin wash, eye flush 20 min; charcoal 1 g/kg only if fully conscious, oral, harmful, within 1 hour, adsorbable; avoid household products, paraffin, petrol, corrosives, iron, lithium, alcohol; no milk, no vomiting.
- Poisons Information Helpline: **0861 555 777**.

- CDST toxidromes: sympathomimetic (sweaty, dilated), anticholinergic (dry, dilated, urinary retention), opioid (pinpoint, respiratory depression), cholinergic (pinpoint, secretions, bradycardia), sedative-hypnotic (slurred speech, depressed LOC).
- Charcoal: no proven outcome benefit; selective use only.
- Mixed overdose: assume the worst; ECG; avoid flumazenil; titrate naloxone; paracetamol level in all deliberate overdoses.
- History is vital but often unreliable — collateral history and containers.

Outcome 60. Comprehensively discuss poisoning in both adults and children in relation to pathophysiology, clinical presentation and management

For each poisoning, the **prehospital management** follows the CDST Poisoning tool (Outcome 62) unless otherwise stated: scene safety, ABCDE, oxygen if SpO_2 <92% (or RR >30 or <9, or GCS ≤14), glucose, decontamination, toxidrome identification, specific treatment within scope, the Poisons Information Helpline (0861 555 777), and transport with 10-minute monitoring (HPCSA PBEC, 2025). Hospital-level treatments are included so that you understand the full pathway, and are labelled as such.

60.1 Stimulant overdose

Agents: amphetamine-type stimulants (methamphetamine "tik", amphetamine, MDMA "ecstasy", methcathinone "cat" and synthetic cathinones), cocaine (powder and crack), prescription stimulants (methylphenidate), and in some definitions caffeine and theophylline.

Pathophysiology: amphetamines cause release (and block reuptake) of noradrenaline, dopamine and serotonin from presynaptic neurons; cocaine blocks the reuptake of noradrenaline, dopamine and serotonin and also **blocks fast sodium channels** (a local anaesthetic effect causing QRS widening and arrhythmias). The result is excessive central and peripheral sympathetic stimulation: CNS excitation, vasoconstriction (including coronary vasospasm), tachycardia, hypertension, increased myocardial oxygen demand, hyperthermia (increased muscle activity and heat production with impaired heat loss due to vasoconstriction), and, in severe cases, rhabdomyolysis, metabolic acidosis, DIC and multi-organ failure. MDMA's serotonergic effects cause serotonin toxicity and hyponatraemia (SIADH plus excessive water intake).

Clinical presentation (sympathomimetic toxidrome): agitation, anxiety, paranoia, hallucinations, psychosis, seizures, **sweating, hyperthermia, dilated pupils, tachycardia and hypertension** (HPCSA PBEC, 2025, p. 55); bruxism (jaw clenching); chest pain (myocardial ischaemia or infarction, aortic dissection, pneumothorax from smoking); palpitations and arrhythmias; headache, stroke and intracranial haemorrhage; abdominal pain (mesenteric ischaemia); **acute behavioural disturbance** with risk of sudden death (RCEM, 2025). Body packers and stuffers may present with sudden severe toxicity when packets rupture.

Children: may ingest adult methamphetamine or prescription stimulants, presenting with agitation, tachycardia, hypertension, hyperthermia and seizures; younger children may present with seizures as the first sign.

Prehospital management (CDST):

- Scene safety; SAPS if violent.
- ABCDE; oxygen if SpO_2 <92%; glucose.
- **Agitation**: CDST Aggression tool — de-escalation, then benzodiazepine sedation (*midazolam 0.2 mg/kg IM, max 15 mg*), or ^ketamine 4 mg/kg IM^ if midazolam

unavailable (HPCSA PBEC, 2025, pp. 46–47). **Benzodiazepines are the treatment of choice** for stimulant toxicity because they reduce CNS excitation, sympathetic outflow, heart rate, blood pressure, seizures and hyperthermia. The RCEM (2025) notes that midazolam may be preferred where transient increases in tachycardia or hypertension would be particularly dangerous (for example possible cocaine-induced psychosis or ischaemic heart disease).

- **Hyperthermia** (temperature ≥38 °C): remove clothing, water spray and fan, or ambulance air-conditioning (HPCSA PBEC, 2025, p. 57); minimise physical struggle and restraint (which increases heat production).
- **Chest pain from suspected stimulant use**: aspirin 300 mg orally; ^diazepam 5 mg or midazolam 1 mg IV slowly^; ^isosorbide dinitrate 5 mg SL or GTN 400 mcg SL^; exclude a cardiac cause (Chest pain tool) (HPCSA PBEC, 2025, p. 57).
- **Widened QRS** (cocaine sodium-channel blockade): ^sodium bicarbonate 8.4% 100 mL IV over 3 min^ (child 2 mL/kg) (HPCSA PBEC, 2025, pp. 56–57).
- **Seizures**: CDST Seizures tool (midazolam 10 mg IM/buccal/IN or 10 mg IV in adults).
- IV fluids for dehydration and to protect the kidneys from rhabdomyolysis.
- Continuous ECG; monitor every 10 minutes; transport.

Hospital: further benzodiazepines; active cooling; avoid beta-blockers alone in cocaine toxicity (unopposed alpha stimulation is a traditional concern); phentolamine or nitrates for hypertension; management of rhabdomyolysis, hyponatraemia (MDMA), ACS and arrhythmias; chlorpromazine for serotonin toxicity with severe hyperthermia (RCEM, 2025); intubation and paralysis for refractory hyperthermia.

60.2 Unknown / mixed drug overdose

The general approach to the unknown and mixed overdose is described in detail in Outcomes 62.4 and 62.5. In summary:

Pathophysiology: the effects reflect the combined, sometimes synergistic (for example opioids plus benzodiazepines plus alcohol) and sometimes opposing (for example opioid plus stimulant) actions of multiple agents, with unpredictable and delayed toxicity (for example paracetamol hepatotoxicity, sustained-release cardiotoxicity).

Presentation: mixed or incomplete toxidromes; altered consciousness; unexplained arrhythmias, seizures, acidosis or hypotension. About half of ICU poisoning admissions involve more than one substance (Hüser et al., 2025).

Children: consider any medicine in the household, including grandparents' medicines (antihypertensives, oral hypoglycaemics, opioids); "one pill can kill" agents in toddlers include TCAs, calcium channel blockers, beta-blockers, sulphonylureas, opioids (including methadone), clonidine and antimalarials.

Management: ABCDE; glucose; empirical naloxone if an opioid picture with respiratory depression (titrated); benzodiazepines for seizures and agitation; sodium bicarbonate for widened QRS (ANT/ECP); no flumazenil; activated charcoal only if CDST criteria are met; ECG; Poisons Information Helpline; transport all unknown or mixed overdoses (HPCSA PBEC, 2025, p. 55); hospital measurement of paracetamol and salicylate levels in all deliberate overdoses.

60.3 Marijuana (cannabis)

Agents: herbal cannabis ("dagga", "weed"), resin (hashish), oils, edibles (cookies, sweets, cakes), vaping products, and **synthetic cannabinoids** ("spice", "K2", "Mojo"), which are much more potent full agonists. Cannabis was legalised for private adult use in South Africa following the Constitutional Court's 2018 judgment in *Minister of Justice and Constitutional Development v Prince*, and the Cannabis for Private Purposes Act 7 of 2024; the availability of edibles increases the risk of accidental ingestion by children.

Pathophysiology: Δ^9-tetrahydrocannabinol (THC) is a partial agonist at cannabinoid CB1 receptors (CNS — mood, perception, memory, coordination, appetite) and CB2 receptors (immune). Inhaled THC acts within minutes and lasts two to four hours; ingested THC has a delayed onset (30 minutes to two hours) and prolonged effects (up to 12 hours or longer), so people may consume more while waiting for effects. Synthetic cannabinoids are full CB1 agonists with much greater toxicity.

Clinical presentation:

- **Adults**: euphoria, relaxation, altered time perception, impaired attention, memory and coordination, conjunctival injection, dry mouth, increased appetite, **tachycardia**, orthostatic hypotension; in higher doses or in naïve users, anxiety, panic, paranoia, psychosis and, rarely, chest pain and arrhythmias (myocardial infarction has been reported in young users). **Cannabinoid hyperemesis syndrome**: cyclical severe nausea, vomiting and abdominal pain in chronic heavy users, relieved by hot showers, which can cause dehydration and electrolyte disturbance.
- **Children** (usually edibles): drowsiness, ataxia, hypotonia, lethargy, mydriasis, tachycardia, and in larger ingestions coma, respiratory depression, apnoea and seizures.
- **Synthetic cannabinoids**: severe agitation, psychosis, seizures, tachycardia, hypertension, chest pain and myocardial ischaemia, hyperthermia, acute kidney injury and death.

Prehospital management: reassurance in a calm, quiet environment for anxiety and panic; ABCDE; glucose; airway support in children with depressed consciousness (lateral position, BVM if needed); benzodiazepines for severe agitation or seizures (CDST); IV fluids for dehydration from hyperemesis; consider co-ingestants (alcohol, methaqualone "white pipe"); transport children and anyone with significant symptoms. There is no antidote.

60.4 Hallucinogens

Agents: lysergic acid diethylamide (LSD), psilocybin ("magic mushrooms"), mescaline (peyote), dimethyltryptamine (DMT), MDMA (which has both stimulant and entactogenic effects), and dissociative agents (phencyclidine (PCP), ketamine, dextromethorphan in high doses). Plant-derived anticholinergic hallucinogens (*Datura stramonium*) are covered in 63.1.

Pathophysiology: classic hallucinogens (LSD, psilocybin, mescaline, DMT) are **serotonin 5-HT2A receptor agonists**, altering perception and cognition; dissociatives (PCP, ketamine) are **NMDA receptor antagonists**, producing a dissociative state, analgesia and sympathetic stimulation.

Clinical presentation:

- **Classic hallucinogens**: visual hallucinations and illusions, synaesthesia ("seeing sounds"), distorted perception of time and self, euphoria or intense fear ("bad trip"), panic, paranoia, impaired judgement (risk of trauma — falls, jumping from heights, walking into traffic); mild sympathomimetic signs (**mydriasis**, tachycardia, hypertension, hyperthermia, tremor, sweating). Life-threatening physical toxicity is uncommon with LSD and psilocybin alone.
- **PCP/ketamine**: dissociation, agitation, unpredictable violence, analgesia (patients may injure themselves without feeling pain), **horizontal, vertical and rotatory nystagmus**, hypertension, tachycardia, hyperthermia, rhabdomyolysis, seizures; with large doses, coma.
- **MDMA**: euphoria, bruxism, hyperthermia, serotonin toxicity, **hyponatraemia** (headache, vomiting, confusion, seizures, cerebral oedema), rhabdomyolysis, DIC.
- **Children**: accidental ingestions are rare but may cause agitation, hallucinations, seizures and hyperthermia.

Prehospital management:

- A calm, dim, quiet environment; "talking down" with reassurance; one provider; prevent injury.
- ABCDE; glucose; temperature.
- Benzodiazepine sedation per CDST for severe agitation or seizures.
- Cooling for hyperthermia.
- Consider hyponatraemia in MDMA users with seizures or confusion (avoid giving large volumes of hypotonic fluids).
- Look for trauma.
- Transport.

60.5 Sedatives and hypnotics

Agents: benzodiazepines (diazepam, lorazepam, alprazolam, clonazepam, midazolam, flunitrazepam), Z-drugs (zolpidem, zopiclone), barbiturates (phenobarbital), methaqualone ("Mandrax"), GHB/GBL, sedating antihistamines, and ethanol (60.29).

Pathophysiology: benzodiazepines and Z-drugs enhance the effect of **GABA at the GABA-A receptor** by increasing the **frequency** of chloride channel opening, causing neuronal hyperpolarisation and CNS depression; barbiturates increase the **duration** of channel opening and, at high doses, open the channel directly, which is why barbiturates cause profound respiratory and cardiovascular depression and have a much narrower safety margin. Alcohol, opioids and other CNS depressants act synergistically.

Clinical presentation (sedative-hypnotic toxidrome): depressed level of consciousness, **slurred speech**, ataxia, nystagmus, and respiratory depression (HPCSA PBEC, 2025, p. 55), with vital signs often near normal in pure benzodiazepine overdose. Pupils are usually normal or small. Hypotension, hypothermia and coma occur with large doses, barbiturates or co-ingestants. Complications include aspiration pneumonia, rhabdomyolysis and pressure injuries

from prolonged immobility, and **"coma blisters"** (bullae, classically with barbiturates). Pure benzodiazepine overdose is rarely fatal; deaths usually involve co-ingestion of alcohol or opioids. GHB causes abrupt coma with abrupt awakening, bradycardia, vomiting and myoclonus.

Children: more sensitive to respiratory depression; ataxia and drowsiness may be the first signs; paradoxical agitation can occur.

Prehospital management:

- ABCDE; airway positioning (lateral), suction, BVM ventilation if inadequate breathing; airway tool if unable to maintain the airway.
- Oxygen if SpO_2 <92%; glucose.
- Temperature (treat hypothermia; CDST cooling if ≥38 °C).
- If agitated (paradoxical reaction or withdrawal) — Aggression tool.
- **Flumazenil is not in the CDST** and is not recommended prehospitally. In hospital, flumazenil (titrated 0.1 mg/min to a maximum of 1 mg) may reverse CNS depression in selected pure benzodiazepine or Z-drug overdoses, but it is **contraindicated** in concomitant antidepressant (TCA) overdose, in patients with epilepsy or on antiepileptic drugs, and in benzodiazepine-dependent patients, because it can precipitate seizures and withdrawal (Hüser et al., 2025).
- Activated charcoal only if fully conscious within one hour (rarely appropriate, because consciousness may deteriorate).
- Hospital: supportive care; intubation if needed (Hüser et al., 2025, note that a conservative approach to intubation in comatose poisoned patients without respiratory failure, shock or vomiting reduced pneumonia and ICU admissions); MDAC and urinary alkalinisation may enhance phenobarbital elimination.
- Transport and monitor every 10 minutes.

60.6 Narcotics, opiates and opioids

Definitions: **opiates** are naturally occurring alkaloids derived from the opium poppy (morphine, codeine); **opioids** are all substances (natural, semi-synthetic and synthetic) that act on opioid receptors (heroin, oxycodone, fentanyl, methadone, tramadol, pethidine); **narcotic** is a legal rather than pharmacological term for controlled drugs.

Pathophysiology: agonism at **mu (μ) opioid receptors** in the CNS causes analgesia, euphoria, sedation, **respiratory depression** (reduced brainstem responsiveness to CO_2 and hypoxia — the cause of death), miosis (via the Edinger–Westphal nucleus), nausea, reduced gut motility, and histamine release (morphine). Death results from hypoventilation, hypoxia and aspiration. Some opioids have additional toxicity: **tramadol** (serotonergic and noradrenergic reuptake inhibition — seizures and serotonin syndrome; less miosis), **methadone** (very long half-life, QT prolongation and torsade de pointes), **pethidine** (serotonin syndrome, seizures from the norpethidine metabolite), **codeine** (variable conversion to morphine — ultra-rapid metabolisers are at risk), **loperamide** in massive doses (cardiotoxicity), and **fentanyl** (high potency, chest wall rigidity with rapid IV injection).

Clinical presentation (opioid toxidrome): depressed level of consciousness, **pinpoint pupils**, **respiratory depression** (RR <12, shallow breathing, apnoea), bradycardia, hypotension and hypothermia (HPCSA PBEC, 2025, p. 55); cyanosis; non-cardiogenic pulmonary oedema; seizures (tramadol); signs of injection (track marks) or patches. Note that the pupils may be dilated if there is severe hypoxia or a co-ingested anticholinergic or stimulant.

Children: highly sensitive — a single tablet of methadone, buprenorphine or a fentanyl patch can be fatal for a toddler; codeine-containing cough syrups; children may present with lethargy, miosis, apnoea, and seizures; delayed and prolonged toxicity with methadone.

Prehospital management (CDST):

- **Airway and ventilation first**: open the airway, BVM ventilation with oxygen (one breath every three to five seconds) — ventilation corrects hypoxia while naloxone is prepared and reduces the risk of pulmonary oedema and agitation on reversal.
- **Naloxone** if respiratory depression (HPCSA PBEC, 2025, p. 56):
  - adult: *naloxone 0.4 mg IV/IM*; if a known opioid abuser, *0.04 mg IV/IM* to limit withdrawal symptoms; reassess every two minutes; if the respiratory rate is still low, give **increasing doses of 0.8 mg, 2 mg, 4 mg** every two minutes up to a **maximum total of 10 mg**; if the respiratory rate remains low, consider other poisoning;
  - child: *naloxone 0.1 mg/kg IV/IM, up to 2 mg*, repeated every two minutes to a maximum cumulative dose of 10 mg;
  - the drug appendix also lists the intranasal route (HPCSA PBEC, 2025, p. 72).
- **Titrate to adequate breathing, not full consciousness**, to avoid precipitating acute withdrawal (agitation, vomiting with aspiration risk, combativeness, and rarely pulmonary oedema or arrhythmias). Hüser et al. (2025) suggest 0.05–0.1 mg IV every two minutes in hospital, with higher initial doses (0.2–0.4 mg) in pre-arrest patients.
- **Naloxone wears off quickly** (duration 30–90 minutes, shorter than most opioids, especially methadone and sustained-release preparations): monitor the respiratory rate closely and give further doses later if needed (HPCSA PBEC, 2025, p. 56). Patients who refuse transport after naloxone reversal are at risk of re-sedation; strongly encourage transport.
- Glucose; temperature; look for co-ingestion and trauma.
- In cardiac arrest, follow standard resuscitation; naloxone is not a priority over CPR and ventilation (Lavonas et al., 2023).
- Transport and monitor every 10 minutes.

60.7 Cardiac medications

The most important cardiac medications in overdose are **beta-blockers**, **calcium channel blockers**, **digoxin** (cardiac glycosides), and **other antihypertensives** (clonidine, ACE inhibitors, angiotensin receptor blockers, diuretics). Antiarrhythmics (flecainide — a sodium-channel blocker; amiodarone) are less common.

Beta-blockers (propranolol, atenolol, metoprolol, carvedilol, sotalol)

- **Pathophysiology**: competitive antagonism of β1 receptors (reduced heart rate, AV conduction and contractility) and β2 receptors (bronchospasm, impaired glycogenolysis — hypoglycaemia, especially in children). **Propranolol** is lipophilic and has **sodium-channel-blocking** (membrane-stabilising) effects — causing QRS widening, seizures and coma; **sotalol** has potassium-channel-blocking effects (QT prolongation, torsade de pointes).
- **Presentation**: bradycardia, AV block, hypotension, cardiogenic shock, cardiac arrest; altered consciousness and seizures (propranolol); bronchospasm; **hypoglycaemia** (particularly children).

Calcium channel blockers (verapamil, diltiazem — non-dihydropyridines; amlodipine, nifedipine — dihydropyridines)

- **Pathophysiology**: blockade of L-type calcium channels in the myocardium (reduced contractility and conduction — mainly verapamil and diltiazem), vascular smooth muscle (vasodilatation — mainly dihydropyridines), and pancreatic β-cells (reduced insulin release — **hyperglycaemia**). Verapamil and diltiazem are the most dangerous; sustained-release preparations cause delayed and prolonged toxicity.
- **Presentation**: hypotension, bradycardia (or reflex tachycardia with dihydropyridines), AV block, cardiogenic and vasodilatory shock, non-cardiogenic pulmonary oedema; mental status is often **preserved** until late (unlike beta-blocker toxicity); **hyperglycaemia** is a marker of severity.

Digoxin and cardiac glycosides

- **Pathophysiology**: inhibition of the **Na^+/K^+-ATPase pump**, increasing intracellular calcium (increased contractility) and vagal tone (slowed AV conduction), with increased automaticity. In **acute** overdose, inhibition of the pump causes **hyperkalaemia** (a marker of severity and mortality). Plant cardiac glycosides (oleander, *Urginea*/*Drimia*, *Cotyledon*) cause the same syndrome (63.1).
- **Presentation**: nausea, vomiting, abdominal pain, confusion, **visual disturbances** (blurred vision, yellow-green halos — xanthopsia), fatigue; virtually **any arrhythmia** — bradycardia, AV block, ventricular ectopy, bigeminy, **paroxysmal atrial tachycardia with block**, bidirectional ventricular tachycardia, VF (Hüser et al., 2025); hyperkalaemia in acute toxicity. **Chronic toxicity** (more common in older people) is precipitated by renal impairment, hypokalaemia, dehydration and interacting drugs.

Clonidine and other antihypertensives

- **Clonidine** (central α2 agonist) causes a picture resembling opioid toxicity: depressed consciousness, miosis, bradycardia, hypotension (sometimes transient initial hypertension), respiratory depression — particularly dangerous in children; it may partially respond to naloxone. Abrupt clonidine withdrawal causes rebound hypertension.
- **ACE inhibitors/ARBs**: hypotension, usually mild and responsive to fluids.
- **Diuretics**: dehydration and electrolyte disturbances.

Children

Small numbers of tablets of calcium channel blockers, beta-blockers, clonidine or digoxin can be life-threatening in toddlers; children are particularly prone to hypoglycaemia with beta-blockers.

Prehospital management (CDST)

- ABCDE; IV access; ECG monitoring and 12-lead ECG; glucose (hypoglycaemia in beta-blocker toxicity; hyperglycaemia in CCB toxicity).
- **Bradycardia**: follow the RCSA bradycardia algorithm (atropine, adrenaline infusion, pacing — though pacing often fails to improve blood pressure because the problem is contractility) (HPCSA PBEC, 2025, p. 57; Hüser et al., 2025).
- **Hypotension** (HPCSA PBEC, 2025, p. 57):
    1. crystalloid 250–500 mL IV over 5–10 min (child 10 mL/kg);
    2. if still hypotensive: ^calcium chloride 10% 10 mL IV over 10 min^ **or** ^calcium gluconate 10% 30 mL IV over 10 min^ under ECG monitoring (child: calcium chloride 10% 0.2 mL/kg or calcium gluconate 10% 0.6 mL/kg at 1 mL/min);
    3. in a child where beta-blocker poisoning is more likely: *glucagon 100 mcg/kg IV over 1–2 min (not exceeding 1 mg/min)*, repeated after 10 min if inadequate (the drug appendix lists glucagon IV for poisoning as within ECT, ANT and ECP scope; HPCSA PBEC, 2025, p. 69);
    4. if still hypotensive: ^adrenaline infusion^ — adult 1 mg in 50 mL (or 4 mg in 200 mL) normal saline, starting at 5 mL/h (1.6 mcg/min), target 6–30 mL/h (2–10 mcg/min), titrated by 1 mL/h every five minutes to SBP 90/MAP 65; child 0.06 mg/kg in 50 mL, starting at 1 mL/h (target 0.02–0.1 mcg/kg/min).
- **Digoxin toxicity**: the CDST has no specific prehospital antidote; treat arrhythmias per RCSA algorithms, avoid calcium (traditional concern about "stone heart" in digoxin toxicity, although recent evidence questions this), correct hypoxia, and transport urgently for **digoxin-specific antibody fragments (Fab)** in hospital.
- Contact the Poisons Information Helpline; transport all cardiac drug overdoses, even if asymptomatic (delayed toxicity with sustained-release preparations).

Hospital management (Hüser et al., 2025)

- **High-dose insulin euglycaemia therapy (HIET)** for CCB and BB cardiogenic shock: insulin bolus 1 U/kg IV, then infusion 1 U/kg/h, titrated up to 10 U/kg/h, with dextrose (starting 0.5 g/kg) to maintain glucose >5.5 mmol/L and close potassium monitoring; onset may take 15–30 minutes, with the full effect taking hours; combine with vasopressors.
- **Glucagon** for beta-blocker toxicity: 50–150 mcg/kg IV bolus (typically 3–5 mg), then 2–5 mg/h infusion if responsive; evidence is weaker and vomiting is common.
- **Digoxin-specific Fab fragments**: for life-threatening arrhythmias or hyperkalaemia; with unknown amounts, two vials initially in acute poisoning or one vial in chronic poisoning, repeated as needed.
- **Sodium bicarbonate** for QRS widening (propranolol, flecainide).

- **Lipid emulsion** as a last resort for refractory toxicity from lipophilic drugs; **VA-ECMO** for refractory cardiogenic shock or arrest.
- Whole bowel irrigation for sustained-release preparations.

60.8 Organophosphates

Agents: organophosphate insecticides (for example chlorpyrifos, dimethoate, malathion, parathion), organophosphate nerve agents (sarin, VX — 60.32), and the closely related **carbamates** (aldicarb — "Two-step", methomyl, carbaryl). In South Africa, cholinesterase inhibitors are the most commonly ingested non-drug chemicals reported to the Tygerberg Poison Information Centre (8.8%), and aldicarb is a toxicovigilance target (Veale, Wium and Müller, 2013); street-sold aldicarb is an important cause of child deaths (Rother, 2010).

Pathophysiology: organophosphates **irreversibly** phosphorylate (inhibit) **acetylcholinesterase**, the enzyme that breaks down acetylcholine at synapses; over time the phosphorylated enzyme undergoes **"ageing"**, after which it cannot be reactivated by oximes. Carbamates inhibit acetylcholinesterase **reversibly**, so their toxicity is usually shorter, and they penetrate the CNS less. Acetylcholine accumulates at:

- **muscarinic receptors** (parasympathetic end-organs, sweat glands) → DUMBELS;
- **nicotinic receptors** at the neuromuscular junction and autonomic ganglia → fasciculations, muscle weakness and paralysis (including respiratory muscles), tachycardia, hypertension;
- **CNS receptors** → agitation, confusion, seizures, coma, central respiratory depression.

Death is usually from **respiratory failure** due to the combination of bronchorrhoea, bronchospasm, respiratory muscle weakness and central respiratory depression (Eddleston et al., 2008).

Clinical presentation (cholinergic toxidrome) (HPCSA PBEC, 2025, p. 55): depressed level of consciousness and seizures; **sweating**; **pinpoint pupils, watery eyes**; **bradycardia**; **drooling**; urinary incontinence; diarrhoea and vomiting; **airway secretions and crackles** on auscultation; plus fasciculations and weakness. Note that tachycardia and mydriasis may occur from nicotinic effects. A **garlic or petroleum odour** is common (from the solvent). **Intermediate syndrome** (24–96 hours after poisoning) causes proximal muscle, neck flexor, cranial nerve and respiratory muscle weakness, and can cause respiratory arrest after initial improvement; **organophosphate-induced delayed polyneuropathy** occurs weeks later.

Children: may present atypically with predominantly CNS signs (lethargy, coma, seizures, hypotonia) and fewer classic muscarinic signs; pinpoint pupils and secretions are important clues.

Prehospital management (CDST):

1. **Scene safety and PPE** — gloves, apron, eye protection and mask; organophosphates are absorbed through skin, and vomitus and clothing are contaminated (HPCSA PBEC, 2025, p. 54; Kaswa, 2024). Ventilate the ambulance.
2. **Decontamination**: remove all clothing (bag it), wash the skin with soap and water, flush eyes.

3. **ABCDE**: suction secretions; oxygen; BVM ventilation if needed; airway tool if unable to maintain the airway — **avoid succinylcholine** for RSI (it is metabolised by plasma cholinesterase, which is also inhibited, causing prolonged paralysis) (HPCSA PBEC, 2025, p. 56).
4. **Atropine (ANT/ECP)** (HPCSA PBEC, 2025, p. 56):
   - adult: ^atropine 2 mg IV^; child: ^0.05 mg/kg IV^;
   - look for a response after five minutes: **clear lung sounds, dilated pupils, increased heart rate, dry mouth**;
   - if no response: ^double the previous atropine dose every three minutes until clinical improvement^;
   - when the patient responds: calculate the total amount of atropine given as boluses, and ^give 20% of this dose per hour via IV infusion^.
   - The goal of atropinisation is to dry secretions and reverse bronchorrhoea and bradycardia, **not** to normalise the pupils alone; the endpoints commonly used in the literature are a clear chest on auscultation, heart rate above 80/min, systolic BP above 80 mmHg, dry axillae and pupils no longer pinpoint (Eddleston et al., 2008). Very large doses (tens to hundreds of milligrams) may be needed. Tachycardia is **not** a contraindication to atropine when bronchorrhoea is present. Watch for atropine toxicity (agitation, hyperthermia, absent bowel sounds, urinary retention).
5. **Seizures**: benzodiazepines (CDST Seizures tool).
6. **Glucose**; IV fluids for hypotension.
7. Contact the Poisons Information Helpline; transport to an emergency centre (ICU likely).

Hospital: continued atropine titration; **oximes** (pralidoxime or obidoxime) to reactivate acetylcholinesterase before ageing — their benefit in organophosphate pesticide poisoning remains uncertain and availability varies (Eddleston et al., 2008); intubation and ventilation (avoiding succinylcholine); benzodiazepines; monitoring for intermediate syndrome.

60.9 Carbon monoxide

Sources: incomplete combustion of carbon-containing fuels — indoor coal, wood and charcoal braziers ("imbawulas"), paraffin heaters and stoves, gas geysers and heaters, generators, vehicle exhausts in enclosed spaces (including deliberate self-harm), and house and shack fires (often combined with cyanide). Methylene chloride (paint stripper) is metabolised to CO.

Pathophysiology (Rose et al., 2017):

- CO binds haemoglobin with about **200–250 times the affinity of oxygen**, forming **carboxyhaemoglobin (COHb)**, which reduces oxygen-carrying capacity;
- it **shifts the oxyhaemoglobin dissociation curve to the left**, impairing oxygen release to tissues;
- it binds **myoglobin** (cardiac and skeletal muscle dysfunction) and **cytochrome c oxidase** (impaired mitochondrial oxidative phosphorylation);

- it causes **lipid peroxidation, inflammation and neuronal injury**, contributing to delayed neurological sequelae;
- the fetus is especially vulnerable (fetal haemoglobin binds CO even more avidly, and fetal COHb levels rise more slowly and fall more slowly).

The **half-life of COHb** is about four to six hours breathing room air, falls to about 60–90 minutes with 100% oxygen at normal pressure, and to about 20–30 minutes with hyperbaric oxygen (Rose et al., 2017).

Clinical presentation: non-specific — **headache** (most common), dizziness, nausea and vomiting, fatigue, confusion, "flu-like" illness in several members of a household (a key clue, especially in winter), chest pain, palpitations, dyspnoea; severe poisoning causes syncope, seizures, coma, myocardial ischaemia, arrhythmias, metabolic acidosis, pulmonary oedema and death. The "**cherry-red**" skin colour is a late and unreliable sign, usually post-mortem. **Pulse oximetry is falsely normal**, because standard two-wavelength oximeters cannot distinguish COHb from oxyhaemoglobin (a CO-oximeter is required). **Delayed neurological sequelae** (cognitive deficits, personality change, parkinsonism, memory impairment) may appear days to weeks after apparent recovery.

Children and pregnancy: children have higher metabolic and respiratory rates and may become symptomatic earlier; pregnant women are at high risk because of fetal vulnerability.

Prehospital management:

- **Scene safety**: do not enter an enclosed space with suspected CO without appropriate breathing apparatus (fire and rescue); ventilate; remove all patients to fresh air; consider multiple casualties (the whole household).
- **100% oxygen via a non-rebreather mask at 15 L/min** (or via BVM/advanced airway if needed) for all suspected CO poisoning, regardless of SpO_2 — the CDST Burns tool lists suspected carbon monoxide or cyanide poisoning (headache, dizziness, vomiting, chest pain, confusion) and fire in an enclosed space as indications for 100% non-rebreather oxygen (HPCSA PBEC, 2025, p. 34).
- Ignore the pulse oximetry reading for decision-making; use a CO-oximeter (SpCO) if available.
- Airway management for decreased consciousness; seizures per CDST; 12-lead ECG (ischaemia, arrhythmias).
- **Consider concomitant cyanide** poisoning in fire victims (60.11).
- Transport; pre-alert the emergency centre.

Hospital: continued 100% oxygen; COHb measurement; **hyperbaric oxygen therapy** may be considered for severe poisoning (loss of consciousness, neurological deficits, seizures, cardiovascular dysfunction, severe acidosis, very high COHb, or pregnancy), because a randomised trial showed reduced cognitive sequelae (Weaver et al., 2002), although access in South Africa is limited.

60.10 Chlorine gas

Sources: swimming pool chemicals (calcium hypochlorite), industrial accidents and water treatment plants, transport spills, and mixing household bleach (sodium hypochlorite) with acids (for example toilet cleaners) or ammonia (producing chloramine gas); chlorine has been used as a chemical weapon (60.32).

Pathophysiology: chlorine is a yellow-green gas with a pungent odour, heavier than air (it accumulates in low-lying areas). It has **intermediate water solubility**, so it injures both the upper and lower airways. On contact with moist mucosa it forms **hydrochloric acid and hypochlorous acid** and generates oxygen free radicals, causing direct cellular injury, inflammation, mucosal oedema, bronchospasm, loss of surfactant, and in severe exposures **acute lung injury and non-cardiogenic pulmonary oedema** (which may be delayed for hours) (Henretig, Kirk and McKay, 2019).

Clinical presentation: immediate burning of the eyes, nose and throat, lacrimation, rhinorrhoea, cough, chest tightness, dyspnoea, wheezing (bronchospasm), stridor (laryngeal oedema), hoarseness, nausea and vomiting; severe exposures cause hypoxia, pulmonary oedema, ARDS and death. Skin and eye burns may occur with high concentrations or liquid chlorine. People with asthma, COPD, children and older adults are most vulnerable.

Prehospital management:

- **Scene safety and HAZMAT**: approach from upwind and uphill; fire and rescue in appropriate PPE to remove patients; mass-casualty triage if multiple exposures.
- Remove the patient from exposure to fresh air; remove contaminated clothing; irrigate eyes and skin with copious water (HPCSA PBEC, 2025, p. 54).
- ABCDE; **high-flow oxygen**; position upright if conscious and dyspnoeic.
- **Bronchospasm**: salbutamol 5 mg nebulised (CDST, all categories) and ipratropium bromide 0.5 mg nebulised; consider hydrocortisone for severe bronchospasm per the CDST Breathing difficulty tool.
- **Stridor/upper airway oedema**: early airway management; ^nebulised adrenaline^ may be considered for upper airway involvement (as in the CDST Burns tool for inhalational injury); RSI by ECPs (Airway tool) for impending airway obstruction; Kaswa (2024) lists chlorine inhalation among exposures that may require rapid sequence intubation and ventilatory support.
- CPAP for pulmonary oedema if available and appropriate (Airway/Breathing tools).
- Transport all symptomatic patients and observe for delayed pulmonary oedema (hospital observation for several hours).
- **Hospital**: nebulised sodium bicarbonate has been used in some centres (limited evidence); supportive care; ventilation for ARDS.

60.11 Cyanide

Sources: **smoke inhalation in house fires** (from burning plastics, polyurethane, wool, silk and synthetic materials — the most common source in practice), industrial exposures (gold mining — cyanide is used extensively in gold extraction in South Africa; electroplating; metal

cleaning; photography), deliberate ingestion of cyanide salts, **sodium nitroprusside** infusion, and plant sources (amygdalin in the kernels of apricots, peaches and bitter almonds; cassava).

Pathophysiology: cyanide binds the ferric (Fe^{3+}) iron of **cytochrome c oxidase (cytochrome a3)** in the mitochondrial electron transport chain, **blocking aerobic respiration**. Cells cannot use oxygen despite adequate delivery, so the body switches to anaerobic metabolism, producing severe **lactic acidosis**. Tissues with high oxygen demand — the brain and heart — fail first. Because oxygen is not extracted, venous blood remains highly oxygenated ("arterialisation of venous blood") (Tintinalli et al., 2020).

Clinical presentation: rapid onset (seconds to minutes after inhalation, minutes to an hour after ingestion of salts, delayed after amygdalin): headache, anxiety, confusion, dizziness, dyspnoea and tachypnoea (early), tachycardia and hypertension (early), followed by **seizures, coma, apnoea, bradycardia, hypotension and cardiac arrest**; severe metabolic (lactic) acidosis; the classic "**bitter almond**" smell (which only some people can detect); skin may appear pink; **cyanosis is a late sign**. In fire victims, a lactate above about 8–10 mmol/L suggests significant cyanide toxicity (Borron et al., 2007).

Children: more vulnerable in house fires; the same presentation.

Prehospital management:

- **Scene safety**: fire and rescue with breathing apparatus; avoid mouth-to-mouth ventilation (use BVM); PPE and decontamination (remove clothing, wash skin) for salt or liquid exposure.
- **100% oxygen** (non-rebreather at 15 L/min or BVM/advanced airway) (HPCSA PBEC, 2025, p. 34); oxygen works synergistically with antidotes.
- ABCDE; seizures per CDST; treat shock; CPR per resuscitation guidelines.
- **Suspect combined CO and cyanide poisoning** in enclosed-space fire victims with altered consciousness, seizures, cardiovascular instability or severe acidosis.
- Transport rapidly; pre-alert the hospital for antidote availability.

Antidotes (hospital — not CDST prehospital scope):

- **Hydroxocobalamin** (Cyanokit®) — the antidote of choice, especially in smoke inhalation: **5 g IV over 15 minutes in adults (70 mg/kg in children)**, repeated once if needed; it binds cyanide to form **cyanocobalamin (vitamin B12)**, which is excreted in the urine; side effects include red discoloration of the skin, mucosa and urine, hypertension, and interference with colorimetric laboratory tests (Borron et al., 2007).
- **Sodium thiosulfate** — a sulphur donor that enhances conversion of cyanide to thiocyanate by the enzyme rhodanese (slower onset; may be given with hydroxocobalamin).
- **Nitrites** (amyl nitrite, sodium nitrite) — induce methaemoglobinaemia, which binds cyanide; **avoided in smoke inhalation** because methaemoglobinaemia further reduces oxygen-carrying capacity in patients who may already have carboxyhaemoglobinaemia.

60.12 Caustic agents ingestion

Note: "caustic" and "corrosive" are often used interchangeably. In this guide, **60.12** focuses on **alkaline caustics** and the general approach, and **60.15** focuses on **acids** and other corrosives; the principles overlap.

Agents: alkalis (pH >11–12) — drain and oven cleaners (**sodium hydroxide**, potassium hydroxide), dishwasher detergents, cement and lime (calcium oxide/hydroxide), ammonia, button batteries (which generate hydroxide ions locally), and concentrated bleach. Common bleach (dilute sodium hypochlorite, 3–6%) is an irritant rather than a strong caustic and usually causes only minor injury unless ingested in large amounts.

Pathophysiology: alkalis cause **liquefactive necrosis** — saponification of fats and denaturation of proteins — which allows the alkali to **penetrate deeply** through tissue layers until neutralised. Injury is most severe in the **oesophagus** (alkalis are often tasteless and odourless, allowing larger volumes to be swallowed). Injury progresses over minutes to hours, with oedema, ulceration and necrosis, and complications including **airway oedema**, **oesophageal and gastric perforation** (mediastinitis, peritonitis), haemorrhage, and later **strictures** (weeks to months) and an increased lifetime risk of oesophageal carcinoma (Tintinalli et al., 2020).

Clinical presentation: oral pain, burns and blisters on the lips, tongue and pharynx, **drooling** and inability to swallow (dysphagia, odynophagia), vomiting (including haematemesis), chest or abdominal pain, **stridor, hoarseness, dyspnoea** (airway involvement — an emergency), and signs of perforation (severe chest/abdominal pain, subcutaneous emphysema, peritonism, shock). **The absence of oral burns does not exclude significant oesophageal injury.**

Children: usually accidental ingestion of small amounts (often from containers without child-resistant caps, or chemicals decanted into cold-drink bottles); drooling, refusal to feed, stridor and vomiting suggest significant injury. **Button batteries** lodged in the oesophagus can cause perforation and catastrophic aortoenteric fistula within hours and are a surgical emergency.

Prehospital management:

- Scene safety and PPE; remove contaminated clothing; irrigate skin and eyes copiously (alkali eye injuries require prolonged irrigation — CDST at least 20 minutes) (HPCSA PBEC, 2025, p. 54).
- **Airway**: assess for stridor, hoarseness, drooling and oropharyngeal oedema; give oxygen; early airway management by the most experienced provider if airway compromise is developing (RSI with a smaller tube; avoid blind nasal intubation; be prepared for a surgical airway). Kaswa (2024) notes that corrosive ingestion can compromise the respiratory tract and may require RSI and ventilatory support.
- **Do not induce vomiting** (re-exposure) (HPCSA PBEC, 2025, p. 54).
- **Do not give activated charcoal** (it does not bind caustics and obscures endoscopy) (HPCSA PBEC, 2025, p. 54).
- **Do not attempt neutralisation** with acids or alkalis (exothermic reaction causing thermal injury).

- **Dilution**: small sips of water may be considered only in an alert patient with no airway compromise, vomiting or signs of perforation, very soon after ingestion — but avoid large volumes (which can induce vomiting); many protocols advise nil by mouth, and the CDST advises avoiding milk. Follow local protocol.
- IV access, fluids for shock; analgesia (CDST Pain tool); antiemetic if within scope.
- Keep the container; transport to a facility with endoscopy and surgical capability.

Hospital: early endoscopy (within 12–24 hours) to grade injury; CT for suspected perforation; surgery for perforation or extensive necrosis; nutritional support; long-term follow-up for strictures.

60.13 Common household items ingestion

Agents and their risks:

Household item	Main toxicity	Key points
Paraffin, petrol, turpentine, lamp oil	Aspiration pneumonitis	See 60.16 and 60.30 — the most important dangerous household poison in South African children
Bleach (sodium hypochlorite 3–6%)	Mucosal irritation, vomiting; rarely burns with concentrated products or large volumes; chlorine/chloramine gas if mixed with acids/ammonia	Usually minor; assess for burns and stridor
Detergents, soaps, shampoos, dishwashing liquid	Nausea, vomiting, diarrhoea; foaming and aspiration	Usually minor
Liquid laundry detergent capsules ("pods")	Vomiting, CNS depression, aspiration, airway injury, corneal injury	More toxic than conventional detergents in children
Dishwasher detergents, drain cleaners, oven cleaners	Caustic injury (alkali)	See 60.12
Toilet cleaners, descalers	Corrosive (acid)	See 60.15
Button batteries	Oesophageal burns and perforation within hours; aortoenteric fistula	Emergency: transport urgently for X-ray and endoscopic removal if in the oesophagus
Magnets (multiple)	Bowel perforation and fistula	Urgent surgical assessment
Methylated spirits / methanol	Toxic alcohol: CNS depression, metabolic acidosis, blindness	See 60.19/60.29

Household item	Main toxicity	Key points
Antifreeze (ethylene glycol)	Metabolic acidosis, renal failure	See 60.19
Mothballs (naphthalene or paradichlorobenzene)	Naphthalene — haemolysis (especially in G6PD deficiency), methaemoglobinaemia, seizures	
Rat poison — anticoagulants ("superwarfarins")	Delayed coagulopathy (24–72 hours)	Single small ingestions in children are usually harmless; large or deliberate ingestions need INR monitoring and vitamin K
Street "rat poison" / aldicarb ("Two-step")	Cholinergic crisis	Life-threatening; see 60.8
Iron supplements	See 60.25	Dangerous in children
Essential oils, camphor, eucalyptus oil, methyl salicylate (oil of wintergreen)	Seizures (camphor, eucalyptus), aspiration, salicylate toxicity (wintergreen — a teaspoon can be lethal to a child)	
Cosmetics, nail polish remover (acetone), hand sanitiser (ethanol/isopropanol)	CNS depression, hypoglycaemia in children (ethanol)	
Silica gel, candles, crayons, chalk, most plants	Generally non-toxic	Reassurance (with poison centre confirmation)
Cigarettes and nicotine liquid (e-cigarettes)	Nicotine toxicity — vomiting, tachycardia then bradycardia, seizures, paralysis; e-liquids are highly concentrated	
Medications (grandparents' or parents')	See relevant sections	Child-resistant storage

Pathophysiology and presentation: as above, depending on the agent — irritation, caustic injury, aspiration, CNS depression, metabolic toxicity.

Children vs adults: children ingest small amounts accidentally (most are minor, but some — paraffin, button batteries, caustics, iron, methyl salicylate, laundry pods, camphor, aldicarb, nicotine liquid — can be dangerous); adults usually ingest household products deliberately, in larger amounts.

Prehospital management:

- Identify the product (take the container; read the label and the Safety Data Sheet if available).
- ABCDE; airway assessment for burns and stridor; oxygen.
- **Do not induce vomiting; avoid milk** (HPCSA PBEC, 2025, p. 54).
- **Do not give activated charcoal for household products** (CDST exclusion) (HPCSA PBEC, 2025, p. 54).
- Skin and eye decontamination.
- Contact the **Poisons Information Helpline (0861 555 777)**: many household exposures in children can safely be managed at home on the advice of the poison centre; the CDST allows a patient who has ingested a non-toxic dose and is well to remain at home with monitoring (HPCSA PBEC, 2025, p. 55) — but not in deliberate ingestions, suspected abuse or neglect, or dangerous products.
- Transport symptomatic patients and those who ingested dangerous products.
- **Prevention advice**: store chemicals in original, labelled containers, out of reach of children, and never in cold-drink bottles.

60.14 Drugs abused for sexual purposes

This outcome covers two related areas: (1) **drug-facilitated sexual assault (DFSA)** — the use of drugs or alcohol to incapacitate a victim so that sexual assault can be committed; and (2) drugs used **recreationally to enhance sexual experience** ("chemsex" and aphrodisiacs), which can cause toxicity.

Drug-facilitated sexual assault

Agents: **alcohol** is by far the most common agent, either alone or with other drugs; others include **benzodiazepines** (including flunitrazepam — "roofies" — midazolam, clonazepam), **Z-drugs** (zolpidem), **GHB/GBL**, **ketamine**, sedating antihistamines, opioids, **scopolamine** ("devil's breath", an anticholinergic), and cannabis (Tintinalli et al., 2020).

Pathophysiology: CNS depressants (GABAergic drugs, alcohol, opioids) cause sedation, disinhibition, muscle relaxation and **anterograde amnesia**; ketamine causes dissociation; scopolamine causes confusion, compliance and amnesia. Many of these drugs are colourless, tasteless and odourless when added to drinks, act quickly, and have short half-lives, so they may be undetectable within hours (GHB within about 6–12 hours in blood and 12 hours in urine).

Clinical presentation: sudden, unexpected intoxication out of proportion to the alcohol consumed; confusion, drowsiness, dizziness, slurred speech, ataxia, nausea, impaired vision; loss of consciousness; **amnesia** for a period of time; waking in an unfamiliar place with disarranged clothing, genital soreness or injuries; respiratory depression and bradycardia (GHB); hallucinations (ketamine). The victim may present hours later with no memory of events.

Legal significance: under the SOA, a person in an "altered state of consciousness, including under the influence of any medicine, drug, alcohol or other substance" who is incapable of appreciating the nature of the sexual act cannot consent (s 1(3)(d)(iii)); see Outcome 54.

Prehospital management:

- ABCDE (airway protection, ventilation, glucose); naloxone if opioid toxicity is suspected (CDST); supportive care.
- Treat as a potential sexual assault: trauma-informed care, evidence preservation (53.3) — in particular, **collect the first urine passed** in a clean container if the patient needs to urinate (drug testing), and preserve vomitus and any drink containers.
- Inform SAPS (with the survivor's consent for adults; mandatory for children and vulnerable persons — SOA s 54).
- Transport to a TCC or designated facility promptly (PEP within 72 hours; early blood and urine samples for toxicology).

Drugs used to enhance sexual activity ("chemsex" and aphrodisiacs)

Agent	Toxicity
Methamphetamine, cocaine, MDMA, mephedrone	Sympathomimetic toxicity (60.1); high-risk sexual behaviour; HIV and STI transmission
GHB/GBL	Rapid coma, respiratory depression, bradycardia, vomiting and aspiration; severe withdrawal syndrome in dependent users (agitation, delirium, seizures)
Alkyl nitrites ("poppers" — amyl/butyl nitrite)	Vasodilatation, hypotension, headache, syncope; **methaemoglobinaemia** (cyanosis unresponsive to oxygen, "chocolate-brown" blood; Hüser et al., 2025, note SpO_2 around 85% despite oxygen); dangerous hypotension with PDE-5 inhibitors
PDE-5 inhibitors (sildenafil, tadalafil)	Hypotension, especially with nitrates (GTN, isosorbide dinitrate, poppers) — **never give nitrates to a patient who has taken a PDE-5 inhibitor recently** (24 hours for sildenafil, 48 hours for tadalafil); priapism
Cantharidin ("Spanish fly")	Severe mucosal blistering, GI haemorrhage, renal failure
Yohimbine and some herbal "sex enhancers"	Sympathomimetic effects; adulteration with undeclared PDE-5 inhibitors

Prehospital management: supportive ABCDE care; benzodiazepines for stimulant toxicity; airway support in GHB coma (patients often wake abruptly and may become agitated); oxygen for methaemoglobinaemia (hospital antidote: methylene blue 1–2 mg/kg IV; Hüser et al., 2025); **avoid nitrates** if PDE-5 inhibitors have been used; fluids for hypotension; transport.

60.15 Corrosive ingestion

Agents: **acids** (pH <2) — toilet bowl cleaners and descalers (hydrochloric acid), battery acid (sulphuric acid), rust removers (phosphoric, oxalic and hydrofluoric acids — 60.24), swimming

pool acids (hydrochloric acid, “spirits of salts”), metal-cleaning products, and some traditional or industrial chemicals; also strong alkalis (60.12) and oxidising agents.

Pathophysiology: acids cause **coagulation necrosis** — protein denaturation forms a protective **eschar** that limits deeper penetration. Because acids are painful and bitter on contact, smaller volumes are usually swallowed. Acids pass rapidly through the oesophagus and cause the most severe injury in the **stomach** (especially the antrum and pylorus, where pooling occurs), with risk of gastric perforation, haemorrhage and later gastric outlet obstruction. **Systemic absorption** of acids can cause metabolic acidosis, haemolysis, renal failure and DIC; hydrofluoric acid causes life-threatening hypocalcaemia and hypomagnesaemia (60.24) (Tintinalli et al., 2020).

Comparison of acid and alkali injury:

Feature	Acid	Alkali
Type of necrosis	Coagulation (eschar limits depth)	Liquefaction (deep penetration)
Taste	Painful, bitter — smaller volumes	Often tasteless — larger volumes
Main site of injury	Stomach	Oesophagus
Systemic effects	Acidosis, haemolysis, renal failure	Less common
Long-term	Gastric outlet obstruction	Oesophageal strictures, carcinoma

Clinical presentation: oral and pharyngeal burns, pain, drooling, dysphagia, odynophagia, vomiting and haematemesis, epigastric pain, stridor and dyspnoea (airway oedema or aspiration), signs of perforation (peritonitis, mediastinitis, shock), and metabolic acidosis (Kussmaul breathing) in severe cases.

Children: usually accidental, small volumes; airway compromise and drooling indicate significant injury.

Prehospital management: as for caustics (60.12):

- PPE; decontaminate skin and eyes (irrigate ≥20 min) (HPCSA PBEC, 2025, p. 54).
- Airway assessment and early definitive airway if compromise (Kaswa, 2024).
- **No induced vomiting, no activated charcoal, no neutralisation, avoid milk** (HPCSA PBEC, 2025, p. 54); nil by mouth (or minimal water per local protocol).
- IV access, fluids for shock, analgesia.
- For hydrofluoric acid, see 60.24 (calcium).
- Transport with the container to a facility with endoscopy and surgery.

60.16 Petroleum products ingestion/inhalation

Agents: **paraffin (kerosene)** — the most important in South Africa; petrol (gasoline); diesel; lamp oil; lighter fluid; turpentine and mineral spirits; paint thinners; mineral oil and

lubricants; and the volatile aromatic and halogenated hydrocarbons used in inhalant abuse (toluene in glue and paint, butane, propane, trichloroethane). Hydrocarbons are discussed more broadly in 60.30.

Pathophysiology:

- **Aspiration pneumonitis** is the main danger after ingestion. Hydrocarbons with **low viscosity**, **low surface tension** and **high volatility** (paraffin, petrol, turpentine, lamp oil) spread readily across the airway and into the lungs, even when only a small amount is swallowed, especially if the patient coughs, gags or vomits. They dissolve and inactivate **surfactant**, causing alveolar collapse, direct inflammation and capillary injury, haemorrhagic pulmonary oedema, ventilation–perfusion mismatch and hypoxia. Pneumatoceles and pneumothorax may develop. GI absorption of most aliphatic hydrocarbons such as paraffin is limited, so systemic toxicity after ingestion is usually less important than pulmonary toxicity.
- **CNS effects**: inhaled volatile hydrocarbons are highly lipid-soluble and cause intoxication (euphoria, disinhibition, slurred speech, ataxia), then CNS depression, seizures and coma; chronic use causes leukoencephalopathy, cerebellar damage and peripheral neuropathy (n-hexane).
- **Cardiac sensitisation** (halogenated and aromatic hydrocarbons): the myocardium is sensitised to catecholamines, causing **ventricular arrhythmias and "sudden sniffing death"**, typically when the user is startled or exerts themselves after inhaling.
- **Skin**: chemical burns and dermatitis with prolonged contact.

Clinical presentation:

- **After ingestion/aspiration**: coughing, choking and gagging at the time of ingestion (suggesting aspiration), followed by tachypnoea, dyspnoea, wheeze, crackles, grunting, cyanosis and hypoxia; respiratory symptoms may be delayed up to six hours. Fever is common (chemical pneumonitis, not necessarily infection). The breath smells of paraffin or petrol. Vomiting; drowsiness and occasionally seizures.
- **After inhalation (volatile substance abuse)**: "huffing" (soaked rag), "bagging" (bag over the face) or "sniffing"; perioral rash or paint around the mouth and nose ("glue sniffer's rash"), chemical odour, intoxication, ataxia, hallucinations, CNS depression, seizures, arrhythmias and sudden death.

Children: paraffin is one of the most common causes of poisoning in South African children (24% of incidents at Red Cross War Memorial Children's Hospital; Balme et al., 2012), usually from cold-drink bottles; toddlers are at high risk of aspiration pneumonitis. Volatile substance abuse is common among adolescents and street children.

Prehospital management:

- Scene safety (flammable and toxic vapours); remove contaminated clothing and wash skin with soap and water (HPCSA PBEC, 2025, p. 54).
- ABCDE; **oxygen** to maintain SpO_2 >92% (or as per CDST indications); position upright if conscious and breathing is laboured.

- **Do not induce vomiting** (increases aspiration) and **do not give activated charcoal** (CDST excludes paraffin and petrol; charcoal does not bind them and increases the risk of vomiting and aspiration) (HPCSA PBEC, 2025, p. 54).
- Do not give milk or oil.
- Bronchospasm: salbutamol nebulised (CDST Breathing difficulty tool).
- Airway support and ventilation for respiratory failure (CPAP or intubation per Airway tool).
- **Volatile substance abuse**: **keep the patient calm and avoid exertion or startling** (catecholamine surge can trigger fatal arrhythmias); **avoid adrenaline** where possible except in cardiac arrest (sensitised myocardium); continuous ECG monitoring.
- Seizures per CDST; glucose.
- Transport: all symptomatic children and adults; asymptomatic children who coughed or choked after ingestion should be observed in hospital for at least six hours (chest X-ray), because pneumonitis may develop later.
- **Hospital**: supportive respiratory care; antibiotics are not routinely indicated (fever is from chemical pneumonitis); steroids are not recommended.

High-yield points — 60.1 to 60.16

- **Stimulants**: benzodiazepines are first-line (CDST midazolam); cool; chest pain → aspirin, benzodiazepine, nitrate; widened QRS → bicarbonate.
- **Opioids**: CDST naloxone 0.4 mg (0.04 mg in known abusers), escalating 0.8/2/4 mg to 10 mg total; child 0.1 mg/kg; titrate to breathing; naloxone wears off quickly.
- **Sedatives**: supportive; no prehospital flumazenil.
- **Cardiac drugs**: fluids → calcium (chloride 10 mL or gluconate 30 mL of 10%) → glucagon (child BB) → adrenaline infusion; hospital HIET; digoxin Fab.
- **Organophosphates**: PPE, decontaminate, atropine 2 mg (child 0.05 mg/kg) doubling every 3 min until chest clear, then 20% of total per hour; no succinylcholine.
- **CO**: 100% O_2; SpO_2 falsely normal; COHb half-life 4–6 h on air, 60–90 min on 100% O_2.
- **Cyanide**: fires and mining; lactic acidosis; 100% O_2; hydroxocobalamin 5 g IV (70 mg/kg child) in hospital.
- **Chlorine**: upwind, fresh air, O_2, bronchodilators, airway, delayed pulmonary oedema.
- **Caustics/corrosives**: alkali = liquefaction, oesophagus; acid = coagulation, stomach; no vomiting, no charcoal, no neutralisation; airway.
- **Household items**: call 0861 555 777; button batteries and paraffin are emergencies; no charcoal for household products.
- **DFSA**: alcohol most common; GHB, benzodiazepines; collect first urine; SOA consent provisions.
- **Petroleum**: aspiration pneumonitis; no vomiting, no charcoal; observe ≥6 h; sudden sniffing death — avoid startling and adrenaline.

60.17 Salicylate poisoning

Sources: aspirin (acetylsalicylic acid) tablets and powders, combination analgesics, **methyl salicylate (oil of wintergreen)** in liniments and muscle rubs (extremely concentrated — 5 mL of 98% methyl salicylate contains about 7 g of salicylate), and bismuth subsalicylate.

Toxic doses: acute ingestions of more than about 150 mg/kg are potentially toxic, more than about 300 mg/kg cause serious toxicity, and more than about 500 mg/kg are potentially lethal (Tintinalli et al., 2020). **Chronic toxicity** (therapeutic overuse in older people) is often more dangerous than the levels suggest and is frequently misdiagnosed as sepsis, delirium or dementia.

Pathophysiology:

1. **Direct stimulation of the respiratory centre** in the medulla → hyperventilation → **respiratory alkalosis** (the early finding in adults).
2. **Uncoupling of oxidative phosphorylation** in mitochondria → increased oxygen consumption, heat production (**hyperthermia**), increased CO_2 production, and anaerobic metabolism → **high anion gap metabolic acidosis** (lactate, ketones, salicylic acid).
3. Interference with the Krebs cycle and carbohydrate metabolism → hypo- or hyperglycaemia; **neuroglycopenia** (low CNS glucose) even with normal blood glucose.
4. **Acidaemia increases toxicity**: in an acidic environment, more salicylate exists in its non-ionised, lipid-soluble form, which crosses the blood–brain barrier — CNS salicylate toxicity is the main cause of death. This is why preventing acidaemia (and avoiding hypoventilation) is critical, and why alkalinisation helps.
5. Increased capillary permeability → **non-cardiogenic pulmonary oedema** and cerebral oedema.
6. Fluid losses (vomiting, sweating, hyperventilation) → dehydration; hypokalaemia.
7. Platelet dysfunction and hypoprothrombinaemia (bleeding).

The classic acid–base picture in adults is a **mixed respiratory alkalosis and high anion gap metabolic acidosis**; young children may present with metabolic acidosis alone (Hüser et al., 2025).

Clinical presentation: nausea, vomiting, **tinnitus** and deafness (early and characteristic), tachypnoea and hyperpnoea (deep, rapid breathing), sweating, **hyperthermia**, tachycardia, dehydration, agitation, confusion, delirium, hallucinations, seizures, coma; pulmonary oedema; hypoglycaemia; and, in severe cases, cardiovascular collapse.

Children: more prone to rapid metabolic acidosis, hypoglycaemia and CNS toxicity; methyl salicylate ingestion can be fatal in small amounts.

Prehospital management:

- ABCDE; oxygen; **glucose** (treat hypoglycaemia — and remember neuroglycopenia).
- IV fluids for dehydration.

- Activated charcoal **if fully conscious and within one hour** (salicylates are adsorbed; Hüser et al., 2025, note that a longer window may apply to large or enteric-coated ingestions and pharmacobezoars).
- **If intubation is required**, it must be done by experienced providers and the patient must be **hyperventilated** to match the pre-intubation minute ventilation — a sudden rise in CO_2 after intubation causes acidaemia, shifts salicylate into the brain and can cause cardiac arrest (Hüser et al., 2025: "Avoid hypoventilation if intubation is necessary").
- Cooling for hyperthermia; benzodiazepines for seizures.
- Transport; Poisons Information Helpline.

Hospital:

- Serial salicylate levels (every two to four hours until falling), blood gas, electrolytes, glucose.
- **Urinary alkalinisation with sodium bicarbonate** — the aim is a urine pH ≥7.5 while avoiding a blood pH >7.5; potassium replacement is essential, because hypokalaemia prevents urinary alkalinisation (Hüser et al., 2025).
- **Haemodialysis** for severe poisoning — for example very high salicylate concentrations, altered mental status, new hypoxaemia requiring oxygen, or acidaemia (pH ≤7.20) despite treatment (EXTRIP recommendations; Juurlink et al., 2015).
- Glucose supplementation even with normal blood glucose in patients with altered mental status.

60.18 Tricyclic poisoning

Agents: tricyclic antidepressants (TCAs) such as **amitriptyline** (commonly prescribed in South Africa for depression, neuropathic pain, migraine and insomnia), imipramine, clomipramine, nortriptyline and dothiepin. TCAs are among the most lethal medications in overdose; toxicity is expected with ingestions of about 10 mg/kg, and more than about 15–20 mg/kg may be life-threatening.

Pathophysiology — TCAs have multiple receptor and channel effects:

1. **Fast sodium channel blockade** (membrane-stabilising, quinidine-like effect) in the myocardium → slowed depolarisation → **QRS widening**, a **terminal R wave in aVR**, right-axis deviation of the terminal QRS, ventricular arrhythmias (wide-complex tachycardia, VT, VF), negative inotropy and hypotension. This is the main cause of death.
2. **Anticholinergic (antimuscarinic) effects** → tachycardia, dilated pupils, dry skin, urinary retention, ileus (delayed absorption), hyperthermia, delirium.
3. **Alpha-1 adrenergic blockade** → vasodilatation and hypotension.
4. **Inhibition of noradrenaline and serotonin reuptake** → early tachycardia and hypertension, then catecholamine depletion.
5. **GABA-A antagonism** and other CNS effects → **seizures**, coma.
6. **Potassium channel blockade** → QT prolongation.

Clinical presentation: rapid deterioration, often within one to two hours of ingestion — a patient who is talking on scene can become comatose, fit and arrest within minutes. Features include anticholinergic signs (tachycardia, mydriasis, dry flushed skin, urinary retention, decreased bowel sounds, agitation, delirium), **decreased level of consciousness and coma**, **seizures**, **hypotension**, and **ECG changes** — sinus tachycardia, **QRS >100 ms** (associated with seizures), **QRS >160 ms** (associated with ventricular arrhythmias), a terminal R wave >3 mm in aVR, and QT prolongation (Hüser et al., 2025).

Children: a few tablets can be fatal; seizures and arrhythmias may be the first signs.

Prehospital management (CDST, HPCSA PBEC, 2025, p. 56):

- ABCDE; oxygen; IV access; **attach the ECG immediately** and repeat 12-lead ECGs to measure the QRS.
- **Sodium bicarbonate (ANT/ECP)**: if signs of toxicity — **QRS widened, ventricular arrhythmia, seizures, or hypotension not responsive to fluids** — give ^sodium bicarbonate 8.4% 100 mL IV over 3 min^ (child: ^2 mL/kg^); reassess after 10 minutes; repeat if toxicity persists; then consult a senior medical advisor or the Poisons Information Helpline.
  - **Mechanism**: the sodium load overcomes sodium channel blockade, and **alkalinisation** (target blood pH about 7.45–7.55) reduces TCA binding to sodium channels (Hüser et al., 2025).
- **If intubated**: ^hyperventilate with 100% oxygen at 20–24 breaths per minute^, aiming for an $ETCO_2$ of 20–25 mmHg (respiratory alkalosis) (HPCSA PBEC, 2025, p. 56).
- **Hypotension**: IV crystalloid; bicarbonate; vasopressors (adrenaline infusion within scope).
- **Seizures**: benzodiazepines (CDST Seizures tool); seizures cause acidosis, which worsens cardiotoxicity.
- **Avoid**: flumazenil, physostigmine (contraindicated in TCA overdose with widened QRS — Hüser et al., 2025), class IA and IC antiarrhythmics (procainamide, flecainide), and activated charcoal in a patient whose consciousness may decline.
- Transport urgently with continuous ECG monitoring; pre-alert.

Hospital: further bicarbonate to keep pH 7.45–7.55, with monitoring for hypernatraemia and hypokalaemia; hypertonic saline or lidocaine for refractory arrhythmias (on expert advice); lipid emulsion and VA-ECMO as rescue therapies (Hüser et al., 2025); ICU monitoring for at least 12–24 hours.

60.19 Ethylene glycol

Sources: antifreeze and coolant for car radiators, de-icing solutions, brake fluid, some solvents and paints. It is sweet-tasting, which increases the risk of ingestion by children and animals. It is closely related to the other **toxic alcohols** — **methanol** (methylated spirits, illicit or contaminated home-brewed alcohol, windscreen washer fluid) and diethylene glycol (a contaminant in medicines).

Pathophysiology: ethylene glycol itself is relatively non-toxic (causing inebriation like ethanol), but it is metabolised by **alcohol dehydrogenase** to glycoaldehyde, then to **glycolic**

acid (the main cause of the **high anion gap metabolic acidosis**), glyoxylic acid and **oxalic acid**, which combines with calcium to form **calcium oxalate crystals** that deposit in the renal tubules (**acute kidney injury**), brain and other tissues, causing **hypocalcaemia** (Kraut and Mullins, 2018). Methanol is metabolised to formaldehyde and **formic acid**, which causes metabolic acidosis and **retinal and optic nerve toxicity** (blindness) and basal ganglia damage. An early **raised osmolal gap** (from the parent alcohol) is followed by a raised anion gap as the toxic metabolites accumulate.

Clinical presentation — classically in three overlapping stages:

1. **Neurological stage (30 minutes to 12 hours)**: apparent drunkenness without the smell of ethanol, nausea, vomiting, ataxia, slurred speech, confusion, seizures, coma.
2. **Cardiopulmonary stage (12–24 hours)**: tachycardia, hypertension or hypotension, tachypnoea and **Kussmaul breathing** (compensation for metabolic acidosis), heart failure, pulmonary oedema, shock.
3. **Renal stage (24–72 hours)**: flank pain, oliguria, haematuria, crystalluria, **acute kidney injury**; hypocalcaemia (tetany, QT prolongation, seizures).

Methanol causes headache, abdominal pain, metabolic acidosis, and **visual disturbances** ("snowfield" vision, blurred vision, blindness) after a latent period of 12–24 hours (longer if ethanol was co-ingested).

Children: accidental ingestion of sweet antifreeze; a small volume can be dangerous.

Prehospital management:

- ABCDE; airway protection; oxygen; glucose; seizures per CDST; IV fluids.
- **Do not give activated charcoal** — alcohols are not adsorbed (HPCSA PBEC, 2025, p. 54 lists alcohol among exclusions).
- Recognise Kussmaul breathing as a sign of metabolic acidosis; if intubation is needed, maintain a high minute ventilation.
- Take the container; contact the Poisons Information Helpline; transport (delayed toxicity — the patient may appear well initially).

Hospital (Kraut and Mullins, 2018; Kaswa, 2024):

- **Alcohol dehydrogenase inhibition**: **fomepizole** (preferred where available) or **ethanol** (which competes for alcohol dehydrogenase — Kaswa, 2024, lists ethyl alcohol as the antidote for methanol), to prevent the formation of toxic metabolites.
- **Haemodialysis** to remove the parent alcohol and toxic acids and to correct acidosis and renal failure — indicated in severe metabolic acidosis, renal failure, visual impairment (methanol) or high levels (Kaswa, 2024; Hüser et al., 2025).
- Sodium bicarbonate for acidosis.
- Cofactors: **thiamine and pyridoxine** (ethylene glycol), **folinic acid/folate** (methanol).
- Calcium only for symptomatic hypocalcaemia or seizures.

60.20 Colchicine

Sources: colchicine tablets (used for gout and familial Mediterranean fever, and pericarditis) and colchicine-containing plants — notably the **flame lily (*Gloriosa superba*)**, which is indigenous to southern Africa and has been implicated in poisonings (including through traditional medicine use), and the autumn crocus (*Colchicum autumnale*).

Pathophysiology: colchicine binds tubulin and **inhibits microtubule polymerisation**, arresting cell division in metaphase. Rapidly dividing cells are affected first (gastrointestinal mucosa, bone marrow, hair follicles), followed by multi-organ dysfunction. It has a narrow therapeutic index; ingestions above about 0.5 mg/kg are associated with severe toxicity and above about 0.8 mg/kg are often fatal (Finkelstein et al., 2010).

Clinical presentation — three phases (Finkelstein et al., 2010):

1. **Gastrointestinal phase (0–24 hours)**: profuse vomiting, diarrhoea (which may be bloody) and abdominal pain, leading to dehydration and hypovolaemia; leucocytosis.
2. **Multi-organ dysfunction phase (24 hours to 7 days)**: **bone marrow suppression** (pancytopenia — infection and bleeding), cardiogenic and distributive shock, arrhythmias and sudden cardiac arrest, ARDS, acute kidney injury, liver injury, rhabdomyolysis, DIC, metabolic acidosis, electrolyte disturbances, seizures, coma and death.
3. **Recovery phase (after about seven days)**: rebound leucocytosis and **alopecia** (hair loss).

Early severe GI symptoms after a colchicine or plant ingestion should be taken very seriously, because the patient may deteriorate days later.

Children: small numbers of tablets can be dangerous; ingestion of flame lily tubers.

Prehospital management: ABCDE; IV fluids for dehydration and shock; glucose; activated charcoal if fully conscious within one hour (colchicine is adsorbed); contact the Poisons Information Helpline; **transport all colchicine ingestions**, even if asymptomatic.

Hospital: there is **no specific antidote** in routine use; very early gastric decontamination (Hüser et al., 2025, list high-dose colchicine among the exceptional cases in which gastric lavage may be considered very early, and early activated charcoal is recommended); MDAC (enterohepatic recirculation); intensive supportive care (fluids, vasopressors, ventilation, granulocyte colony-stimulating factor for bone marrow suppression, ECMO for refractory shock).

60.21 Paracetamol

Background: paracetamol (acetaminophen) is the single most common medicine involved in poisoning exposures reported to the Tygerberg Poison Information Centre (15.8%) (Veale, Wium and Müller, 2013), and is a leading cause of acute liver failure worldwide. It is widely available over the counter, alone and in combination products (for example with codeine).

Toxic dose: a single acute ingestion of **more than about 150 mg/kg** (or 7.5–10 g in an adult) is potentially hepatotoxic (Hüser et al., 2025); risk is increased in chronic alcohol use, malnutrition, chronic liver disease and enzyme-inducing drugs (some anti-TB drugs and anticonvulsants). Staggered ingestions and repeated supratherapeutic ingestions are also dangerous.

Pathophysiology: at therapeutic doses, about 90% of paracetamol is conjugated (glucuronidation and sulphation) to non-toxic metabolites, and about 5–10% is oxidised by cytochrome P450 (mainly CYP2E1) to the highly reactive metabolite **NAPQI** (N-acetyl-p-benzoquinone imine), which is immediately detoxified by conjugation with **glutathione**. In overdose, the conjugation pathways become saturated, more NAPQI is produced, **glutathione stores are depleted** (below about 30% of normal), and free NAPQI binds to hepatocyte proteins, causing **centrilobular (zone 3) hepatic necrosis**. Renal tubular necrosis can also occur. **N-acetylcysteine (NAC)** works by replenishing glutathione (acting as a glutathione precursor), and is most effective if started within eight hours of ingestion (Tintinalli et al., 2020).

Clinical presentation — four stages:

1. **Stage 1 (0–24 hours)**: often **asymptomatic**, or mild nausea, vomiting, malaise and sweating; this is why paracetamol overdose is easily underestimated.
2. **Stage 2 (24–72 hours)**: right upper quadrant pain and tenderness; rising transaminases (ALT/AST), prolonged INR; nausea and vomiting may improve.
3. **Stage 3 (72–96 hours)**: **fulminant hepatic failure** — jaundice, coagulopathy and bleeding, hypoglycaemia, encephalopathy, lactic acidosis, renal failure, and death.
4. **Stage 4 (4 days to 2 weeks)**: recovery (with complete hepatic regeneration in survivors) or death.

Children: young children are relatively less susceptible after single accidental ingestions (smaller amounts, greater sulphation capacity), but toxicity occurs with large ingestions and with repeated supratherapeutic dosing (for example caregivers giving too much syrup or combining products).

Prehospital management:

- ABCDE; glucose (hypoglycaemia in late stages).
- **Activated charcoal** if fully conscious and within one hour (CDST) — paracetamol is well adsorbed.
- Establish the **time of ingestion**, the amount (calculate mg/kg), the formulation and any co-ingestants.
- **Always transport** a potentially toxic paracetamol ingestion and all deliberate overdoses, even if the patient looks well — this is one of the most important teaching points, because the patient is asymptomatic during the window in which treatment is most effective.
- Contact the Poisons Information Helpline.

Hospital:

- **Serum paracetamol level at four hours or later** after ingestion, plotted on the treatment nomogram to decide on NAC (not interpretable before four hours, or with staggered ingestions or unknown times, which are treated empirically).
- **NAC**: the traditional 21-hour IV regimen is **150 mg/kg over one hour, then 50 mg/kg over four hours, then 100 mg/kg over 16 hours** (Hüser et al., 2025); abbreviated two-bag regimens such as the Scottish and Newcastle Acetylcysteine Protocol (SNAP) reduce adverse reactions. NAC is most effective within eight hours, but should be given even if the patient presents late or in liver failure.
- Monitoring of liver function, INR, renal function, glucose and acid–base status; referral to a liver unit for fulminant hepatic failure (transplant criteria).
- Psychosocial assessment before discharge.

60.22 Theophylline

Background: theophylline and aminophylline (methylxanthines) are used for asthma and COPD; their use has declined, but they remain available, often in sustained-release formulations. Theophylline has a narrow therapeutic index (therapeutic levels about 10–20 mg/L), and toxicity occurs with acute overdose, chronic accumulation (older people, liver disease, heart failure), or drug interactions (ciprofloxacin, erythromycin and other enzyme inhibitors).

Pathophysiology: **adenosine receptor antagonism** (CNS stimulation, seizures, arrhythmias), **phosphodiesterase inhibition** (increased cAMP — tachycardia, vasodilatation), and **catecholamine release** with **β-adrenergic stimulation** (tachyarrhythmias, hypokalaemia from intracellular potassium shift, hyperglycaemia, metabolic acidosis, hypotension from β2 vasodilatation).

Clinical presentation: **persistent, intractable vomiting** (a hallmark — Kaswa, 2024), haematemesis, abdominal pain; agitation, tremor, insomnia; **sinus tachycardia** and **tachyarrhythmias** (atrial fibrillation, multifocal atrial tachycardia, SVT, ventricular ectopy, VT, VF — Kaswa, 2024, notes arrhythmia in theophylline toxicity); hypotension (acute overdose); **hypokalaemia**, hyperglycaemia, hypophosphataemia, metabolic acidosis; and **seizures**, which are often prolonged, refractory to treatment, and indicate a poor prognosis (Kaswa, 2024, lists theophylline among causes of seizures requiring airway support). Sustained-release preparations cause delayed peak toxicity.

Children: accidental ingestions can cause severe toxicity; chronic toxicity also occurs.

Prehospital management:

- ABCDE; oxygen; IV access; continuous ECG monitoring.
- **Seizures**: benzodiazepines (CDST Seizures tool); expect refractory seizures and prepare for airway management.
- Antiemetic if within scope; IV fluids for hypotension; glucose.
- Activated charcoal if fully conscious and within one hour.
- Transport; Poisons Information Helpline.

Hospital: **multiple-dose activated charcoal** (theophylline is a classic MDAC indication — Hüser et al., 2025); **haemodialysis** for severe toxicity (very high levels, seizures, life-threatening arrhythmias, or shock) (EXTRIP); potassium replacement (with caution, because hypokalaemia is due to a shift); beta-blockade (esmolol or propranolol) for tachyarrhythmias and hypotension due to β2 stimulation, used cautiously in patients with bronchospasm; benzodiazepines and then phenobarbital for seizures (phenytoin is ineffective); whole bowel irrigation for sustained-release preparations.

60.23 Drugs of abuse

The individual drug classes are covered in 52.8 (DSM-5-TR disorders) and in 60.1 (stimulants), 60.3 (cannabis), 60.4 (hallucinogens), 60.5 (sedatives), 60.6 (opioids), 60.16 (inhalants), 60.29 (ethanol) and 60.14 (GHB and drugs used for sexual purposes). The table below summarises the drugs of abuse most relevant to South African prehospital practice.

Drug (South African street names)	Class / mechanism	Key toxicity	Prehospital management (CDST)
Alcohol ("dop")	GABA-A enhancement, NMDA antagonism	CNS depression, aspiration, hypoglycaemia, hypothermia, trauma, withdrawal	Airway, glucose, thiamine, temperature, look for injury
Cannabis ("dagga", "zol", "weed")	CB1 agonist	Anxiety, psychosis, tachycardia; children — coma; hyperemesis	Reassurance, supportive care, benzodiazepines if severe
Methamphetamine ("tik", "crystal")	Monoamine release	Sympathomimetic toxidrome, psychosis, ABD, stroke, MI, hyperthermia	Benzodiazepine sedation, cooling, ECG
Methaqualone ("Mandrax", "buttons"; smoked with cannabis as a "white pipe")	Sedative-hypnotic (GABA-A)	CNS depression and coma, but unusually with **increased muscle tone, hyperreflexia and myoclonus**; respiratory depression with co-ingestants	Airway support, supportive care
Heroin and heroin-based mixtures ("nyaope", "whoonga", "unga"), codeine, tramadol	Mu-opioid agonists	Opioid toxidrome, respiratory depression, pulmonary oedema; adulterants	Ventilation, naloxone (titrated)
Cocaine ("coke",	Monoamine reuptake	Sympathomimetic	Benzodiazepines,

Drug (South African street names)	Class / mechanism	Key toxicity	Prehospital management (CDST)
"rock" for crack)	inhibition, sodium-channel blockade	toxicity, chest pain, arrhythmias, wide QRS	aspirin and nitrates for chest pain, bicarbonate for wide QRS
MDMA ("ecstasy", "E", "Molly")	Serotonin and catecholamine release	Hyperthermia, serotonin toxicity, hyponatraemia	Cooling, benzodiazepines, caution with fluids
Methcathinone ("cat") and synthetic cathinones	Stimulant	Sympathomimetic toxicity, psychosis	As for stimulants
LSD, psilocybin	5-HT2A agonists	Perceptual disturbance, panic, trauma	Calm environment, benzodiazepines
Ketamine, PCP	NMDA antagonists	Dissociation, agitation, nystagmus, hypertension	Calm environment, benzodiazepines
GHB/GBL	GABA-B agonist	Abrupt coma, respiratory depression, vomiting; severe withdrawal	Airway support
Inhalants (glue, petrol, thinners, butane)	CNS depressants; cardiac sensitisation	Intoxication, coma, sudden sniffing death	Calm, avoid startling, ECG, avoid adrenaline except in arrest
Prescription drugs (benzodiazepines, opioids, methylphenidate)	Various	Depending on class	Depending on class

General principles: street drugs are of unknown purity and frequently adulterated; polysubstance use is the norm; trauma, hypoglycaemia, hypothermia, aspiration, infection and withdrawal frequently coexist; and the patient's own report of what they took is often inaccurate (Hüser et al., 2025). Management is based on the **toxidrome** and supportive care (Outcome 62).

60.24 Hydrofluoric acid

Sources: glass etching and frosting, rust and stain removers, wheel and aluminium cleaners, brick and masonry cleaners, semiconductor manufacturing, petroleum refining, and some household rust-removal products. Hydrofluoric acid (HF) is a **weak acid** but an extremely dangerous poison.

Pathophysiology: because HF is a weak acid, it remains largely **non-ionised** and **penetrates deeply** through the skin and tissues. Once in the tissues, it dissociates, and the free **fluoride ion** binds **calcium and magnesium**, forming insoluble salts, causing:

- **local tissue destruction** (liquefactive necrosis, bone erosion) and **severe pain** (from calcium depletion and potassium efflux affecting nerves), often **out of proportion to visible signs**, and often delayed by hours with dilute solutions (<20%);
- **systemic toxicity**: **hypocalcaemia**, **hypomagnesaemia** and **hyperkalaemia**, leading to QT prolongation, ventricular arrhythmias, **ventricular fibrillation** and sudden death — even with relatively small burns of concentrated acid (for example burns involving more than about 1–2% of body surface area with concentrated HF, or any ingestion or significant inhalation).

Clinical presentation:

- **Dermal**: severe throbbing pain (sometimes with minimal visible injury initially), erythema, blanching, then grey-white discoloration, blistering and necrosis; fingertip exposures through gloves are common in occupational settings.
- **Eye**: severe pain, corneal damage, blindness.
- **Inhalation**: airway irritation, bronchospasm, pulmonary oedema.
- **Ingestion**: GI burns plus rapid and often fatal systemic hypocalcaemia.
- **Systemic**: tetany, carpopedal spasm, prolonged QT, arrhythmias, cardiac arrest.

Children: household rust removers; children have a larger surface area relative to weight and are at greater risk of systemic toxicity.

Prehospital management:

- **PPE** (HF penetrates standard gloves; use appropriate chemical-resistant gloves); remove contaminated clothing.
- **Irrigate copiously with water** (skin at least 20 minutes, ideally longer; CDST chemical burns — cooling with running water for at least one hour; eyes at least 20 minutes) (HPCSA PBEC, 2025, pp. 34, 54).
- **Topical calcium gluconate gel (2.5%)** applied to dermal burns and massaged into the area (often with a glove filled with gel for hand burns) — the fluoride binds the calcium in the gel; this is available in industrial first-aid kits and in hospitals.
- ABCDE; continuous **ECG monitoring** (QT prolongation, peaked T waves, arrhythmias).
- **Systemic calcium**: Kaswa (2024) lists **calcium gluconate** as the antidote for hydrofluoric acid. In the CDST, IV calcium (^calcium gluconate 10%^ or ^calcium chloride 10%^) is within ANT/ECP scope for poisoning (HPCSA PBEC, 2025, p. 68); for suspected systemic HF toxicity (large or concentrated burns, ingestion, inhalation, arrhythmias, tetany), discuss IV calcium with a senior medical advisor or the Poisons Information Helpline, and give it if instructed or if the patient is peri-arrest with signs of hypocalcaemia.
- Analgesia (persistent pain indicates ongoing fluoride activity and the need for more calcium).
- Do not induce vomiting; for ingestion, transport urgently.

- Transport all HF exposures other than trivial dilute skin contact; pre-alert.

Hospital: repeated topical calcium; **subcutaneous infiltration of calcium gluconate** or **intra-arterial calcium** for digital burns unresponsive to gel; **IV calcium and magnesium** with frequent electrolyte monitoring; treatment of hyperkalaemia; haemodialysis in severe systemic toxicity; surgical debridement.

60.25 Iron

Sources: iron supplements (ferrous sulphate, ferrous fumarate, ferrous gluconate — which contain about 20%, 33% and 12% elemental iron respectively), prenatal vitamins and multivitamins. Iron tablets are a classic cause of serious accidental poisoning in young children, because they look like sweets and are found in homes with pregnant or postpartum women.

Toxic dose (elemental iron): **less than 20 mg/kg** — usually asymptomatic; **20–60 mg/kg** — GI symptoms; **more than 60 mg/kg** — potentially serious systemic toxicity (Hüser et al., 2025, list >60 mg/kg as the threshold for deferoxamine consideration); more than about 100–200 mg/kg — potentially lethal (Tintinalli et al., 2020).

Pathophysiology:

- **Direct corrosive effect** on the GI mucosa → haemorrhagic gastroenteritis, fluid and blood loss, and later scarring.
- Once transferrin is saturated, **free iron** circulates and enters cells, where it **disrupts mitochondrial oxidative phosphorylation** and catalyses free-radical formation → **metabolic (lactic) acidosis**, shock (hypovolaemic, distributive and cardiogenic), coagulopathy, and **hepatic necrosis** (the liver receives the first-pass iron load).

Clinical presentation — classically five stages (which may overlap; Tintinalli et al., 2020):

1. **Gastrointestinal stage (0–6 hours)**: vomiting (including haematemesis), diarrhoea (including bloody), abdominal pain; **the absence of vomiting within six hours makes serious toxicity unlikely**.
2. **Latent or "quiescent" stage (6–24 hours)**: apparent improvement of GI symptoms, but ongoing cellular toxicity — **a dangerous false reassurance**; subtle tachycardia and acidosis may be present.
3. **Shock and metabolic acidosis stage (12–48 hours)**: hypotension, profound metabolic acidosis, coagulopathy, lethargy, coma, seizures, multi-organ failure.
4. **Hepatotoxicity (2–3 days)**: hepatic necrosis and failure.
5. **Bowel obstruction (2–6 weeks)**: gastric outlet or small bowel strictures from scarring.

Children: the main at-risk group; even a small number of adult-strength tablets may be dangerous for a toddler.

Prehospital management:

- ABCDE; IV access; **IV fluids** for hypovolaemia (aggressive fluid resuscitation is often required); glucose.

- **Activated charcoal does not bind iron** — do not give it (CDST exclusion) (HPCSA PBEC, 2025, p. 54); do not induce vomiting.
- Establish the product, number of tablets and elemental iron content (mg/kg), and time of ingestion; take the container.
- **Transport** any child with a potentially toxic ingestion (>20 mg/kg elemental iron) or any symptoms, and all deliberate ingestions, even if they appear to have improved (latent stage).
- Contact the Poisons Information Helpline.

Hospital: serum iron level (at about four to six hours); abdominal X-ray (iron tablets are radio-opaque); **whole bowel irrigation** for large ingestions with tablets visible on X-ray (Kaswa, 2024; Hüser et al., 2025); **deferoxamine** (a chelator that binds free iron to form ferrioxamine, excreted in the urine, which may turn a reddish "vin rosé" colour) at **15 mg/kg/h IV**, maximum about 6–8 g/day, for severe toxicity — systemic symptoms, metabolic acidosis, shock, or very high iron levels (Hüser et al., 2025); monitoring for hypotension and, with prolonged infusions, ARDS.

60.26 Oral hypoglycaemic medications

Agents:

- **Sulphonylureas** (glibenclamide, gliclazide, glimepiride) — stimulate insulin release from pancreatic β-cells by closing ATP-sensitive potassium channels;
- **meglitinides** (repaglinide) — similar mechanism, shorter-acting;
- **biguanides** — **metformin** — reduce hepatic glucose production and increase peripheral glucose uptake; rarely cause hypoglycaemia alone, but can cause **metformin-associated lactic acidosis (MALA)** in overdose or with renal failure;
- **DPP-4 inhibitors** (sitagliptin), **SGLT2 inhibitors** (dapagliflozin, empagliflozin — risk of euglycaemic diabetic ketoacidosis), **thiazolidinediones** (pioglitazone) — low risk of hypoglycaemia;
- **insulin** (injected) for comparison.

Pathophysiology and presentation:

- **Sulphonylureas** cause **profound and prolonged hypoglycaemia** (up to 24–72 hours, especially with long-acting agents and in renal impairment), which may be **delayed** for up to 8–16 hours after ingestion (or longer with modified-release preparations), and may recur after apparently successful treatment because glucose administration stimulates further insulin release. Symptoms: adrenergic (sweating, tremor, palpitations, anxiety, hunger) and neuroglycopenic (confusion, abnormal behaviour — often mistaken for intoxication or a psychiatric emergency — slurred speech, focal deficits mimicking stroke, seizures, coma). **A single sulphonylurea tablet can cause serious hypoglycaemia in a toddler.**
- **Metformin overdose**: nausea, vomiting, abdominal pain, then **severe lactic acidosis** (Kussmaul breathing, hypotension, altered consciousness), hypothermia, and multi-organ failure; hypoglycaemia is uncommon.

Children: high risk with sulphonylureas; may present with seizures or coma; hypoglycaemia may be delayed.

Prehospital management (CDST glucose management, HPCSA PBEC, 2025, pp. 47, 54):

- Check glucose in every patient with altered consciousness, seizures or abnormal behaviour.
- **If glucose <3.5 mmol/L**:
  - conscious and able to swallow: oral glucose; or smear **25 g oral glucose** inside the cheek/gums (child: <5 years 5 g, 5–11 years 10 g, >12 years 15 g);
  - otherwise: ***dextrose 50% 20 mL IV*** (consider dilution with 20 mL of sterile water or normal saline) (child: **dextrose 10% 3 mL/kg IV**);
  - no IV access: ***glucagon 1 mg IM*** (child 0.5 mg) — note that glucagon is less effective in patients with depleted glycogen stores (malnutrition, alcohol) and may be less effective in sulphonylurea toxicity;
  - **recheck glucose after 10 minutes** and repeat if still <3.5 mmol/L;
  - give **thiamine 100 mg IV/IM** if there is concern about vitamin deficiency (malnourishment, chronic alcohol use) (HPCSA PBEC, 2025, p. 73).
- After correcting hypoglycaemia, give oral carbohydrate if the patient can eat, and monitor glucose closely.
- Activated charcoal if fully conscious within one hour.
- **Transport all sulphonylurea overdoses and all children** who may have ingested a sulphonylurea, even if the glucose is currently normal (delayed and recurrent hypoglycaemia); **refusal of transport after treatment of hypoglycaemia caused by an oral agent is unsafe**.
- Metformin overdose with acidosis: ABCDE, fluids, transport.

Hospital: dextrose infusion with frequent glucose monitoring; **octreotide** (a somatostatin analogue that inhibits insulin release) for sulphonylurea-induced recurrent hypoglycaemia; observation for at least 24 hours (longer with modified-release preparations); **haemodialysis** for severe metformin-associated lactic acidosis.

60.27 Lithium

Background: lithium carbonate is a mood stabiliser used for bipolar disorder (52.4). It has a **narrow therapeutic index** (therapeutic serum levels about 0.6–1.2 mmol/L), is not metabolised, and is excreted almost entirely by the kidneys, so toxicity is precipitated by dehydration (vomiting, diarrhoea, heat, reduced fluid intake), renal impairment, low-sodium diets, and drugs that reduce lithium clearance (**NSAIDs, ACE inhibitors, ARBs, thiazide diuretics**).

Types of toxicity (Decker et al., 2015):

1. **Acute toxicity** (a single large ingestion in a lithium-naïve person): GI symptoms dominate initially; neurotoxicity may be delayed as lithium slowly distributes into the brain; serum levels may be very high with relatively few neurological signs at first.

2. **Chronic toxicity** (gradual accumulation in a patient on long-term lithium): **neurological symptoms dominate**, often at lower serum levels, because brain levels are already high; it is common in older people and is often precipitated by intercurrent illness or new medications.
3. **Acute-on-chronic toxicity** (an overdose in a patient taking lithium regularly): the most dangerous.

Pathophysiology: lithium interferes with sodium and potassium transport, second-messenger systems (inositol phosphate and cAMP), and neurotransmission; it accumulates in the brain, causing neurotoxicity, and in the kidney causes **nephrogenic diabetes insipidus** (polyuria, dehydration — which further reduces lithium excretion).

Clinical presentation:

- **GI**: nausea, vomiting, diarrhoea.
- **Neurological**: fine tremor becoming **coarse tremor**, ataxia, dysarthria, **confusion**, agitation, **hyperreflexia, clonus, myoclonus, fasciculations**, choreoathetosis, seizures, coma; **SILENT** (Syndrome of Irreversible Lithium-Effectuated Neurotoxicity — persistent cerebellar dysfunction and cognitive impairment) after severe toxicity.
- **Cardiac**: T-wave flattening or inversion, bradycardia, sinus node dysfunction, QT prolongation (usually not life-threatening).
- **Renal**: polyuria (nephrogenic DI), acute kidney injury.
- Lithium can also contribute to serotonin syndrome when combined with serotonergic drugs.

Children: accidental ingestion of a parent's lithium; toxicity follows the acute pattern.

Prehospital management:

- ABCDE; glucose; seizures per CDST.
- **IV normal saline** to correct dehydration and promote renal lithium excretion (cautious in heart failure).
- **Activated charcoal does not bind lithium — do not give it** (CDST exclusion) (HPCSA PBEC, 2025, p. 54).
- Ask about NSAIDs, ACE inhibitors and diuretics, and about recent illness, vomiting or diarrhoea.
- Transport; Poisons Information Helpline; bring all medications.

Hospital (Decker et al., 2015):

- Serial serum lithium levels (repeated every four to six hours because of delayed absorption, especially with sustained-release preparations; Hüser et al., 2025), electrolytes, renal function, ECG.
- **Whole bowel irrigation** for large ingestions of sustained-release lithium (Kaswa, 2024; Hüser et al., 2025).
- **Haemodialysis** (lithium is small, water-soluble, not protein-bound and therefore highly dialysable): EXTRIP recommends haemodialysis in severe poisoning — for example lithium >4.0 mmol/L with impaired kidney function, or the presence of a

decreased level of consciousness, seizures or life-threatening dysrhythmias irrespective of the level — and suggests it when the level is >5.0 mmol/L, when there is significant confusion, or when the expected time to reach <1.0 mmol/L with optimal management is more than 36 hours.

- Stop lithium and interacting drugs.

60.28 Psychiatric medications

TCAs (60.18) and lithium (60.27) are covered separately. This section covers **antipsychotics**, **SSRIs/SNRIs and other antidepressants**, **anticonvulsant mood stabilisers**, and the important drug-induced syndromes (**serotonin syndrome** and **neuroleptic malignant syndrome**).

Antipsychotics

Agents: first-generation (haloperidol, chlorpromazine, flupenthixol, zuclopenthixol) and second-generation (olanzapine, risperidone, quetiapine, clozapine, aripiprazole) antipsychotics.

Pathophysiology and presentation in overdose:

- **Dopamine D2 blockade** → extrapyramidal effects (acute dystonia, akathisia, parkinsonism), and rarely **neuroleptic malignant syndrome**.
- **Alpha-1 blockade** → vasodilatation, **hypotension**, reflex tachycardia (especially quetiapine, chlorpromazine, clozapine). Hüser et al. (2025) note that hypotension due to alpha-antagonism may respond well to phenylephrine or vasopressin.
- **Antihistamine (H1) effects → sedation, CNS depression, coma** (quetiapine, olanzapine, clozapine).
- **Antimuscarinic effects** → anticholinergic toxidrome (olanzapine, quetiapine, clozapine, chlorpromazine).
- **Potassium channel blockade → QT prolongation and torsade de pointes** (haloperidol, droperidol, quetiapine); some cause sodium channel blockade (thioridazine, chlorpromazine).
- **Seizures** (clozapine, chlorpromazine, olanzapine).
- **Hypothermia or hyperthermia**.
- **Clozapine**: hypersalivation, seizures, myocarditis, agranulocytosis.

Children: a single tablet of some antipsychotics can cause sedation, dystonia or hypotension in a toddler.

Management: ABCDE; airway protection for CNS depression; IV fluids for hypotension (vasopressors if refractory); continuous ECG (QT); benzodiazepines for seizures; activated charcoal only if CDST criteria are met; transport. **Acute dystonia** is treated in hospital (or by protocol) with an anticholinergic (biperiden or benztropine) or diphenhydramine; **torsade de pointes** with IV magnesium sulphate and correction of hypokalaemia (hospital/ACLS).

SSRIs, SNRIs and other antidepressants

- **SSRIs** (fluoxetine, sertraline, citalopram, escitalopram, paroxetine): relatively safe in isolated overdose — nausea, vomiting, drowsiness, tremor, tachycardia; **seizures** and

QT prolongation with **citalopram and escitalopram**; risk of **serotonin syndrome**, especially with combinations.

- **SNRIs** (venlafaxine, duloxetine): more toxic than SSRIs — seizures, tachycardia, hypotension, **QRS widening** and arrhythmias with large venlafaxine overdoses, and serotonin syndrome.
- **Mirtazapine**: sedation, usually benign.
- **Trazodone**: sedation, hypotension (alpha-blockade), priapism.
- **Bupropion**: **seizures** (common, may be delayed with sustained-release forms), tachycardia, QRS widening and cardiotoxicity (Hüser et al., 2025).
- **MAOIs** (rarely used): delayed, severe sympathomimetic and serotonergic toxicity, hypertensive crises with tyramine-containing foods or interacting drugs.

Serotonin syndrome (serotonin toxicity)

- **Cause**: excess serotonergic activity in the CNS, usually from combinations of serotonergic drugs (SSRIs, SNRIs, MAOIs, TCAs, tramadol, pethidine, fentanyl, MDMA, linezolid, lithium, St John's wort, ondansetron in combinations) or overdose (Boyer and Shannon, 2005).
- **Onset**: rapid — usually within hours (often within six hours) of a dose change or overdose.
- **Triad**: (1) **altered mental status** — agitation, anxiety, confusion; (2) **autonomic instability** — tachycardia, hypertension, hyperthermia, diaphoresis, mydriasis, diarrhoea; (3) **neuromuscular hyperactivity** — **clonus** (spontaneous, inducible or ocular), **hyperreflexia**, tremor, rigidity (especially of the lower limbs). Diagnosed with the **Hunter criteria** (Dunkley et al., 2003).
- **Management**: stop serotonergic drugs; ABCDE; **benzodiazepines** for agitation, tremor and hyperthermia (CDST Aggression tool); **active cooling** for hyperthermia (CDST: ≥38 °C — remove clothing, water spray, fan or air-conditioning); IV fluids; transport. Hospital: **cyproheptadine** (a 5-HT2A antagonist, via nasogastric tube) or IV olanzapine (Hüser et al., 2025); chlorpromazine for severe serotonin toxicity with hyperthermia in addition to benzodiazepines (RCEM, 2025); intubation and paralysis for temperature above about 41 °C. **Avoid physical restraint** (worsens hyperthermia and lactic acidosis) and avoid fentanyl (RCEM, 2025).

Neuroleptic malignant syndrome (NMS)

- **Cause**: dopamine D2 blockade (antipsychotics, metoclopramide) or abrupt withdrawal of dopaminergic drugs (levodopa) — an idiosyncratic reaction, not dose-dependent.
- **Onset**: gradual, over days (usually within the first two weeks of starting or increasing the drug).
- **Features**: **"lead-pipe" muscle rigidity**, **hyperthermia**, **altered mental status** (confusion, stupor, mutism), **autonomic instability** (labile BP, tachycardia, diaphoresis, incontinence), **elevated creatine kinase** (rhabdomyolysis), leucocytosis; **bradyreflexia** and absence of clonus distinguish it from serotonin syndrome.
- **Management**: stop the causative drug; ABCDE; **active cooling**; IV fluids (rhabdomyolysis, AKI); benzodiazepines for agitation and rigidity; transport. Hospital:

bromocriptine (dopamine agonist), amantadine, **dantrolene** (muscle relaxant) (Hüser et al., 2025); ICU care; ECT for refractory cases.

Feature	Serotonin syndrome	NMS
Drug	Serotonergic agents	Dopamine antagonists
Onset	Hours	Days
Neuromuscular	**Clonus, hyperreflexia, tremor**	**Lead-pipe rigidity, bradyreflexia**
Pupils	Dilated	Normal
Bowel sounds	Increased, diarrhoea	Normal/decreased
Resolution	Usually within 24 hours	Days to two weeks
Specific treatment	Cyproheptadine	Bromocriptine, dantrolene

Anticonvulsant mood stabilisers

- **Valproate**: CNS depression and coma, respiratory depression, hypotension, metabolic acidosis, **hyperammonaemia** (encephalopathy, cerebral oedema), hypernatraemia, hepatotoxicity, pancreatitis, bone marrow suppression; hospital treatment includes **L-carnitine** for hyperammonaemia and haemodialysis for severe toxicity.
- **Carbamazepine**: ataxia, **nystagmus**, dysarthria, drowsiness, coma, **seizures** (paradoxically), anticholinergic effects, respiratory depression, hypotension, **sodium channel blockade** (QRS widening, arrhythmias), hyponatraemia; delayed and erratic absorption (forms bezoars); **MDAC** is indicated (Hüser et al., 2025); haemodialysis for severe toxicity; sodium bicarbonate for QRS widening.
- **Lamotrigine**: seizures, QRS widening in large overdoses.

60.29 Ethanol

Sources: alcoholic beverages (beer, wine, spirits, traditional sorghum beer “umqombothi”, illicit home-brewed alcohol), and non-beverage ethanol in **hand sanitisers, mouthwashes, perfumes, colognes and cooking extracts** (a particular risk for children).

Pathophysiology: ethanol is rapidly absorbed (mainly from the small intestine), crosses the blood–brain barrier, and enhances **GABA-A** activity and inhibits **NMDA glutamate** receptors, causing CNS depression. It is metabolised mainly in the liver by **alcohol dehydrogenase** (to acetaldehyde) and aldehyde dehydrogenase (to acetate), at roughly constant (zero-order) rates; this metabolism generates NADH, which **inhibits gluconeogenesis**, causing **hypoglycaemia** (especially in children and in malnourished or fasting adults), and contributes to lactic acidosis and **alcoholic ketoacidosis**. Ethanol also causes vasodilatation (heat loss — **hypothermia**), diuresis (inhibition of ADH), and gastric irritation.

Clinical presentation (correlates approximately with blood alcohol concentration, modified by tolerance):

- mild: euphoria, disinhibition, impaired judgement and coordination;
- moderate: slurred speech, ataxia, nystagmus, labile mood, aggression, nausea and vomiting;

- severe: stupor, coma, **respiratory depression**, **aspiration** of vomitus, hypotension, **hypothermia**, **hypoglycaemia**, and death.

Associated problems include **trauma** (head injury — "never assume a drunk patient is just drunk"), violence, road traffic collisions, drowning, co-ingestion of other drugs, alcoholic ketoacidosis, Wernicke's encephalopathy, withdrawal and delirium tremens (52.8.3).

Children: are particularly prone to **hypoglycaemia, hypothermia, seizures and coma** after relatively small amounts (for example drinking from adults' glasses, or ingesting hand sanitiser, mouthwash or perfume).

Toxic alcohols: methanol and ethylene glycol (60.19) and isopropanol (rubbing alcohol — CNS depression, haemorrhagic gastritis, ketosis without acidosis) must be considered when a patient appears intoxicated without the smell of ethanol, or has an unexplained metabolic acidosis.

Prehospital management:

- ABCDE: **airway protection** (lateral/recovery position, suction; airway tool if unable to maintain the airway); oxygen; BVM if hypoventilating.
- **Glucose** — treat hypoglycaemia (CDST); **thiamine 100 mg IV/IM** in chronic alcohol users or malnourished patients (HPCSA PBEC, 2025, pp. 42, 73).
- **Temperature** — prevent and treat hypothermia.
- **Full secondary survey for injuries**, especially head injury; GCS that does not improve or deteriorates must be attributed to head injury or other causes until proven otherwise.
- **No activated charcoal** (not adsorbed) (HPCSA PBEC, 2025, p. 54).
- Agitation and aggression: CDST Aggression tool (with caution — benzodiazepines add to respiratory depression).
- Withdrawal seizures: CDST Seizures tool.
- Capacity: an intoxicated patient may lack capacity to refuse care; do not leave an intoxicated patient who cannot protect their airway or who is at risk.
- Transport (children always).

60.30 Hydrocarbons

Definition and classification: hydrocarbons are organic compounds composed mainly of carbon and hydrogen, derived from petroleum, coal, wood or plants. They are classified as:

1. **Aliphatic (straight- or branched-chain)**: paraffin (kerosene), petrol, diesel, lighter fluid, mineral spirits, mineral oil, lubricating oils, butane and propane.
2. **Aromatic (benzene ring)**: **benzene** (bone marrow toxicity, leukaemia), **toluene** (glue, paint thinners — CNS toxicity, **renal tubular acidosis with hypokalaemia**), xylene.
3. **Halogenated**: carbon tetrachloride (hepatotoxicity and nephrotoxicity), trichloroethane and trichloroethylene (cardiac sensitisation), **methylene chloride** (paint strippers — metabolised to **carbon monoxide**).
4. **Terpenes (plant-derived)**: turpentine, pine oil, eucalyptus oil, camphor (seizures).

Pathophysiology — four main toxic effects (Tintinalli et al., 2020):

1. **Pulmonary (aspiration pneumonitis)** — the most common and important, determined by low **viscosity**, low **surface tension** and high **volatility** (60.16).
2. **CNS** — intoxication, depression, seizures and coma, especially with inhaled volatile, aromatic and halogenated hydrocarbons; chronic encephalopathy with abuse.
3. **Cardiac** — **myocardial sensitisation to catecholamines** with halogenated and aromatic hydrocarbons → ventricular arrhythmias and **sudden sniffing death**.
4. **Other organ toxicity** — hepatotoxicity (carbon tetrachloride), nephrotoxicity, haematological toxicity (benzene), peripheral neuropathy (n-hexane), carbon monoxide production (methylene chloride), and skin and eye burns.

Clinical presentation: as in 60.16 — coughing, choking, tachypnoea, wheeze, crackles and hypoxia after ingestion and aspiration (with fever and possibly delayed onset over six hours); CNS depression; arrhythmias; and characteristic odours.

Children: paraffin ingestion is common in South African children (Balme et al., 2012), with a high risk of aspiration.

Prehospital management: as in 60.16 — decontamination (remove clothing, wash skin), oxygen, airway support, bronchodilators for bronchospasm, **no induced vomiting and no activated charcoal** (HPCSA PBEC, 2025, p. 54), avoidance of exertion and adrenaline in inhalant abusers (except cardiac arrest), continuous ECG, seizure management, and transport; observe asymptomatic children who coughed or choked for at least six hours in hospital. For methylene chloride, treat as carbon monoxide poisoning (100% oxygen).

60.31 NSAID

Agents: ibuprofen (the most commonly ingested), diclofenac, naproxen, **mefenamic acid**, indomethacin, ketoprofen, and others; aspirin is covered under salicylates (60.17). In South Africa, ibuprofen, diclofenac and mefenamic acid are widely available.

Pathophysiology: NSAIDs inhibit **cyclooxygenase (COX-1 and COX-2)**, reducing prostaglandin synthesis. In overdose this causes **GI irritation and bleeding** (loss of protective prostaglandins), **renal effects** (reduced renal blood flow — acute kidney injury, especially with dehydration, ACE inhibitors and diuretics; hyperkalaemia), and, with very large ingestions, **CNS depression**, **seizures** (particularly with **mefenamic acid**), and **metabolic acidosis**; NSAIDs also reduce lithium clearance (60.27).

Clinical presentation: most ibuprofen overdoses are mild or asymptomatic — nausea, vomiting, epigastric pain, drowsiness, headache, tinnitus. **Ibuprofen ingestions below about 100 mg/kg are usually asymptomatic**, whereas **more than about 400 mg/kg** can cause serious toxicity — coma, seizures, metabolic acidosis, hypotension, acute kidney injury, respiratory depression and, rarely, death (Tintinalli et al., 2020). Mefenamic acid overdose frequently causes seizures. GI bleeding and AKI may occur with large ingestions or in vulnerable people (older adults, dehydration).

Children: accidental ingestion of ibuprofen syrup is common and usually benign; calculate the mg/kg dose.

Prehospital management:

- ABCDE; glucose.
- **Seizures** (mefenamic acid, massive ibuprofen): CDST Seizures tool.
- **Activated charcoal** if fully conscious and within one hour of a potentially toxic ingestion (CDST).
- IV fluids if dehydrated or hypotensive.
- Calculate mg/kg; contact the Poisons Information Helpline; per CDST, a non-toxic dose in a well patient may be managed at home with advice, but deliberate ingestions require transport and psychosocial assessment (HPCSA PBEC, 2025, p. 55).

Hospital: supportive care; renal function and electrolytes; acid–base monitoring; proton pump inhibitors for GI symptoms; there is no specific antidote.

60.32 Chemical warfare agents

Chemical warfare agents (CWAs) are toxic chemicals used deliberately to kill, injure or incapacitate. They may be encountered in terrorism, military conflict, or as industrial chemicals with similar effects (chlorine, ammonia, cyanide, organophosphates). Riot-control agents (pepper spray) are used routinely by SAPS and private security, and are a common prehospital exposure.

Principles of response (CBRN incidents)

1. **Recognise** a chemical incident: multiple casualties with similar symptoms, unexplained odours or clouds, dead animals, reports of explosions with unusual symptoms.
2. **Scene safety**: approach from **upwind and uphill**; do not enter the **hot zone** (contaminated area) without appropriate PPE and training; establish **hot, warm (decontamination) and cold (treatment) zones**; inform control and request HAZMAT/fire services and SAPS.
3. **Decontamination** before treatment in the cold zone (except for immediate life-saving interventions by appropriately protected personnel): **remove clothing** (which removes the majority of contamination), then wash with copious water (soap and water) (HPCSA PBEC, 2025, p. 54; Henretig, Kirk and McKay, 2019).
4. **Mass-casualty triage** (for example START/SALT).
5. **Antidotes** where available, and supportive ABCDE care.
6. **Notify** the receiving hospitals early to avoid secondary contamination of emergency departments.

Classes of chemical warfare agents

Class	Examples	Mechanism	Features	Management
Nerve agents	Tabun (GA), sarin (GB), soman (GD), VX, Novichok agents	Organophosphates — irreversible acetylcholinesterase inhibition	**Cholinergic crisis**: miosis, rhinorrhoea, bronchorrhoea, bronchospasm,	Decontamination; **atropine** (titrated until secretions dry — as in 60.8,

Class	Examples	Mechanism	Features	Management
		(very rapid ageing with soman)	salivation, sweating, vomiting, diarrhoea, fasciculations, weakness, seizures, apnoea, death within minutes	CDST ^atropine 2 mg IV^ doubling every 3 min); **oximes** (pralidoxime) early, before ageing; **benzodiazepines** (diazepam/midazolam) for seizures; ventilation (avoid succinylcholine); military autoinjectors combine atropine and pralidoxime
Vesicants (blister agents)	Sulphur mustard, nitrogen mustard, lewisite, phosgene oxime	Alkylating agents (mustard) — DNA damage; arsenical (lewisite)	Mustard: **delayed** (2–24 h) erythema and blistering, eye injury (conjunctivitis, blindness), airway injury, bone marrow suppression; lewisite: **immediate** pain	Immediate decontamination (effects are delayed but damage begins within minutes); eye irrigation; burn-type care; airway management; **dimercaprol (BAL)** for lewisite (hospital)
Pulmonary (choking) agents	**Chlorine**, **phosgene**, diphosgene, chloropicrin, ammonia	Airway and alveolar injury (60.10); phosgene (smell of freshly cut hay/grass) causes **delayed non-cardiogenic**	Cough, dyspnoea, chest tightness, wheeze, then pulmonary oedema	Remove from exposure; oxygen; **rest** (exertion worsens pulmonary oedema with phosgene); bronchodilators;

Class	Examples	Mechanism	Features	Management
		pulmonary oedema (up to 24–48 h)		observe for 24–48 h
Blood agents (systemic asphyxiants)	**Hydrogen cyanide**, cyanogen chloride, arsine	Inhibition of cytochrome oxidase (60.11); arsine causes haemolysis	Rapid dyspnoea, seizures, coma, cardiovascular collapse, lactic acidosis	100% oxygen; hydroxocobalamin (hospital); supportive care
Incapacitating agents	BZ (3-quinuclidinyl benzilate), fentanyl derivatives (for example carfentanil used in the 2002 Moscow theatre siege)	Anticholinergic (BZ); opioid agonists	Anticholinergic delirium (BZ); opioid toxidrome	Supportive care; physostigmine (hospital) for BZ; **naloxone** (CDST) for opioid agents
Riot-control agents (lacrimators)	**Oleoresin capsicum (OC, "pepper spray")**, CS gas ("tear gas"), CN	TRPV1/TRPA1 activation — intense sensory irritation	Burning eyes, blepharospasm, lacrimation, rhinorrhoea, coughing, chest tightness, skin burning; usually self-limiting within 30–60 minutes; bronchospasm in asthmatics; corneal abrasion	Remove to fresh air; remove contaminated clothing; **irrigate eyes with water or saline**; do not rub; bronchodilators for bronchospasm; reassure; assess for injuries from restraint (relevant to ABD — RCEM, 2025, notes that patients may have been exposed to PAVA spray or Tasers before arrival)

High-yield points — 60.17 to 60.32

- **Salicylates**: tinnitus, hyperventilation, mixed respiratory alkalosis + high anion gap metabolic acidosis, hyperthermia; never hypoventilate after intubation; urinary alkalinisation; haemodialysis.
- **TCAs**: rapid deterioration; QRS >100 ms → seizures; >160 ms → VT; CDST sodium bicarbonate 8.4% 100 mL (child 2 mL/kg), hyperventilate if intubated ($ETCO_2$ 20–25 mmHg); no flumazenil or physostigmine.
- **Ethylene glycol / methanol**: osmolal then anion gap; calcium oxalate → AKI (EG); blindness (methanol); no charcoal; fomepizole/ethanol, haemodialysis.
- **Colchicine**: GI phase → multi-organ failure → alopecia; flame lily (*Gloriosa superba*); no antidote; transport all.
- **Paracetamol**: >150 mg/kg; NAPQI; asymptomatic first 24 h; level at ≥4 h; NAC 150/50/100 mg/kg; transport all.
- **Theophylline**: intractable vomiting, tachyarrhythmias, hypokalaemia, refractory seizures; MDAC, haemodialysis.
- **HF acid**: pain out of proportion; hypocalcaemia, hypomagnesaemia, hyperkalaemia, VF; irrigate, calcium gluconate gel, IV calcium.
- **Iron**: >60 mg/kg serious; five stages including a deceptive latent phase; no charcoal; WBI; deferoxamine.
- **Oral hypoglycaemics**: sulphonylureas — prolonged, delayed and recurrent hypoglycaemia; CDST dextrose 50% 20 mL / 10% 3 mL/kg / glucagon; transport all; octreotide in hospital.
- **Lithium**: narrow therapeutic index; chronic toxicity — neurotoxicity; no charcoal; saline; haemodialysis (EXTRIP).
- **Psychiatric drugs**: antipsychotics — sedation, hypotension, QT; serotonin syndrome (clonus, hours) vs NMS (rigidity, days).
- **Ethanol**: airway, glucose, thiamine, temperature, look for head injury; children — hypoglycaemia.
- **Hydrocarbons**: aspiration pneumonitis, CNS, cardiac sensitisation, organ toxicity.
- **NSAIDs**: usually benign; >400 mg/kg ibuprofen serious; mefenamic acid seizures.
- **Chemical warfare agents**: upwind, zones, undress and decontaminate; nerve agents → atropine, oximes, benzodiazepines; phosgene → rest, delayed oedema.

Outcome 63. Discuss poisonous plants and mushrooms indigenous to South Africa commonly involved in serious poisoning incidents for both adults and children

Context

South Africa has one of the richest floras in the world, and many plants are used in traditional medicine. Plant poisonings (biological toxins) account for part of the 12.6% of poisoning exposures attributed to biological toxins in the Tygerberg Poison Information Centre data (Veale, Wium and Müller, 2013). Plant poisoning occurs through accidental ingestion by children (attractive berries, seeds and flowers), deliberate recreational use (for example *Datura* for hallucinogenic effects), misidentification of edible plants and mushrooms, contamination of grain, and — importantly — the use of plants in **traditional medicines** (63.2) (Van Wyk, Van Heerden and Van Oudtshoorn, 2002).

63.1 Discuss common South African plants and mushrooms implicated in poisoning incidents

Important poisonous plants

Plant (common names)	Toxin and mechanism	Clinical features	Management
Datura stramonium and *D. ferox* (thorn apple, "malpitte", "olieboom", jimsonweed) — a widespread weed in South Africa; seeds, leaves and flowers	**Tropane alkaloids** (atropine/hyoscyamine and scopolamine/hyoscine) — **antimuscarinic**	**Anticholinergic toxidrome**: delirium, agitation, visual hallucinations, mydriasis and blurred vision, dry flushed hot skin, hyperthermia, tachycardia, urinary retention, absent bowel sounds; seizures and coma in severe cases; prolonged duration (up to 24–48 hours); used recreationally by adolescents and in traditional medicine; fatal poisoning documented in South Africa (Steenkamp et al., 2004)	ABCDE; calm environment; CDST Aggression tool (benzodiazepines) for agitation; cooling (≥38 °C); urinary catheter in hospital; activated charcoal if fully conscious within 1 h (and later on expert advice because anticholinergics delay gastric emptying — Hüser et al., 2025); hospital: physostigmine for severe agitated delirium unresponsive to benzodiazepines, with ECG monitoring (contraindicated with QRS widening)

Plant (common names)	Toxin and mechanism	Clinical features	Management
			(Hüser et al., 2025)
Oleander (*Nerium oleander* — pink/white oleander; *Thevetia peruviana* — yellow oleander) — common garden shrubs	**Cardiac glycosides** (oleandrin, thevetin) — Na^+/K^+-ATPase inhibition (like digoxin)	Nausea, vomiting, abdominal pain, diarrhoea; bradycardia, AV block, arrhythmias; hyperkalaemia; cardiac arrest	As for digoxin (60.7): ECG, RCSA bradycardia algorithm, transport; hospital — digoxin-specific Fab fragments, MDAC
Slangkop and related bulbs (*Drimia / Urginea* species — "slangkop", "bushman's poison bulb") and ***Bowiea volubilis*** (climbing onion — "igibisila"), and **plakkie / "krimpsiekte" plants** (*Cotyledon*, *Kalanchoe*, *Tylecodon*)	**Cardiac glycosides** (bufadienolides)	Vomiting, abdominal pain, bradycardia, arrhythmias, hyperkalaemia, cardiovascular collapse; important causes of livestock and human poisoning, including through traditional remedies	As for cardiac glycosides
Callilepis laureola (Impila, "ox-eye daisy")	**Atractyloside**-type compounds — inhibit mitochondrial ADP/ATP transport	**Hepatic necrosis and renal tubular necrosis** (hepatorenal failure), hypoglycaemia, convulsions, coma, death — particularly in children given traditional remedies, including enemas (Popat et al., 2001)	Supportive care, glucose, fluids; transport; hospital — renal and hepatic support
Ricinus communis (castor oil plant) — seeds	**Ricin** — inhibits ribosomal protein synthesis (seeds must be chewed to release ricin)	Delayed (hours) severe vomiting, bloody diarrhoea, dehydration, hypovolaemic shock, hepatic and renal failure	Supportive; fluids; transport
Boophone disticha ("gifbol", "bushman poison bulb",	**Isoquinoline (Amaryllidaceae) alkaloids**	Hallucinations, agitation, confusion, coma, respiratory	Supportive; airway; transport

Plant (common names)	Toxin and mechanism	Clinical features	Management
"incotha")	(buphanidrine, distichine)	depression, bradycardia or tachycardia, hypotension; used in traditional medicine	
Senecio* species** (ragwort, "geelopslag") and ***Crotalaria — contaminating wheat or used as herbal teas	**Pyrrolizidine alkaloids** — hepatotoxic metabolites	**Hepatic veno-occlusive disease** (sinusoidal obstruction syndrome): abdominal pain, ascites, hepatomegaly, liver failure ("bread poisoning" from contaminated flour; "Senecio poisoning")	Supportive; transport
Gloriosa superba (flame lily) — tubers	**Colchicine**	Severe GI phase, then multi-organ failure, bone marrow suppression (60.20)	As for colchicine
Dieffenbachia (dumb cane), *Zantedeschia* (arum lily), *Philodendron*, *Alocasia*/*Colocasia* (elephant's ear)	**Insoluble calcium oxalate crystals** (raphides)	Immediate burning pain and swelling of the lips, tongue and mouth, drooling, dysphagia; rarely airway obstruction	Rinse mouth, cold fluids; airway assessment; usually self-limiting
Nicotiana glauca (wild tobacco, "tree tobacco")	**Anabasine** (nicotinic agonist)	Vomiting, salivation, muscle weakness, fasciculations, paralysis, respiratory failure — fatalities from mistaken use as spinach	Airway and ventilation; supportive
Euphorbia species (milkweeds, "melkbos")	Irritant latex (diterpene esters)	Skin and eye burns (severe keratoconjunctivitis), GI irritation if ingested	Irrigate eyes and skin; supportive
Lantana camara (tickberry) — unripe berries	Triterpenes	GI symptoms, rarely CNS depression in children	Supportive

Plant (common names)	Toxin and mechanism	Clinical features	Management
Apricot, peach and plum kernels, bitter almonds, cassava	**Cyanogenic glycosides** (amygdalin) — release cyanide	Delayed cyanide toxicity (60.11)	Oxygen; hospital cyanide antidotes

Poisonous mushrooms

Mushroom poisoning usually results from misidentification during foraging, accidental ingestion by children, or deliberate use of hallucinogenic mushrooms.

Mushroom	Toxin	Features	Management
Amanita phalloides (**death cap**) — an introduced species found in the Western Cape, growing under oak and pine trees; responsible for most mushroom deaths	**Amatoxins** (α-amanitin) — inhibit **RNA polymerase II**, stopping protein synthesis in hepatocytes and renal tubular cells; not destroyed by cooking	**Delayed onset 6–24 hours** after ingestion (a key diagnostic clue): severe vomiting, abdominal pain and **profuse watery diarrhoea** → dehydration; then a **"false recovery"** (24–72 hours); then **fulminant hepatic failure** (days 2–5), renal failure, coagulopathy, encephalopathy and death	Transport; aggressive IV fluids; glucose; hospital — **MDAC**, **silibinin** (milk thistle extract) and/or high-dose **penicillin G**, N-acetylcysteine, intensive liver support, **liver transplantation** in severe cases. **Rule**: GI symptoms beginning more than six hours after eating wild mushrooms should be treated as amatoxin poisoning until proven otherwise
Amanita muscaria (fly agaric) and ***A. pantherina*** (panther cap)	**Ibotenic acid and muscimol** (GABA agonist and glutamate agonist)	Within 30 minutes to 2 hours: confusion, agitation, hallucinations, ataxia, alternating drowsiness and excitation, myoclonus, seizures, coma	Supportive; benzodiazepines for agitation and seizures; airway
Chlorophyllum molybdites (false parasol, green-spored lepiota — common	GI irritants	Early onset (within 1–3 hours) of vomiting, abdominal cramps and	Fluids; supportive

Mushroom	Toxin	Features	Management
on lawns) and many other "little brown mushrooms"		diarrhoea; usually self-limiting	
Inocybe and ***Clitocybe*** species	**Muscarine**	Early cholinergic (muscarinic) syndrome: sweating, salivation, lacrimation, bronchorrhoea, bradycardia, miosis, vomiting, diarrhoea	Atropine for significant secretions/bradycardia (ANT/ECP per CDST cholinergic dosing); supportive
Psilocybe species ("magic mushrooms")	Psilocybin/psilocin (5-HT2A agonists)	Hallucinations, anxiety, mydriasis, tachycardia (60.4)	Calm environment; benzodiazepines

Prehospital approach to plant and mushroom poisoning

1. Identify the plant or mushroom: bring a sample (in a paper bag, not plastic), photographs, the remains, and any vomitus.
2. Establish the time of ingestion and onset of symptoms (early versus delayed onset for mushrooms).
3. ABCDE; glucose; toxidrome recognition (anticholinergic — *Datura*; cholinergic — muscarinic mushrooms, *Nicotiana*; cardiac glycoside — bradycardia and arrhythmias).
4. Activated charcoal if fully conscious and within one hour (or later for delayed-onset toxins on expert advice).
5. Contact the Poisons Information Helpline (0861 555 777), which can assist with plant identification.
6. Transport symptomatic patients, all suspected amatoxin, cardiac glycoside, *Callilepis*, colchicine or ricin ingestions, and all children who have ingested unknown plants or mushrooms with any symptoms.

63.2 Discuss the role of traditional medicine in poisoning incidents, with reference to pesticide involvement, hepatotoxicity and cardiovascular collapse

Context

Traditional medicine (often called "muthi" in South Africa) is widely used, alongside or instead of Western medicine, for physical, spiritual and social problems, including protection from harm, cleansing, treating illness, and fertility. Traditional health practitioners (izangoma, izinyanga, amagqirha) are recognised under the **Traditional Health Practitioners Act 22 of 2007**. Most traditional remedies are used safely, but poisoning can result from inherently toxic plants, incorrect dosing, the wrong plant being used (misidentification), adulteration and contamination (including with pesticides, heavy metals and pharmaceuticals), and the route of administration (for example **enemas**, which allow rapid absorption and can cause bowel injury) (Stewart et al., 1999; Kaswa, 2024).

The CDST explicitly directs providers to ask about **"herbal, chemicals, pesticides, traditional medications"** in every poisoning (HPCSA PBEC, 2025, p. 54).

Pesticide involvement

- Pesticides — especially **organophosphates and carbamates** (including illegally decanted **aldicarb**, "Two-step") — are sometimes added to, or used as, traditional remedies, or used for protective rituals, or sold by informal traders alongside traditional products (Rother, 2010).
- Accidental or deliberate ingestion causes a **cholinergic crisis** (60.8), which may not initially be recognised if the history is of "muthi" rather than "poison".
- Pesticides stored in unlabelled containers in households and traditional practitioners' premises contribute to accidental child poisoning (Davies, Hlela and Rother, 2023, cited by Kaswa, 2024).
- **Prehospital implication**: a patient who has taken "muthi" and presents with pinpoint pupils, secretions, bradycardia, sweating, vomiting, diarrhoea and fasciculations must be treated as a **cholinergic poisoning** — PPE, decontamination, ^{atropine} per CDST, airway management.

Hepatotoxicity

- ***Callilepis laureola*** (Impila) is the best-known South African traditional remedy causing **acute hepatic and renal tubular necrosis**, with high mortality, particularly in children, often administered as an enema (Popat et al., 2001).
- **Pyrrolizidine alkaloids** (from *Senecio*, *Crotalaria* and other plants used in herbal teas and remedies) cause **hepatic veno-occlusive disease**.
- **Atractyloside**-containing plants (for example *Callilepis*) cause hepatorenal failure and hypoglycaemia.
- Traditional remedies may interact with **antiretrovirals** and **anti-TB drugs**, increasing the risk of drug-induced liver injury, or may contain hepatotoxic pharmaceuticals.
- **Clinical features**: vomiting, abdominal pain, jaundice, hypoglycaemia, altered consciousness, coagulopathy, renal failure.
- **Prehospital implication**: check glucose (hypoglycaemia is common), treat dehydration, and transport early; liver failure may develop over days.

Cardiovascular collapse

- **Cardiac glycoside-containing plants** used in traditional medicine — *Drimia/Urginea* (slangkop), ***Bowiea volubilis*** (climbing onion), *Cotyledon*, and oleander — cause bradyarrhythmias, AV block, ventricular arrhythmias, hyperkalaemia and cardiovascular collapse.
- **Anticholinergic plants** (*Datura*) cause tachyarrhythmias, hyperthermia and collapse.
- ***Boophone disticha*** causes CNS depression, bradycardia and hypotension.
- **Enemas** and purgatives cause profuse diarrhoea, **dehydration, hypovolaemic shock and electrolyte disturbances** (hypokalaemia, hyperkalaemia), and can cause **bowel perforation** and chemical colitis; caustic or oxidising substances (for example **potassium dichromate**, sometimes found in traditional remedies) cause **acute kidney injury**, haemolysis and cardiovascular collapse (Stewart et al., 1999).

- **Prehospital implication**: ECG monitoring, IV fluids, RCSA bradycardia algorithm, CDST management of shock; transport urgently.

Approach to suspected traditional medicine poisoning

1. **Respectful, non-judgemental inquiry**: patients and families may be reluctant to disclose traditional medicine use for fear of criticism. Ask specifically: "Have you (or your child) taken any traditional or herbal medicines, or had an enema, in the last few days?"
2. **Obtain the remedy** (the container, plant material or powder) for identification, and the name and contact of the traditional health practitioner if possible.
3. **Toxidrome-based management** (cholinergic, anticholinergic, cardiac glycoside, hepatotoxic, renal).
4. **Glucose** in every case.
5. **Contact the Poisons Information Helpline**.
6. **Transport** — toxicity may be delayed.
7. **Children**: consider whether the child is in need of care and protection only where there is wilful neglect or abuse; most families act out of love and cultural belief, and education is the appropriate response.

High-yield points — Outcome 63

- *Datura*: anticholinergic toxidrome ("hot, dry, red, blind, mad, full"); benzodiazepines; physostigmine in hospital.
- Cardiac glycoside plants (oleander, *Drimia*, *Bowiea*, *Cotyledon*): bradycardia, AV block, hyperkalaemia; digoxin Fab in hospital.
- *Callilepis laureola* (Impila): hepatorenal failure and hypoglycaemia, especially in children.
- *Amanita phalloides*: GI symptoms >6 h after ingestion, false recovery, liver failure.
- Traditional medicine: pesticides (cholinergic crisis), hepatotoxicity (Impila, pyrrolizidine alkaloids), cardiovascular collapse (glycosides, enemas, dichromate).

Outcome 64. Discuss envenomation by snakes, spiders, scorpions and poisonous marine animals

64.1 Discuss snake bite in the Southern African and global context

Global context

The WHO estimates that each year **1.8–2.7 million people** are envenomed by snakes, with **81,000–138,000 deaths** and about **400,000** people left with permanent disabilities (amputations, contractures, blindness, chronic kidney disease); snakebite envenoming was added to the WHO list of **neglected tropical diseases** in 2017, and the WHO strategy (2019) aims to halve deaths and disabilities by 2030 (WHO, 2019; Gutiérrez et al., 2017). The burden falls mainly on rural populations in sub-Saharan Africa, South and Southeast Asia and Latin America — agricultural workers, herders and children — with limited access to antivenom.

Southern African context

South Africa has about 170 snake species and subspecies, of which fewer than 20 are considered potentially deadly to humans; most bites occur in rural and peri-urban areas, at night or in summer, on the lower limbs, and the **puff adder** (cytotoxic) and **Mozambique spitting cobra** (cytotoxic) are responsible for a large proportion of significant bites, while **mambas** and **non-spitting cobras** cause the most deaths (Müller et al., 2012). The majority of bites are by non-venomous snakes or are "dry bites" (no venom injected), but every bite must be assessed.

Classification of South African snake venoms by clinical syndrome (Müller et al., 2012)

Syndrome	Snakes	Pathophysiology	Clinical features
Cytotoxic (progressive painful swelling)	**Puff adder** (*Bitis arietans*), Gaboon adder (*B. gabonica*), **Mozambique spitting cobra** (*Naja mossambica*), black-necked spitting cobra, rinkhals (mixed), night adders, stiletto (burrowing) snakes	Proteolytic enzymes, phospholipases and metalloproteinases destroy tissue and blood vessels; fluid shifts into the limb	Immediate severe pain; **progressive swelling** (which may involve the whole limb within hours), blistering, bruising, **tissue necrosis** (especially spitting cobras — often with relatively little pain); hypovolaemic shock from fluid loss into the limb (puff adder); compartment-like syndrome (true compartment syndrome is uncommon); mild coagulopathy

Syndrome	Snakes	Pathophysiology	Clinical features
			(Gaboon adder)
Neurotoxic (progressive weakness)	**Black mamba** (*Dendroaspis polylepis*), **green mamba** (*D. angusticeps*), Jameson's mamba; **non-spitting cobras** — **Cape cobra** (*Naja nivea*), **snouted cobra** (*N. annulifera*), forest cobra; rinkhals (mixed); **berg adder** and other small adders (atypical neurotoxicity)	**Postsynaptic** (α-neurotoxins — cobras, block acetylcholine receptors) and **presynaptic/dendrotoxins and fasciculins** (mambas — increase acetylcholine release, then fatigue) causing neuromuscular blockade	Often minimal local signs (cobras may cause some swelling); paraesthesia; sweating, salivation, nausea; **ptosis** (an early sign), diplopia, blurred vision; **bulbar paralysis** — dysphagia, dysarthria, drooling, inability to protrude the tongue; descending flaccid paralysis; **respiratory muscle paralysis and death** — may occur within 30 minutes to a few hours with mamba bites; mamba bites may cause fasciculations; **berg adder** causes ptosis, ophthalmoplegia, loss of taste and smell, and occasionally respiratory failure, with local swelling
Haemotoxic (bleeding)	**Boomslang** (*Dispholidus typus*), **vine/twig snake** (*Thelotornis* species)	Procoagulant venom activates prothrombin → consumption coagulopathy (**venom-induced consumption coagulopathy**, VICC) → incoagulable blood	**Delayed onset** (often hours to more than 24 hours) — a minor bite may seem harmless initially; then bleeding from the bite site, gums, nose, old wounds, venepuncture sites; haematuria, haematemesis, intracranial

Syndrome	Snakes	Pathophysiology	Clinical features
			haemorrhage; renal failure; boomslang bites are uncommon (back-fanged, shy snakes, bites mostly in snake handlers)
Venom in the eyes (spitting cobras, rinkhals)	—	Direct irritation	Intense pain, blepharospasm, conjunctivitis, corneal erosions; risk of blindness if untreated

Prehospital management of snakebite (CDST and Southern African guidance)

First aid and what to do:

1. **Scene safety**: move away from the snake; do not try to catch or kill it (a photograph from a safe distance is helpful; a dead snake can still envenomate by reflex).
2. **Reassure** the patient and **keep them still** (movement increases lymphatic spread of venom); lie the patient down.
3. **Primary survey (ABCDE)**: airway, breathing (assess for respiratory muscle weakness — ability to count aloud, cough, lift head; RR, SpO_2), circulation (shock), disability (ptosis, bulbar signs, GCS).
4. **Expose the bite site and cover it with a sterile dressing; remove any constrictive clothing or jewellery** (rings, watches, bracelets, shoes) before swelling develops (HPCSA PBEC, 2025, p. 54).
5. **Immobilise the bitten limb** with a splint and keep it at about heart level.
6. **Pressure immobilisation bandage (PIB)**: applied firmly (like a sprained-ankle bandage) from the bite site up the entire limb with splinting — **recommended only for bites by known or suspected neurotoxic snakes (mambas, non-spitting cobras)** when transport is prolonged; it slows lymphatic spread of venom. **It must not be used for cytotoxic bites** (puff adder, spitting cobras), because confining the venom worsens local tissue destruction (Müller et al., 2012). Once applied, it should not be removed until the patient is in a facility with antivenom.
7. **Mark the edge of the swelling** with a pen and the time, every 30–60 minutes, to track progression.
8. **IV access** in the **unbitten limb**; IV fluids for shock (CDST shock tool).
9. **Analgesia** (CDST Pain tool — paracetamol, opioids, ketamine for ANT/ECP; **avoid NSAIDs**, which increase bleeding risk; avoid intramuscular injections in patients with possible coagulopathy).
10. **Neurotoxic envenomation with respiratory failure**: **BVM ventilation and, for ECPs, advanced airway management (Airway tool)** — patients who are ventilated

survive neurotoxic paralysis even before antivenom is given; this is the single most important prehospital intervention for mamba and cobra bites.

11. **Venom in the eyes**: **irrigate immediately and copiously** with water or saline (at least 20 minutes; CDST eye irrigation), do not rub, and transport for ophthalmological assessment.
12. **Anaphylaxis** to venom (rare) or to antivenom: RCSA anaphylaxis algorithm (adrenaline 0.5 mg IM 1:1000 in adults).
13. **Transport** rapidly to a facility that can give **antivenom** and manage complications (ICU, ventilation); pre-alert.
14. Contact the Poisons Information Helpline or a recognised snakebite expert for advice.

What NOT to do: do **not** cut, suck or incise the wound; do **not** apply an **arterial tourniquet** (causes ischaemia, worse necrosis and amputations); do **not** apply ice, electric shocks, chemicals or traditional remedies to the wound; do **not** give alcohol; do **not** allow the patient to walk if it can be avoided.

In hospital

- Observation for at least 12–24 hours (longer for boomslang because of delayed coagulopathy), with repeated assessment of swelling, neurological signs and coagulation (20-minute whole blood clotting test, INR, fibrinogen, platelets).
- **Antivenom** when indicated (64.3).
- Supportive care: ventilation, fluids, analgesia, wound care (surgical debridement of necrosis; fasciotomy only if compartment syndrome is confirmed by measured pressures), tetanus prophylaxis, and antibiotics only for established infection.

64.2 Discuss spider bites and scorpion stings in the Southern African and global context

Spider bites

Almost all spiders are venomous, but only a few are medically important. Globally, the most important are the **widow spiders** (*Latrodectus* — latrodectism), the **recluse spiders** (*Loxosceles* — loxoscelism), the Australian **funnel-web spiders** (*Atrax*, *Hadronyche* — severe neurotoxicity) and the Brazilian **wandering spiders** (*Phoneutria*) (Isbister and Fan, 2011). Many lesions blamed on "spider bites" are actually infections (for example community-acquired MRSA) or other conditions.

Southern African medically important spiders:

Spider	Venom type	Clinical features	Management
Button spiders (*Latrodectus* — the **black button spiders**, for example *L. indistinctus*, are more dangerous than the **brown button**	**Neurotoxic** — **α-latrotoxin** causes massive release of neurotransmitters (acetylcholine, noradrenaline) at nerve endings →	The bite may be barely noticed (a pinprick); within 30–60 minutes: **severe pain** spreading from the bite to the regional lymph nodes	ABCDE; reassure; immobilise; analgesia (opioids per CDST Pain tool); benzodiazepines for muscle spasm (within scope);

Spider	Venom type	Clinical features	Management
spider, *L. geometricus*)	**latrodectism**	and then generalised; **painful muscle cramps and rigidity**, especially of the **abdomen** (board-like rigidity mimicking an acute abdomen), back, chest and legs; **sweating** (often localised), restlessness and anxiety, hypertension, tachycardia, nausea and vomiting, ptosis and facial grimacing ("facies latrodectismica"), priapism; rarely respiratory difficulty; children and older people more severely affected	transport; hospital — **button spider antivenom** (South African Vaccine Producers) for severe latrodectism
Violin (recluse) spiders (*Loxosceles*)	**Cytotoxic** — sphingomyelinase D → dermonecrosis	Painless or mildly painful bite; over hours to days, a blistering, necrotic lesion (often with a central dark eschar surrounded by a pale ischaemic halo and erythema — "red, white and blue"), which may ulcerate and take weeks to heal; rarely **systemic loxoscelism** (haemolysis, fever, renal failure)	Wound care, analgesia, tetanus, surgical review; no antivenom available in South Africa
Sac spiders (*Cheiracanthium*)	Cytotoxic	Painful bite with local swelling, blistering and small necrotic lesions,	Wound care; supportive

Spider	Venom type	Clinical features	Management
		often two fang marks	
Six-eyed sand spiders (*Sicarius*)	Highly cytotoxic (rarely bites humans)	Severe necrosis and potentially systemic effects (haemorrhage)	Supportive; hospital
Baboon spiders (tarantulas — Theraphosidae)	Mildly venomous; urticating hairs	Painful bite (mechanical); local reaction; eye irritation from hairs	Analgesia; eye irrigation

Scorpion stings

Global context: scorpion stings are common in North Africa, the Middle East, India, Mexico, and South America, with more than one million stings and several thousand deaths a year, mostly in children; dangerous scorpions belong mainly to the family **Buthidae**.

Southern African context: dangerous scorpions belong to the genus ***Parabuthus*** (family Buthidae) — especially ***P. transvaalicus*** and ***P. granulatus*** — found in the drier western and northern parts of the country; *Parabuthus* species can also spray venom (eye exposure). A useful field rule is that scorpions with **thick tails and small, thin pincers are dangerous** (Buthidae, for example *Parabuthus*), whereas those with **thin tails and large, robust pincers are less venomous** (for example *Opistophthalmus* — burrowing scorpions, which cause painful but not dangerous stings) (Müller et al., 2012).

Pathophysiology: *Parabuthus* venom is **neurotoxic** — it prolongs sodium channel opening in nerve membranes, causing excessive release of neurotransmitters (both sympathetic and parasympathetic) and neuromuscular hyperactivity.

Clinical features:

- **Local**: burning pain at the sting site, often with **paraesthesia and hyperaesthesia** (worsened by tapping — "tap test"); little swelling.
- **Systemic**: spreading paraesthesia, restlessness, involuntary muscle jerks and **muscle pain and cramps**, **dysphagia** and difficulty swallowing saliva (drooling), **hypersalivation**, sweating, ptosis, blurred vision, slurred speech, **urinary retention**, priapism, tachycardia and hypertension (sometimes bradycardia), and, in severe cases, generalised muscle weakness, **respiratory distress and respiratory failure**. Children and older people are most at risk of severe envenomation.

Management:

- ABCDE; reassure; keep still.
- **Local pain**: analgesia (CDST Pain tool); local cold application; local anaesthetic infiltration in hospital.
- **Systemic envenomation**: oxygen; airway management and **ventilatory support** for respiratory failure (the most important life-saving measure); IV fluids; benzodiazepines for agitation and muscle spasm with care (respiratory depression); transport.

- **Hospital**: **SAVP scorpion antivenom** for severe systemic envenomation (Müller et al., 2012).
- Eye exposure: irrigate.

64.3 Discuss the use, role and complications associated with antivenom for snake and spider bites

What antivenom is

Antivenom (antivenin) consists of **immunoglobulin fragments** (in South Africa, purified equine $F(ab')_2$ fragments) produced by immunising horses with venom; it **neutralises circulating (unbound) venom** but cannot reverse tissue damage that has already occurred or neurotoxins already bound to receptors (WHO, 2016). In South Africa, antivenoms are produced by the **South African Vaccine Producers (SAVP)**:

- **SAIMR polyvalent snake antivenom** — raised against the venom of ten species: **puff adder, Gaboon adder, rinkhals, green mamba, black mamba, Jameson's mamba, Cape cobra, forest cobra, snouted cobra and Mozambique spitting cobra**. It is **not effective for berg adder and other small adders**, and its effectiveness against the Mozambique spitting cobra is weaker; it is **not effective for boomslang or vine snake** bites. Cross-protection for other spitting cobras is uncertain.
- **Boomslang monovalent antivenom** — for boomslang envenomation with coagulopathy (there is **no antivenom for the vine/twig snake**).
- **Scorpion antivenom** (*Parabuthus*).
- **Spider (button spider) antivenom** (*Latrodectus*).

Supply of SAVP antivenom has been constrained at times, so it may be held at selected referral centres; the poison centres can advise on availability.

Role and indications

Antivenom is **the only specific treatment** for systemic envenomation, but it is expensive, scarce and can cause severe reactions, so it is given **only when clearly indicated**, not for every bite. Indications for polyvalent antivenom in Southern Africa include (Müller et al., 2012; WHO, 2016):

- **progressive neurotoxicity** — ptosis progressing to bulbar or respiratory weakness (cobra, mamba);
- **progressive, rapidly spreading swelling** (for example swelling extending beyond the wrist or ankle within a few hours of a bite on the hand or foot, or involving more than half of the limb), or swelling of a digit after a bite on that digit — particularly with puff adder and Gaboon adder bites;
- **shock** or cardiovascular instability attributable to envenomation;
- **coagulopathy or spontaneous bleeding** (boomslang — monovalent antivenom);
- **bites by a positively identified dangerous snake** with evolving signs, especially in children.

Dose and administration (hospital-level; follow the SAVP package insert and expert advice): antivenom is given **intravenously** (never intramuscularly into the bitten site), diluted in normal saline over about 30 minutes, or as a slow IV injection, in a setting where anaphylaxis can be treated. The **dose is based on the amount of venom injected, not on body weight, so children receive the same dose as adults**. Typical initial doses of polyvalent antivenom are in the range of several vials (for example 5–10 vials, 50–100 mL), with larger initial and repeated doses (up to about 20 vials) in severe mamba envenomation, and further doses according to clinical response (Müller et al., 2012). **Antivenom should be given as soon as possible** once indicated, but can still be effective many hours (or even days, for coagulopathy) after the bite.

Complications (adverse reactions)

1. **Early (acute) reactions** — occurring within minutes to about three hours of starting the infusion: urticaria, itching, fever, rigors, nausea, vomiting, tachycardia, bronchospasm, angioedema, hypotension and **anaphylaxis** (which can be fatal). These are mostly non-IgE-mediated (anaphylactoid, complement activation) and occur in a substantial proportion of patients receiving equine antivenoms (WHO, 2016). **Management**: stop or slow the infusion; **adrenaline 0.5 mg IM (1:1000)** in adults (0.01 mg/kg in children) per the RCSA anaphylaxis algorithm; IV fluids; antihistamines and corticosteroids as adjuncts; restart cautiously once the reaction has resolved if antivenom is still needed. **Preparation**: adrenaline drawn up, airway equipment ready, IV access, monitoring. Some guidelines suggest prophylactic low-dose subcutaneous adrenaline before antivenom in high-risk settings (WHO, 2016).
2. **Pyrogenic reactions** — fever and rigors (from endotoxin contamination), within one to two hours.
3. **Late reactions — serum sickness** — occurring 5–24 days after antivenom: fever, urticaria, arthralgia, lymphadenopathy, proteinuria, and rarely neuritis or glomerulonephritis; treated with antihistamines and a short course of corticosteroids.

Other limitations: species-specificity (antivenom does not cover all snakes); limited availability, cost and cold-chain requirements; it does not reverse established necrosis; and in neurotoxic envenomation it may not reverse paralysis once neurotoxins are bound — **ventilation remains essential**.

Prehospital relevance: antivenom is a hospital intervention in South Africa and is not part of the CDST; the ECP's role is to recognise envenomation syndromes, provide airway and ventilatory support, manage shock and pain, apply the correct first aid (immobilisation; PIB for neurotoxic bites only), treat anaphylaxis, and transport rapidly to a facility that holds antivenom.

64.4 Discuss poisonous marine animals in the Southern African and global context

Marine envenomations and poisonings are encountered along South Africa's long coastline, particularly in KwaZulu-Natal (warm Indian Ocean waters) and during the summer holiday season.

Envenomation by stinging animals

Animal	Southern African relevance	Venom / mechanism	Clinical features	Management
Bluebottle (*Physalia* — Portuguese man-o'-war / bluebottle)	Very common on South African beaches, especially with onshore winds	Nematocysts (stinging cells) on tentacles inject venom	Immediate intense burning pain, linear red welts; occasionally systemic symptoms (nausea, muscle cramps), rarely anaphylaxis	Remove from water; remove adhered tentacles (with gloves or by rinsing with **seawater** — not fresh water, which may trigger nematocyst discharge); **hot water immersion (about 45 °C) for 20 minutes** is more effective than ice for pain relief (Loten et al., 2006); vinegar is **not** recommended for *Physalia* (it may trigger further nematocyst discharge in this species); analgesia; treat anaphylaxis
Box jellyfish (*Chironex fleckeri*) and Irukandji jellyfish	Mainly northern Australia and the Indo-Pacific (not a significant South African hazard)	Potent cardiotoxic and neurotoxic venom	Severe pain, cardiovascular collapse and death within minutes (box jellyfish); Irukandji syndrome (delayed severe pain, hypertension, pulmonary	**Vinegar** to inactivate undischarged nematocysts (Australian practice), CPR, box jellyfish antivenom (Australia)

Animal	Southern African relevance	Venom / mechanism	Clinical features	Management
			oedema)	
Venomous fish — **stonefish** (*Synanceia verrucosa*, found on KwaZulu-Natal reefs), scorpionfish, lionfish, catfish, and **stingrays**	Stepping on stonefish in shallow reef waters; stingray injuries in shallow sandy water	**Heat-labile** protein venoms injected through spines	**Excruciating pain** out of proportion to the wound, swelling, local necrosis; systemic effects (stonefish): nausea, sweating, hypotension, arrhythmias, pulmonary oedema, collapse; stingray: deep lacerations and puncture wounds with barb fragments, possible penetrating chest or abdominal injury	**Hot water immersion (as hot as tolerated, about 45 °C, test on an unaffected limb first) for 30–90 minutes** (denatures heat-labile venom and relieves pain); analgesia (opioids per CDST); remove visible spines; irrigate; tetanus; X-ray for retained fragments; antibiotics for marine-contaminated wounds in hospital; stonefish antivenom (from Australia) is rarely available in South Africa
Sea urchins	Common on rocky shores	Spines (some venomous)	Pain, embedded spines, granulomas	Hot water immersion; spine removal
Cone snails (*Conus*)	Found in KwaZulu-Natal (Indo-Pacific species)	**Conotoxins** — neuromuscular blockade	Local pain or numbness, then paralysis and **respiratory failure**	Pressure immobilisation (as for neurotoxic snakebite); **ventilatory support**; no

Animal	Southern African relevance	Venom / mechanism	Clinical features	Management
				antivenom
Blue-ringed octopus (*Hapalochlaena*)	Indo-Pacific and Australia (not established in South Africa)	**Tetrodotoxin** — sodium channel blockade	Painless bite, then rapid paralysis and respiratory failure with preserved consciousness	Pressure immobilisation; ventilation until the toxin wears off

Marine poisoning by ingestion (seafood toxins)

Syndrome	Source	Toxin	Features	Management
Scombroid (histamine fish poisoning)	Improperly stored fish (tuna, mackerel and others)	Histamine	See Outcome 65	Antihistamines
Paralytic shellfish poisoning (PSP)	Mussels, oysters and clams contaminated during **harmful algal blooms ("red tides")** — a recurring problem on the **South African west coast**	**Saxitoxins** (sodium channel blockers), heat-stable	Within 30 minutes to a few hours: perioral and limb **paraesthesia** and numbness, floating sensation, ataxia, dysarthria, dysphagia, weakness, then **respiratory paralysis**	Supportive; **ventilation** (patients recover as the toxin is cleared); activated charcoal if early and conscious; report to environmental health; avoid harvesting shellfish during red-tide warnings
Tetrodotoxin poisoning	**Pufferfish** ("blaasop" in South Africa), fugu	**Tetrodotoxin** (sodium channel blocker), heat-stable	Perioral paraesthesia, numbness, vomiting, ascending paralysis, hypotension, bradycardia, respiratory failure, with preserved consciousness	Supportive; **ventilation**; no antidote

Syndrome	Source	Toxin	Features	Management
Ciguatera	Large reef fish (barracuda, grouper, snapper) from tropical waters	Ciguatoxins (sodium channel activators), heat-stable	GI symptoms, then neurological symptoms — paraesthesia, **reversal of hot and cold sensation**, pruritus, bradycardia, hypotension; may persist for weeks	Supportive; IV fluids; atropine for symptomatic bradycardia

High-yield points — Outcome 64

- WHO: 1.8–2.7 million snakebite envenomings and 81,000–138,000 deaths a year; NTD since 2017.
- SA syndromes: **cytotoxic** (puff adder, spitting cobras — painful progressive swelling, necrosis), **neurotoxic** (mambas, Cape and snouted cobras — ptosis, bulbar paralysis, respiratory failure), **haemotoxic** (boomslang, vine snake — delayed bleeding).
- First aid: reassure, keep still, remove rings, immobilise, mark swelling; PIB **only for neurotoxic bites**; **no** tourniquet, cutting, sucking, ice; **ventilate** neurotoxic paralysis; irrigate venom in the eyes.
- SAVP polyvalent covers 10 species (puff adder, Gaboon adder, rinkhals, 3 mambas, Cape, forest, snouted cobras, Mozambique spitting cobra); not berg adder, boomslang or vine snake; boomslang monovalent; same dose for children; give IV with adrenaline ready; early anaphylactic reactions, pyrogenic reactions, serum sickness at 5–24 days.
- Button spider (*Latrodectus*) — neurotoxic latrodectism (abdominal rigidity, sweating); violin spider — necrosis.
- *Parabuthus* scorpions (thick tail, thin pincers) — neurotoxic; respiratory failure in children and older people.
- Marine: hot water immersion for fish venoms and bluebottles; ventilation for paralysis (PSP, tetrodotoxin, cone snail); red tide on the west coast.

Outcome 65. Discuss the management of a patient with scombroid food poisoning

Definition and cause

Scombroid fish poisoning (histamine fish poisoning) is a form of food poisoning caused by eating fish that contains high levels of **histamine** (and other biogenic amines, such as cadaverine and putrescine, which enhance histamine's effects). It is named after the **Scombridae** family of fish (tuna, mackerel, bonito, skipjack), but also occurs with non-scombroid fish such as sardines, anchovies, herring, mahi-mahi (dorado), bluefish, amberjack and marlin (Hungerford, 2010). It is one of the most common forms of fish-related food poisoning worldwide, and outbreaks typically affect several people who ate the same fish.

Pathophysiology

These fish naturally contain high levels of free **histidine** in their muscle. When fish is **not kept adequately chilled** after it is caught (inadequate refrigeration during storage, transport or handling), bacteria on and in the fish (for example *Morganella morganii*, *Klebsiella*, *Proteus*, *Hafnia*, *Photobacterium*) produce the enzyme **histidine decarboxylase**, which converts histidine to **histamine** (Hungerford, 2010). Key facts:

- **Histamine is heat-stable** — it is **not destroyed by cooking, canning, freezing or smoking** once it has formed.
- The fish usually **looks, smells and tastes normal**, although some people notice a sharp, peppery, metallic or bitter taste.
- The ingested histamine (together with the other amines, which inhibit the gut enzymes that normally break down histamine — diamine oxidase and histamine N-methyltransferase) is absorbed and acts on **H1 and H2 receptors**, producing an illness that **mimics an allergic reaction** but is **not an allergy** (it is a toxic reaction; no IgE is involved, and the person is not allergic to fish).
- Patients taking **isoniazid** (common in South Africa because of TB treatment) or monoamine oxidase inhibitors are at risk of more severe reactions, because these drugs inhibit histamine-metabolising enzymes.

Clinical presentation

- **Onset**: rapid — typically **10–30 minutes** (range a few minutes to about two hours) after eating the fish.
- **Skin**: **flushing** of the face, neck and upper body ("sunburn-like"), **urticaria**, pruritus, sweating.
- **Oral**: burning or tingling in the mouth, **peppery or metallic taste**, perioral swelling.
- **Neurological**: **throbbing headache**, dizziness, palpitations, anxiety.
- **Gastrointestinal**: nausea, vomiting, **abdominal cramps, diarrhoea**.
- **Cardiovascular**: **tachycardia**, palpitations, **hypotension** (vasodilatation).
- **Respiratory** (less common): bronchospasm, chest tightness — more likely in people with asthma.

- Severe cases (rare): significant hypotension, arrhythmias, respiratory distress.
- **Duration**: usually self-limiting within **12–24 hours** (often a few hours), with full recovery.
- **Diagnostic clue**: **several people who ate the same fish develop similar symptoms** at about the same time.

Differential diagnosis

- **Fish or seafood allergy (IgE-mediated anaphylaxis)** — the most important differential: allergy usually affects one person, the person often has a known allergy, angioedema and severe bronchospasm are more prominent, and the reaction recurs with every exposure to that fish. Scombroid affects several people, is linked to a particular batch of fish, and does not recur with fresh fish.
- Ciguatera, paralytic shellfish poisoning and tetrodotoxin poisoning (64.4) — prominent neurological features (paraesthesia, paralysis).
- Other food poisonings (bacterial gastroenteritis — longer incubation).
- Monosodium glutamate reaction.

Prehospital management

1. **Scene and history**: what fish was eaten, when, how it was stored and prepared, how many people are affected; known allergies; medications (isoniazid, MAOIs); asthma.
2. **Primary survey (ABCDE)**: airway (angioedema, stridor), breathing (bronchospasm, SpO_2), circulation (BP, HR, perfusion), disability, exposure (flushing, urticaria).
3. **Oxygen** if SpO_2 <92% or respiratory distress.
4. **If features of anaphylaxis** (airway swelling, bronchospasm, hypotension) — which cannot always be distinguished from severe scombroid in the field — treat as **anaphylaxis** according to the RCSA anaphylaxis algorithm referenced in the CDST: **adrenaline 0.5 mg IM (1:1000)** in adults (child 0.01 mL/kg of 1:1000, up to the age-appropriate maximum per the CDST formulary) (HPCSA PBEC, 2025, pp. 68, 74), high-flow oxygen, IV crystalloid for hypotension, and **salbutamol 5 mg nebulised** for bronchospasm (HPCSA PBEC, 2025, p. 73). Hydrocortisone 200 mg IV (child 5 mg/kg) may be considered per the CDST Breathing difficulty tool.
5. **IV access** and **IV crystalloid** for hypotension or dehydration from vomiting and diarrhoea (CDST shock tool).
6. **Antihistamines** are the specific treatment: an **H1 antagonist** (for example promethazine, chlorphenamine or diphenhydramine) combined with an **H2 antagonist** (for example famotidine or cimetidine) relieves flushing, urticaria, headache, hypotension and GI symptoms (Hungerford, 2010). These are hospital or primary-care medications; they are not listed in the CDST drug appendix for emergency care providers, so the prehospital priority is supportive care and treatment of anaphylaxis-like features, with antihistamines given at the receiving facility (or according to your service's protocol if they are within your scope).
7. **Analgesia** for headache (paracetamol per CDST); antiemetics if within scope.
8. **Monitoring**: vital signs every 10 minutes; ECG if palpitations, arrhythmias or significant hypotension.

9. **Transport**: patients with hypotension, bronchospasm, airway involvement, significant symptoms, comorbidities (asthma, cardiac disease, isoniazid use), or when anaphylaxis cannot be excluded. Mild cases often resolve within hours, but should be assessed at a facility.
10. **Public health**: scombroid poisoning usually occurs in clusters; **food-borne illness outbreaks are notifiable medical conditions in South Africa**, so the facility should notify the local **environmental health** officials, and the remaining fish should be kept (refrigerated) for testing of histamine levels. Advise patients that they are **not allergic to fish** and can eat fresh, properly stored fish in future (unless an allergy is confirmed).

Prevention

Rapid chilling of fish after catching (ideally to 0–4 °C) and maintenance of the cold chain through storage, transport and retail; buying fish from reputable suppliers; and discarding fish that has been left unrefrigerated. Cooking does not make contaminated fish safe.

High-yield points — Outcome 65

- Cause: bacterial decarboxylation of histidine to **histamine** in improperly chilled fish (tuna, mackerel, sardines); **heat-stable** — cooking does not prevent it.
- Onset 10–30 minutes: flushing, urticaria, peppery/metallic taste, headache, palpitations, GI symptoms, hypotension; resolves within 12–24 hours; clusters of diners.
- Not an allergy (no IgE); isoniazid increases severity.
- Prehospital: ABCDE; treat anaphylaxis features with **adrenaline 0.5 mg IM**, oxygen, fluids, salbutamol; transport.
- Specific treatment: **H1 + H2 antihistamines** (facility level).
- Notify environmental health; keep the fish.

APPENDIX A — Tutorial questions with verified model answers

Each question below is stated **exactly as it appears in your tutorial memorandum**, followed by a verified model answer written in full sentences. The answers build on the lecturer's memo, add South African prehospital context (in particular the CDST), and correct the memo where it differs from current evidence. Where a full answer is already given in the main guide, the section is cross-referenced so that you can revise it in depth.

A.1 Behavioural Emergencies — Introduction — Tutorial 1

Question 1

"Evaluate the role of psychological assessment in emergency medical contexts, including how mental status examinations (MSE) and cognitive screening inform acute management and disposition decisions."

Psychological assessment in the emergency setting serves to identify immediately dangerous states (suicidality, homicidality, acute psychosis, delirium, intoxication and withdrawal), to distinguish organic from functional causes of abnormal behaviour, to assess decision-making capacity, and to determine a safe disposition (Boland, Verduin and Ruiz, 2021). The Mental State Examination provides a structured, objective description of appearance, behaviour, speech, mood, affect, thought form and content (including suicidal and homicidal ideation), perception, cognition, insight and judgement at a single point in time (Outcome 40.1). Cognitive screening tools such as the MMSE (Folstein, Folstein and McHugh, 1975) and the MoCA (Nasreddine et al., 2005) quantify orientation, attention, memory, language and visuospatial function, and impaired orientation and attention point towards delirium, dementia or intoxication rather than a primary psychiatric illness, which in turn directs the clinician towards urgent medical investigation (glucose, oxygenation, head injury, infection, toxins). Delirium specifically requires a dedicated tool such as the CAM, because a single MMSE cannot capture acute onset, fluctuation and inattention (Inouye et al., 1990). These assessments inform acute management (for example whether sedation is needed, whether the patient needs medical rather than psychiatric care, and whether the patient may refuse treatment) and disposition (emergency centre versus psychiatric facility; voluntary versus involuntary care under the MHCA; safe discharge with follow-up). Their limitations in the emergency setting are that they require a cooperative patient, take time, are influenced by language, education and culture (a major limitation in South Africa), and cannot replace a decision-specific capacity assessment or a suicide risk assessment (Outcome 40). In prehospital practice, the CDST requires the provider never to assume a psychiatric cause and to check glucose, oxygen saturation and blood pressure in every agitated patient (HPCSA PBEC, 2025, p. 47), and Stander, Hodkinson and Dippenaar (2021) caution against using the GCS as a substitute for capacity assessment.

Question 2

"Discuss the phenomenology and clinical significance of Specific Phobia, Social Anxiety Disorder, and OCD as part of the anxiety spectrum. How would you differentiate these conditions diagnostically and what are the implications for management in emergency medical care settings?"

Specific phobia is marked fear or anxiety about a specific object or situation (for example needles, blood, heights, animals) that is out of proportion to the actual danger, is actively avoided or endured with intense distress, and persists for six months or longer; the blood-injection-injury subtype is clinically significant because it can cause vasovagal syncope during procedures (APA, 2022). **Social anxiety disorder** is marked fear of social situations in which the person may be scrutinised by others, with fear of acting in a way that will be negatively evaluated (humiliation, embarrassment, rejection). **Obsessive-compulsive disorder** — which DSM-5-TR now places in a separate chapter of obsessive-compulsive and related disorders rather than among the anxiety disorders — consists of obsessions (recurrent, intrusive, unwanted thoughts, urges or images that cause anxiety and that the person tries to neutralise) and/or compulsions (repetitive behaviours or mental acts performed to reduce the anxiety or according to rigid rules), which take more than an hour a day or cause significant impairment, with insight that may be good, poor or absent (APA, 2022). **Differentiation** rests on the focus of the fear: a circumscribed object or situation (specific phobia), social evaluation (social anxiety disorder), or intrusive thoughts with neutralising rituals that are not limited to one feared stimulus (OCD). All three must be distinguished from substance- and medical-induced anxiety (for example stimulant use, hyperthyroidism). **Emergency implications**: specific phobias may cause panic or syncope during injections, cannulation or confinement in an ambulance, so procedures should be explained, the patient allowed to lie down, and distraction used; social anxiety may limit the history the patient will give in front of others, so a private setting is helpful; OCD rituals (for example handwashing, checking) may delay cooperation, and these can usually be accommodated if safe while maintaining infection control. Comorbid depression and suicidality must be screened for. Long-term management consists of CBT with exposure (phobias and OCD, with exposure and response prevention for OCD) and SSRIs (OCD and social anxiety disorder); benzodiazepines are rarely appropriate and only for short-term, acute anxiolysis (Craske and Stein, 2016).

Question 3

“Critically assess the interplay between OCD symptomatology and stigma in clinical and community settings. How can healthcare professionals counteract stigma to improve engagement and adherence to treatment?”

OCD symptoms are often secret, embarrassing and ego-dystonic (for example intrusive violent, sexual or blasphemous thoughts), which makes people fearful of being judged as dangerous or “mad”, while the trivialisation of OCD in popular culture (“I’m so OCD about neatness”) minimises the suffering involved. Stigma operates as **public stigma** (prejudice and discrimination from others), **self-stigma** (internalised shame) and **anticipated stigma** (fear of discrimination), and results in concealment of symptoms, delayed help-seeking (often for many years), reduced adherence to treatment, social withdrawal and worse outcomes (Tutorial memo). In clinical settings, stigma may also arise from staff who are impatient with rituals or who misinterpret intrusive thoughts as intent. In the South African context, mental illness stigma is reinforced by limited mental health literacy and resources, and negative attitudes have been documented even among prehospital providers towards patients with mental health crises (Evans, Geduld and Stassen, 2018). Health professionals can counteract stigma by adopting a compassionate, non-judgemental approach; using person-first, non-stigmatising language; providing psychoeducation that explains OCD as a common, treatable neurobiological

and cognitive condition and that intrusive thoughts do not reflect a desire to act; normalising and explaining evidence-based treatments (exposure and response prevention, SSRIs) and their side effects; involving families; providing culturally sensitive care; linking patients with peer support; and advocating for mental health parity and public awareness. In emergency care, respectful accommodation of rituals where safe, and avoidance of dismissive comments, preserve trust and future engagement.

Question 4

"Evaluate pharmacological and psychotherapeutic treatment options for Generalised Anxiety Disorder (GAD) and Panic Disorder, including first-line medications, psychotherapy approaches, and considerations for comorbidity and safety."

First-line pharmacological treatments for both GAD and panic disorder are **SSRIs** (for example sertraline, escitalopram, paroxetine) and **SNRIs** (for example venlafaxine, duloxetine), which take several weeks to work, may initially increase anxiety (so low starting doses are used), and should be continued for months after remission (Craske and Stein, 2016). Pregabalin and buspirone are alternatives for GAD. **Benzodiazepines** provide rapid anxiolysis but are limited to short-term use because of tolerance, dependence, withdrawal, cognitive impairment, falls in older people, and dangerous synergy with alcohol and opioids. The best-supported psychological treatment for both disorders is **cognitive behavioural therapy (CBT)** — cognitive restructuring of catastrophic thoughts, with interoceptive and in vivo **exposure** being particularly important in panic disorder — and mindfulness-based and applied relaxation therapies are additional options; psychotherapy and medication are comparably effective, and combination may be used in severe cases. **Comorbidity** is common (depression, other anxiety disorders, substance and alcohol use), and treatment should be integrated; **safety** considerations include screening for suicidality (anxiety disorders increase suicide risk, especially with comorbid depression), avoiding benzodiazepines in patients with substance use disorders, monitoring for serotonin syndrome with combinations of serotonergic drugs (60.28), and remembering that SSRIs can increase suicidal thinking in young people early in treatment. In emergency practice, a "panic attack" is a diagnosis of exclusion: the ECP must exclude ACS, PE, arrhythmias, asthma, hypoglycaemia, thyrotoxicosis and stimulant toxicity with vital signs, SpO_2, glucose and a 12-lead ECG before attributing symptoms to anxiety (Outcome 42).

Question 5

"Explain the DSM-5-TR diagnostic criteria for bipolar I disorder and contrast it with bipolar II disorder, emphasising the clinical implications for treatment and prognosis."

Bipolar I disorder requires **at least one lifetime manic episode** — a distinct period of abnormally and persistently elevated, expansive or irritable mood **and** increased goal-directed activity or energy, lasting **at least seven days** (or any duration if hospitalisation is necessary), present most of the day nearly every day, with **three or more** of the following (**four** if the mood is only irritable): inflated self-esteem or grandiosity, decreased need for sleep, pressured speech, flight of ideas or racing thoughts, distractibility, increased goal-directed activity or psychomotor agitation, and excessive involvement in risky activities; the episode causes marked impairment, requires hospitalisation or has psychotic features, and is not due to a substance or medical condition (APA, 2022). Major depressive episodes commonly occur but

are **not required** for bipolar I. **Bipolar II** requires at least one **hypomanic episode** (the same symptoms for at least **four consecutive days**, with an observable change in functioning but **without** marked impairment, hospitalisation or psychosis) **and** at least one **major depressive episode**, with **no history of mania**. Clinically, bipolar I carries a higher risk of psychosis, severe functional impairment, dangerous behaviour during mania and hospitalisation, whereas bipolar II is dominated by recurrent and often prolonged depressive episodes, frequently misdiagnosed as unipolar depression. Both carry a **high suicide risk**, particularly during depressive and mixed episodes. Treatment includes mood stabilisers (lithium, valproate, carbamazepine, lamotrigine) and atypical antipsychotics, with psychotherapy and psychoeducation; antidepressants are used cautiously and only with a mood stabiliser because of the risk of switching to mania or rapid cycling (Vieta et al., 2018). Emergency implications include managing manic agitation (CDST sedation if dangerous), exclusion of stimulant use and thyrotoxicosis, recognition of **lithium toxicity** (60.27), and suicide risk assessment (52.4).

Question 6

"Analyse the potential consequences of DAMA (discharge against medical advice) in psychiatric emergencies and relate to prehospital decision-making."

Patients who leave psychiatric care against medical advice show reduced treatment benefit, worse psychiatric, medical, psychosocial and socioeconomic functioning, decreased access to outpatient services, overuse of emergency services and earlier readmission (Brook et al., 2006, cited by Evans, Geduld and Stassen, 2018), and psychiatric DAMA is a specific risk factor for **suicide** (Kuo et al., 2010, cited by Evans, Geduld and Stassen, 2018). In prehospital care, the equivalent is the **refusal of hospital treatment (RHT)** on scene. Evans, Geduld and Stassen (2018) found that South African providers frequently accepted refusals from suicidal patients, believed that patients could never be transported without consent, and believed that a signed refusal absolved them of responsibility; several had personally experienced a patient's death after leaving them on scene. The consequences of an inappropriate prehospital refusal therefore include death or serious harm to the patient, moral distress and trauma for the provider, legal liability (negligence) and disciplinary action, and repeated calls. Prehospital decision-making must therefore include: a careful, documented **capacity assessment**; a structured **suicide and violence risk assessment**; recognition that a refusal driven by mental illness is not a valid autonomous choice; application of **MHCA s 9(1)(c)** (care without consent due to mental illness where delay may cause death, serious harm to self or others, or serious property damage) and the **CDST obligation to transport** a patient who is a danger to themselves or may be a suicide risk (HPCSA PBEC, 2025, p. 48); involvement of family, SAPS (s 40) and senior advice; use of the least restrictive means; and, where a competent patient lawfully refuses, documentation of the information given, safety-netting and referral to community resources (Outcomes 45–47 and 50).

Question 7

"Critically assess the ethical tensions embedded in the HPCSA Guidelines for the Withholding and Withdrawing of Treatment when applied to emergency prehospital care of suicidal patients."

A full answer is given in Outcome 45.1. In summary, Booklet 7 was written to guide decisions about withholding or withdrawing life-prolonging treatment in patients with terminal illness or little hope of recovery, and it explicitly prohibits killing, active euthanasia and assisted suicide (HPCSA, 2023, para 2.2). Its emphasis on **respect for autonomy** — that a competent adult's refusal must be respected even if it results in death (para 8.1.2) — creates a tension with the duty of **beneficence** to prevent a suicide, but the same paragraph contains the critical exception: the refusal is not binding where the patient's "state of mind is such that he/she is incapable of taking or participating in the decision", which is frequently the case in a suicidal crisis. **Non-maleficence** is engaged because preventing suicide may require coercive measures (restraint, sedation, involuntary transport) that themselves cause harm and must therefore be proportionate. The guideline's emphasis on **consultation**, second opinions, involvement of family and proxies, and documentation (paras 3.5, 3.7, 5.2, 12) is difficult to achieve under prehospital time pressure, uncertainty and limited information, but paragraph 7.1 resolves this for emergencies by directing practitioners to start treatment that may be of benefit until a clearer assessment can be made, with the patient informed later (para 7.2). A suicide note is not a valid **advance directive**, which presupposes a competent person planning for terminal illness, and a directive cannot compel care outside acceptable standards (para 9.1.4). **Resource** considerations (para 10) can never justify withholding emergency care from a suicidal patient. The ECP should therefore treat, assess capacity and risk, transport when the MHCA or CDST criteria apply, consult and document thoroughly, and respect autonomy only where capacity is clearly present.

Question 8

"Explain how the Mental Health Care Act of 2002 influences prehospital decision-making when a patient who has attempted suicide refuses transport. Include potential safeguards for both patient rights and provider safety."

The MHCA (Act 17 of 2002) permits a health care provider — which includes the ECP — to provide care without consent when, **due to mental illness**, any delay may result in death or irreversible harm to the health of the user, the user inflicting serious harm to themselves or others, or serious damage to property (s 9(1)(c)); a written report must be made to the Review Board and care without consent may not continue beyond **24 hours** unless a Chapter V application is made (s 9(2)). The formal involuntary pathway requires a written application (usually by a relative, or by a health care provider if relatives are unwilling, incapable or unavailable), examination by two mental health care practitioners and a 72-hour assessment (ss 32–34) — which cannot be done on scene, so the prehospital role is emergency assessment and **transport for evaluation** (Evans, Geduld and Stassen, 2018). SAPS has the statutory duty to apprehend a person who, due to mental illness, is likely to inflict serious harm to themselves or others and to take them to a State health establishment (s 40). The CDST obliges the provider to transport a patient who is a danger to themselves or may be a suicide risk, and to request SAPS instruction for mechanical restraint (HPCSA PBEC, 2025, p. 48). **Safeguards for patient rights** include: a documented capacity and risk assessment; use of the least restrictive intervention (de-escalation, persuasion, family involvement, voluntary oral sedation before parenteral sedation and restraint); respect for dignity and privacy (MHCA s 8); avoidance of deception; brief, safe, proportionate restraint only to facilitate sedation; continuous monitoring; transport to an appropriate State emergency centre; clear documentation of the

legal basis so that the facility can comply with s 9(2); and confidentiality. **Safeguards for provider safety** include staging until SAPS secures the scene, removal of weapons and means of self-harm, never working alone, adequate numbers for restraint (five plus one), SAPS escort in the ambulance, securing the vehicle doors, PPE, senior medical consultation, debriefing and incident reporting (Outcomes 43 and 46).

Question 9

"Critically appraise the tension between the duty to relieve suffering and the obligation not to prolong life when considering palliative/terminal care in the emergency setting."

The HPCSA states that "it is not the health practitioner's duty to prolong life at all cost", but that it is their responsibility "to know when to initiate and when to cease attempts at prolonging life, while ensuring that patients receive appropriate relief from distress" (HPCSA, 2023, para 4.2), and that practitioners "shall refrain from employing unusual methods of treatment which are of no benefit to the patient" (para 2.3). The WMA Declaration on Terminal Illness, annexed to the guideline, affirms the patient's right to refuse treatment and to request palliative measures "which may have the additional effect of accelerating the dying process", while prohibiting assisted suicide (Annexure 1, para 6). The tension in emergency care arises because the ECP often meets a dying patient without knowing the prognosis, the patient's wishes or the existence of an advance directive, under time pressure and with distressed family; aggressive resuscitation may then prolong suffering without benefit, whereas withholding treatment in error may cause avoidable death. The HPCSA resolves the emergency dilemma by directing practitioners to start treatment that may be of benefit when there is uncertainty, until a clearer assessment can be made (para 7.1), while respecting valid advance refusals (para 9.2.1), consulting family to determine best interests where the patient lacks capacity (para 3.4), respecting cultural practices around death and dying (para 4.4), documenting decisions (para 12), and ensuring that withholding treatment never means withholding comfort care (para 2.3). In prehospital practice, the CDST includes a **"Palliative end-of-life care"** tool that supports symptom control (pain, breathlessness, agitation, secretions, nausea) and linkage with palliative care teams (HPCSA PBEC, 2025), and the ECP should seek senior advice and documentation (for example a valid do-not-resuscitate decision) where available. Ethical practice is therefore patient-centred, honest with families, avoids futile interventions, and prioritises dignity and the relief of suffering.

Question 10

"What are the key differences between de-escalation and physical containment strategies in violent patient management, and under what circumstances should each be employed according to PHC guidelines?"

De-escalation is a non-coercive, collaborative set of verbal and environmental techniques aimed at calming the patient, preserving dignity and the therapeutic relationship, and avoiding restraint; it is the **first-line and continuous** approach in every agitated patient (RCEM, 2025; Richmond et al., 2012). **Physical containment** (restraint) is a coercive intervention that restricts movement by force or devices, carries significant physical and psychological risks, and is a **last resort**, used only when there is an imminent risk of serious harm to the patient or

others and less restrictive measures have failed or are unsafe (Outcome 41.1). In South African prehospital practice, the CDST (HPCSA PBEC, 2025, pp. 46–48) sets out the sequence: first try to calm the patient (approach from a safe distance, talk in a quiet area, listen, be calm and reassuring, show kindness and empathy, set clear boundaries; avoid turning your back, direct eye contact, unnecessary touch, arguing with delusions, attempting to reason, or prolonged conversation); if the patient remains severely agitated, offer **voluntary sedation**; only if the patient refuses and remains dangerous, **restrain physically with five people plus a sixth to give medication, briefly and only to facilitate sedation**, contact SAPS, and apply mechanical restraint only on SAPS instruction; give rapid sedation; then monitor every 10 minutes and check distal pulses in restrained limbs. Containment must be proportionate, performed by trained staff, never involve pressure on the neck, chest or back, be documented (indication, duration, method, people involved) and be followed by a debrief and safe disposition. (The memo's reference to "PHC guidelines" — the primary health care standard treatment guidelines — is consistent with these principles; for prehospital ECP practice, the CDST is the applicable national guidance.)

Question 11

"Discuss communication strategies and environmental modifications that can reduce aggression in a violent patient, including room layout, lighting, and staff positioning, with reference to best practices in South Africa."

Communication strategies include having only **one person speak**; introducing yourself and explaining that your aim is to keep the patient safe; using a calm, quiet, slow voice and short, simple sentences; allowing time to respond; active listening with clarifying statements; identifying and validating the patient's wants and feelings; agreeing where possible or "agreeing to disagree" (fogging); offering choices, acts of kindness and oral medication; setting clear limits as matter-of-fact information rather than threats; avoiding sarcasm, accusations, arguing or challenging delusions; and using a trusted interpreter or family member where helpful (RCEM, 2025; HPCSA PBEC, 2025, p. 46). Environmental modifications include a dedicated or adapted low-stimulus area that is quiet and not too warm; adequate lighting without glare (dimmed where possible for confused older patients); **exits** positioned so that staff cannot be trapped, with doors that open outwards; removal of furniture and objects that could be used as weapons or to barricade an exit; absence of ligature points; constant observation; and an easy way to summon help (RCEM, 2025). **Staff positioning**: stay at least two arm-lengths away, at an angle rather than face-on, with hands visible, never turning your back and never allowing yourself to be cornered, with both you and the patient having an exit (HPCSA PBEC, 2025, p. 46). In South African prehospital practice, this translates into switching off sirens, radios and televisions; moving bystanders away; obtaining collateral history before approaching; requesting SAPS if the risk is significant; and securing the ambulance (attendant seated nearest the door). Staff should be trained in de-escalation, and the CDST provides the national framework.

Question 12

"Identify the immediate medical assessments and potential differential diagnoses to consider when a violent patient presents with altered mental status or intoxication, and outline the initial management plan."

The immediate assessment is scene safety followed by a rapid primary survey once the patient is contained: airway, breathing (SpO_2 — give oxygen if <92%), circulation (BP — manage shock if <90/60), disability (GCS, pupils, **glucose — treat if <3.5 mmol/L**, seizures, lateralising signs), and exposure (temperature, injuries — especially head injury — track marks, signs of self-harm, toxidromes) (HPCSA PBEC, 2025, p. 47). The **differential diagnoses** include intoxication (alcohol, cannabis, methamphetamine, cocaine, sedatives, hallucinogens), withdrawal (alcohol — delirium tremens; benzodiazepines), **hypoglycaemia**, **hypoxia**, **head injury**, infection (sepsis, meningitis, encephalitis, HIV-related CNS infections), seizures and post-ictal states, metabolic disturbances (hyponatraemia, uraemia, hepatic encephalopathy), thyrotoxicosis, heat stroke, anticholinergic toxicity, serotonin syndrome, neuroleptic malignant syndrome, and acute psychiatric illness (RCEM, 2025). The initial management plan is: safety and SAPS; verbal de-escalation; voluntary or rapid sedation per the CDST if dangerous (*midazolam 0.2 mg/kg IM, max 15 mg; 0.1 mg/kg if >60 years*; or lorazepam 4 mg IM or ^ketamine 4 mg/kg IM^); primary survey and correction of hypoxia, hypotension and hypoglycaemia; thiamine 100 mg IV/IM in alcohol users (HPCSA PBEC, 2025); seizure management per the CDST; treatment of specific toxidromes (for example naloxone, cooling); IV access once contained; collateral history; monitoring every 10 minutes; documentation; and transport to an emergency centre. The CDST explicitly instructs: “Do not assume aggression/agitation is due to a psychiatric condition” (HPCSA PBEC, 2025, p. 47).

Question 13

“Outline the pharmacological considerations for managing agitation or aggression in violent patients, including restraints, if used, and the role of sedatives, analgesics, and antipsychotics under PHC guidelines.”

Medication should be used only as part of a stepwise plan after de-escalation, preferably offered voluntarily first (oral/buccal/IN), with parenteral rapid sedation reserved for patients who remain dangerous (RCEM, 2025; HPCSA PBEC, 2025). In South African prehospital practice the CDST agents are **benzodiazepines** — *midazolam* (7.5 mg PO, 10 mg buccal/IN, 2 mg IV voluntary; 0.2 mg/kg IM rapid sedation, max 15 mg, halved if >60 years; repeat 0.1 mg/kg IM, max 5 mg), *lorazepam* (2 mg PO, 4 mg IM) and diazepam (10 mg PO) — and, for ANT/ECP, ^ketamine 4 mg/kg IM^ (HPCSA PBEC, 2025, pp. 46–47). Benzodiazepines are preferred for stimulant toxicity and alcohol or sedative withdrawal, but cause respiratory depression (especially with alcohol and opioids), paradoxical agitation, and worsening delirium in older people (Desai, Kulkarni and Verma, n.d.). **Antipsychotics** (haloperidol, droperidol, olanzapine) are used in emergency departments and psychiatric settings for psychosis-related agitation; they are not in the CDST. Evidence shows that haloperidol alone causes frequent acute dystonia and is difficult to justify without an adjunct such as promethazine, while droperidol, olanzapine and IM ziprasidone are effective alternatives, and lorazepam combined with an antipsychotic enhances efficacy (Saucedo Uribe et al., 2025); antipsychotics carry risks of QT prolongation, extrapyramidal effects, hypotension and NMS, and are contraindicated in Parkinson’s disease and Lewy body dementia (RCEM, 2025). **Ketamine** has the fastest onset and preserves airway reflexes and haemodynamic stability, but causes hypersalivation, laryngospasm, emergence phenomena, hypertension and tachycardia, and prehospital use is associated with higher intubation rates (RCEM, 2025). **Analgesia** should be given for pain, which often drives agitation (CDST Pain tool), but the CDST warns: “If airway unsecured, avoid

high analgesic doses as these have respiratory depressant effects" (HPCSA PBEC, 2025, p. 47). **Restraint**, if used, must be brief, only to facilitate sedation, with five people plus one, never prone with weight on the torso, and mechanical restraint only on SAPS instruction. **Monitoring** after medication (BP, HR, RR, SpO_2, GCS every 10 minutes; ECG; ideally $ETCO_2$) and documentation of drug, dose, route, time and response are mandatory (HPCSA PBEC, 2025, p. 47; RCEM, 2025).

A.2 Behavioural Emergencies — Vulnerable Patient Populations — Tutorial 2

Question 1

"Discuss the particular vulnerabilities and risks faced by children with disabilities in the South African and wider Southern African context, as articulated in Mesthrie's analysis and the Disability Rights framework. What policy measures should be strengthened or introduced to protect this group within prevention and early intervention mandates?"

Children with disabilities face a disproportionately high risk of abuse, neglect and exploitation because of dependence on caregivers for intimate care, communication barriers that make disclosure difficult (and lead to their accounts being disbelieved), social isolation, stigma and harmful beliefs about disability, institutional care, poverty, and lack of accessible services; and they are often excluded from prevention programmes and from participation in decisions about their protection. Mesthrie (2025) situates this within extremely high levels of violence against children in South Africa (a child homicide rate nearly twice the global average; 45% of households reporting violence against children), emphasises that the National Child Care and Protection Policy itself acknowledges "lack of adequate support for children with disabilities" as a past gap, and argues that the right to protection is inseparable from the right to be heard and to participate. The disability rights framework (the UN Convention on the Rights of Persons with Disabilities and the African Disability Protocol) requires equal protection, accessibility, reasonable accommodation and inclusion. Policy measures that should be strengthened include: explicit inclusion of disability in all prevention and early intervention norms and standards under the Children's Act; mandatory training for social workers, police, health workers (including EMS) and educators on disability-specific communication and trauma-informed practice; accessible reporting mechanisms (sign language interpretation, pictorial tools, SMS and alternative formats); safeguarding standards in schools, health facilities and residential care; family support to reduce caregiver burden and stigma (for example respite care and the care dependency grant); child participation mechanisms adapted to disability; robust disaggregated data collection; and coordinated multisectoral oversight (Tutorial memo; Mesthrie, 2025). For the ECP, this means maintaining a high index of suspicion, communicating through trusted persons and accessible methods, documenting the child's own expressions, and reporting suspected abuse (Children's Act s 110; SOA s 54; DVA ss 2A–2B).

Question 2

"Critically assess Article 19 of the UN Convention on the Rights of the Child (CRC) in the context of South Africa's Children's Act 38 of 2005 and the National Childcare and Protection Policy. How do prevention and early intervention strategies reflect a child-

rights based approach, and where do gaps remain in aligning domestic policy with international obligations?"

Article 19 obliges States Parties to "take all appropriate legislative, administrative, social and educational measures to protect the child from all forms of physical or mental violence, injury or abuse, neglect or negligent treatment, maltreatment or exploitation, including sexual abuse", including social programmes to support children and carers, and procedures for prevention, identification, reporting, referral, investigation, treatment and follow-up (Mesthrie, 2025). General Comment 13 interprets this through a **child rights-based approach** that treats children as rights-holders, with the best interests of the child, non-discrimination, the right to life, survival and development, and the right to be heard as guiding principles. South Africa's **Children's Act** shifted from removal of children to a developmental, family-strengthening model: chapter 7 provides for protection (including s 110 reporting and the 24-hour notification duty on police) and chapter 8 for prevention and early intervention programmes aimed at preserving families, building caregiver capacity and avoiding removal (s 144), with norms and standards (Annexure B) that require high-risk identification, developmental assessment, and child participation in assessments. The **National Child Care and Protection Policy (2021)** adopts a broad definition of children in need of care and protection, recognises vulnerability factors such as poverty and caregiver substance abuse, prioritises targeted prevention and community-based interventions such as "cash plus care", and acknowledges past gaps (Mesthrie, 2025). **Gaps** identified by Mesthrie (2025) include insufficient attention to meaningful **child participation** across the Act and Policy; the absence of an explicit **gendered** approach despite the gendered nature of violence; a narrow family-focused conception of prevention that neglects schools and communities; weak **implementation, funding and monitoring**; norms and standards whose justiciability is uncertain; and the absence of an independent child rights ombudsman or coordinating framework recommended by the UN Committee. The overall conclusion is that the legal framework largely complies with the CRC in form, but that translating it into effective practice remains the main challenge.

Question 3

"Describe how the narratives of street children and the social work responses in Harare illuminate the limits of purely protectionist policies. What lessons could be translated to South Africa's approach to preventing streetism and safeguarding vulnerable children?"

The memo summarises Chikoko's work on street children in Harare, which shows that children move to the streets because of poverty, abuse, family breakdown, orphanhood and social exclusion, and that the street offers them survival strategies, peer support and a degree of agency. Purely **protectionist** responses — removing children from the streets through police round-ups or institutional placement without addressing the causes, and without the children's participation — tend to fail, because children return to the streets, may experience the interventions as punitive, and remain exposed to the same structural drivers. Effective responses are holistic and rights-based: they address basic needs (food, shelter, health care), support families and reunify children only where it is safe, provide education and livelihood opportunities, offer community-based drop-in centres and safe spaces, integrate child protection with health and education services, and involve children in decisions about interventions. For South Africa, where at least 250,000 homeless children and young people live

on the streets (Mesthrie, 2025), the lessons are to prioritise prevention by reducing poverty and family stress, strengthen family support and safe reunification, invest in community-based and child-friendly services, ensure child participation consistent with the CRC and Children's Act, link services across sectors, and monitor outcomes. For EMS, street children present with injuries, substance use (including inhalants), sexual exploitation and mental health crises, and require non-judgemental care, attention to consent and confidentiality, and referral to social services and designated child protection organisations (Children's Act s 150 lists children who live or work on the street or beg as children in need of care and protection).

Question 4

"Discuss the role of benzodiazepines in the management of agitation in older adults, including contexts where they are appropriate and where they are discouraged, and the potential risks involved."

Benzodiazepines have a limited but important role in older adults: they are **appropriate** for agitation due to **alcohol or benzodiazepine withdrawal**, for **seizures**, and in some **toxicological** situations (for example stimulant toxicity), where GABAergic mechanisms are central (Desai, Kulkarni and Verma, n.d.). They are **discouraged** as routine first-line agents for undifferentiated agitation, delirium or dementia-related distress, because the 2023 Beers Criteria warn that they increase the risk of cognitive impairment, delirium, falls, fractures and psychomotor impairment, and advise avoiding them in patients with or at risk of delirium (American Geriatrics Society Beers Criteria Update Expert Panel, 2023). Risks include respiratory depression and hypoventilation (particularly in frail patients and with other sedatives), worsening delirium and prolonged confusion, falls and fractures, paradoxical agitation, excessive sedation with aspiration risk, and cumulative toxicity because of reduced clearance. **Midazolam** has raised particular safety concern in older adults (Casey et al., cited by Desai, Kulkarni and Verma, n.d.; Saucedo Uribe et al., 2025). In South African prehospital practice, midazolam is nevertheless the main sedative in the CDST, which reflects this risk by **halving the dose in patients older than 60 years** (0.1 mg/kg IM rapid sedation; 0.05 mg/kg IM repeat dose) (HPCSA PBEC, 2025, p. 47). The safe approach is to use non-pharmacological measures first, identify the cause (syndrome-directed treatment), use the lowest effective dose, avoid stacking repeat doses, monitor closely (airway, SpO_2, BP, GCS every 10 minutes), hand over clearly what was given and why, and plan deprescribing after the crisis.

Question 5

"Explain the essential components of a cause-oriented assessment framework for acute agitation in older adults in the emergency department, and discuss how this approach informs pharmacologic decision-making."

A cause-oriented assessment treats behaviour change in an older adult as **medically caused until proven otherwise** (Desai, Kulkarni and Verma, n.d.). Its components are layered: (1) immediate **physiology and safety** — hypoxia, hypoglycaemia, shock, acute coronary syndrome, stroke, intracranial haemorrhage, sepsis and severe pain; (2) the **baseline cognitive status and dementia subtype** (from collateral history about baseline cognition, onset, fluctuation and previous episodes — essential for distinguishing delirium from dementia); (3) **precipitating stressors** — environmental overstimulation, unfamiliar

surroundings, sleep disruption, toileting needs, urinary retention, constipation, hunger, thirst, missing hearing aids or glasses; and (4) **iatrogenic contributors** — anticholinergics, opioid escalation, sedative withdrawal, akathisia from dopamine blockers, corticosteroids, polypharmacy, tethers and immobilisation. This approach informs pharmacological decisions by matching the agent to the syndrome: **benzodiazepines** for withdrawal or seizures; avoidance of **dopamine antagonists** (haloperidol, droperidol) in Parkinson's disease or Lewy body dementia, where quetiapine is preferred if an antipsychotic is needed; low-dose **olanzapine** or haloperidol for dangerous agitation due to psychosis or dementia-related distress when non-drug measures fail; **analgesia** rather than sedation when pain is the driver; **oxygen** rather than sedation when hypoxia is the driver; and treatment of the underlying illness in delirium, with medication used only as a short-term bridge for safety. It also emphasises low starting doses, delayed repeat dosing, attention to QT and respiratory risk, rapid reassessment to avoid masking the underlying illness, and discontinuation of psychotropics after the crisis (Desai, Kulkarni and Verma, n.d.).

Question 6

"Identify and discuss at least three common etiologies of agitation in older adults that require distinct pharmacologic considerations, with examples of appropriate therapeutic responses."

(1) **Delirium due to acute medical illness** (for example urinary tract infection, pneumonia, hypoxia, metabolic disturbance, medication toxicity): the priority is to identify and treat the cause and to use non-pharmacological measures (reorientation, family presence, sensory aids, quiet environment); medication is reserved for dangerous behaviour, using the lowest dose of a low-risk agent, and benzodiazepines are avoided because they worsen delirium (Desai, Kulkarni and Verma, n.d.; Inouye, Westendorp and Saczynski, 2014). (2) **Dementia-related behavioural disturbance**, including **Lewy body dementia and Parkinson's disease**: non-pharmacological care (addressing pain, toileting, environment, caregiver presence) comes first; if an antipsychotic is unavoidable, low-dose olanzapine may be used in Alzheimer-type dementia, while patients with Lewy body or Parkinson's disease are extremely sensitive to dopamine antagonists, so haloperidol and droperidol are contraindicated and quetiapine is preferred (Desai, Kulkarni and Verma, n.d.; RCEM, 2025). (3) **Alcohol or benzodiazepine withdrawal**: benzodiazepines are the appropriate treatment, at the minimum effective dose with careful monitoring, together with thiamine (CDST: thiamine 100 mg IV/IM) and treatment of dehydration and electrolyte disturbances. Further examples include (4) **pain** (for example an occult hip fracture or ischaemic limb), where analgesia rather than sedation is the correct response, and (5) **medication-induced akathisia**, where the offending dopamine blocker should be stopped rather than another antipsychotic added. Understanding the syndrome therefore determines whether the right treatment is an analgesic, oxygen, a benzodiazepine, a specific antipsychotic, or no drug at all.

Question 7

"Critically assess the role of ketamine in the management of acute agitation and discuss why its routine use in older adults is cautioned against in the literature provided."

Ketamine is an NMDA receptor antagonist that produces dissociative sedation with a rapid onset (minutes after IM injection), relative preservation of airway reflexes and respiratory drive, and haemodynamic stability; it is associated with shorter times to adequate sedation than benzodiazepines or antipsychotics, and the RCEM (2025) recommends IM ketamine (4 mg/kg) as a first-line agent for severe acute behavioural disturbance in emergency departments, while the CDST includes ^ketamine 4 mg/kg IM^ for aggression (ANT/ECP) when midazolam is unavailable or a more rapid onset is required (HPCSA PBEC, 2025, p. 47). Its disadvantages include hypersalivation, laryngospasm, vomiting, emergence phenomena, hypertension and tachycardia (which may be synergistic with stimulants), and high rates of subsequent intubation in prehospital series (RCEM, 2025). In older adults, Desai, Kulkarni and Verma (n.d.) caution that the evidence for ketamine comes from general (mostly younger) emergency populations, with major guidance such as the 2024 ACEP severe agitation policy excluding patients older than 65 years, and that older adults "face greater cardiovascular variability, aspiration concern, and post-sedation diagnostic complexity" — ketamine may precipitate myocardial ischaemia or arrhythmias in patients with coronary disease, may cause aspiration in frail patients with impaired swallowing, and its dissociative and emergence effects can mimic or worsen delirium, obscuring the diagnosis. Ketamine "should not be imported into routine geriatric agitation protocols simply because it performs well in younger profoundly agitated patients"; it is better reserved for exceptional cases with close monitoring and the ability to manage the airway and haemodynamic changes.

Question 8

"Evaluate the ethical and legal considerations involved in the use of restraint in ABD, including the Mental Capacity Act and the aim of least restrictive practice."

Important note: the **Mental Capacity Act 2005** referred to in the RCEM (2025) guideline is **United Kingdom legislation**, and it does not apply in South Africa. The equivalent South African legal bases are explained below. Patients with severe acute behavioural disturbance (ABD) are likely to lack capacity to make treatment decisions, which should be formally assessed and documented, and may require emergency treatment, including restraint, under the appropriate legislation (RCEM, 2025). In South Africa, the lawful bases for restraint and treatment without consent are **MHCA s 9(1)(c)** (where, due to mental illness, delay may result in death or irreversible harm, serious harm to self or others, or serious property damage), **NHA s 7(1)(e)** (emergency treatment where delay may result in death or irreversible damage to health and the patient has not refused), the common-law defence of **necessity**, and SAPS powers under **MHCA s 40** (including the use of necessary and proportionate constraining measures, s 40(8)). Ethically, restraint must be **necessary** (imminent risk), a **last resort** after de-escalation and offers of oral sedation, **proportionate** and **least restrictive**, applied for the **shortest possible time**, performed safely by sufficient trained staff (never with weight on the chest or neck), never punitive, respectful of dignity (Constitution s 10; MHCA s 8), and fully documented (capacity assessment, indication, duration, method, police involvement, sedation

and adverse events) (RCEM, 2025; HPCSA PBEC, 2025). The RCEM (2025) notes that continued exertion under restraint may worsen hyperthermia, acidosis and catecholamine excess in ABD, that there is no risk-free method of restraint, and that prolonged restraint should prompt rapid tranquillisation; the CDST similarly limits restraint to a brief period "only to facilitate sedation" and permits mechanical restraint only on SAPS instruction (HPCSA PBEC, 2025, p. 46). Police involvement is appropriate when staff are injured, when healthcare support is not available in time, or when there is a risk of serious injury (RCEM, 2025).

Question 9

"Discuss the importance of capnography and physiological monitoring during rapid tranquilisation and the rationale for maintaining a high-dependency environment after sedation."

Most sedative agents used for rapid tranquillisation have been associated with apnoea, airway obstruction or the need for intubation, and respiratory adverse events are more common with benzodiazepines (RCEM, 2025). Patients with ABD are often physiologically fragile (dehydrated, acidotic, hyperthermic, catecholamine-depleted, possibly with co-ingestants or head injury), cannot report symptoms, and may deteriorate suddenly into peri-arrest. **Capnography ($ETCO_2$)** detects hypoventilation, apnoea and airway obstruction earlier than pulse oximetry — particularly when supplemental oxygen is being given, which delays desaturation — and the RCEM (2025) includes $ETCO_2$ in the routine monitoring of sedated ABD patients alongside ECG, pulse oximetry and blood pressure. Continuous monitoring also detects hypotension, arrhythmias (including QT-related arrhythmias after antipsychotics), hyperthermia, and the evolution of the underlying cause. A **high-dependency (resuscitation) environment** after sedation ensures immediate access to airway equipment, suction, oxygen, a bag-valve mask, drugs (including flumazenil where appropriate), and staff skilled in advanced airway management, and allows correction of electrolyte, glucose and acid–base abnormalities, cooling, and investigation for rhabdomyolysis and DIC (RCEM, 2025). In the prehospital setting, the CDST requires BP, HR, RR, SpO_2 and GCS every 10 minutes after sedation, with immediate reassessment if parameters deteriorate, and the RCEM notes that the practitioner giving rapid tranquillisation must be capable of managing its complications (HPCSA PBEC, 2025, p. 47; RCEM, 2025).

Question 10

"Describe the de-escalation framework recommended by RCEM guidelines and provide examples of practical techniques to implement verbal and environmental de-escalation."

The RCEM (2025) recommends a ten-domain model adapted from the Project BETA consensus (Richmond et al., 2012): (1) respect personal space (identify exits, stay out of arm's reach); (2) do not be provocative (non-confrontational body language, hands visible, do not challenge, insult or argue); (3) establish verbal contact (one staff member speaks, introduce yourself, explain why you are there, reassure the patient that you aim to keep them safe); (4) be concise (short sentences, give time to respond, repeat as needed); (5) identify wants and feelings (identify expectations, empathise); (6) listen closely (clarifying statements); (7) agree, or agree to disagree (fogging — agree with the truth, in principle, or with the odds); (8) set clear limits

(matter-of-fact, not threatening); (9) offer choices and optimism (acts of kindness, offer oral sedative medication); and (10) debrief the patient and staff (explain why intervention was necessary, restore the therapeutic relationship, identify improvements). **Practical verbal techniques** include a calm introduction, explaining each step before doing it, using the patient's name, offering a blanket or (if sedation is not planned) a drink, involving a calm family member, and avoiding argument about delusions. **Environmental techniques** include a dedicated low-stimulus area that is quiet and not too warm, outward-opening doors and accessible exits, removal of potential weapons and ligature points, constant observation, and an easy way to call for help (RCEM, 2025). If sedation or anaesthesia is likely, food or drink should not be offered. The CDST "Try to / Avoid" list (HPCSA PBEC, 2025, p. 46) provides the South African prehospital equivalent (Outcome 43.2).

Question 11

"Critically appraise the use of sedation agents beyond first-line choices, including indications, risks, and how to manage potential complications such as respiratory depression or paradoxical agitation."

In the RCEM (2025) guideline, the first-line agents for severe ABD are IM ketamine (4 mg/kg) and IM droperidol (5–10 mg); if these are unavailable, **midazolam** (5–10 mg IM), **lorazepam** (4 mg IM) or **haloperidol** (5 mg IM, with or without lorazepam 2 mg IM) may be considered, with **olanzapine** and further benzodiazepines used for maintenance and chlorpromazine for serotonin toxicity with severe psychosis or hyperthermia. (Note that in South African prehospital practice the order is different: the CDST uses midazolam as the first-line parenteral agent, with lorazepam or ketamine as alternatives; HPCSA PBEC, 2025, p. 47.) **Midazolam** is rapid but causes more respiratory adverse events and is of particular concern in older adults; it may be preferred where sympathomimetic effects are dangerous (cocaine, ischaemic heart disease). **Lorazepam** is effective with fewer extrapyramidal effects than antipsychotics, and its combination with haloperidol improves sedation without more adverse effects (RCEM, 2025; Saucedo Uribe et al., 2025). **Haloperidol** has a slower onset, frequent acute dystonia and QT prolongation, and is contraindicated in Parkinson's disease and Lewy body dementia; adding promethazine reduces dystonia (Saucedo Uribe et al., 2025). **Olanzapine** is effective with fewer movement disorders, but serious adverse events after IM olanzapine have been reported mainly in medically ill patients who also received benzodiazepines or other antipsychotics, so combining IM olanzapine with parenteral benzodiazepines requires great caution (Desai, Kulkarni and Verma, n.d.). Selection must consider cardiovascular status, co-ingested substances (serotonin syndrome, stimulants, alcohol), age, the anticipated need for airway control, and monitoring capability. **Management of complications**: for **respiratory depression** — stimulate, position the airway (head tilt–chin lift or jaw thrust, recovery position), give oxygen, ventilate with a bag-valve mask, insert airway adjuncts, use flumazenil cautiously only in pure benzodiazepine sedation without contraindications (not in the CDST), and proceed to advanced airway management if needed; for **paradoxical agitation** with benzodiazepines — stop further benzodiazepine doses, reassess for an organic cause (hypoxia, hypoglycaemia, pain, urinary retention), and consider an alternative agent class; for **acute dystonia** — an anticholinergic (biperiden or benztropine) in hospital; for **hypotension** — fluids and position; for **QT prolongation and torsade de pointes** — magnesium and

correction of potassium; and if sedation fails after full doses, escalate to critical care, review the differential diagnosis, and avoid uncontrolled dose stacking (RCEM, 2025).

A.3 Toxicology — Patient Approach and Management — Tutorial 1

Question 1

"Critically discuss the major toxidromes commonly encountered in acute poisoning and outline how their recognition guides immediate management, including any specific antidotes or supportive measures. Discuss how you would distinguish between them clinically."

A full answer, including the CDST toxidrome table, mnemonics and a comparison table, is given in Outcome 62.3. In summary, the major toxidromes are: **sympathomimetic** (agitation, hallucinations, seizures, sweating, hyperthermia, dilated pupils, tachycardia, hypertension — managed with benzodiazepines, cooling, aspirin and nitrates for stimulant chest pain, and sodium bicarbonate for widened QRS); **anticholinergic** (agitation and delirium, flushed, dry, warm skin, hyperthermia, dilated pupils, tachycardia, urinary retention, reduced bowel sounds — managed with benzodiazepines, cooling, sodium bicarbonate if QRS is widened, and physostigmine in hospital for refractory delirium); **opioid** (depressed consciousness, pinpoint pupils, respiratory depression, bradycardia, hypotension, hypothermia — managed with ventilation and titrated naloxone: CDST 0.4 mg, or 0.04 mg in known opioid abusers, escalating to a total of 10 mg; child 0.1 mg/kg); **cholinergic** (depressed consciousness, seizures, sweating, pinpoint pupils, watery eyes, bradycardia, drooling, incontinence, vomiting and diarrhoea, bronchorrhoea — managed with PPE, decontamination, ^atropine 2 mg IV^ doubled every three minutes until the chest is clear, then 20% of the total per hour, and oximes in hospital); **sedative-hypnotic** (depressed consciousness, slurred speech, respiratory depression — managed supportively, with airway and ventilatory support; flumazenil is not used prehospitally and is contraindicated in mixed overdoses); and **serotonin syndrome** (agitation, autonomic instability, clonus and hyperreflexia — benzodiazepines, cooling, cyproheptadine in hospital) (HPCSA PBEC, 2025, pp. 55–57; Hüser et al., 2025). Clinically, they are distinguished by **mental status, pupil size, heart rate, temperature, skin moisture, bowel sounds, secretions and neuromuscular signs**: pinpoint pupils suggest opioids or cholinergic agents (secretions and sweating point to cholinergic); dilated pupils suggest sympathomimetic (sweaty skin) or anticholinergic (dry skin) toxicity; clonus suggests serotonin toxicity. *Correction to the memo*: the memo lists "mydriasis suggests stimulants or antidepressant overdose"; this is correct for TCAs (anticholinergic effects), but the key differentiator between sympathomimetic and anticholinergic toxicity is sweating versus dry skin. Toxidromes have limitations in mixed ingestions, incomplete presentations and poisons with delayed toxicity (paracetamol, iron, toxic alcohols) (Hüser et al., 2025).

Question 2

"What are the key components and limitations of decontamination strategies in acute poisoning, and how should clinicians decide between skin/eye decontamination, gastric decontamination, and post-absorptive elimination methods? Include evidence-based time windows and contraindications."

Skin and eye decontamination is simple, effective and always indicated for external exposures: protect yourself with PPE, remove clothing, wash skin with soap and water, and irrigate eyes with running water or saline for at least 20 minutes (HPCSA PBEC, 2025, p. 54); it should precede other treatment for organophosphates, corrosives and hydrofluoric acid (with calcium gel for HF). **Gastric decontamination**: **single-dose activated charcoal** (CDST 1 g/kg, adult 50–100 g) is most effective within **one hour** of ingestion and is given prehospitally only if the patient is fully conscious, the ingestion was oral, the substance is harmful, and it is adsorbable — not for household products, paraffin, petrol, corrosives, iron, lithium or alcohol (HPCSA PBEC, 2025, p. 54); the window may be longer for massive ingestions, sustained-release formulations or anticholinergic drugs (Hüser et al., 2025); it is contraindicated with an unprotected airway or reduced consciousness (aspiration risk), bowel obstruction or perforation, and corrosive ingestion. **Whole bowel irrigation** with polyethylene glycol (25–40 mL/kg/h until the effluent is clear) is used for sustained-release or enteric-coated drugs, substances not bound by charcoal (iron, lithium), and body packers, but is contraindicated in GI haemorrhage, obstruction or ileus, haemodynamic instability, and an unprotected airway (Kaswa, 2024). **Gastric lavage** is no longer routine and is reserved for exceptional, very early, life-threatening ingestions (for example high-dose colchicine or cardiotoxicants, or very early ingestions of non-adsorbed toxins), because of the risks of aspiration, hypoxia and perforation; it is contraindicated for hydrocarbons, corrosives and an unprotected airway (Hüser et al., 2025; Kaswa, 2024). **Induced emesis** (ipecac) is not recommended (Höjer et al., 2013; HPCSA PBEC, 2025). **Cathartics are not recommended**; *correction to the memo*: sorbitol is a **cathartic** (causing diarrhoea and dehydration, especially in infants), not an emetic. **Post-absorptive (enhanced) elimination** includes **MDAC** for drugs with enterohepatic or entero-enteric recirculation (carbamazepine, phenobarbital, dapsone, quinine, theophylline), **urinary alkalinisation** for weak acids (salicylates, methotrexate — urine pH ≥7.5), and **extracorporeal elimination** (haemodialysis) for small, water-soluble, poorly protein-bound toxins with a small volume of distribution (lithium, salicylates, toxic alcohols, theophylline, metformin, valproate), guided by EXTRIP recommendations (Hüser et al., 2025). **Decision-making** depends on the route, time since exposure, the toxin's properties, the expected severity, and the patient's airway and haemodynamic stability; no decontamination technique has been shown to reduce morbidity or mortality in clinical trials, so all should be used selectively and must never compromise the airway or delay resuscitation (Hüser et al., 2025).

Question 3

"Analyse the role of diagnostic testing in poisoning alongside clinical assessment. Which laboratory tests are most informative, and how should clinicians balance laborator y results with urgent therapeutic needs? Provide examples of how test results could alter management."

Diagnostic testing complements, but never replaces, clinical assessment: "treat the patient first, not the laboratory results" (Kaswa, 2024). The most informative point-of-care tests are **blood glucose** (hypoglycaemia mimics poisoning and occurs with alcohol, sulphonylureas, salicylates, beta-blockers), the **12-lead ECG** (QRS >100 ms suggests sodium channel blockade and seizure risk; QTc ≥500 ms suggests a risk of torsade de pointes; bradycardia and AV block suggest beta-blockers, calcium channel blockers or digoxin) and **blood gas analysis** (acidosis, hypercapnia, methaemoglobin, carboxyhaemoglobin, lactate) (Hüser et al., 2025). Kaswa (2024)

lists the standard tests for significant poisoning as full blood count, prothrombin time, electrolytes, pulse oximetry, urea and creatinine, 12-lead ECG, glucose, arterial blood gas, calcium, toxicology screen, urinalysis and liver function (in paracetamol toxicity). The **anion gap and osmolal gap** help identify toxic alcohols and other acid-producing toxins, and **quantitative drug levels** (paracetamol, salicylate, lithium, iron, digoxin, theophylline, carbamazepine, valproate) guide treatment and, when repeated every four to six hours, show ongoing absorption and guide enhanced elimination (Hüser et al., 2025). **Urine drug screens** have limited value because of false positives and negatives and their inability to detect many drugs (Z-drugs, GHB, fentanyl analogues) (Hüser et al., 2025). In unstable patients or clear toxidromes, treatment proceeds on clinical grounds while awaiting results. **Examples** of how results change management: a paracetamol level above the nomogram line at four hours mandates NAC; a widened QRS prompts sodium bicarbonate; hyperkalaemia in acute digoxin toxicity prompts digoxin Fab; a high lithium level with neurological signs prompts haemodialysis; a raised osmolal and anion gap prompts fomepizole/ethanol and haemodialysis for toxic alcohols; a salicylate level and blood gas guide urinary alkalinisation or haemodialysis; methaemoglobinaemia prompts methylene blue; and a raised CK prompts aggressive fluids for rhabdomyolysis. Normal results do not exclude poisoning, particularly early after ingestion of agents with delayed toxicity.

Question 4

“Critically evaluate the decision-making process surrounding the use of antidotes in acute poisoning. What factors influence the choice and timing of antidote administration, and how should clinicians reconcile antidote availability with patient urgency and potential adverse effects?”

Antidotes exist for only a limited number of poisons, and resuscitation and supportive care remain the mainstay of treatment (Hüser et al., 2025). The decision to give an antidote depends on: (1) the **probability** that a specific toxin is responsible (history, toxidrome, ECG, levels); (2) the **severity** or expected severity of poisoning; (3) the **evidence of benefit** of the antidote; (4) the **time window** (for example NAC is most effective within eight hours of paracetamol ingestion; oximes must be given before organophosphate “ageing”); (5) the **risks** of the antidote (for example naloxone precipitating withdrawal, agitation and vomiting; flumazenil precipitating seizures in mixed overdoses or benzodiazepine dependence; physostigmine causing seizures and bradycardia, and contraindicated with QRS widening; antivenom causing anaphylaxis; sodium bicarbonate causing hypernatraemia and alkalosis); (6) the provider’s **scope** and the setting (prehospital CDST antidotes are naloxone, atropine, sodium bicarbonate, calcium, glucagon, dextrose and thiamine, with adrenaline for cardiovascular collapse — HPCSA PBEC, 2025); and (7) **availability**. Timing is crucial for antidotes with narrow windows or immediate life-saving effects (naloxone for opioid respiratory depression; atropine for cholinergic crisis; sodium bicarbonate for TCA cardiotoxicity; dextrose for hypoglycaemia; hydroxocobalamin for cyanide). Antidotes should be **titrated to clinical endpoints** (for example naloxone to adequate breathing; atropine to a dry chest) rather than given as fixed large doses. *Correction to the memo*: the memo suggests administering antidotes “within the first hour when possible” as a general rule, citing Kaswa (2024); in reality the time windows are antidote-specific (for example NAC is effective up to eight hours and beyond; antivenom may be effective for coagulopathy days after a bite), and the decision should be based on the specific

toxin. Where an antidote is unavailable, aggressive supportive care must continue while advice is sought from the **Poisons Information Helpline (0861 555 777)** and transfer to a facility holding the antidote is arranged; in unknown poisonings, antidotes with a favourable risk–benefit profile (dextrose, naloxone titrated, thiamine) may be given empirically, while high-risk antidotes (flumazenil, physostigmine) should be avoided.

Question 5

"Argue for and against the indications for gastrointestinal decontamination in pharmaceutical poisoning, with reference to activated charcoal and whole bowel irrigation, including timing and formulary considerations."

For activated charcoal: it adsorbs most pharmaceuticals, reducing absorption substantially when given early (most effective within one hour, but potentially useful later for massive ingestions, sustained-release preparations and anticholinergic drugs that delay gastric emptying); it is cheap, widely available and safe in an alert patient; and MDAC can enhance the elimination of drugs with enterohepatic recirculation (carbamazepine, phenobarbital, dapsone, quinine, theophylline) (Hüser et al., 2025). The usual adult dose is 50 g (CDST 1 g/kg, adult 50–100 g), aiming for a 10:1 charcoal-to-drug ratio, with MDAC doses of 12.5 g every one to four hours in adults (Hüser et al., 2025; HPCSA PBEC, 2025). **Against activated charcoal**: no clinical trial has shown a reduction in morbidity or mortality; most patients present after one hour; it carries a risk of aspiration in patients whose consciousness may decline (TCAs, seizure-inducing drugs); it does not bind iron, lithium, potassium, alcohols, hydrocarbons, corrosives or many household products; vomiting is common; and it may delay transport or other treatment. **For whole bowel irrigation**: it is the main option for sustained-release or enteric-coated pharmaceuticals, toxins not bound by charcoal (iron, lithium), and drug packets (Kaswa, 2024; Hüser et al., 2025). **Against whole bowel irrigation**: there is no outcome evidence; it requires large volumes (25–40 mL/kg/h) via a nasogastric tube, a protected airway and a cooperative or intubated patient; and it is contraindicated in obstruction, perforation, ileus, GI haemorrhage and haemodynamic instability. Gastric lavage and cathartics are not routinely recommended (Hüser et al., 2025). **Formulary considerations** include the availability of charcoal on ambulances (the CDST lists it within all scopes), avoidance of sorbitol-containing preparations (especially in children), the availability of polyethylene glycol, and the need to consult the poison centre. **Conclusion**: gastrointestinal decontamination should be selective — reserved for alert or intubated patients with potentially serious ingestions of suitable agents within an appropriate time window — and must never compromise the airway or delay resuscitation.

Question 6

"Analyse the diagnostic approach to a poisoned patient in the ICU, including the reliability of history, collateral information, toxidromes, ECG interpretation, and the role of laboratory and imaging studies in guiding management."

After ABCDE stabilisation, the diagnostic approach comprises history, examination, ECG, laboratory tests and imaging (Hüser et al., 2025). **History** is often unreliable (altered consciousness, intentional concealment, unknown street drug contents), so **collateral information** from EMS, family, friends and bystanders is essential — including containers and the scene findings that the ECP is uniquely placed to provide (Outcome 62.6). **Toxidromes** help

identify the likely class and anticipate the course, but may be obscured in mixed ingestions with antagonistic effects (Hüser et al., 2025). An **ECG** should be obtained in all suspected poisonings: changes in the PR interval (beta-blockers, calcium channel blockers, digoxin), QRS (sodium channel blockers — TCAs, diphenhydramine, local anaesthetics, flecainide), QT (antipsychotics, methadone, antiemetics, diphenhydramine) and ST segment indicate the toxin and risk; QRS >100 ms and QTc ≥500 ms are associated with life-threatening arrhythmias and seizures, and serial ECGs track the course (Hüser et al., 2025). **Laboratory tests** detect differential diagnoses (hypoglycaemia) and complications (acidosis, hypercapnia, methaemoglobinaemia, organ dysfunction, rhabdomyolysis); **quantitative levels** (paracetamol, lithium, salicylate) identify toxins, assess severity and, repeated at four- to six-hour intervals, show ongoing absorption and guide enhanced elimination; urine immunoassays are of limited value; and retaining a blood sample for later analysis can be useful (Hüser et al., 2025). **Imaging** is reserved for selected cases: CT brain for altered consciousness with trauma or focal signs; abdominal X-ray, CT or ultrasound for radio-opaque tablets (iron), pharmacobezoars or drug packets, which may guide whole bowel irrigation, gastroscopy or surgery; and imaging must never delay specific treatment (for example sodium bicarbonate in salicylate poisoning) (Hüser et al., 2025). Clinicians should avoid attributing altered mental status solely to intoxication, because trauma, infection and metabolic disturbances often coexist, and should consult a poison centre early.

Question 7

“Explain the indications for extracorporeal membrane oxygenation (ECMO) in toxin-induced cardiogenic shock, including patient selection criteria, timing, and multidisciplinary coordination required for a successful ECMO pathway.”

Refractory cardiovascular failure is a leading cause of death in pharmaceutical poisoning, and severe cardiotoxicity develops rapidly with hypotension, high-grade AV block, asystole or pulseless ventricular arrhythmias after overdoses of calcium channel blockers, beta-blockers, membrane-stabilising agents and other cardiotoxicants (for example bupropion) (Hüser et al., 2025). **Veno-arterial ECMO (VA-ECMO)** takes over cardiac function during refractory cardiogenic shock or cardiac arrest until the toxin is cleared and myocardial function recovers, minimising myocardial work, maintaining organ perfusion, and preserving hepatic metabolism and renal and biliary elimination of the toxin; support is usually needed only briefly, until plasma concentrations fall (Hüser et al., 2025). **Indications**: persistent cardiogenic shock or cardiac arrest due to a cardiotoxicant **despite** optimal conventional treatment — fluids, vasopressors and inotropes, and antidotes (calcium, HIET, glucagon, sodium bicarbonate, lipid emulsion as appropriate). Pacing is useful only when inotropic function is preserved, and an intra-aortic balloon pump is of little use because it requires an intrinsic rhythm. **Selection criteria** include a reversible toxic cause, no contraindication to anticoagulation, the absence of irreversible organ damage or terminal comorbidity, and access to an established ECMO programme. **Timing**: VA-ECMO should be initiated **early** in refractory shock, before multi-organ failure or prolonged arrest develops, because predictors of refractoriness differ for each class of cardiotoxicant (Hüser et al., 2025). **Veno-venous ECMO** supports respiratory failure (for example aspiration or inhalational injury), with survival rates approaching 90% reported (Hüser et al., 2025). **Coordination** requires an “ECMO rescue chain” with a rapid activation pathway, involving intensivists, emergency physicians, cardiothoracic surgeons or cannulation

teams, perfusionists, cardiologists, toxicologists or poison centres, nursing staff, and transport teams; ILE should be used cautiously when ECMO is anticipated because lipid can occlude the oxygenator (Hüser et al., 2025). The evidence base consists largely of retrospective case series, but supports reasonable use in selected patients. In South Africa, ECMO is available only at a small number of tertiary centres, so early recognition and early referral are essential.

Question 8

"Evaluate the use of antidotes in pharmaceutical poisoning, focusing on sodium bicarbonate for sodium channel blocker toxicity and digoxin-specific antibodies for digoxin-like poisoning, including dosing considerations and monitoring."

Sodium bicarbonate (8.4%) is the antidote of choice for sodium channel blocker poisoning, particularly TCA overdose, and is indicated for ventricular arrhythmias, QRS prolongation >100 ms, hypotension or seizures; it works by increasing extracellular sodium (overcoming channel blockade) and by alkalinisation (target blood pH about 7.5), which reduces TCA binding to sodium channels (Hüser et al., 2025). The hospital dose is 1–2 mL/kg of 8.4% as an IV bolus, repeated as needed or followed by an infusion (20–30 mEq/h), aiming for a pH of about 7.5 rather than complete normalisation of the QRS; blood gases should be checked within an hour and every two to four hours, with monitoring for hypernatraemia, hypokalaemia and metabolic alkalosis (Hüser et al., 2025). In South African prehospital practice, the CDST dose is ^sodium bicarbonate 8.4% 100 mL IV over 3 minutes^ in adults (2 mL/kg in children), reassessed after 10 minutes and repeated once if toxicity persists, with hyperventilation (20–24 breaths/min, $ETCO_2$ 20–25 mmHg) if intubated (HPCSA PBEC, 2025, p. 56). In salicylate and methotrexate poisoning, more dilute bicarbonate (1.4%) is used for urinary alkalinisation (urine pH ≥7.5, avoiding blood pH >7.5). **Digoxin-specific antibody (Fab) fragments** bind free digoxin (and other cardiac glycosides, including plant glycosides from oleander and *Drimia*), forming complexes excreted by the kidneys; they are indicated for life-threatening arrhythmias (for example ventricular arrhythmias, high-grade AV block, paroxysmal atrial tachycardia with block) and hyperkalaemia in acute poisoning (Hüser et al., 2025). When the amount ingested or the serum level is unknown, two vials initially are suggested for acute poisoning or one vial for chronic poisoning, repeated as needed according to clinical response; doses can also be calculated from the amount ingested or the serum concentration (Hüser et al., 2025). Monitoring includes continuous ECG, potassium (which can fall rapidly after Fab), renal function, and clinical response; serum digoxin levels after Fab are unreliable because most assays measure both bound and free digoxin (Hüser et al., 2025). Digoxin Fab is not available prehospitally, so the ECP manages arrhythmias, avoids hypoxia and transports urgently.

Question 9

"Propose a structured treatment algorithm for a patient presenting with suspected pharmaceutical poisoning, integrating initial resuscitation, diagnostic steps, GI decontamination, toxicant-specific therapies, and escalation to ECMO or other rescue strategies."

Step 1 — Scene safety and initial resuscitation (ABCDE): airway (selective intubation — a conservative strategy in comatose patients without respiratory failure, shock or vomiting reduces pneumonia; Hüser et al., 2025), breathing (oxygen if SpO_2 <92%; ventilation),

circulation (IV access, crystalloid 250–500 mL boluses for SBP <90; vasopressors), disability (GCS, pupils, **glucose**; benzodiazepines for seizures), exposure (temperature — cool if ≥38 °C; injuries) (HPCSA PBEC, 2025, pp. 54–57). **Step 2 — Early empirical antidotes**: naloxone (titrated) for opioid respiratory depression; dextrose for hypoglycaemia; thiamine in at-risk patients; sodium bicarbonate for widened QRS; atropine for cholinergic crisis; calcium, glucagon and adrenaline infusion for beta-blocker/calcium channel blocker shock (CDST). **Step 3 — Diagnosis**: history and collateral information, containers, toxidrome identification, 12-lead ECG, blood gas, electrolytes, renal and liver function, paracetamol and salicylate levels in all deliberate overdoses, specific drug levels, and imaging where indicated (Hüser et al., 2025). **Step 4 — Consult the Poisons Information Helpline (0861 555 777)**. **Step 5 — GI decontamination** (if indicated and safe): single-dose activated charcoal (50 g adult; CDST 1 g/kg) within about one hour (longer for sustained-release or massive ingestions) in an alert or intubated patient with an adsorbable toxin; MDAC for enterohepatic drugs; whole bowel irrigation for sustained-release preparations, iron, lithium or packets (Hüser et al., 2025). **Step 6 — Toxicant-specific therapy**: NAC (paracetamol), sodium bicarbonate (sodium channel blockers, salicylates), HIET and glucagon (calcium channel blockers and beta-blockers), digoxin Fab (glycosides), deferoxamine (iron), methylene blue (methaemoglobinaemia), physostigmine (central anticholinergic syndrome without QRS widening), fomepizole/ethanol (toxic alcohols), octreotide (sulphonylureas), and hydroxocobalamin (cyanide) (Hüser et al., 2025). **Step 7 — Enhanced elimination**: urinary alkalinisation (salicylates), extracorporeal treatment (haemodialysis) for dialysable toxins (lithium, salicylates, toxic alcohols, theophylline, metformin, valproate) or refractory acidosis and organ failure, guided by EXTRIP; haemodynamic instability caused by a dialysable toxin is not a contraindication to haemodialysis (Hüser et al., 2025). **Step 8 — Rescue strategies**: intravenous lipid emulsion for local anaesthetic systemic toxicity (first-line) and as a last resort for other lipophilic cardiotoxicants; **VA-ECMO** for refractory cardiogenic shock or arrest, initiated early through an established ECMO pathway (Hüser et al., 2025). **Step 9 — Continuous monitoring, reassessment, documentation and psychosocial assessment** before discharge in deliberate self-poisoning (Kaswa, 2024).

Question 10

“Describe high-dose insulin euglycaemia therapy (HIET) for calcium channel blocker and beta-blocker poisonings, including dosing, monitoring, and potential limitations.”

HIET has become a cornerstone in the management of severe calcium channel blocker and beta-blocker poisoning with cardiogenic shock, because it directly increases myocardial contractility — in shock, the stressed myocardium switches from fatty acid to carbohydrate metabolism, and insulin improves myocardial glucose uptake and use; experimental and clinical data suggest it is haemodynamically superior to catecholamines, particularly in verapamil and diltiazem overdose (Hüser et al., 2025). **Dosing**: a bolus of short-acting insulin **1 U/kg IV**, followed by an infusion starting at **1 U/kg/h**, rapidly titrated (every 15–30 minutes) up to **10 U/kg/h** according to response, together with **dextrose** — starting with about 0.5 g/kg and adjusted to maintain blood glucose **above 5.5 mmol/L (100 mg/dL)**; dextrose may be needed for several hours after insulin is stopped (Hüser et al., 2025). **Monitoring**: frequent blood glucose (initially every 15–30 minutes), **potassium** (insulin drives potassium into cells, so hypokalaemia must be monitored and corrected), phosphate and magnesium, blood

pressure, cardiac output or echocardiographic function, and urine output, in an ICU. **Onset**: improvement may be delayed for 15–30 minutes, and the full effect may take several hours, so HIET should be started **early** when CCB or BB toxicity is suspected and combined with vasopressors (Hüser et al., 2025). **Limitations**: risks of hypoglycaemia, hypokalaemia and hypophosphataemia; the need for large volumes of concentrated dextrose and central access; intensive nursing and laboratory monitoring; delayed onset; variable response; and limited availability outside ICU. HIET is not part of the CDST and is a hospital intervention; prehospitally, the ECP gives fluids, calcium, glucagon (in children with likely beta-blocker toxicity) and an adrenaline infusion within scope (HPCSA PBEC, 2025, p. 57).

REFERENCES

Harvard style. Documents that you supplied are marked with an asterisk (*). Sources cited only through another author (shown in the text as "cited by" or "citing") are listed under the source that was read.

Documents supplied for this module

*Desai, K., Kulkarni, A. and Verma, N. (n.d.) 'Pharmacologic and nonpharmacologic management of acute agitation in older adults across emergency and inpatient clinical settings', *Journal of Contemporary Studies in Science, Technology, and Applied Research* (Albion Journal), pp. 1–15.

*EMC404S (2026) *Behavioural Emergencies — Introduction — Tutorial 1 memorandum*. Emergency Medical Care IV (EMC404S).

*EMC404S (2026) *Behavioural Emergencies — Vulnerable Patient Populations — Tutorial 2 memorandum*. Emergency Medical Care IV (EMC404S).

*EMC404S (2026) *Toxicology — Patient Approach and Management — Tutorial 1 memorandum*. Emergency Medical Care IV (EMC404S).

*Evans, K., Geduld, H. and Stassen, W. (2018) 'Attitudes of prehospital providers on transport decision-making in the management of patients with a suicide attempt refusing care: a survey based on the Mental Health Care Act of 2002', *South African Journal of Psychiatry*, 24, a1156. doi:10.4102/sajpsychiatry.v24i0.1156.

*Health Professions Council of South Africa (HPCSA) (2023) *Guidelines for the withholding and withdrawing of treatment. Booklet 7*. Revised March 2023. Pretoria: HPCSA.

*Health Professions Council of South Africa, Professional Board for Emergency Care (HPCSA PBEC) (2025) *Clinical Decision Support Tools for Emergency Care Providers, South Africa 2024. Version 1.3, October 2025*. Pretoria: HPCSA (developed by the Knowledge Translation Unit, University of Cape Town).

*Hüser, C., Bethlehem, C., Dünser, M.W., Genbrugge, C., Ghannoum, M., Hoegberg, L.C.G., Van der Linden, L., Roedl, K., Sabbe, M., Sikma, M.A., Vodovar, D. and Mégarbane, B. on behalf of the ESICM Trauma, Emergency Medicine Section (2025) 'Critical care management of the patient with pharmaceutical poisoning', *Intensive Care Medicine*, 51, pp. 2378–2391. doi:10.1007/s00134-025-08176-6.

*Kaswa, R. (2024) 'An approach to the management of acute poisoning in emergency settings', *South African Family Practice*, 66(1), a5841. doi:10.4102/safp.v66i1.5841.

*Mesthrie, S. (2025) 'Interpreting the child's right to protection from violence in the South African context in line with international law obligations', *Journal of Human Rights Practice*, 17, pp. 43–57. doi:10.1093/jhuman/huae032.

*Royal College of Emergency Medicine (RCEM) (2025) *Acute behavioural disturbance in emergency departments. Best Practice Guideline, Version 3 (May 2025)*. London: RCEM.

*Saucedo Uribe, E., Bravo Rodríguez, C.A., Navarrete Juárez, M.E., Medrano Juárez, S., Huereca Lucio, R., Rojas Guzmán, K.E. and Lozano Carrillo, L.C. (2025) '"Pharmacological management of acute agitation in psychiatric patients": an umbrella review', *BMC Psychiatry*, 25, 273. doi:10.1186/s12888-024-06426-3.

*Tilley, D., Christopher, L.D., Farrar, T. and Naidoo, N. (2026) 'Typologies of suicidality and suicide presenting to a prehospital South African Emergency Medical Service: a retrospective cross-sectional analysis [version 3; peer review: 3 approved]', *F1000Research*, 14, 1201. doi:10.12688/f1000research.171712.3.

South African legislation and case law

Cannabis for Private Purposes Act 7 of 2024.

Children's Act 38 of 2005.

Constitution of the Republic of South Africa, 1996.

Criminal Law (Sexual Offences and Related Matters) Amendment Act 32 of 2007 (as amended, including by Act 13 of 2021).

Criminal Procedure Act 51 of 1977.

Domestic Violence Act 116 of 1998 (as amended by the Domestic Violence Amendment Act 14 of 2021, in operation from 14 April 2023).

Health Professions Act 56 of 1974.

Inquests Act 58 of 1959.

Mental Health Care Act 17 of 2002 (as amended by the Mental Health Care Amendment Act 12 of 2014), and General Regulations (GN R1467 of 2004).

National Health Act 61 of 2003.

Older Persons Act 13 of 2006.

Prevention of and Treatment for Substance Abuse Act 70 of 2008.

Protection of Personal Information Act 4 of 2013.

Traditional Health Practitioners Act 22 of 2007.

Castell v De Greef 1994 (4) SA 408 (C).

Minister of Justice and Constitutional Development v Prince 2018 (6) SA 393 (CC).

Teddy Bear Clinic for Abused Children v Minister of Justice and Constitutional Development 2014 (2) SA 168 (CC).

Books, guidelines and reports

American Psychiatric Association (APA) (2022) *Diagnostic and statistical manual of mental disorders, fifth edition, text revision (DSM-5-TR)*. Washington, DC: American Psychiatric Association Publishing.

Boland, R., Verduin, M.L. and Ruiz, P. (eds) (2021) *Kaplan & Sadock's synopsis of psychiatry*. 12th edn. Philadelphia: Wolters Kluwer.

National Association of Emergency Medical Technicians (NAEMT) (2021) *Advanced medical life support: an assessment-based approach*. 3rd edn. Burlington, MA: Jones & Bartlett Learning.

National Department of Health (2003) *National management guidelines for sexual assault care*. Pretoria: National Department of Health.

National Department of Health (NDoH), Statistics South Africa (Stats SA), South African Medical Research Council (SAMRC) and ICF (2019) *South Africa Demographic and Health Survey 2016*. Pretoria and Rockville, MD: NDoH, Stats SA, SAMRC and ICF.

Tintinalli, J.E., Ma, O.J., Yealy, D.M., Meckler, G.D., Stapczynski, J.S., Cline, D.M. and Thomas, S.H. (eds) (2020) *Tintinalli's emergency medicine: a comprehensive study guide*. 9th edn. New York: McGraw-Hill Education.

Van Wyk, B.-E., Van Heerden, F. and Van Oudtshoorn, B. (2002) *Poisonous plants of South Africa*. Pretoria: Briza Publications.

World Health Organization (WHO) (2013) *Responding to intimate partner violence and sexual violence against women: WHO clinical and policy guidelines*. Geneva: WHO.

World Health Organization (WHO) (2014) *Preventing suicide: a global imperative*. Geneva: WHO.

World Health Organization (WHO) (2016) *Guidelines for the management of snakebites*. 2nd edn. New Delhi: WHO Regional Office for South-East Asia.

World Health Organization (WHO) (2019) *Snakebite envenoming: a strategy for prevention and control*. Geneva: WHO.

World Health Organization (WHO) (2021) *Violence against women prevalence estimates, 2018*. Geneva: WHO.

Journal articles

Abrahams, N., Mathews, S., Martin, L.J., Lombard, C. and Jewkes, R. (2013) 'Intimate partner femicide in South Africa in 1999 and 2009', *PLoS Medicine*, 10(4), e1001412.

American Geriatrics Society Beers Criteria Update Expert Panel (2023) 'American Geriatrics Society 2023 updated AGS Beers Criteria® for potentially inappropriate medication use in older adults', *Journal of the American Geriatrics Society*, 71(7), pp. 2052–2081.

Antinori, A., Arendt, G., Becker, J.T., et al. (2007) ‘Updated research nosology for HIV-associated neurocognitive disorders’, *Neurology*, 69(18), pp. 1789–1799.

Balme, K.H., Roberts, J.C., Glasstone, M., Curling, L. and Mann, M.D. (2012) ‘The changing trends of childhood poisoning at a tertiary children’s hospital in South Africa’, *South African Medical Journal*, 102(3), pp. 142–146.

Bloem, B.R., Okun, M.S. and Klein, C. (2021) ‘Parkinson’s disease’, *The Lancet*, 397(10291), pp. 2284–2303.

Borron, S.W., Baud, F.J., Barriot, P., Imbert, M. and Bismuth, C. (2007) ‘Prospective study of hydroxocobalamin for acute cyanide poisoning in smoke inhalation’, *Annals of Emergency Medicine*, 49(6), pp. 794–801.

Boyer, E.W. and Shannon, M. (2005) ‘The serotonin syndrome’, *New England Journal of Medicine*, 352(11), pp. 1112–1120.

Bush, G., Fink, M., Petrides, G., Dowling, F. and Francis, A. (1996) ‘Catatonia. I. Rating scale and standardized examination’, *Acta Psychiatrica Scandinavica*, 93(2), pp. 129–136.

Calver, L., Stokes, B. and Isbister, G.K. (2011) ‘Sedation assessment tool to score acute behavioural disturbance in the emergency department’, *Emergency Medicine Australasia*, 23(6), pp. 732–740.

Craske, M.G. and Stein, M.B. (2016) ‘Anxiety’, *The Lancet*, 388(10063), pp. 3048–3059.

Curry, S.J., Krist, A.H., Owens, D.K., et al. (US Preventive Services Task Force) (2018) ‘Screening for intimate partner violence, elder abuse, and abuse of vulnerable adults: US Preventive Services Task Force final recommendation statement’, *JAMA*, 320(16), pp. 1678–1687.

Davies, B., Hlela, M.B.K.M. and Rother, H.-A. (2023) ‘Child and adolescent mortality associated with pesticide toxicity in Cape Town, South Africa, 2010–2019: a retrospective case review’, *BMC Public Health*, 23(1), 792.

Decker, B.S., Goldfarb, D.S., Dargan, P.I., et al. (EXTRIP Workgroup) (2015) ‘Extracorporeal treatment for lithium poisoning: systematic review and recommendations from the EXTRIP workgroup’, *Clinical Journal of the American Society of Nephrology*, 10(5), pp. 875–887.

Dunkley, E.J.C., Isbister, G.K., Sibbritt, D., Dawson, A.H. and Whyte, I.M. (2003) ‘The Hunter Serotonin Toxicity Criteria: simple and accurate diagnostic decision rules for serotonin toxicity’, *QJM*, 96(9), pp. 635–642.

Eddleston, M., Buckley, N.A., Eyer, P. and Dawson, A.H. (2008) ‘Management of acute organophosphorus pesticide poisoning’, *The Lancet*, 371(9612), pp. 597–607.

Finkelstein, Y., Aks, S.E., Hutson, J.R., et al. (2010) ‘Colchicine poisoning: the dark side of an ancient drug’, *Clinical Toxicology*, 48(5), pp. 407–414.

Folstein, M.F., Folstein, S.E. and McHugh, P.R. (1975) ‘“Mini-mental state”: a practical method for grading the cognitive state of patients for the clinician’, *Journal of Psychiatric Research*, 12(3), pp. 189–198.

Gutiérrez, J.M., Calvete, J.J., Habib, A.G., Harrison, R.A., Williams, D.J. and Warrell, D.A. (2017) ‘Snakebite envenoming’, *Nature Reviews Disease Primers*, 3, 17063.

Henretig, F.M., Kirk, M.A. and McKay, C.A. (2019) ‘Hazardous chemical emergencies and poisonings’, *New England Journal of Medicine*, 380(17), pp. 1638–1655.

Höjer, J., Troutman, W.G., Hoppu, K., et al. (2013) ‘Position paper update: ipecac syrup for gastrointestinal decontamination’, *Clinical Toxicology*, 51(3), pp. 134–139.

Horak, J., Venter, W.D.F., Wattrus, C., et al. (2023) ‘Southern African HIV Clinicians Society 2023 guideline for post-exposure prophylaxis: updated recommendations’, *Southern African Journal of HIV Medicine*, 24(1), a1522. doi:10.4102/sajhivmed.v24i1.1522.

Hungerford, J.M. (2010) ‘Scombroid poisoning: a review’, *Toxicon*, 56(2), pp. 231–243.

Inouye, S.K., van Dyck, C.H., Alessi, C.A., Balkin, S., Siegal, A.P. and Horwitz, R.I. (1990) ‘Clarifying confusion: the confusion assessment method. A new method for detection of delirium’, *Annals of Internal Medicine*, 113(12), pp. 941–948.

Inouye, S.K., Westendorp, R.G.J. and Saczynski, J.S. (2014) ‘Delirium in elderly people’, *The Lancet*, 383(9920), pp. 911–922.

Isbister, G.K. and Fan, H.W. (2011) ‘Spider bite’, *The Lancet*, 378(9808), pp. 2039–2047.

Juurlink, D.N., Gosselin, S., Kielstein, J.T., et al. (EXTRIP Workgroup) (2015) ‘Extracorporeal treatment for salicylate poisoning: systematic review and recommendations from the EXTRIP workgroup’, *Annals of Emergency Medicine*, 66(2), pp. 165–181.

Kraut, J.A. and Mullins, M.E. (2018) ‘Toxic alcohols’, *New England Journal of Medicine*, 378(3), pp. 270–280.

Lachs, M.S. and Pillemer, K.A. (2015) ‘Elder abuse’, *New England Journal of Medicine*, 373(20), pp. 1947–1956.

Lavonas, E.J., Akpunonu, P.D., Arens, A.M., et al. (2023) ‘2023 American Heart Association focused update on the management of patients with cardiac arrest or life-threatening toxicity due to poisoning’, *Circulation*, 148(16), pp. e149–e184.

Lord, C., Elsabbagh, M., Baird, G. and Veenstra-Vanderweele, J. (2018) ‘Autism spectrum disorder’, *The Lancet*, 392(10146), pp. 508–520.

Loten, C., Stokes, B., Worsley, D., Seymour, J.E., Jiang, S. and Isbister, G.K. (2006) ‘A randomised controlled trial of hot water (45 °C) immersion versus ice packs for pain relief in bluebottle stings’, *Medical Journal of Australia*, 184(7), pp. 329–333.

Maas, A.I.R., Menon, D.K., Manley, G.T., et al. (2022) ‘Traumatic brain injury: progress and challenges in prevention, clinical care, and research’, *The Lancet Neurology*, 21(11), pp. 1004–1060.

Malhi, G.S. and Mann, J.J. (2018) ‘Depression’, *The Lancet*, 392(10161), pp. 2299–2312.

McCutcheon, R.A., Reis Marques, T. and Howes, O.D. (2020) 'Schizophrenia — an overview', *JAMA Psychiatry*, 77(2), pp. 201–210.

Müller, G.J., Modler, H., Wium, C.A., Veale, D.J.H. and Marks, C.J. (2012) 'Snake bite in southern Africa: diagnosis and management', *Continuing Medical Education*, 30(10), pp. 362–381.

Nasreddine, Z.S., Phillips, N.A., Bédirian, V., et al. (2005) 'The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment', *Journal of the American Geriatrics Society*, 53(4), pp. 695–699.

Patterson, W.M., Dohn, H.H., Bird, J. and Patterson, G.A. (1983) 'Evaluation of suicidal patients: the SAD PERSONS scale', *Psychosomatics*, 24(4), pp. 343–349.

Popat, A., Shear, N.H., Malkiewicz, I., et al. (2001) 'The toxicity of *Callilepis laureola*, a South African traditional herbal medicine', *Clinical Biochemistry*, 34(3), pp. 229–236.

Posner, K., Brown, G.K., Stanley, B., et al. (2011) 'The Columbia–Suicide Severity Rating Scale: initial validity and internal consistency findings from three multisite studies with adolescents and adults', *American Journal of Psychiatry*, 168(12), pp. 1266–1277.

Richmond, J.S., Berlin, J.S., Fishkind, A.B., et al. (2012) 'Verbal de-escalation of the agitated patient: consensus statement of the American Association for Emergency Psychiatry Project BETA De-escalation Workgroup', *Western Journal of Emergency Medicine*, 13(1), pp. 17–25.

Rose, J.J., Wang, L., Xu, Q., et al. (2017) 'Carbon monoxide poisoning: pathogenesis, management, and future directions of therapy', *American Journal of Respiratory and Critical Care Medicine*, 195(5), pp. 596–606.

Rother, H.-A. (2010) 'Falling through the regulatory cracks: street selling of pesticides and poisoning among urban youth in South Africa', *International Journal of Occupational and Environmental Health*, 16(2), pp. 202–213.

Scheltens, P., De Strooper, B., Kivipelto, M., et al. (2021) 'Alzheimer's disease', *The Lancet*, 397(10284), pp. 1577–1590.

Schuckit, M.A. (2014) 'Recognition and management of withdrawal delirium (delirium tremens)', *New England Journal of Medicine*, 371(22), pp. 2109–2113.

Shalev, A., Liberzon, I. and Marmar, C. (2017) 'Post-traumatic stress disorder', *New England Journal of Medicine*, 376(25), pp. 2459–2469.

Sherin, K.M., Sinacore, J.M., Li, X.Q., Zitter, R.E. and Shakil, A. (1998) 'HITS: a short domestic violence screening tool for use in a family practice setting', *Family Medicine*, 30(7), pp. 508–512.

Sohal, H., Eldridge, S. and Feder, G. (2007) 'The sensitivity and specificity of four questions (HARK) to identify intimate partner violence: a diagnostic accuracy study in general practice', *BMC Family Practice*, 8, 49.

Stander, C., Hodkinson, P. and Dippenaar, E. (2021) 'Prehospital care providers' understanding of responsibilities during a behavioural emergency', *South African Journal of Psychiatry*, 27, a1545. doi:10.4102/sajpsychiatry.v27i0.1545.

Steenkamp, P.A., Harding, N.M., van Heerden, F.R. and van Wyk, B.-E. (2004) 'Fatal *Datura* poisoning: identification of atropine and scopolamine by high performance liquid chromatography/photodiode array/mass spectrometry', *Forensic Science International*, 145(1), pp. 31–39.

Stewart, M.J., Moar, J.J., Steenkamp, P. and Kokot, M. (1999) 'Findings in fatal cases of poisoning attributed to traditional remedies in South Africa', *Forensic Science International*, 101(3), pp. 177–183.

Sullivan, J.T., Sykora, K., Schneiderman, J., Naranjo, C.A. and Sellers, E.M. (1989) 'Assessment of alcohol withdrawal: the revised Clinical Institute Withdrawal Assessment for Alcohol scale (CIWA-Ar)', *British Journal of Addiction*, 84(11), pp. 1353–1357.

Treasure, J., Duarte, T.A. and Schmidt, U. (2020) 'Eating disorders', *The Lancet*, 395(10227), pp. 899–911.

Veale, D.J.H., Wium, C.A. and Müller, G.J. (2013) 'Toxicovigilance I: a survey of acute poisonings in South Africa based on Tygerberg Poison Information Centre data', *South African Medical Journal*, 103(5), pp. 293–297.

Vieta, E., Berk, M., Schulze, T.G., et al. (2018) 'Bipolar disorders', *Nature Reviews Disease Primers*, 4, 18008.

Weaver, L.K., Hopkins, R.O., Chan, K.J., et al. (2002) 'Hyperbaric oxygen for acute carbon monoxide poisoning', *New England Journal of Medicine*, 347(14), pp. 1057–1067.